



图像与视频处理

Image and Video Processing

第四讲：图像增强-空域滤波

人工智能与机器人研究所

Institute of Artificial Intelligence and Robotics

兰旭光



目录

- 强度转换
 - 图像反转
 - Log转换
 - Gamma 转换
 - 分段线性变换
- 直方图处理
 - 直方图均衡化
 - 直方图匹配
 - 局部直方图处理
- 空间滤波器
 - 空间相关性和卷积
 - 平滑的线性滤波器和空间掩码
 - 非线性滤波器
 - 锐化滤波器



目录

- 强度转换
 - 图像反转
 - Log转换
 - Gamma 转换
 - 分段线性变换
- 直方图处理
 - 直方图均衡化
 - 直方图匹配
 - 局部直方图处理
- 空间滤波器
 - 空间相关性和卷积
 - 平滑的线性滤波器和空间掩码
 - 非线性滤波器
 - 锐化滤波器



空间域与变换域

- 空间域

图像平面本身，直接处理图像平面的强度值

- 变换域

处理变换系数，而不是直接处理图像平面的强度值

空间域变换计算效率更高



空间域处理

$$g(x, y) = T[f(x, y)]$$

$f(x, y)$: input image 输入图像

$g(x, y)$: output image 输出图像

T : an operator on f defined over
a neighborhood of point (x, y)

对于f的基于点(x,y)邻域的操作子



空间域处理

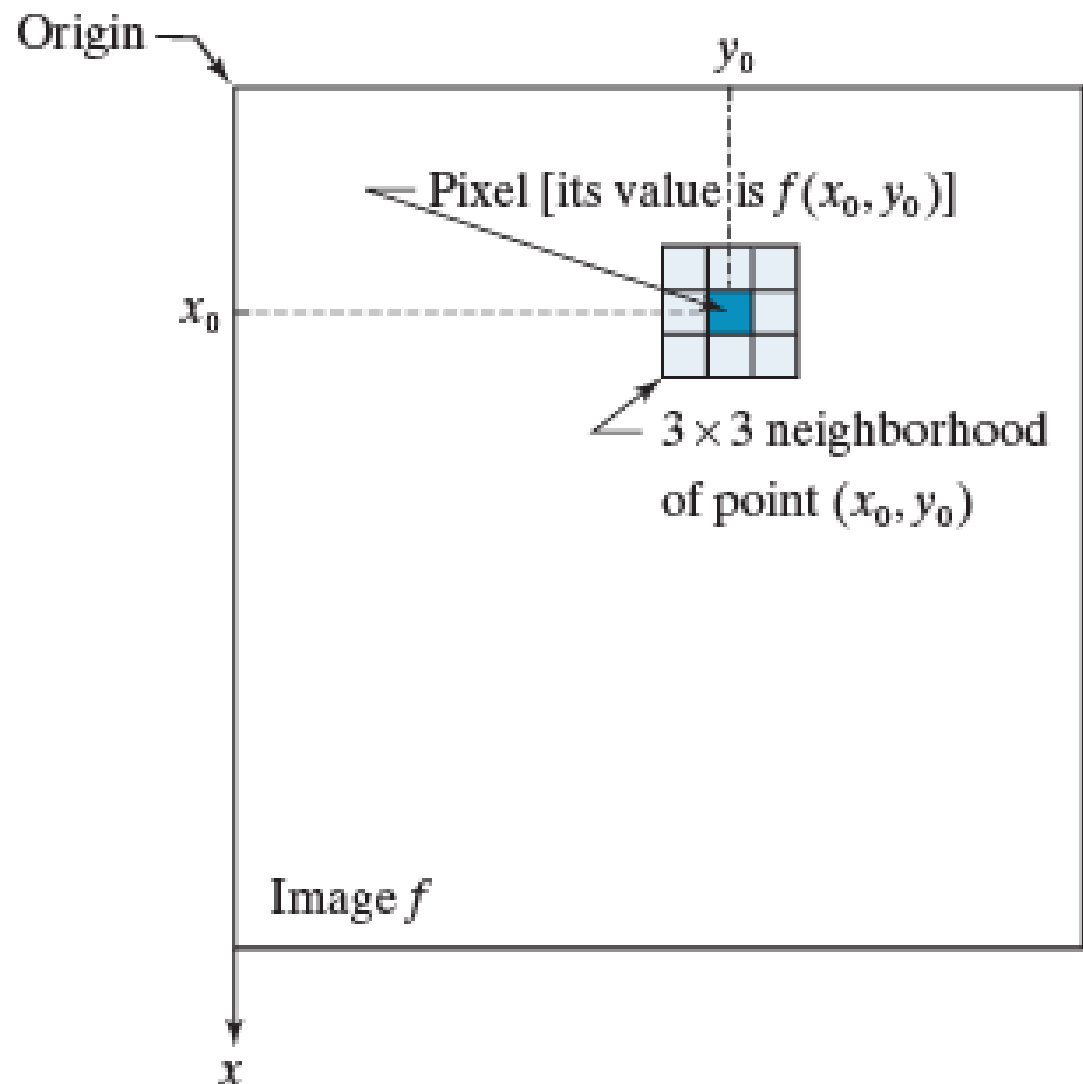


FIGURE 3.1

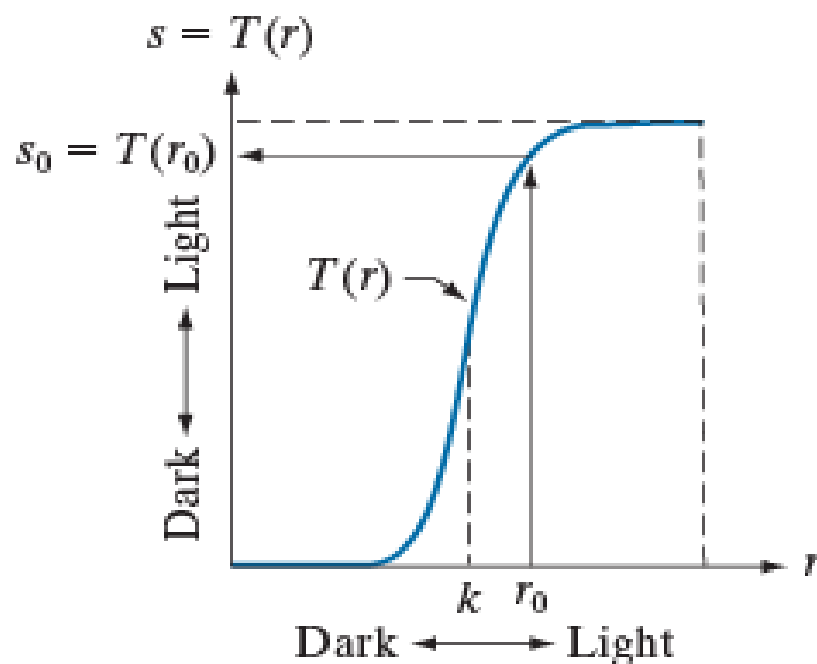
A 3×3 neighborhood about a point (x, y) in an image in the spatial domain. The neighborhood is moved from pixel to pixel in the image to generate an output image.



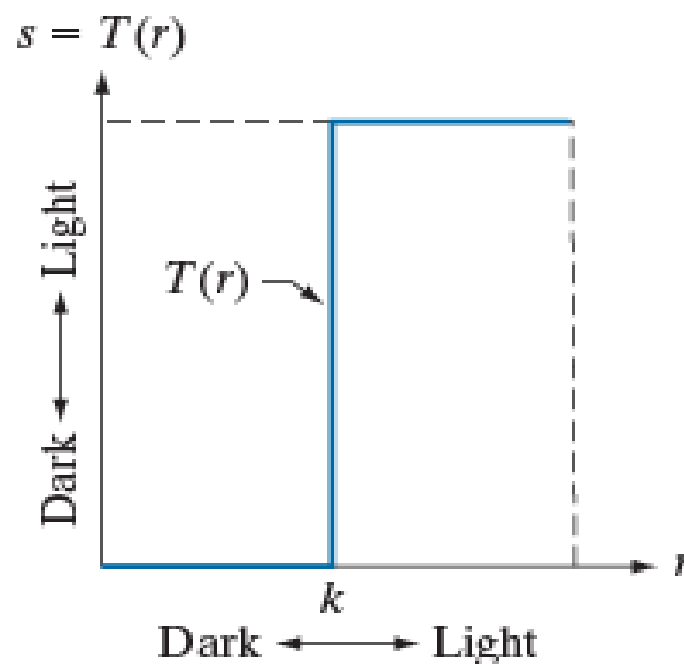
空间域处理

Intensity transformation function

$$s = T(r) \quad \text{强度转换公式}$$



对比度拉伸



阈值化函数.

a b

FIGURE 3.2
Intensity transformation functions.
(a) Contrast-stretching function.
(b) Thresholding function.



一些基本的强度转换函数

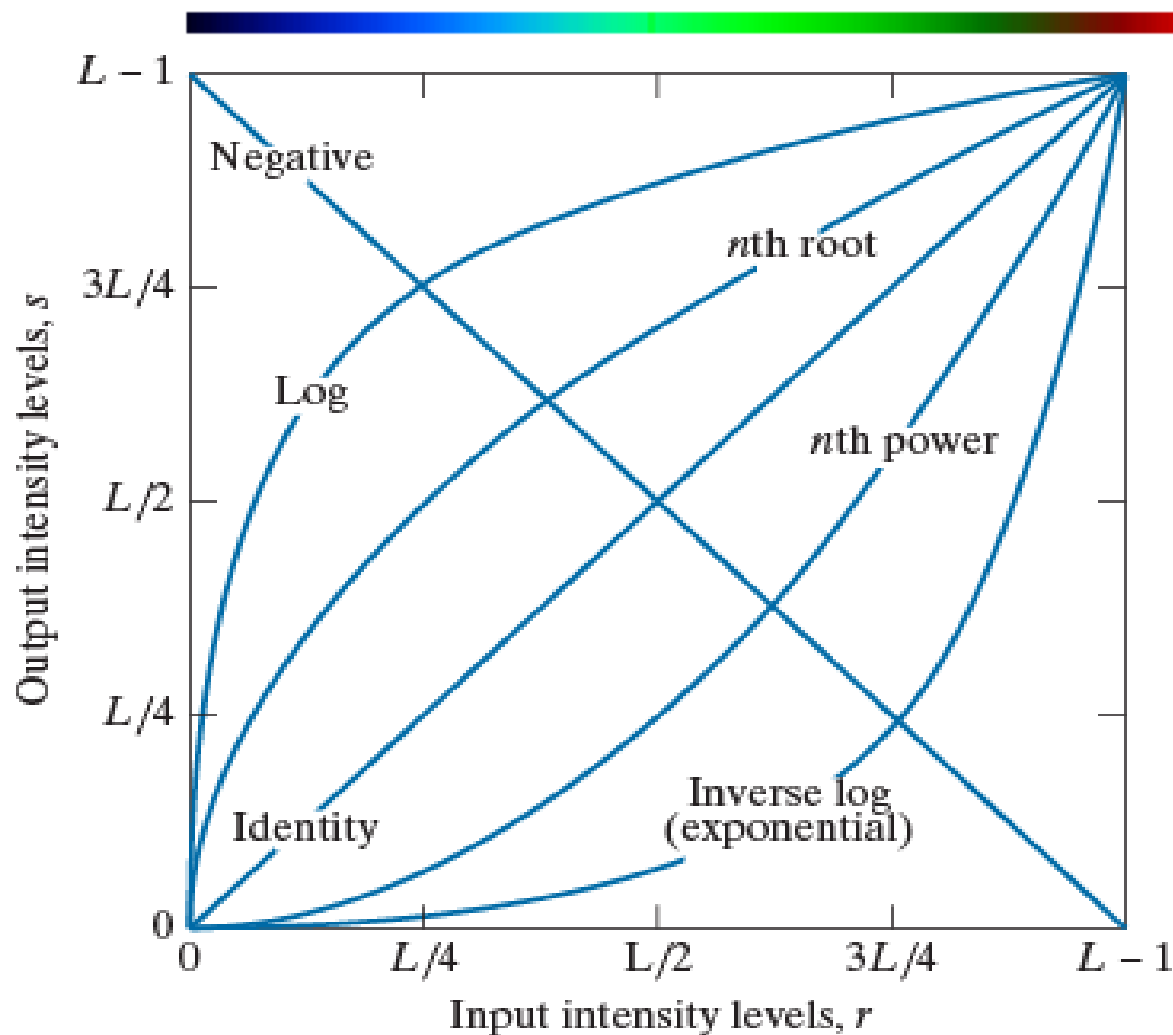


FIGURE 3.3 Some basic intensity transformation functions. All curves were scaled to fit in the range shown.

增强是对图像进行处理的过程，使其结果比原始图像更适合于特定的应用。



图像反转

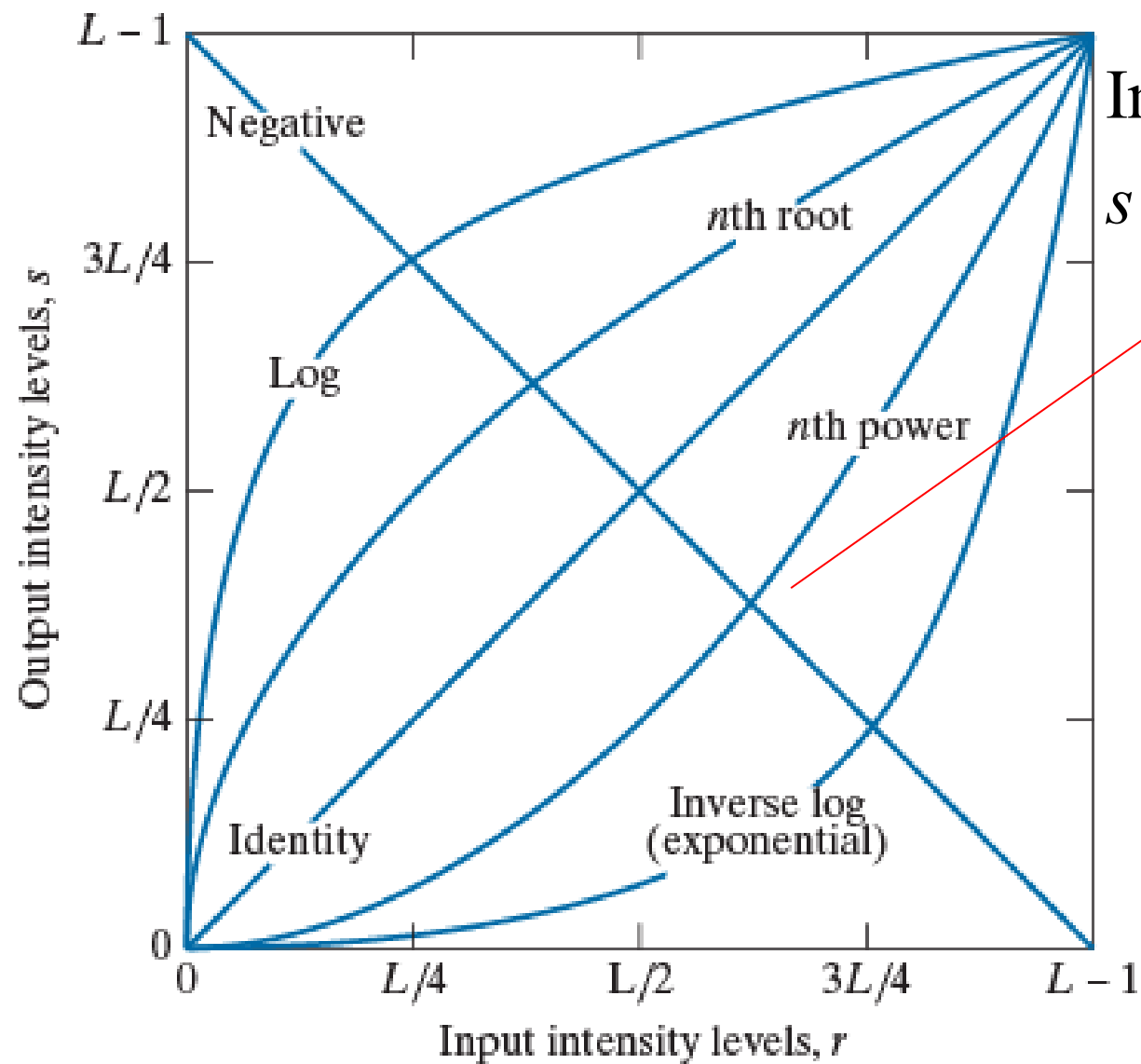


Image negatives

$$s = L - 1 - r$$



实例：图像反转



a b

FIGURE 3.4

(a) Original digital mammogram.
(b) Negative image obtained using the negative transformation in Eq. (3.2-1).
(Courtesy of G.E. Medical Systems.)

小病变

增强嵌入黑暗区域的白色或灰色细节

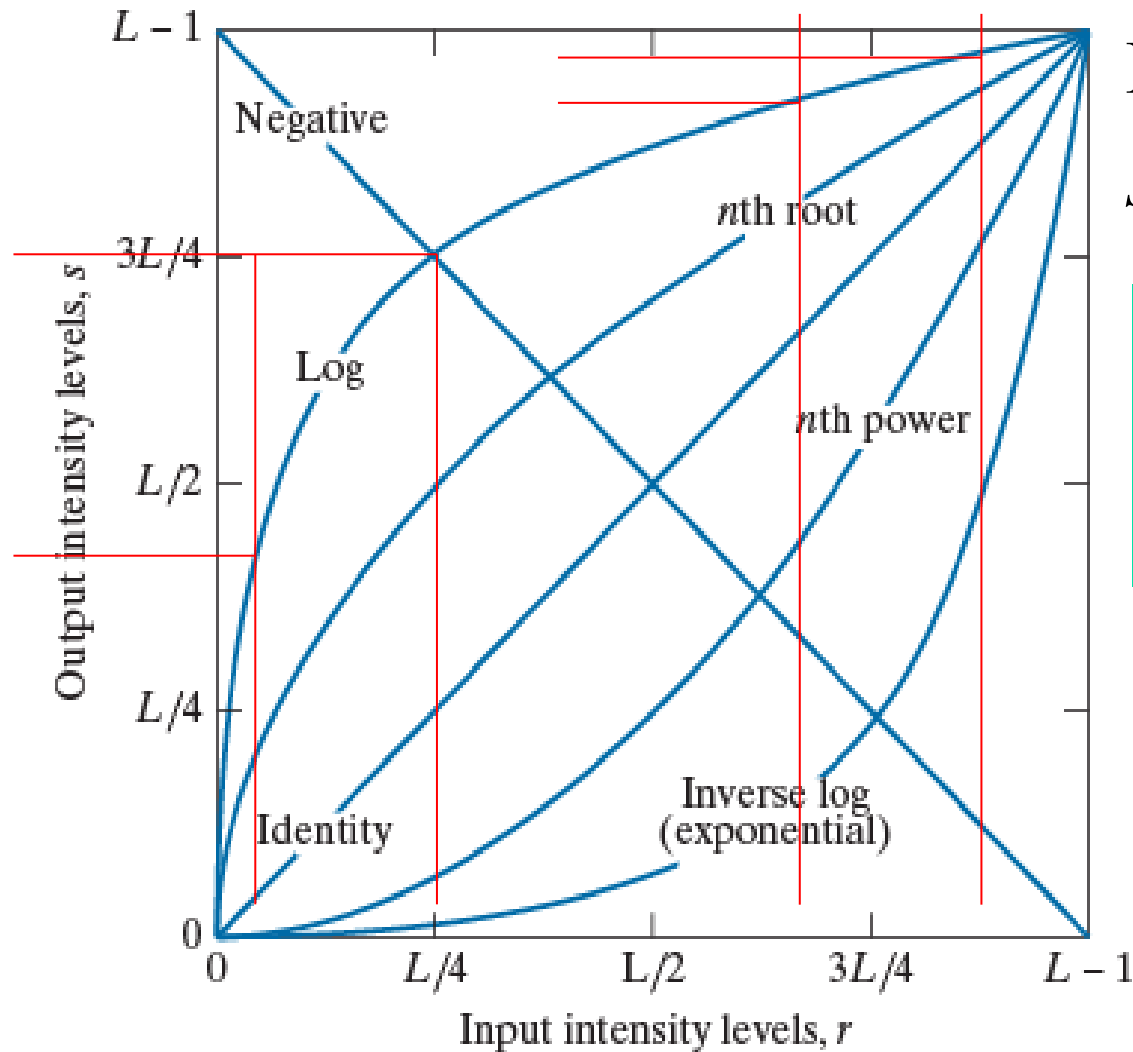


目录

- 强度转换
 - 图像反转
 - Log转换
 - Gamma 转换
 - 碎片化线性变换
- 直方图处理
 - 直方图均衡化
 - 直方图匹配
 - 局部直方图处理
- 空间滤波器
 - 空间相关性和卷积
 - 平滑的线性滤波器和空间掩码
 - 非线性滤波器
 - 锐化滤波器



Log 变换



Log Transformations

$$s = c \log(1 + r)$$

映射一个狭窄的范围
低强度值进入更广泛
的范围输出值

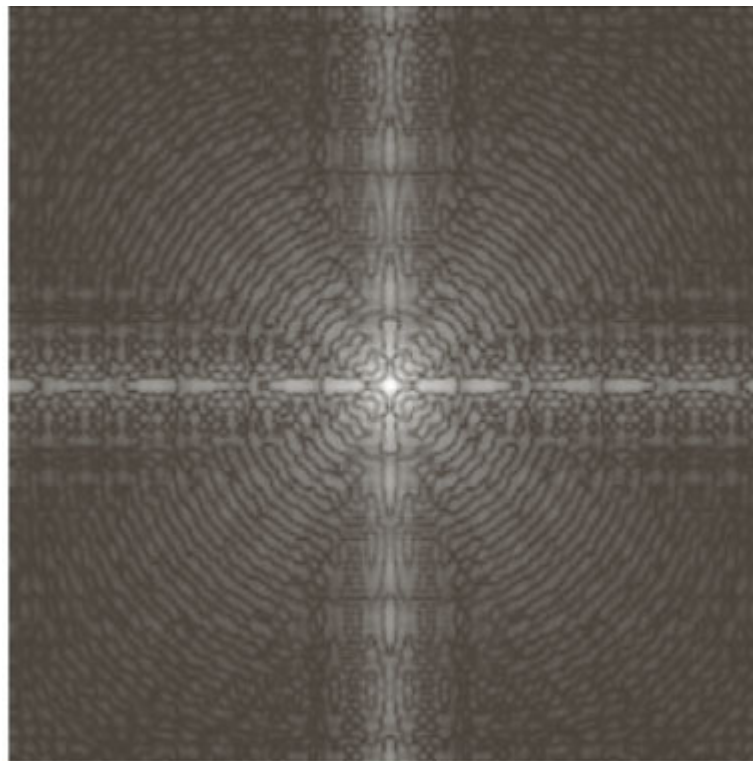
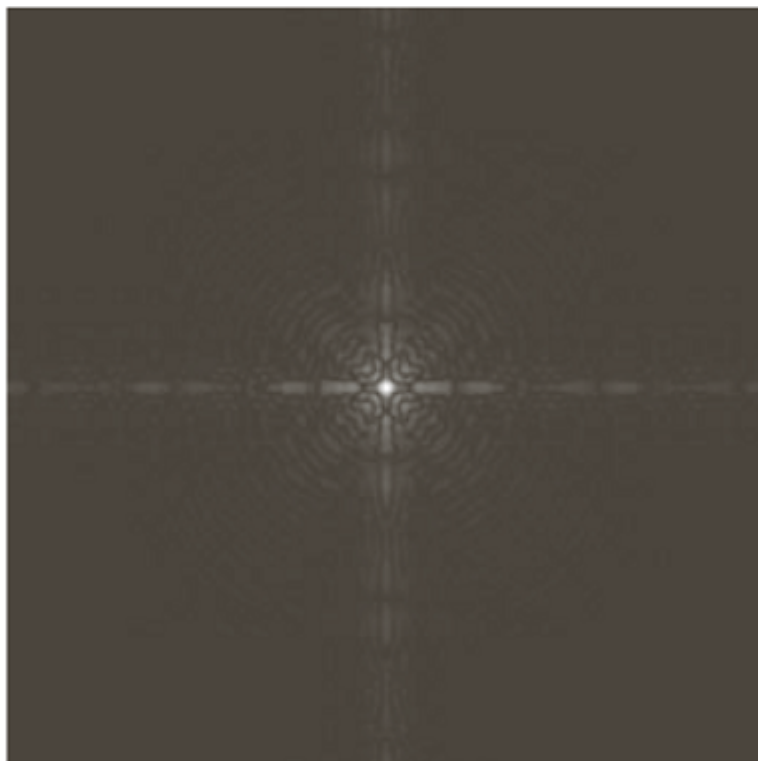
扩大暗像素值
同时压缩更高值



实例：Log 变换

0- 1.5×10^6

0-6.2



a b

FIGURE 3.5

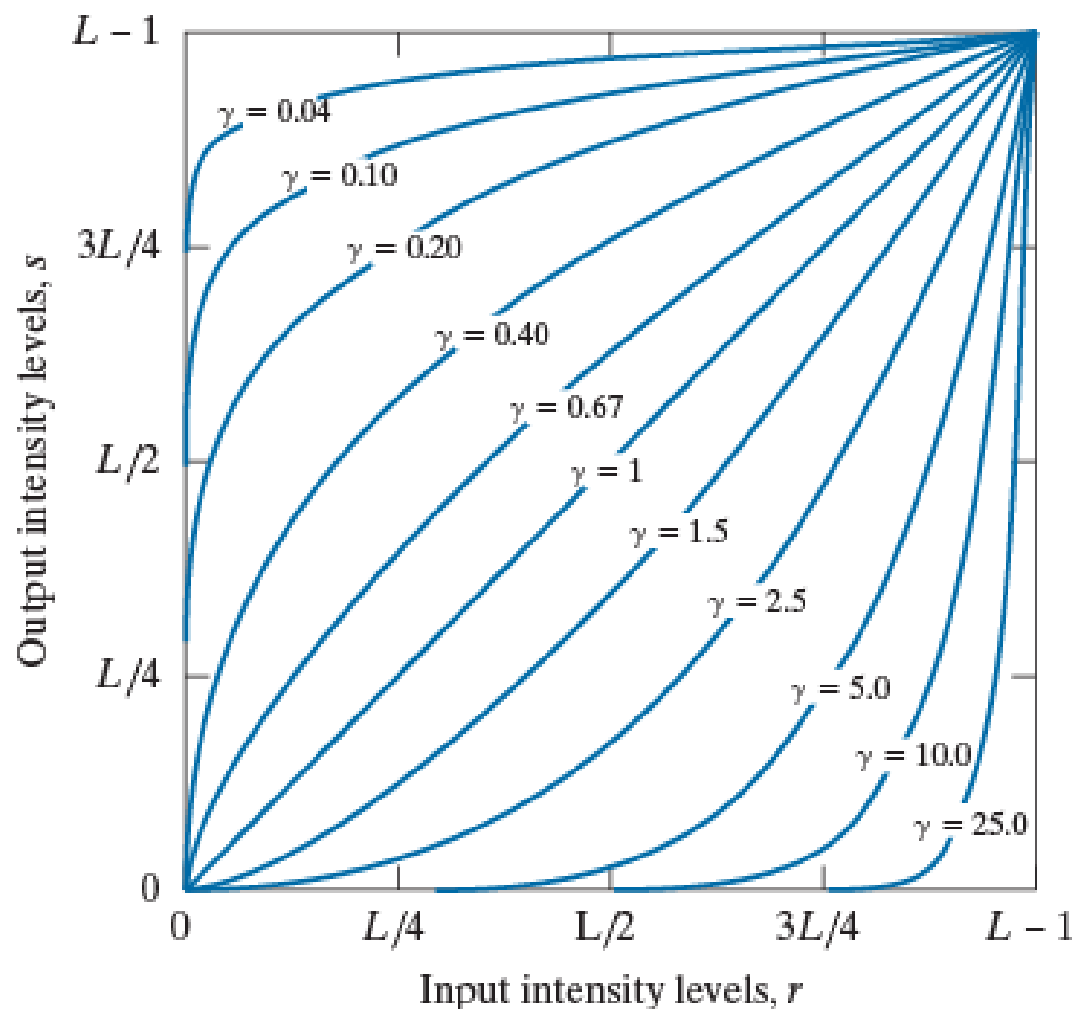
(a) Fourier spectrum.

(b) Result of applying the log transformation in Eq. (3.2-2) with $c = 1$.

变化大的图像的动态范围



幂律 (Gamma) 变换



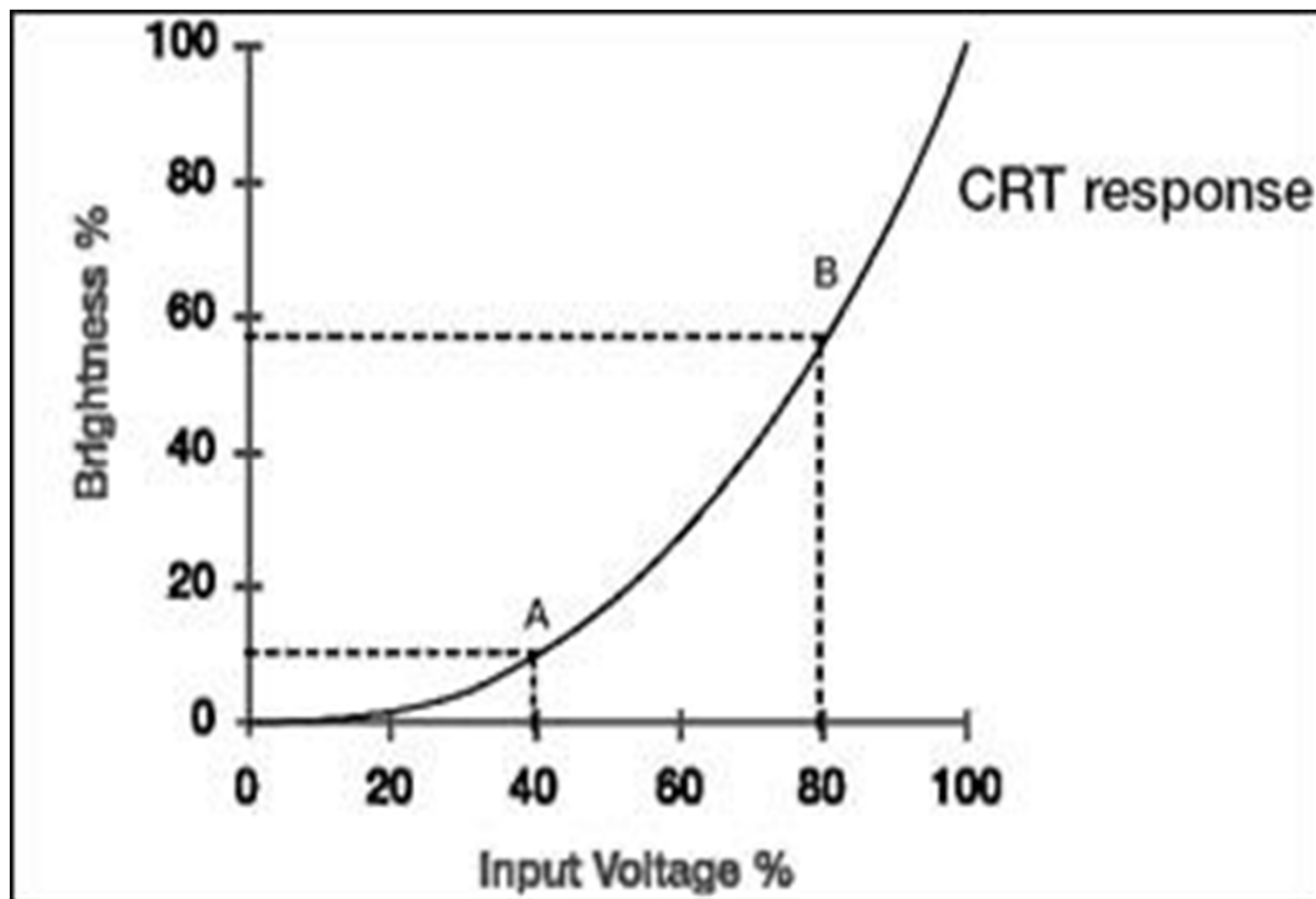
$$s = cr^\gamma$$

FIGURE 3.6 Plots of the equation $s = cr^\gamma$ for various values of γ ($c = 1$ in all cases). All curves were scaled to fit in the range shown.

所有的曲线都按比例调整，以适应所示范围。



图像显示：伽玛校正

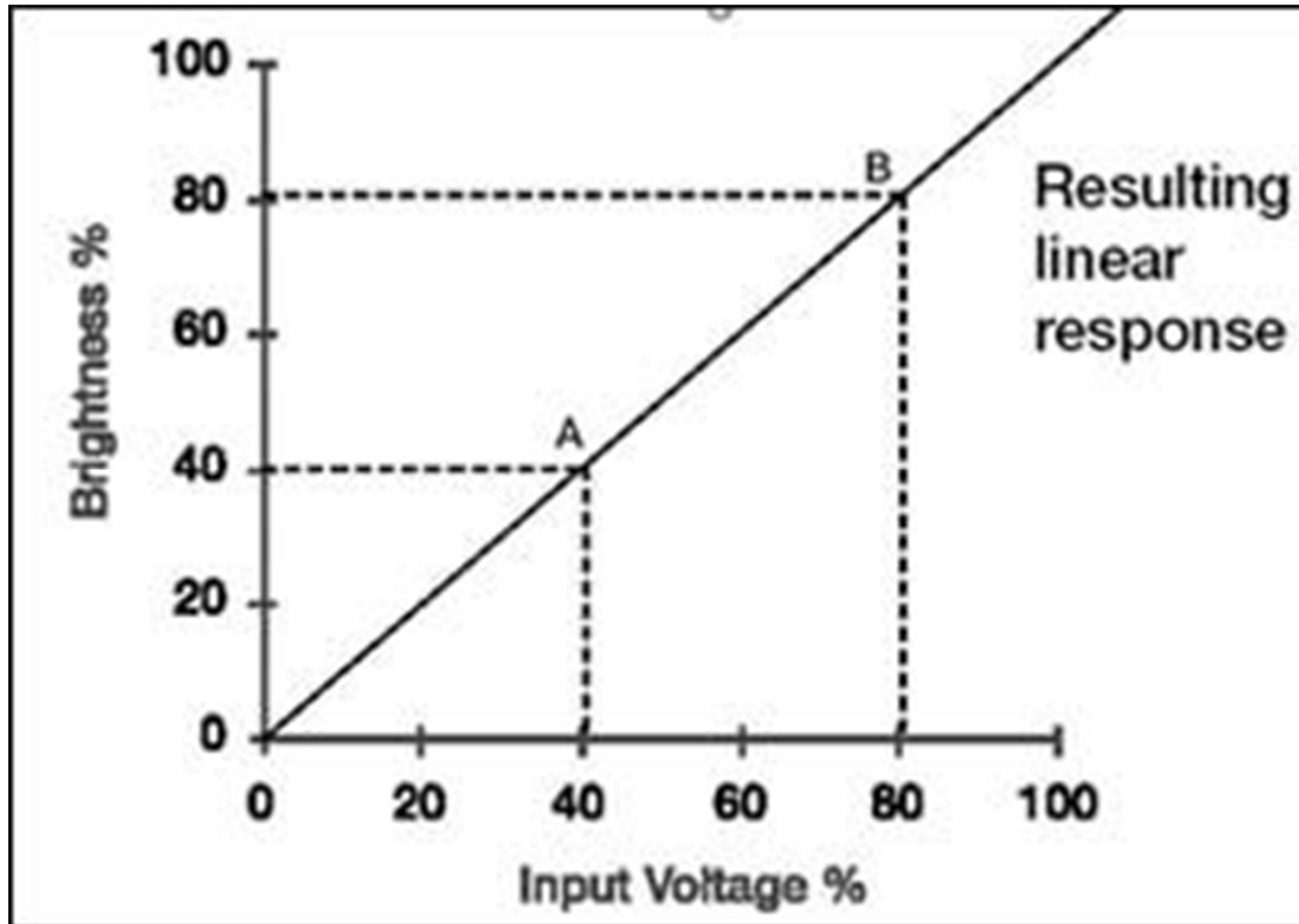


阴极射线管（CRT）设备的强度对电压的反应是一个功率函数，其指数大约从1.8到2.5不等

CRT显示器的亮度响应曲线图

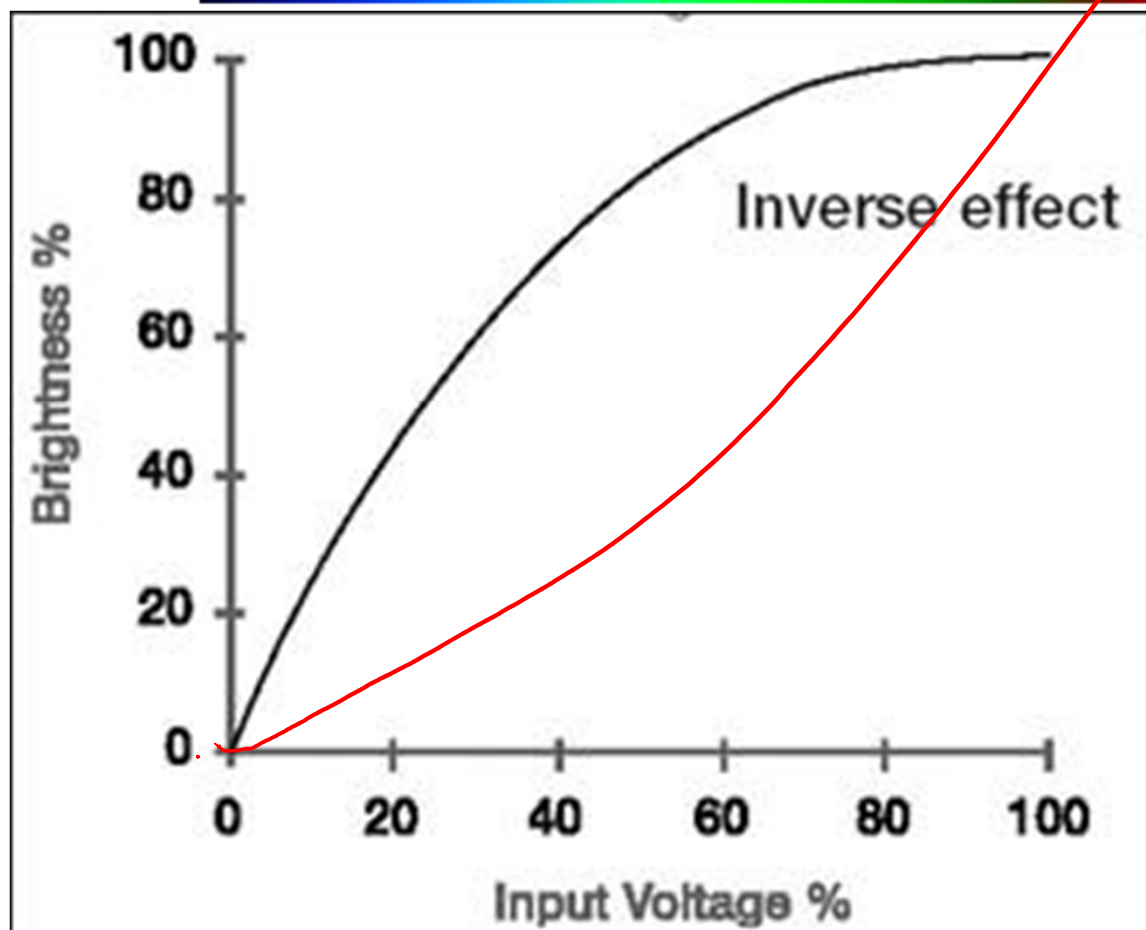


理想状态下的曲线





按图进行曲线补偿

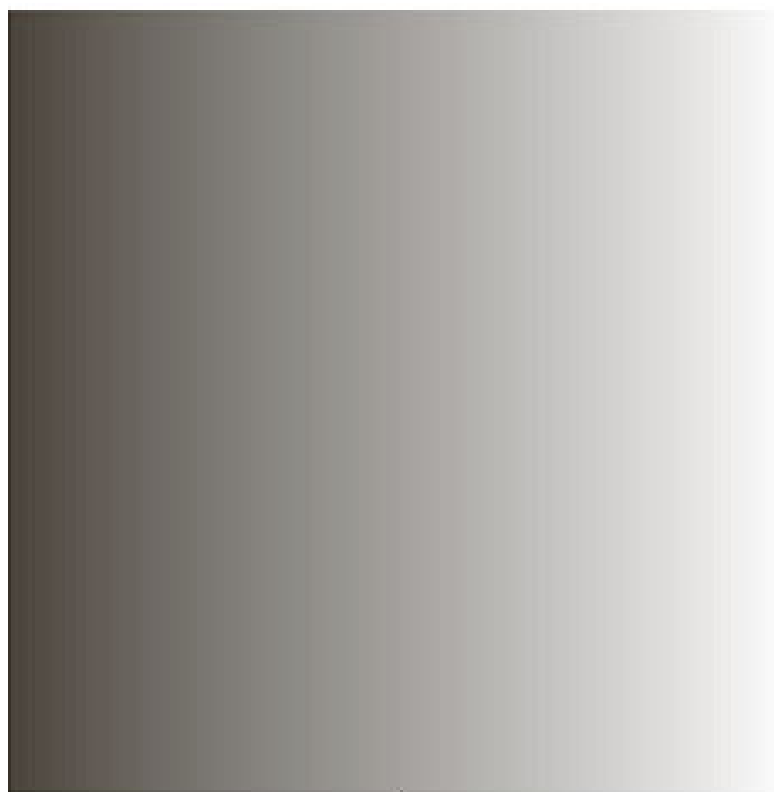


$$corrected = 255 * \left(\frac{uncorrected}{255} \right)^{\gamma}$$

fig. 3: Gamma-corrected colors



示例: Gamma 变换



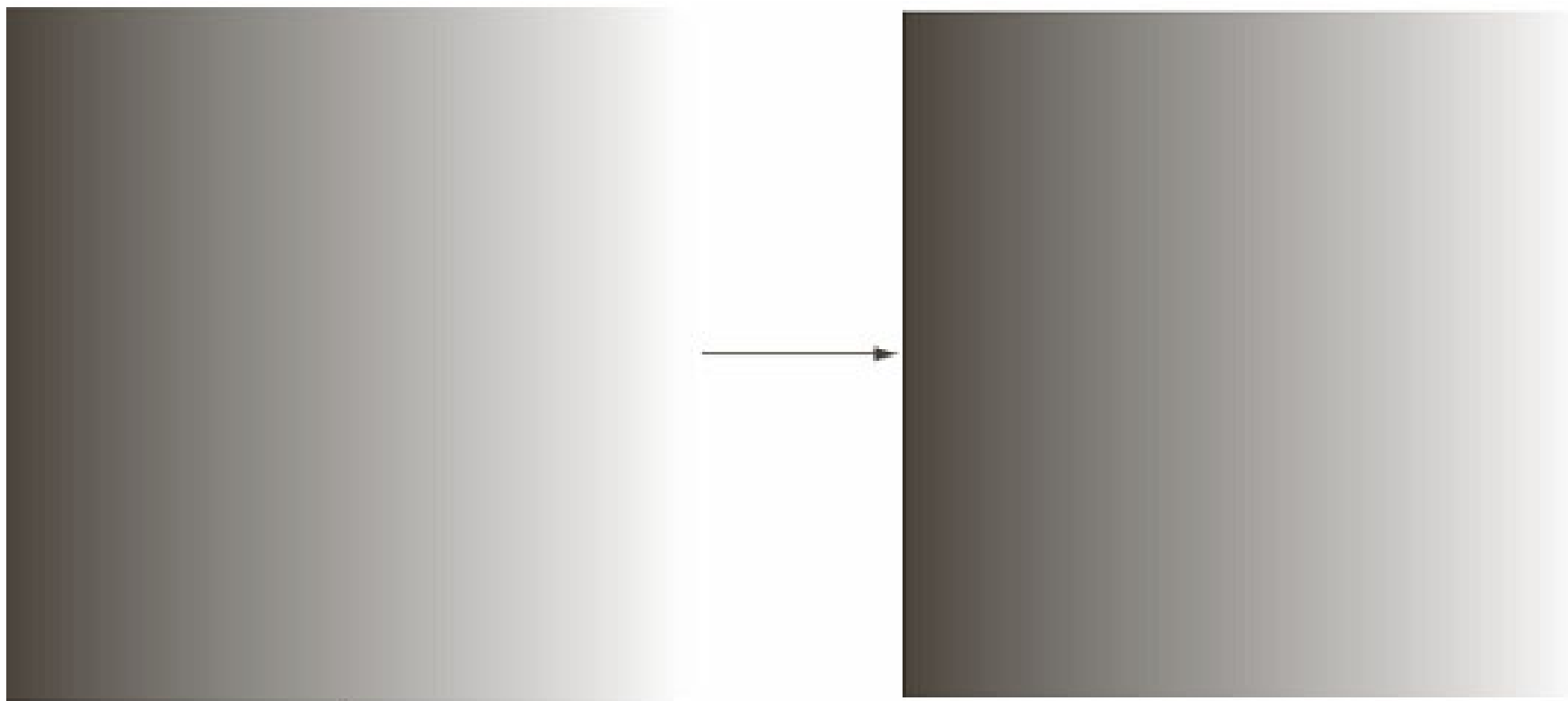
原始图像



在显示器上观看
无伽玛



示例: Gamma 变换



伽马校正的图像

$$S = r^{1/2.5}$$

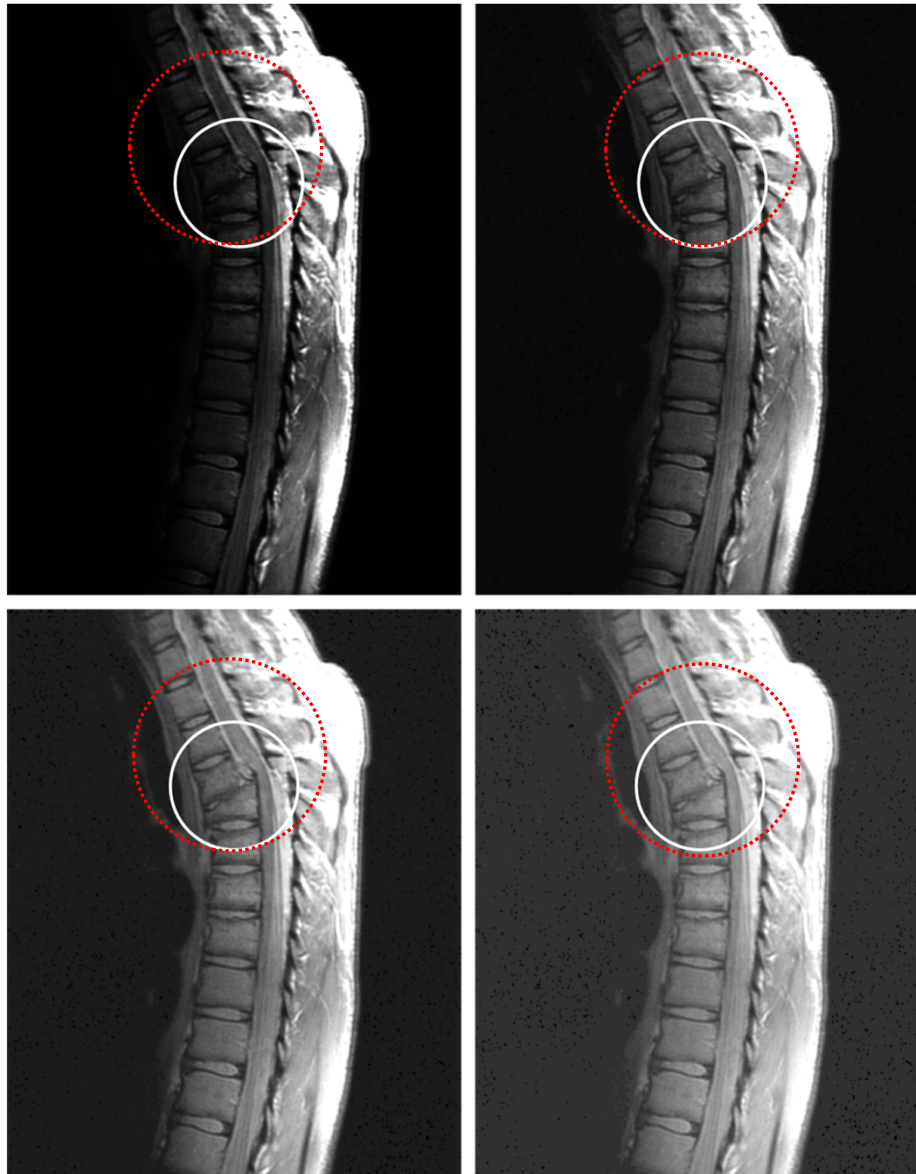
在显示器上观看
无伽玛







示例: Gamma 变换



a	b
c	d

FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine.

(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and

$\gamma = 0.6, 0.4,$ and 0.3 , respectively.

(Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)



示例: Gamma 变换



a	b
c	d

FIGURE 3.9

(a) Aerial image.
(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 3.0, 4.0,$ and 5.0 , respectively. (Original image for this example courtesy of NASA.)



目录

- 强度转换
 - 图像反转
 - Log转换
 - Gamma 转换
 - 分段线性变换
- 直方图处理
 - 直方图均衡化
 - 直方图匹配
 - 局部直方图处理
- 空间滤波器
 - 空间相关性和卷积
 - 平滑的线性滤波器和空间掩码
 - 非线性滤波器
 - 锐化滤波器



分段线性变换

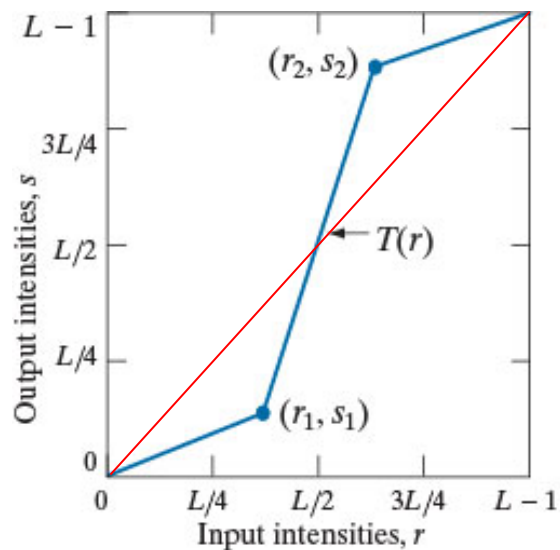


■ 对比度拉伸

— 扩大图像中的强度等级范围，使其跨越记录介质或显示设备的全部强度范围。

■ 强度级切片

— 突出图像中特定的强度范围往往是有意义的



对比度拉伸
a) 变换函数的形式。
b) 低对比度图像
c) 对比度拉伸的结果
d) 阈值处理的结果
原始图像由
Roger Heady博士提供，澳大利亚国立大学生物科学研究学院，堪培拉，澳大利亚。

$$\begin{aligned}(r_1, s_1) &= (r_{\min}, 0) \\ (r_2, s_2) &= (r_{\max}, L-1)\end{aligned}$$

$$\begin{aligned}(r_1, s_1) &= (m, 0) \\ (r_2, s_2) &= (m, L-1)\end{aligned}$$



力一系列图像中时间的函数

[illegible]

mation of the type illustrated in Fig. 1. (b) Result of the gray scale. (c) Result of the contrast enhancement, so that grays in the area of the face of Dr. Thomas R. Gest, University of



Bit-平面切割

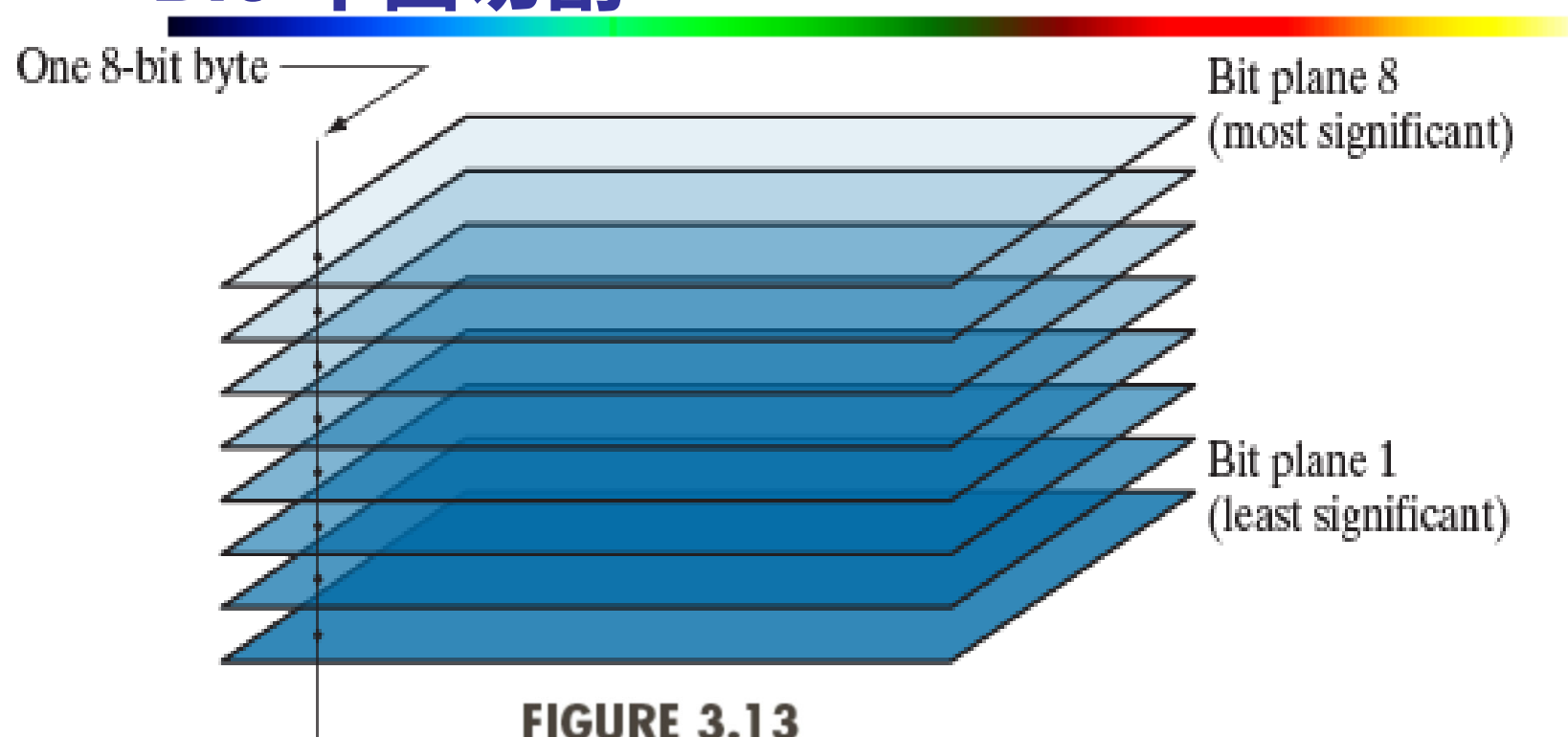


FIGURE 3.13
Bit-plane
representation of
an 8-bit image.



Bit-平面切割



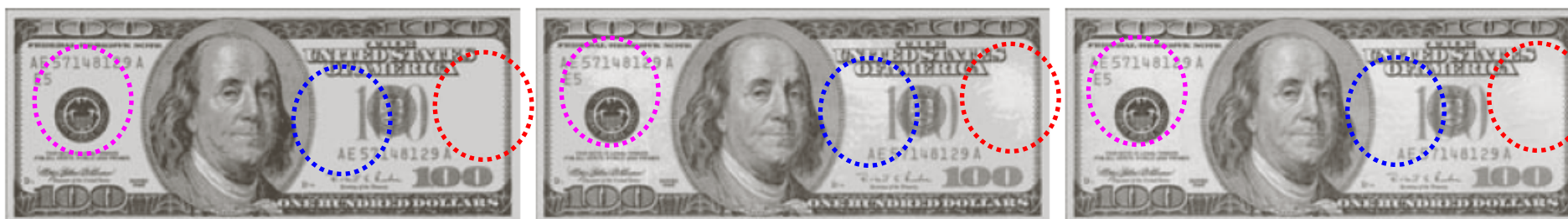
a	b	c
d	e	f
g	h	i

194
11000010

FIGURE 3.14 (a) An 8-bit gray-scale image of size 550×1192 pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image..

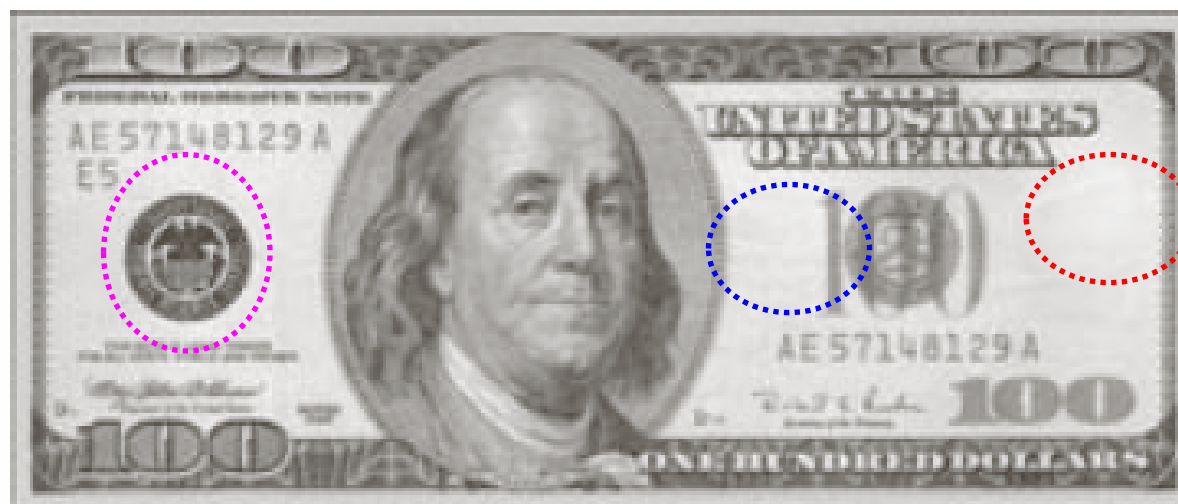


Bit-plane Slicing



a b c

FIGURE 3.15 Images reconstructed using (a) bit planes 8 and 7; (b) bit planes 8, 7, and 6; and (c) bit planes 8, 7, 6, and 5. Compare (c) with Fig. 3.14(a).





目录

- 强度转换
 - 图像反转
 - Log转换
 - Gamma 转换
 - 碎片化线性变换
- 直方图处理
 - 直方图均衡化
 - 直方图匹配
 - 局部直方图处理
- 空间滤波器
 - 空间相关性和卷积
 - 平滑的线性滤波器和空间掩码
 - 非线性滤波器
 - 锐化滤波器



直方图处理

- Histogram Equalization直方图均衡化
- Histogram Matching直方图匹配
- Local Histogram Processing局部直方图处理
- Using Histogram Statistics for Image Enhancement 使用直方图统计进行图像增强



直方图处理

Histogram $h(r_k) = n_k$

r_k is the k^{th} intensity value 第K个强度值

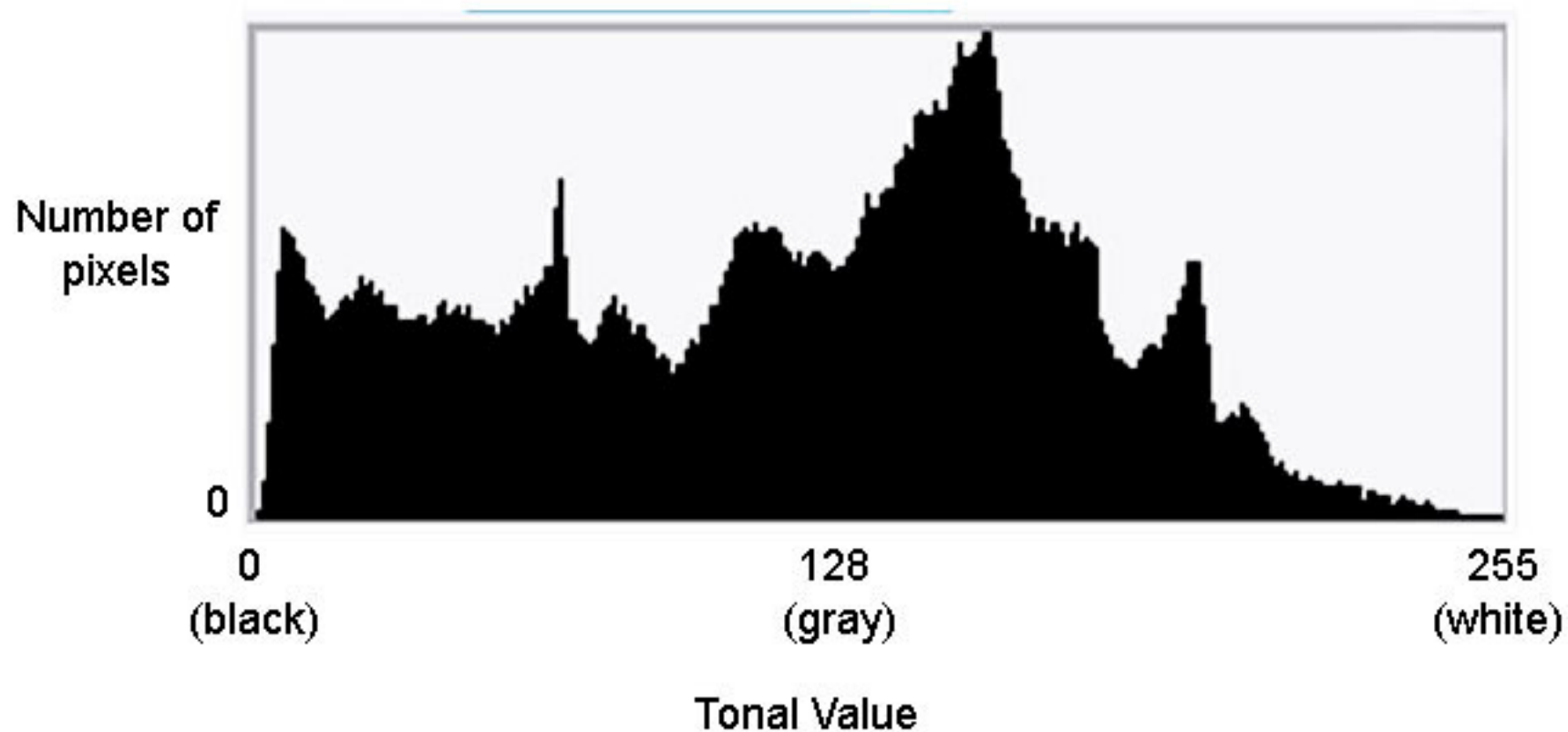
n_k is the number of pixels in the image with intensity r_k
第K个强度值总共的像素数量

Normalized histogram $p(r_k) = \frac{n_k}{MN}$

n_k : the number of pixels in the image of
size $M \times N$ with intensity r_k



什么是直方图?





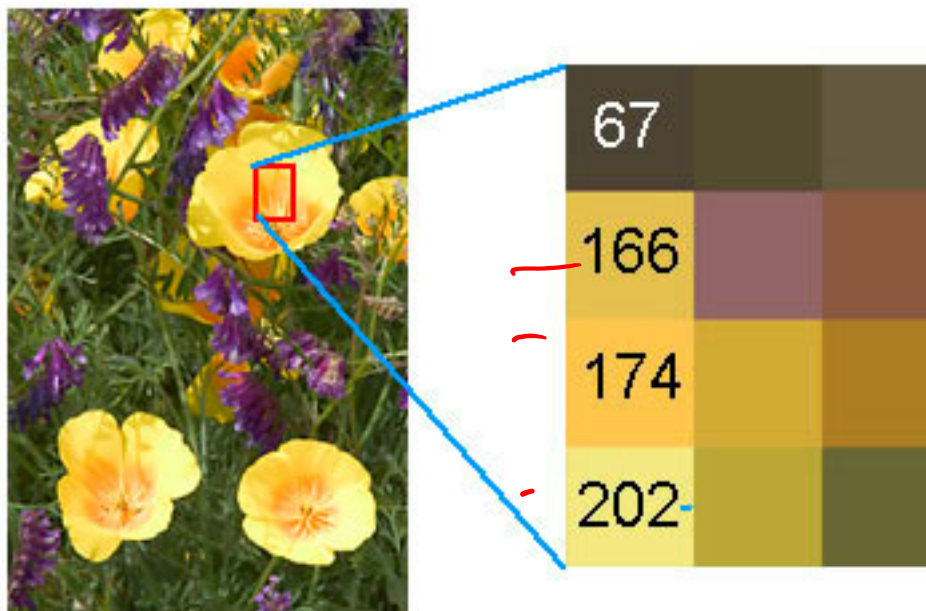
什么是直方图?



图像由像素组成



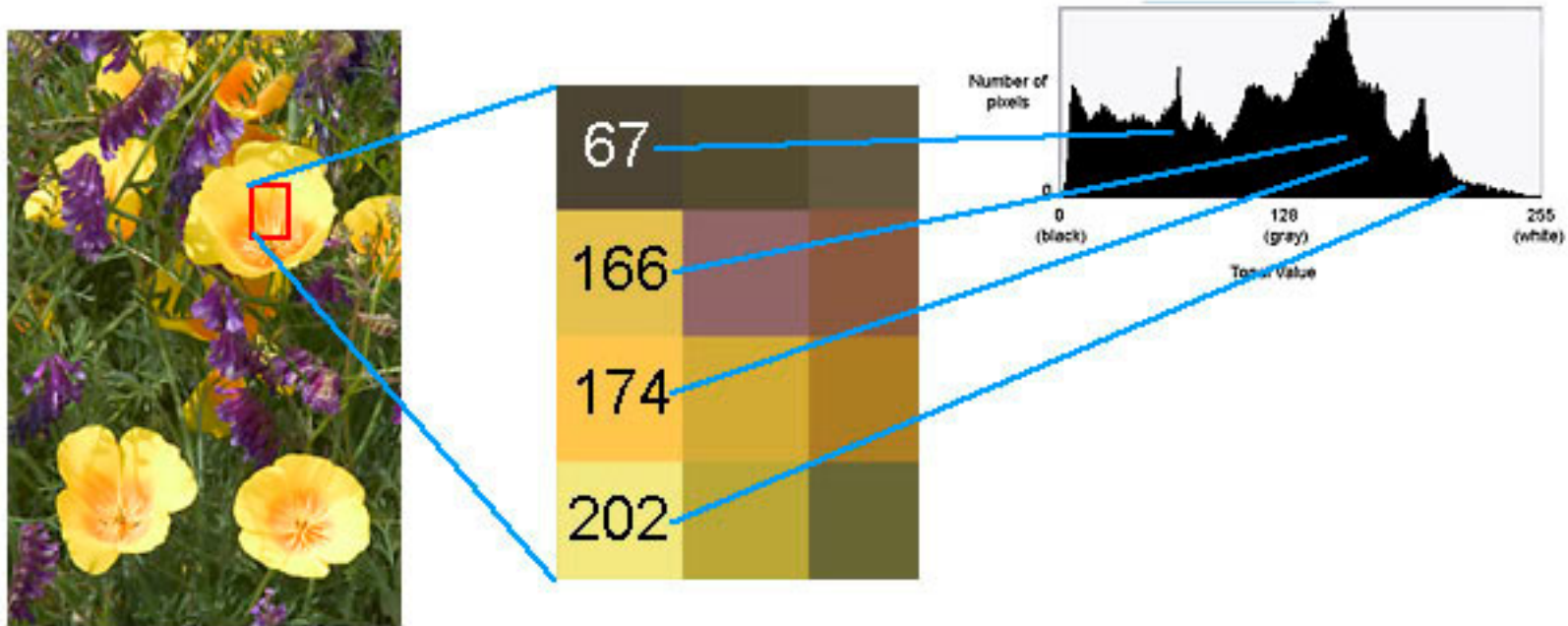
什么是直方图?



每个像素都有一个红值、一个蓝值和一个绿值。复合值是红色、绿色和蓝色的平均值。



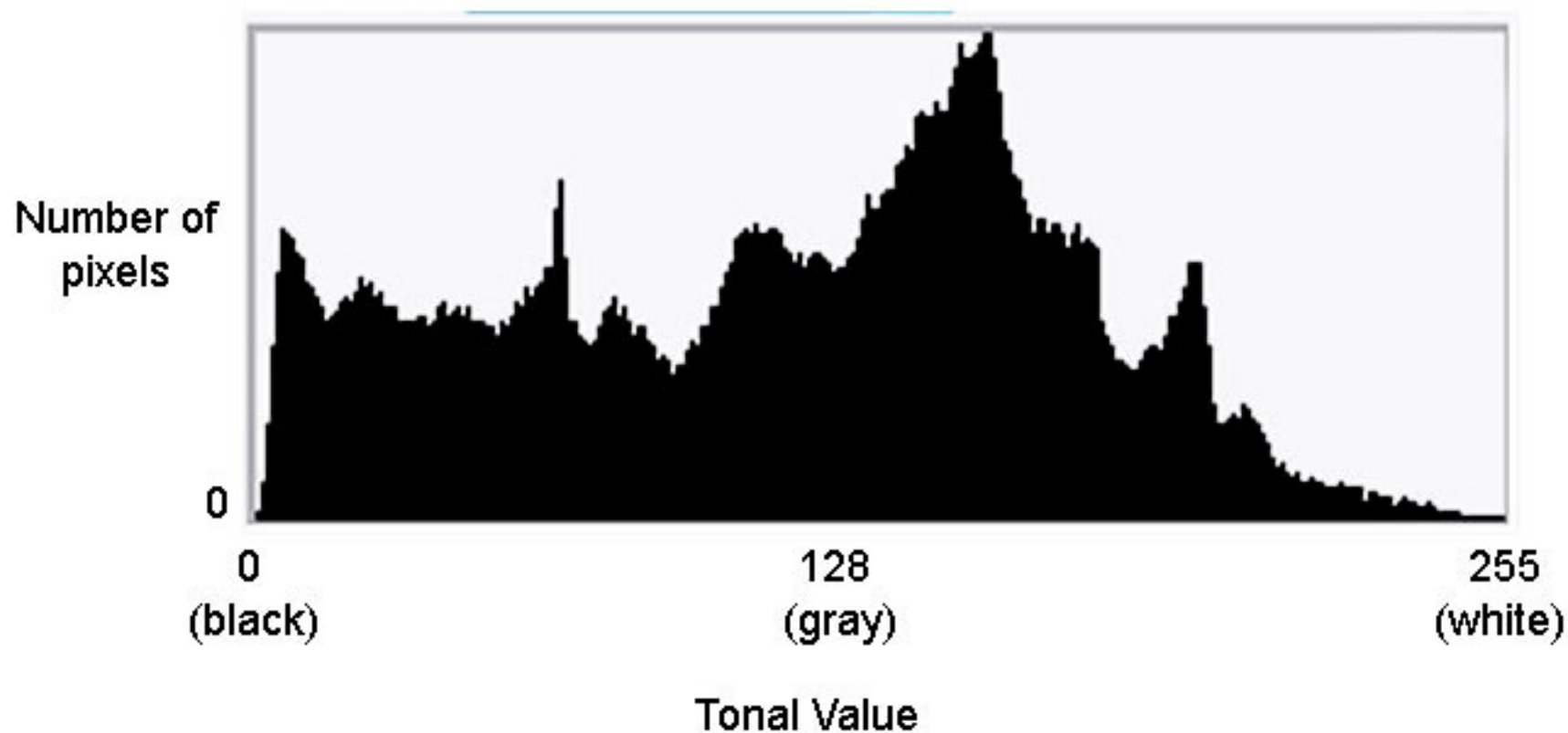
什么是直方图?



像素值被绘制在柱状图上。图像中每个像素的值都被绘制出来。



什么是直方图?

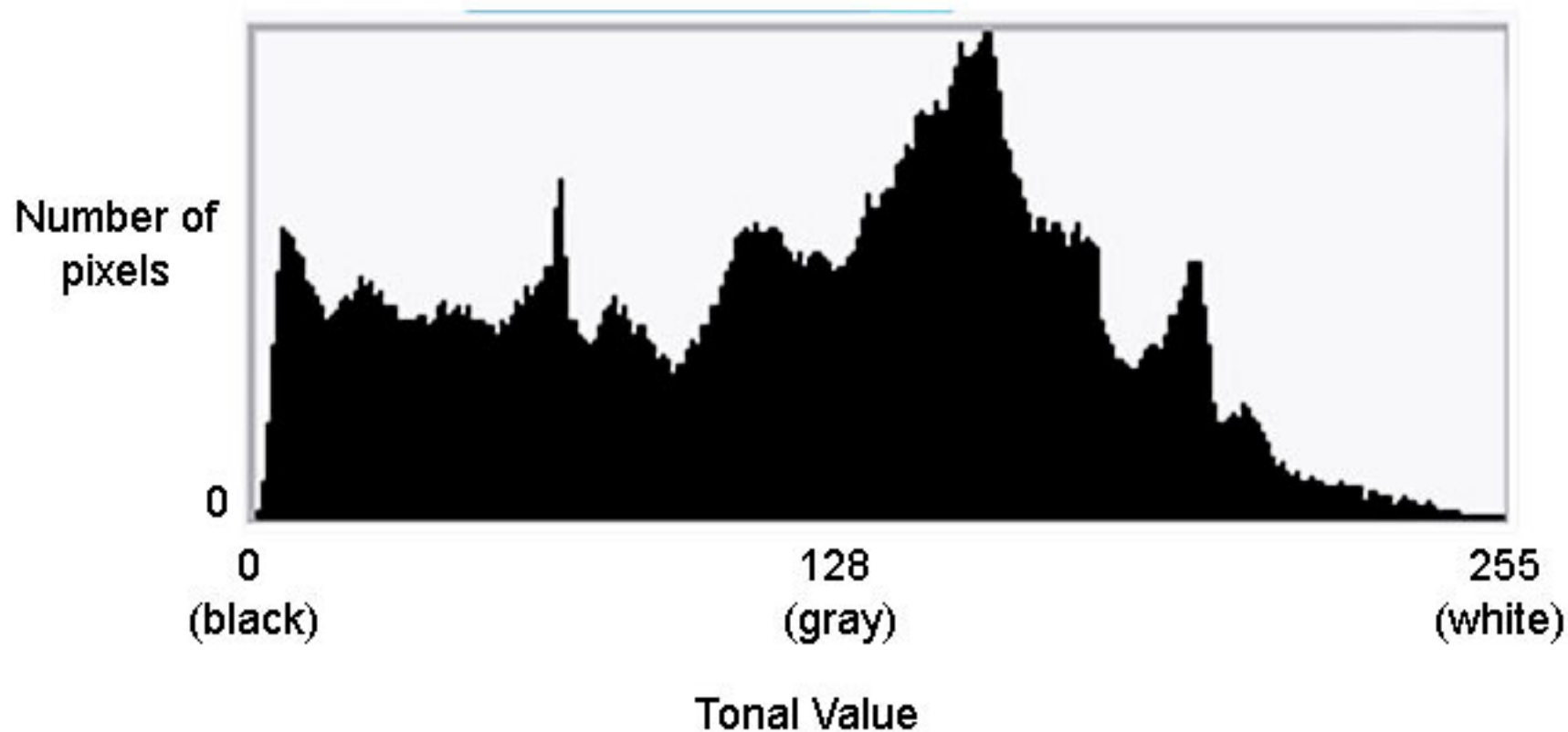


横轴代表从0到255的色调值。

8位文件有256个色调值。12位，4096。16位，65536。



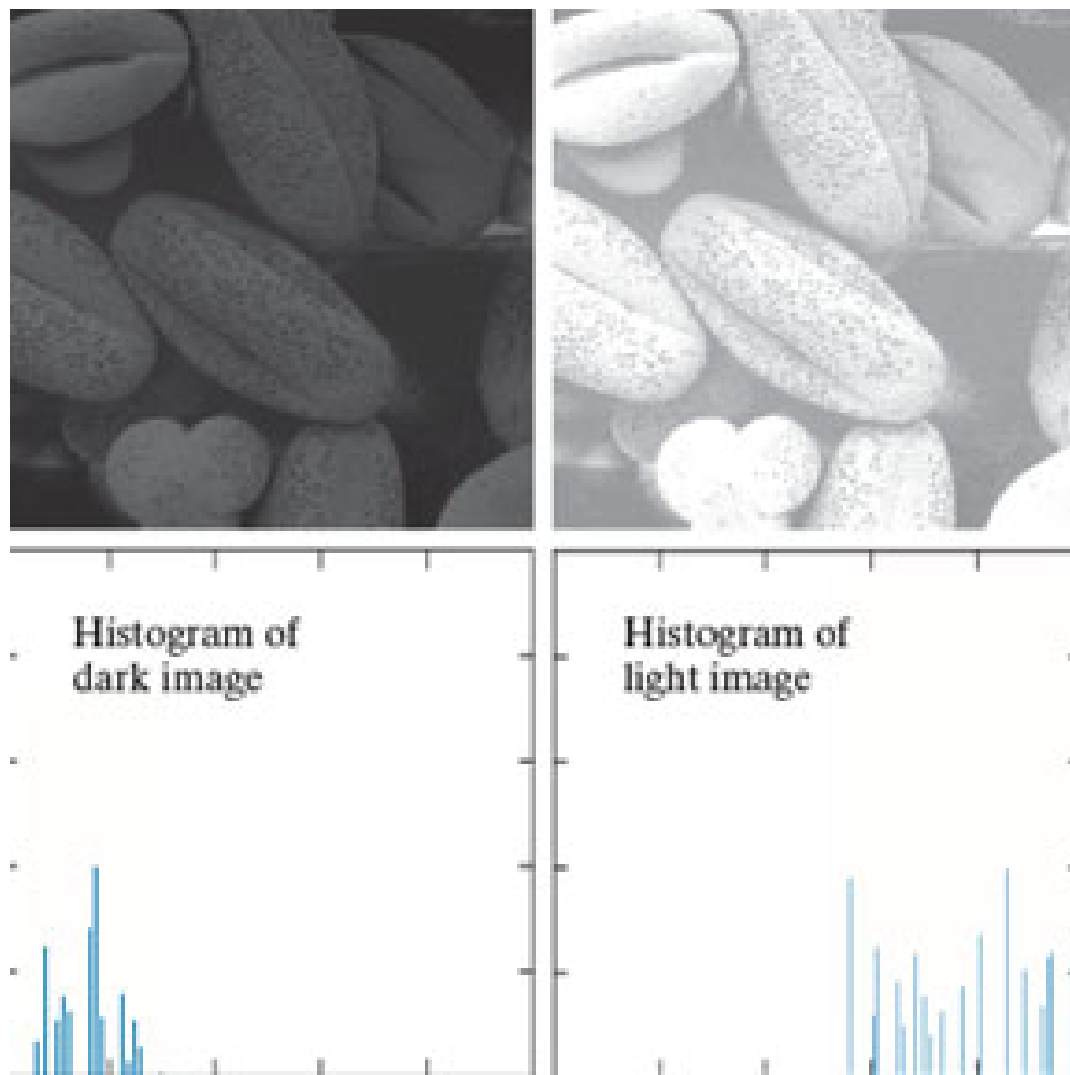
什么是直方图？



垂直轴代表每个色调值的像素数。条形越高，该值的像素就越多。

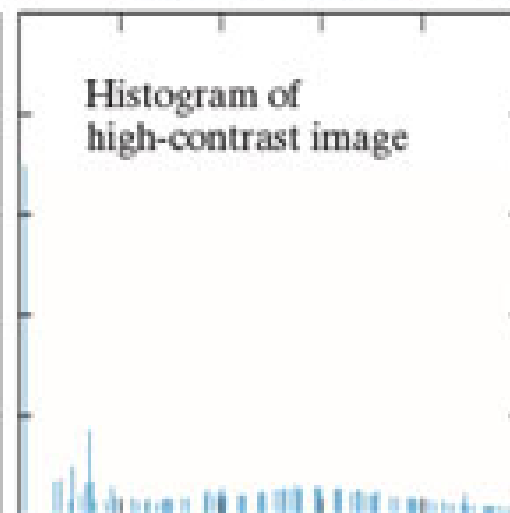
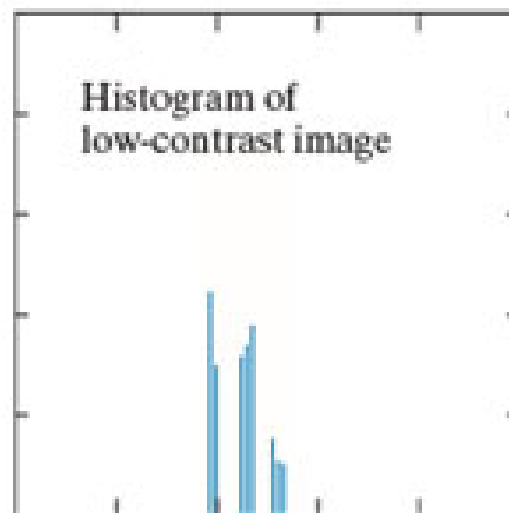
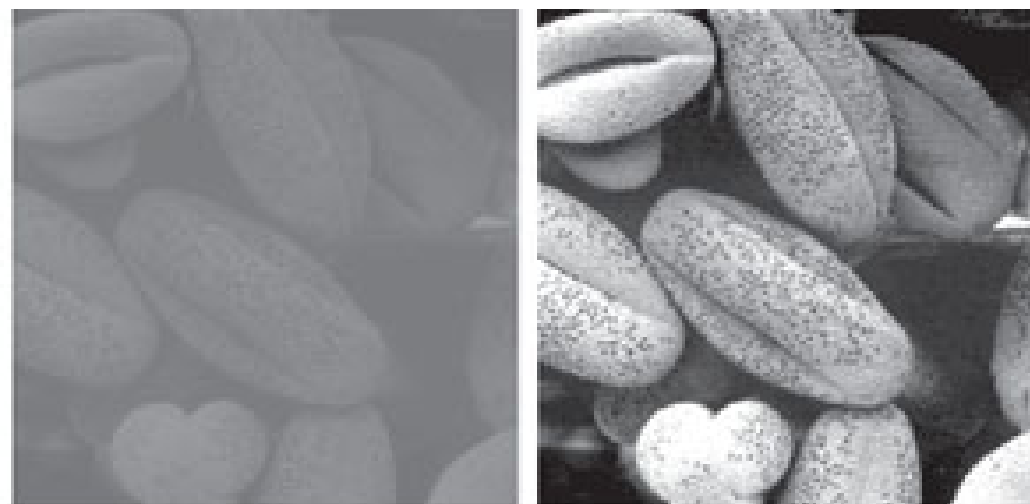


直方图处理

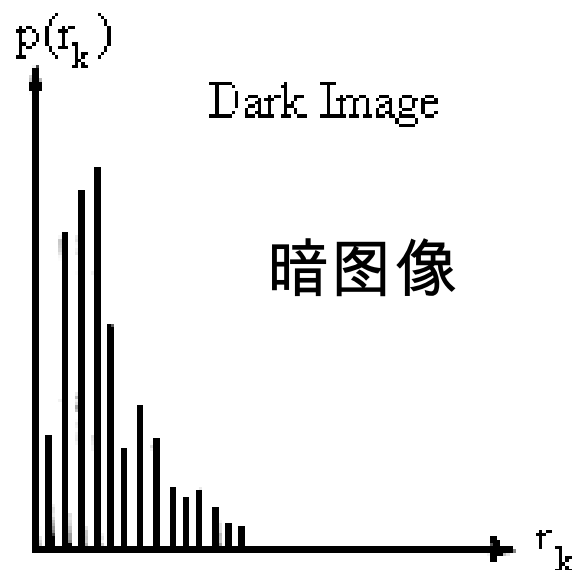




直方图处理



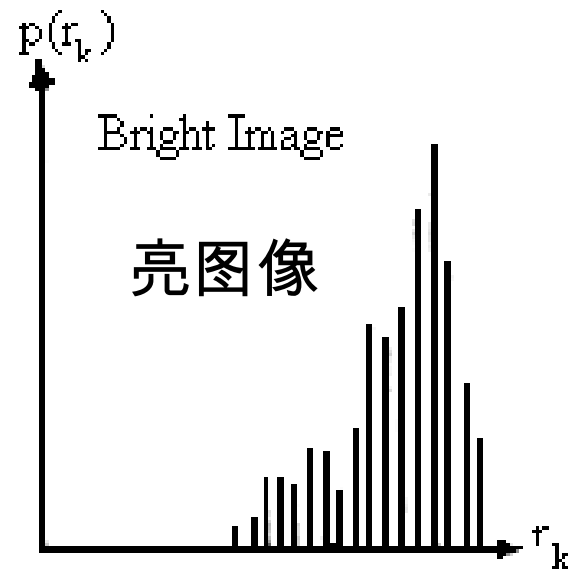
Uniform



Dark Image

暗图像

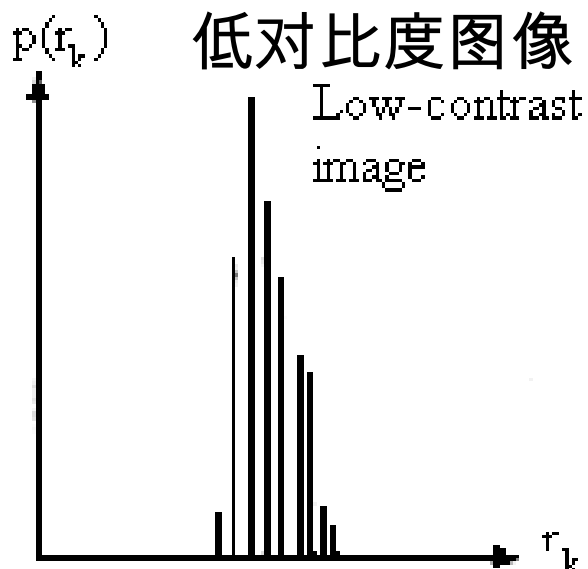
(a)



Bright Image

亮图像

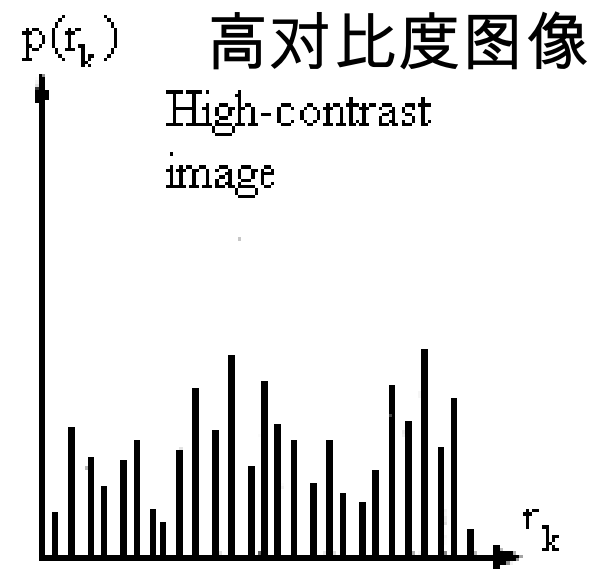
(b)



低对比度图像

Low-contrast
image

(c)



高对比度图像

High-contrast
image

(d)



(a)



(b)



(c)

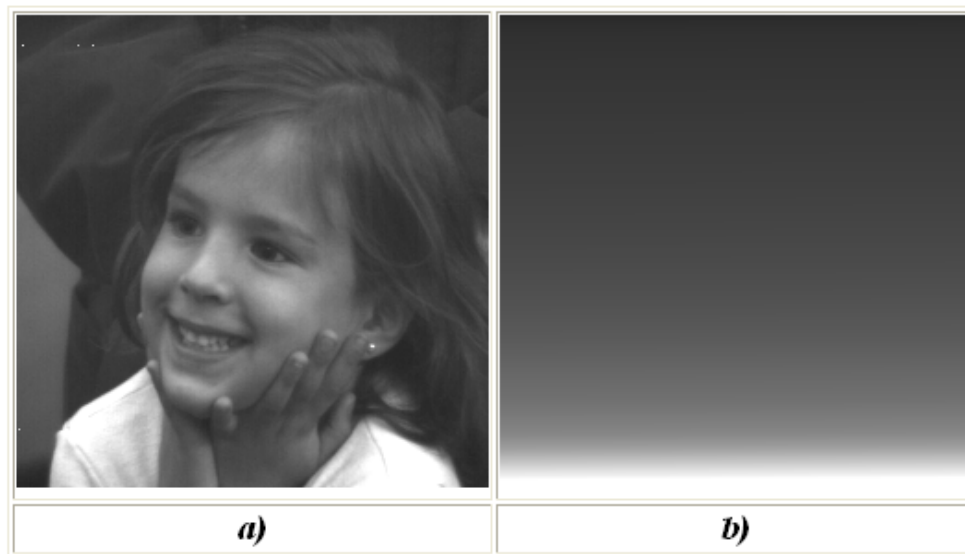
(d)

(e)

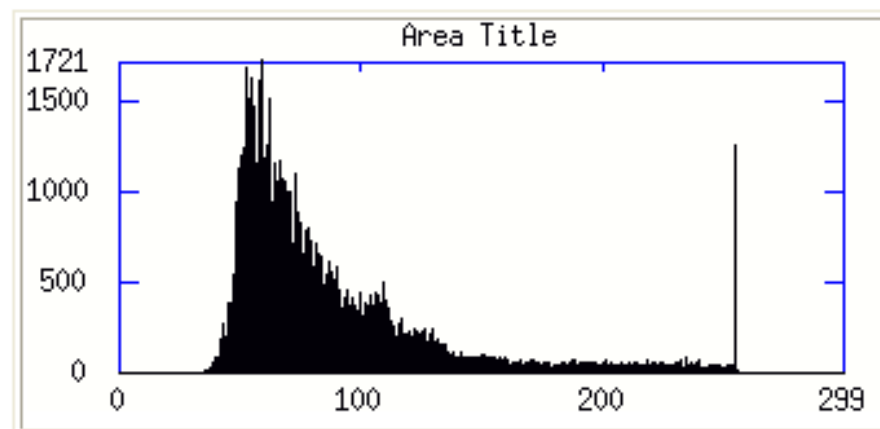
(f)



两张不同的图像--相同的直方图



a)Original image; b)Sorted version





图像处理中的经验性观察

- 更清晰 (sharper) 的图像看起来更好
- 同样的噪音在统一的背景区域比边缘区域的噪音更明显
- 同样的黑暗区域的噪音看起来比明亮区域的噪音更糟糕
- 同样数量的人工噪音看起来比自然的噪音更糟糕
- 具有非自然方面的图像吸引更多观众的注意力

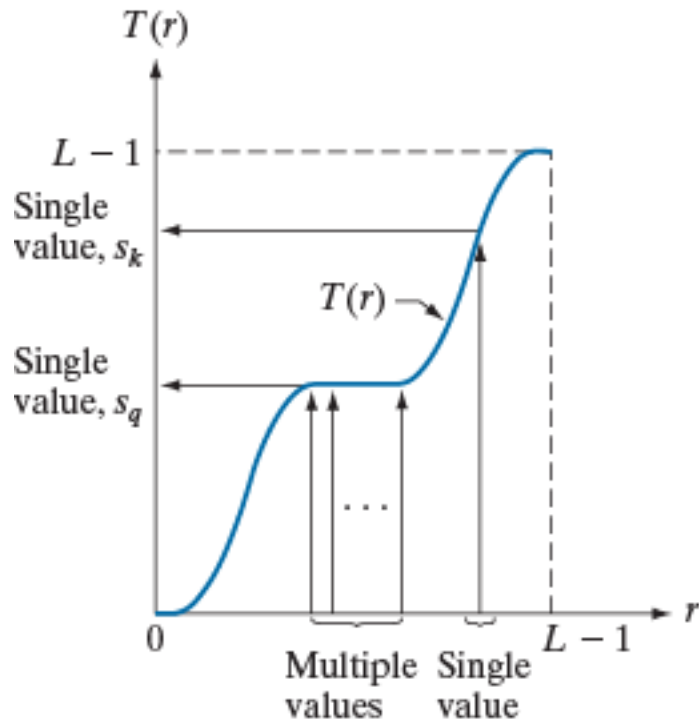


严格单调增

a. $T(r)$ is a $\quad$ monotonically increasing function

in the interval $0 \leq r \leq L-1$;

b. $0 \leq T(r) \leq L-1$ for $0 \leq r \leq L-1$.



a b

FIGURE 3.17

(a) Monotonically increasing function, showing how multiple values can map to a single value.

(b) Strictly monotonically increasing function. This is a one-to-one mapping, both ways.



直方图均衡

$$s = T(r) \quad 0 \leq r \leq L-1$$

严格单调增

- a. $T(r)$ is a monotonically increasing function in the interval $0 \leq r \leq L-1$;
- b. $0 \leq T(r) \leq L-1$ for $0 \leq r \leq L-1$.

连续可微

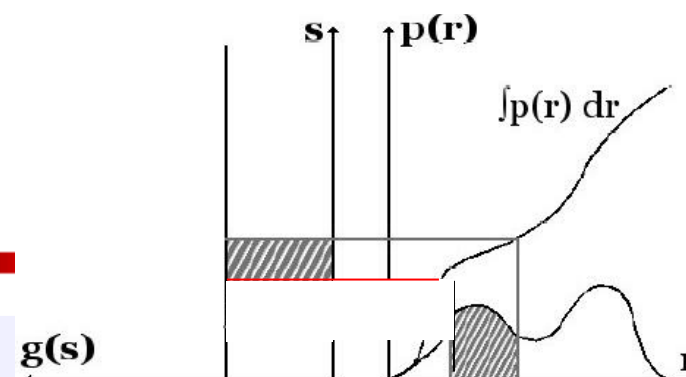
$T(r)$ is continuous and differentiable.

$$p_s(s)ds = p_r(r)dr$$



直方图均衡

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$



单调增[0~L-1]

$$\frac{ds}{dr} = \frac{dT(r)}{dr} = (L-1) \frac{d}{dr} \left[\int_0^r p_r(w) dw \right]$$

$$= (L-1) p_r(r)$$

$$p_s(s) = \frac{p_r(r) dr}{ds} = p_r(r) \Big/ \left(\frac{ds}{dr} \right) = p_r(r) \Big/ ((L-1) p_r(r)) = \frac{1}{L-1}$$

变换后的变量s的 $P_s(s)$ 在定义的区间内是一个均匀的密度



直方图均衡

图像中的强度等级可以被视为区间 $[0, L-1]$ 的随机变量。
让 $p_r(r)$ 和 $p_s(s)$ 表示 随机变量 r 和 s 的概率密度函数 (PDF) .

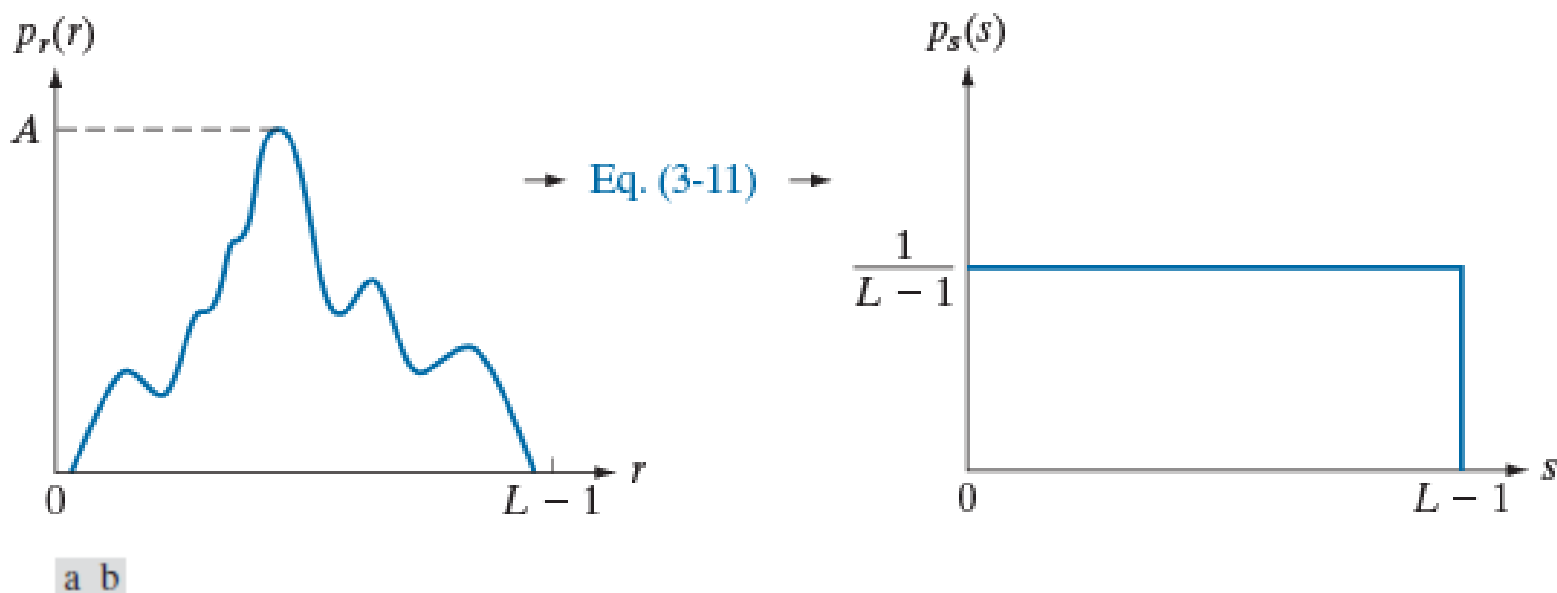


FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying Eq. (3-11) to the input PDF. The resulting PDF is always uniform, independently of the shape of the input.



示例

假设图像中的（连续）强度值有PDF

$$p_r(r) = \begin{cases} \frac{2r}{(L-1)^2}, & \text{for } 0 \leq r \leq L-1 \\ 0, & \text{otherwise} \end{cases}$$

找到用于均衡图像直方图的变换函数



示例

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$

$$p_s(s) = \frac{p_r(r) dr}{ds} = p_r(r) / \left(\frac{ds}{dr} \right) = p_r(r) / ((L-1)p_r(r)) = \frac{1}{L-1}$$



直方图均衡

Continuous case:

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$

Discrete values:

$$\begin{aligned} s_k &= T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j) \\ &= (L-1) \sum_{j=0}^k \frac{n_j}{MN} = \frac{L-1}{MN} \sum_{j=0}^k n_j \quad k=0,1,\dots, L-1 \end{aligned}$$



示例: 直方图均衡

假设一个 3-bit 图像 ($L=8$) 大小 64×64 pixels ($MN = 4096$) 拥有灰度值分布如下, 获取直方图均衡化转换函数, 并给出每个 s_k 的 $p_s(s_k)$ 。

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02



示例： 直方图均衡

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

Discrete values:

$$s_k = T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j)$$

$$s_0 = T(r_0) = 7 \sum_{j=0}^0 p_r(r_j) = 7 \times 0.19 = 1.33$$

→ 1

790

$$s_1 = T(r_1) = 7 \sum_{j=0}^1 p_r(r_j) = 7 \times (0.19 + 0.25) = 3.08$$

→ 3

1023

850

$$s_2 = 4.55 \rightarrow 5$$

$$s_3 = 5.67 \rightarrow 6$$

656+329

$$s_4 = 6.23 \rightarrow 6$$

$$s_5 = 6.65 \rightarrow 7$$

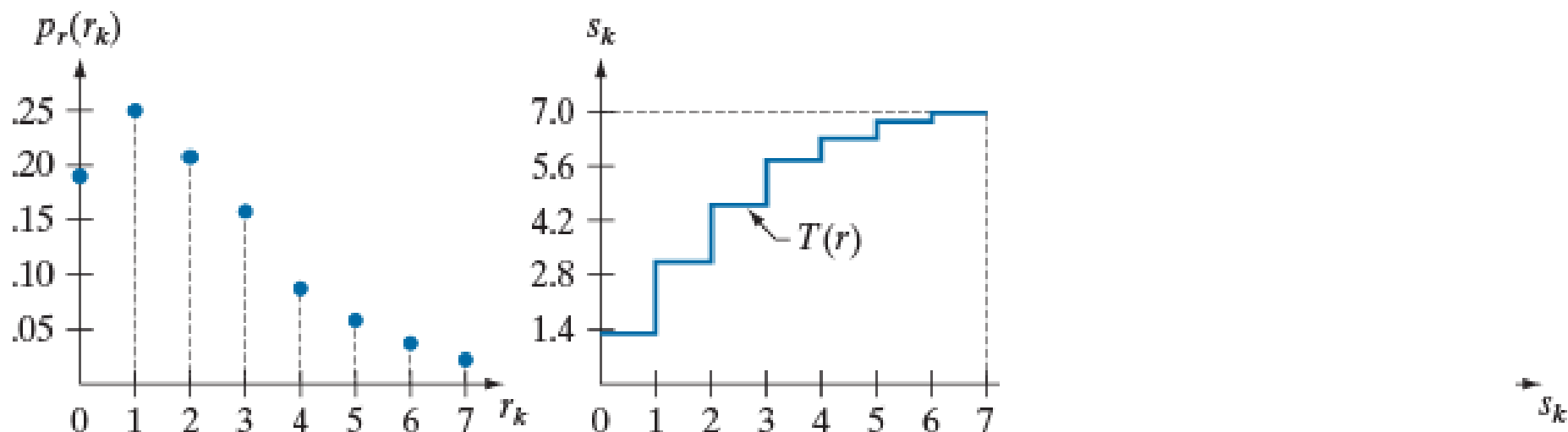
$$s_6 = 6.86 \rightarrow 7$$

$$s_7 = 7.00 \rightarrow 7$$

245+122+81



示例：直方图均衡



a b c

FIGURE 3.19

Histogram
equalization.

(a) Original
histogram.

(b) Transformation
function.

(c) Equalized
histogram.

图3.19

直方图均衡

(a)原始直方图

(b)变换函数

(c)均衡直方图



原始图像

直方图

均衡图像

均衡直方图

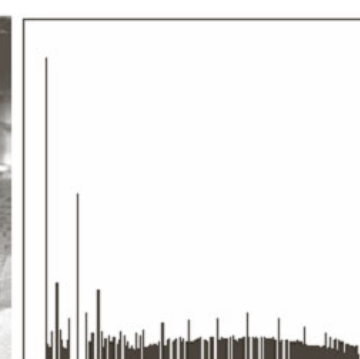
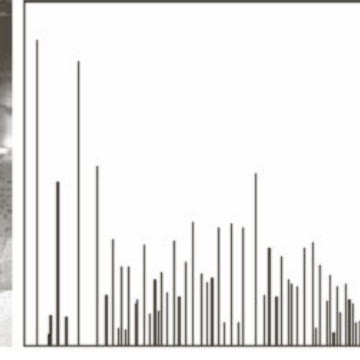
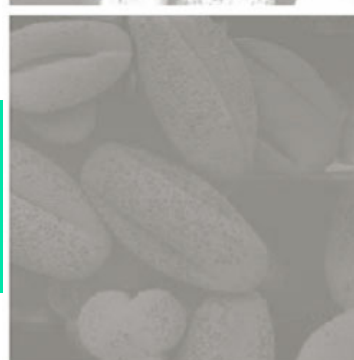
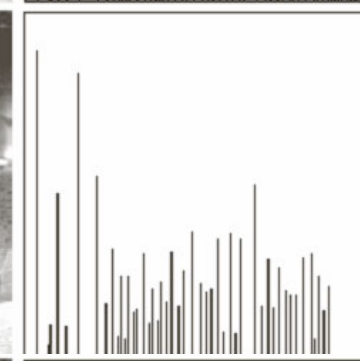
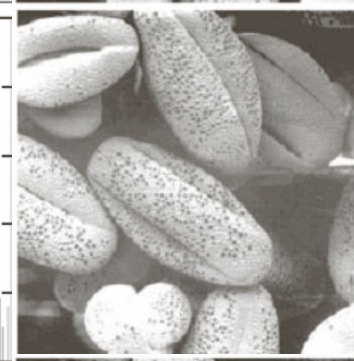
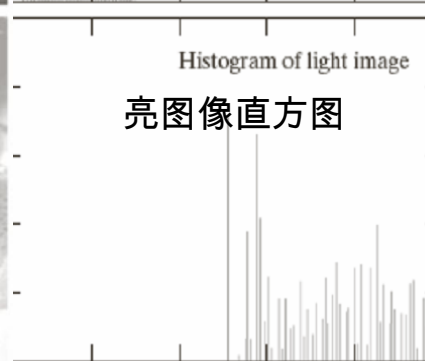
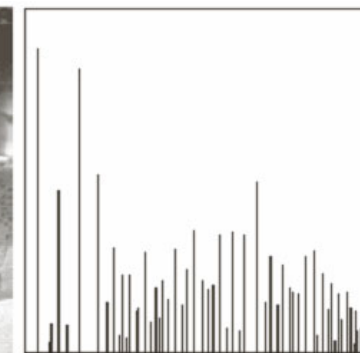
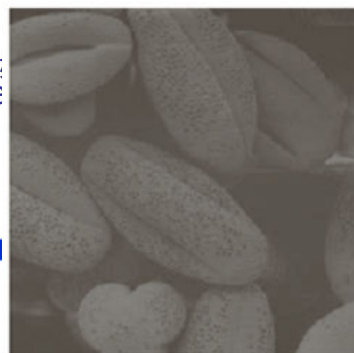




图3.21直方图变换函数均衡。
利用图3.20左列图像的直方图(从上到下)得到(1)到(4)的变换，通过等式(3.3-8)。

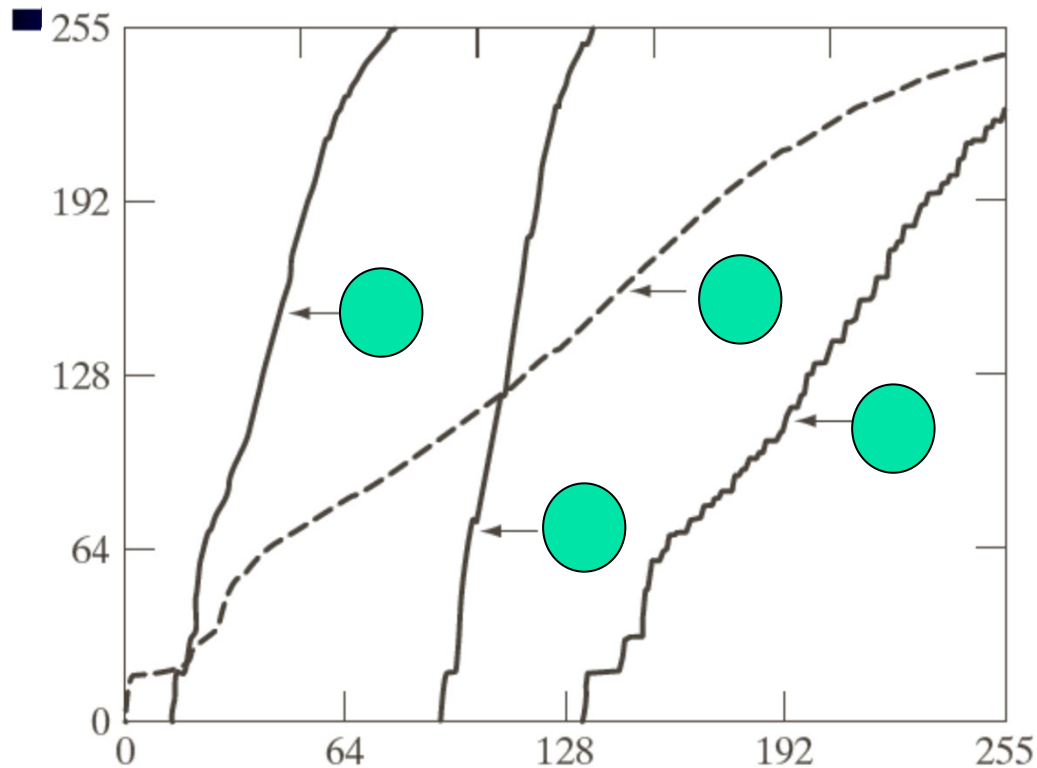


FIGURE 3.21
Transformation functions for histogram equalization. Transformations (1) through (4) were obtained from the histograms of the images (from top to bottom) in the left column of Fig. 3.20 using Eq. (3.3-8).

Transformation 1, 暗的灰度扩展到整个灰度级;
2, 亮的部分亮度值扩展到整个灰度级
3, 偏灰的亮度扩展到整个灰度级
4- approximate linear



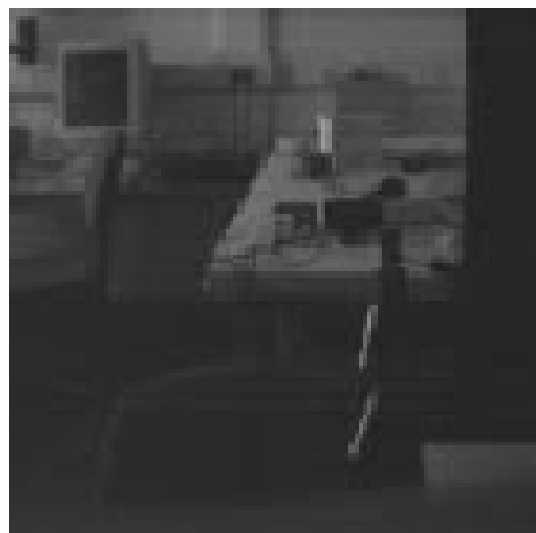
离散形式的直方图均衡

$$s_k = T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j)$$

- 注意，由于直方图均衡的离散近似，新直方图不是平坦的。
- 注意，经过直方图均衡的图像的灰度级被展开，并且总是达到白色。
- 此过程增加了灰度级的动态范围，并增加了图像对比度



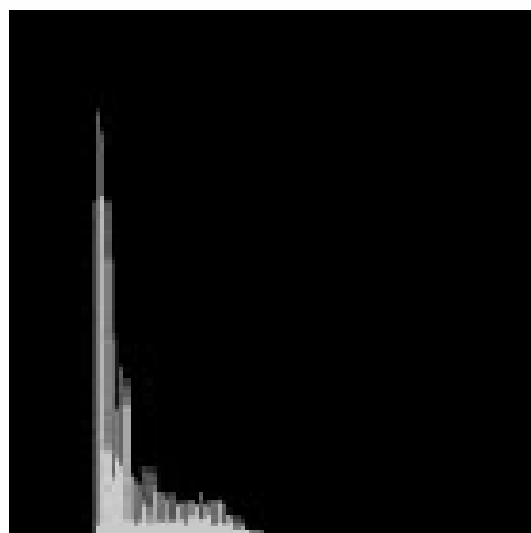
直方图均衡



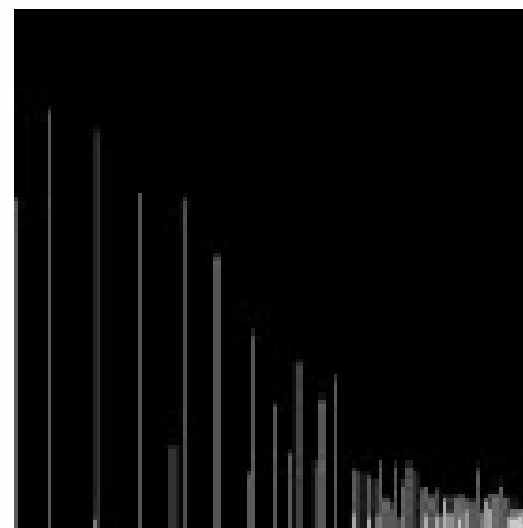
(a)



(b)



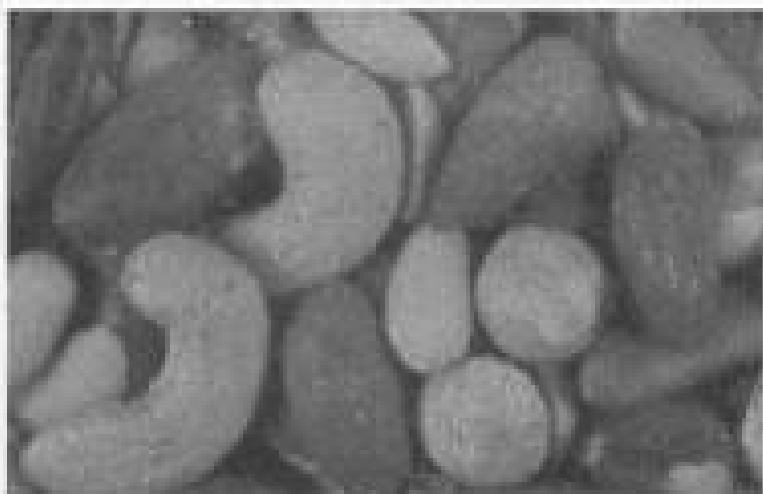
(c)



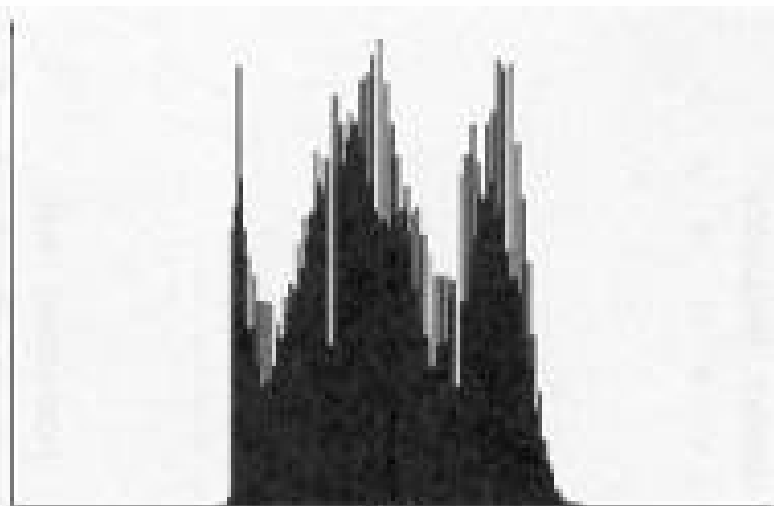
(d)



直方图均衡



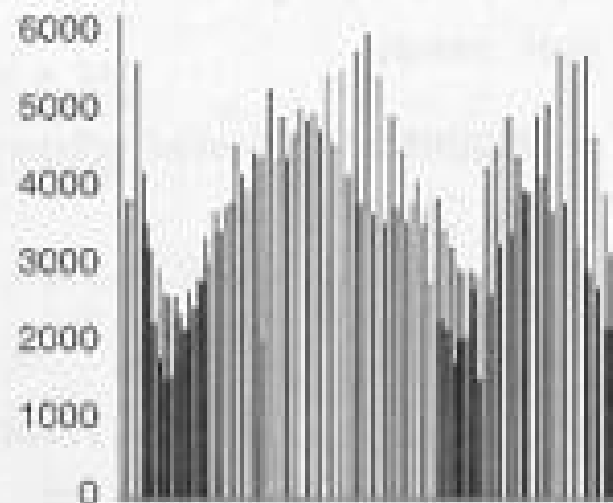
(a)



(b)



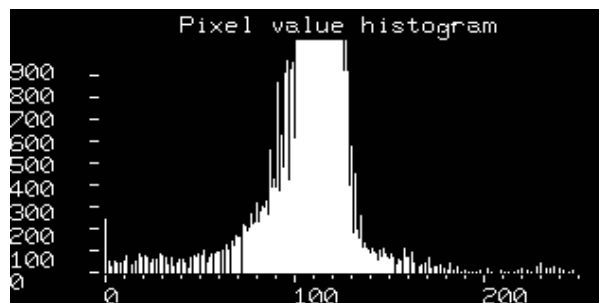
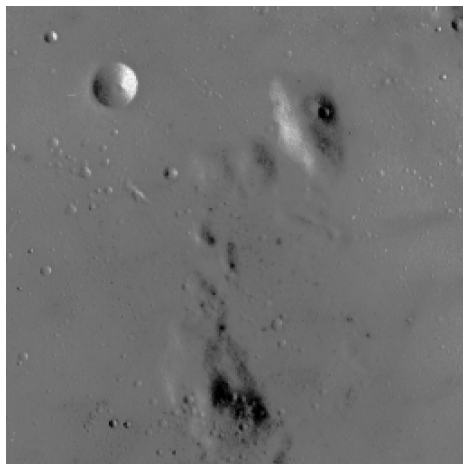
(c)



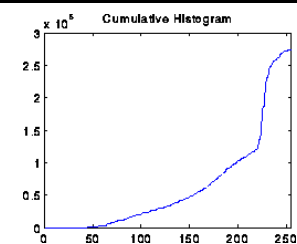
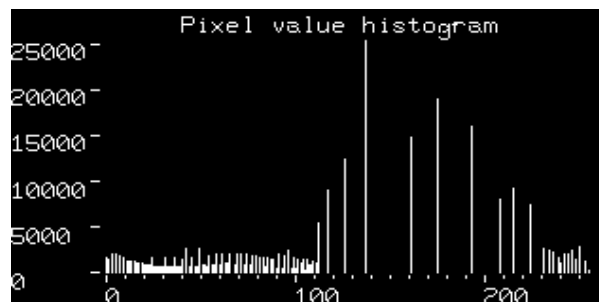
(d)



直方图均衡



像素价柱状图





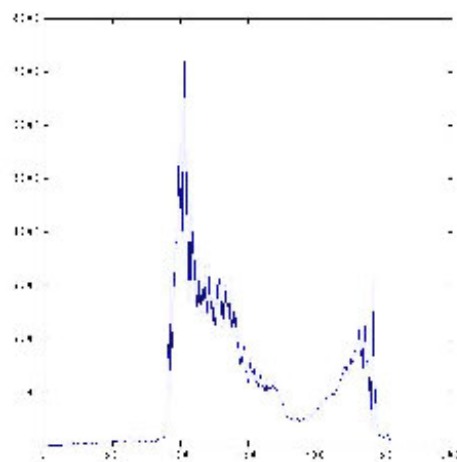
直方图均衡



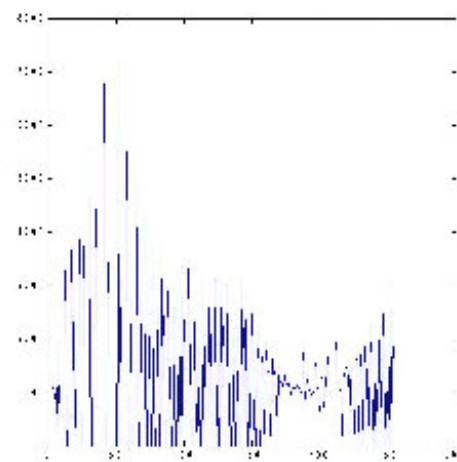
Original Image
原始图像



After Histogram Equalisation
直方图均衡化之后



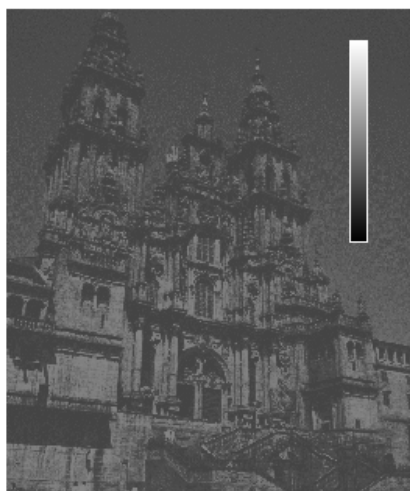
Original Histogram
原始直方图



Histogram after equalisation
均衡化后的直方图

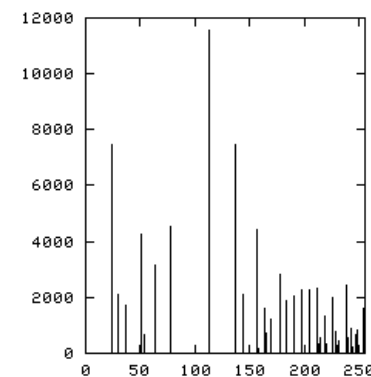


直方图均衡



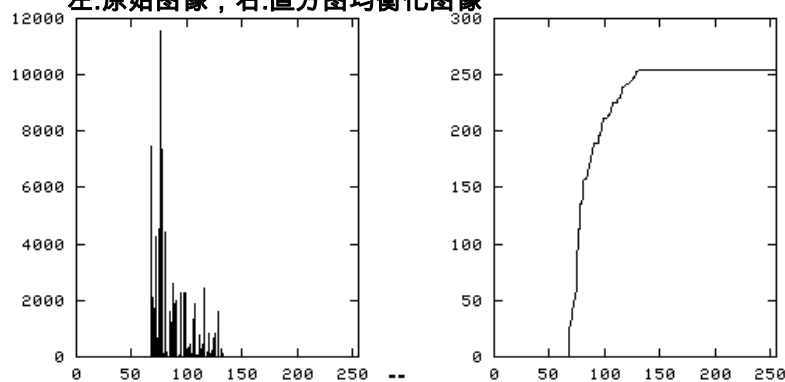
Left: original image, Right: histogram equalized image

左:原始图像, 右:直方图均衡化图像



Left: histogram equalized image, Right: histogram of the histogram equalized image

左:直方图均衡化图像, 右:直方图均衡化图像的直方图



Left: histogram of the original image, Right: normalized cumulative histogram of the original image

左:原图像直方图, 右:原图像归一化累积直方图

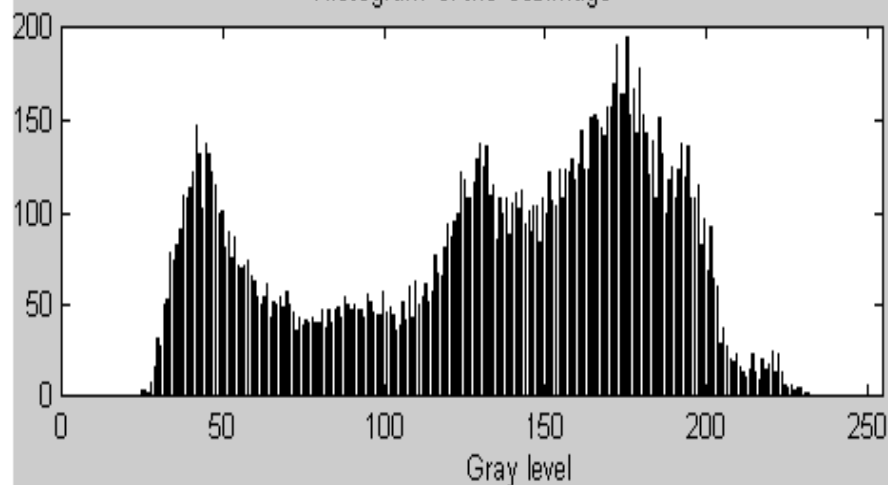


直方图均衡



子图像的直方图

Histogram of the subimage

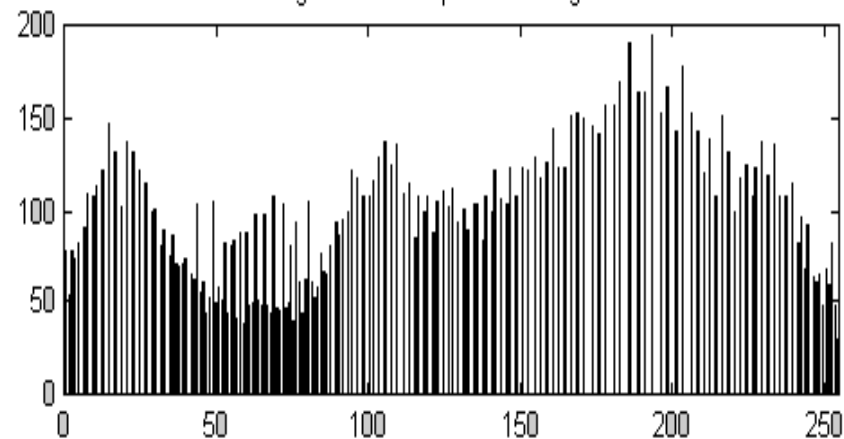


灰度



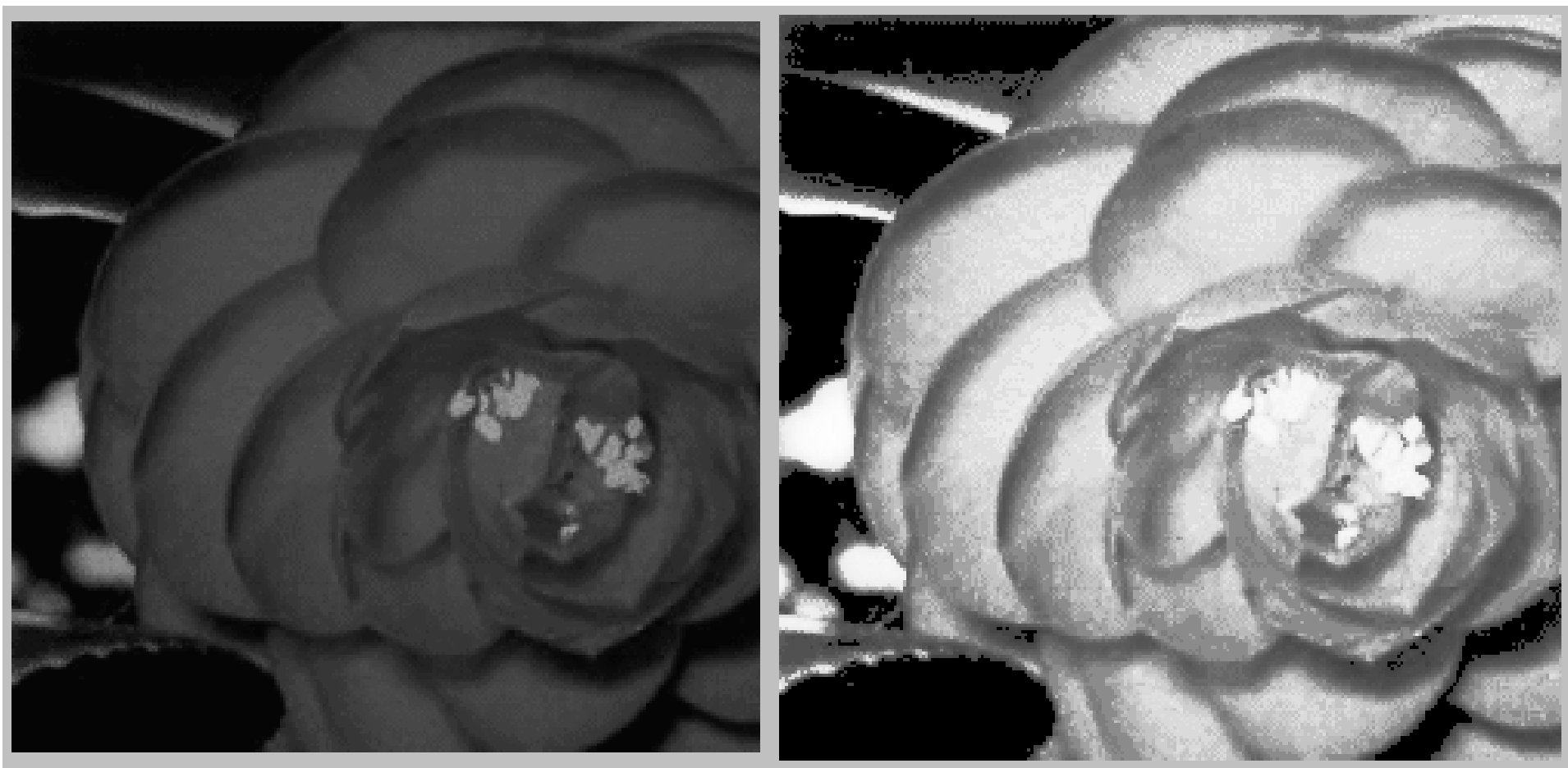
均衡化图像的直方图

Histogram of the equalized image



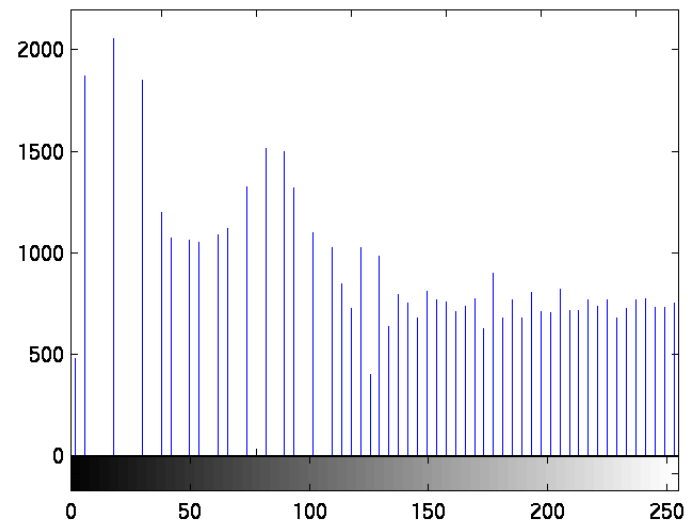
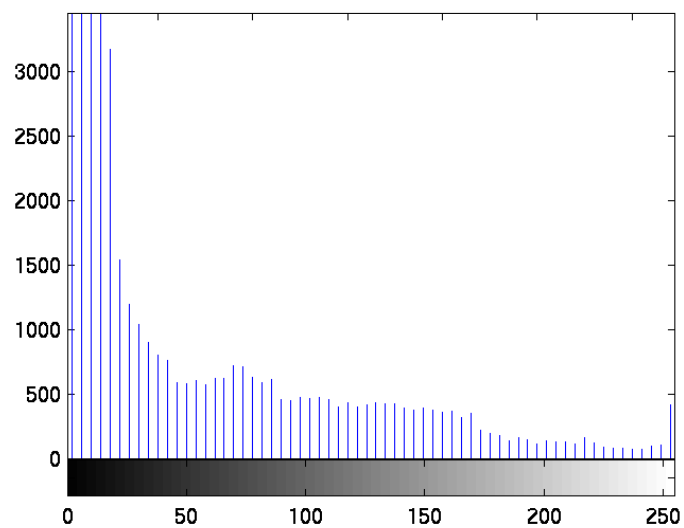


直方图均衡





直方图均衡





图像与视频处理
Image and Video Processing

问题

直方图均衡总是好的吗？

不



示例：直方图匹配

原始图像

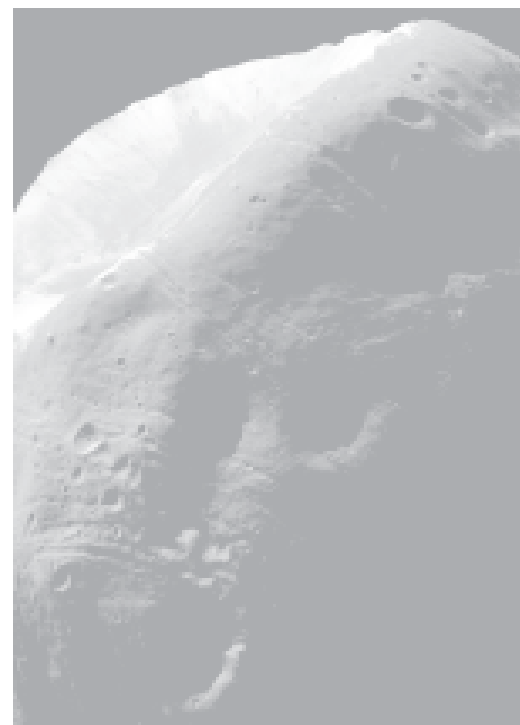
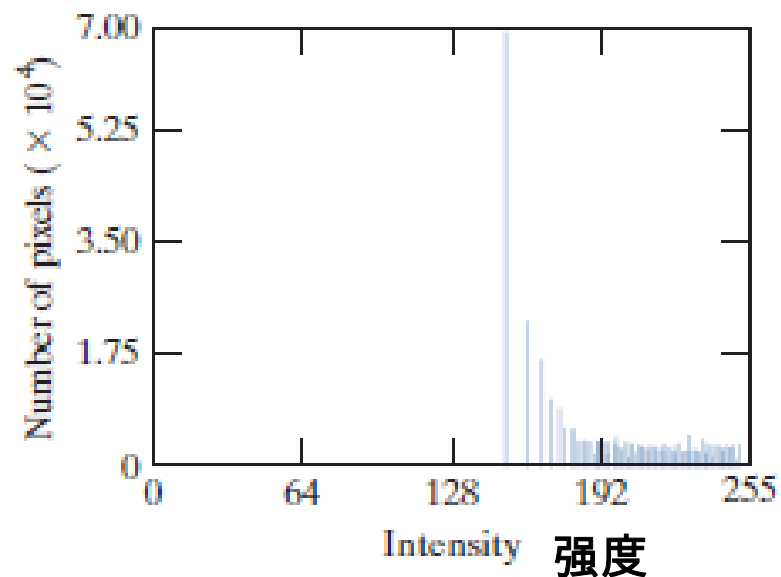
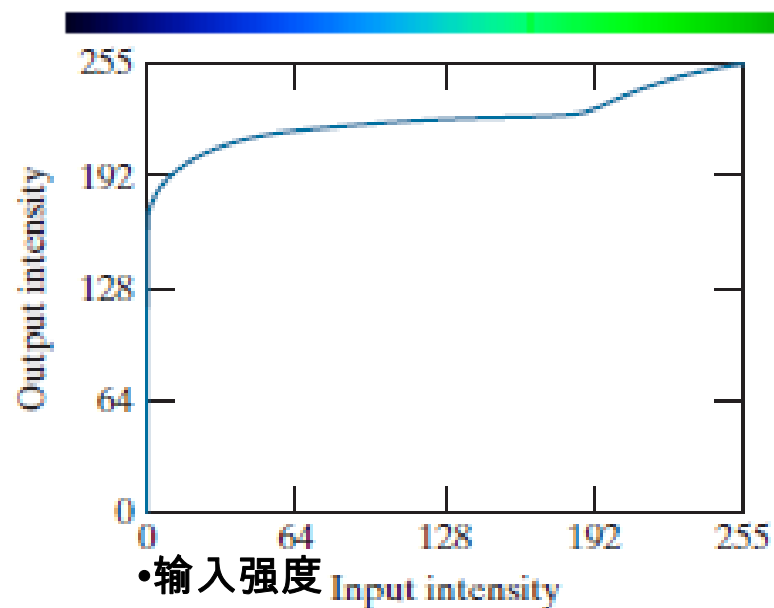


均衡图像





示例：直方图匹配



a b
c

(a)直方图变换函数均衡
(b)组gam均衡化图像
(c)是图像b的直方图。

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$



Content

- 强度变换
 - 图像负片
 - 对数变换
 - 伽马变换
 - 分段线性变换
- 直方图处理
 - 直方图均衡
 - 直方图匹配
 - 局部直方图处理
- 空间过滤器
 - 空间相关性和卷积
 - 平滑线性过滤器和空间遮罩
 - 非线性滤波器
 - 锐化过滤器



直方图匹配

直方图匹配（直方图规格）

— 生成具有指定直方图的处理图像

设 $p_r(r)$ 和 $p_z(z)$ 表示变量 r 和 z 的连续概率密度函数。

$p_z(z)$ 是指定的概率密度函数。

设 s 为概率随机变量

定
$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

$$G(z) = (L - 1) \int_0^z p_z(t) dt = s$$



直方图匹配

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

$$G(z) = (L - 1) \int_0^z p_z(t) dt = s$$

$$z = G^{-1}(s) = G^{-1}[T(r)]$$



直方图匹配：步骤

- 从输入图像中获取 $p_r(r)$ ，然后获取 s 的值

$$s = (L-1) \int_0^r p_r(w) dw$$

- 使用指定的PDF并获得转换函数 $G(z)$

$$G(z) = (L-1) \int_0^z p_z(t) dt = s$$

- 从 s 到 z 的映射

$$z = G^{-1}(s)$$



直方图匹配：示例

假设连续的强度值，假设图像具有强度PDF

$$p_r(r) = \begin{cases} \frac{2r}{(L-1)^2}, & \text{for } 0 \leq r \leq L-1 \\ 0, & \text{otherwise} \end{cases}$$

找到一个变换函数，它将产生一个图像的强度PDF为

$$p_z(z) = \begin{cases} \frac{3z^2}{(L-1)^3}, & \text{for } 0 \leq z \leq (L-1) \\ 0, & \text{otherwise} \end{cases}$$



直方图匹配：示例

求输入图像的直方图均衡变换

$$s = T(r) = (L-1) \int_0^r p_r(w) dw = (L-1) \int_0^r \frac{2w}{(L-1)^2} dw = \frac{r^2}{L-1}$$

查找指定直方图的直方图均衡转换

$$G(z) = (L-1) \int_0^z p_z(t) dt = (L-1) \int_0^z \frac{3t^2}{(L-1)^3} dt = \frac{z^3}{(L-1)^2} = s$$

变换函数为

$$z = \left[(L-1)^2 s \right]^{1/3} = \left[(L-1)^2 \frac{r^2}{L-1} \right]^{1/3} = \left[(L-1) r^2 \right]^{1/3}$$



直方图匹配：离散案例

$$z_q = G^{-1}(s_k)$$



直方图匹配：离散案例

- 从输入图像中得到 $p_r(r_j)$ ，然后得到 s_k 的值，四舍五入到整数范围 $[0, L-1]$ 。

$$s_k = T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j) = \frac{(L-1)}{MN} \sum_{j=0}^k n_j$$

- 使用指定的PDF并获得转换函数 $G(z_q)$ ，将值舍入到整数范围 $[0, L-1]$ 。

$$G(z_q) = (L-1) \sum_{i=0}^q p_z(z_i) = s_k$$

- 从 s_k 到 z_q 的映射

$$z_q = G^{-1}(s_k)$$



示例：直方图匹配

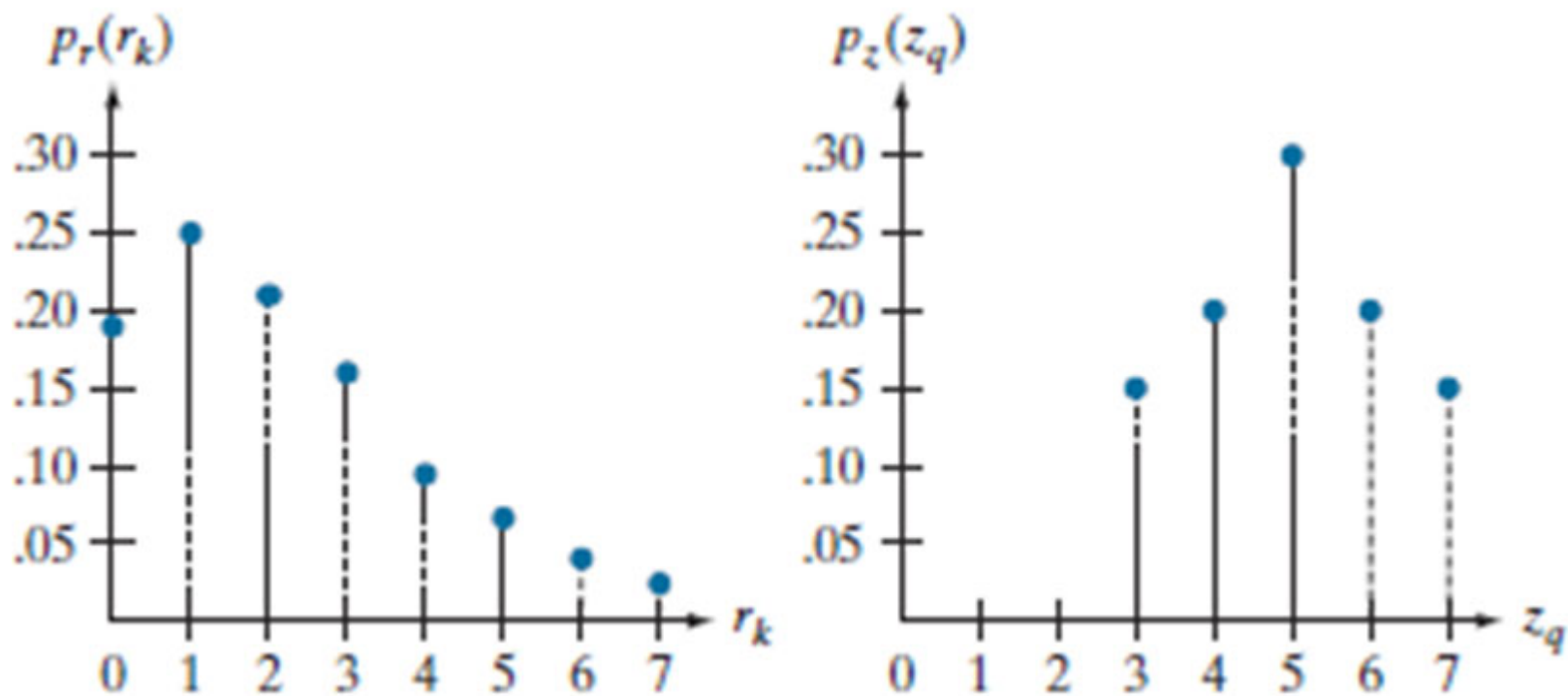
假设一张尺寸为 64×64 像素($MN = 4096$)的3位图像($L=8$)，其强度分布如下表(左侧)所示。获取直方图变换函数，用指定的直方图制作输出图像，如右表所示。

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

z_q	Specified $p_z(z_q)$
$z_0 = 0$	0.00
$z_1 = 1$	0.00
$z_2 = 2$	0.00
$z_3 = 3$	0.15
$z_4 = 4$	0.20
$z_5 = 5$	0.30
$z_6 = 6$	0.20
$z_7 = 7$	0.15



示例：直方图匹配





示例：直方图匹配

$$s_k = T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j)$$

得到缩放后的直方图均衡值，

$$s_0 = 1, s_1 = 3, s_2 = 5, s_3 = 6, s_4 = 6, \\ s_5 = 7, s_6 = 7, s_7 = 7.$$

计算变换函数G的所有值，

$$G(z_0) = 7 \sum_{j=0}^0 p_z(z_j) = 0.00 \rightarrow 0$$

$$\begin{array}{ll} G(z_1) = 0.00 \rightarrow 0 & G(z_2) = 0.00 \rightarrow 0 \\ G(z_3) = 1.05 \rightarrow 1 & \mathbf{s_0} \quad G(z_4) = 2.45 \rightarrow 2 \quad \mathbf{s_1} \\ G(z_5) = 4.55 \rightarrow 5 & \mathbf{s_2} \quad G(z_6) = 5.95 \rightarrow 6 \quad \mathbf{s_3} \\ G(z_7) = 7.00 \rightarrow 7 & \mathbf{s_5} \quad \mathbf{s_6} \quad \mathbf{s_7} \quad \mathbf{s_4} \end{array}$$

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

z_q	Specified $p_z(z_q)$	Actual $p_z(z_k)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19
$z_4 = 4$	0.20	0.25
$z_5 = 5$	0.30	0.21
$z_6 = 6$	0.20	0.24
$z_7 = 7$	0.15	0.11



示例：直方图匹配

$$s_0 = 1, s_1 = 3, s_2 = 5, s_3 = 6, s_4 = 6,$$
$$s_5 = 7, s_6 = 7, s_7 = 7.$$

r_k	s_k	$\rightarrow$	z_q
0	1	$\rightarrow$	3
1	3	$\rightarrow$	4
2	5	$\rightarrow$	5
3	6	$\rightarrow$	6
4	7	$\rightarrow$	7
5			
6			
7			

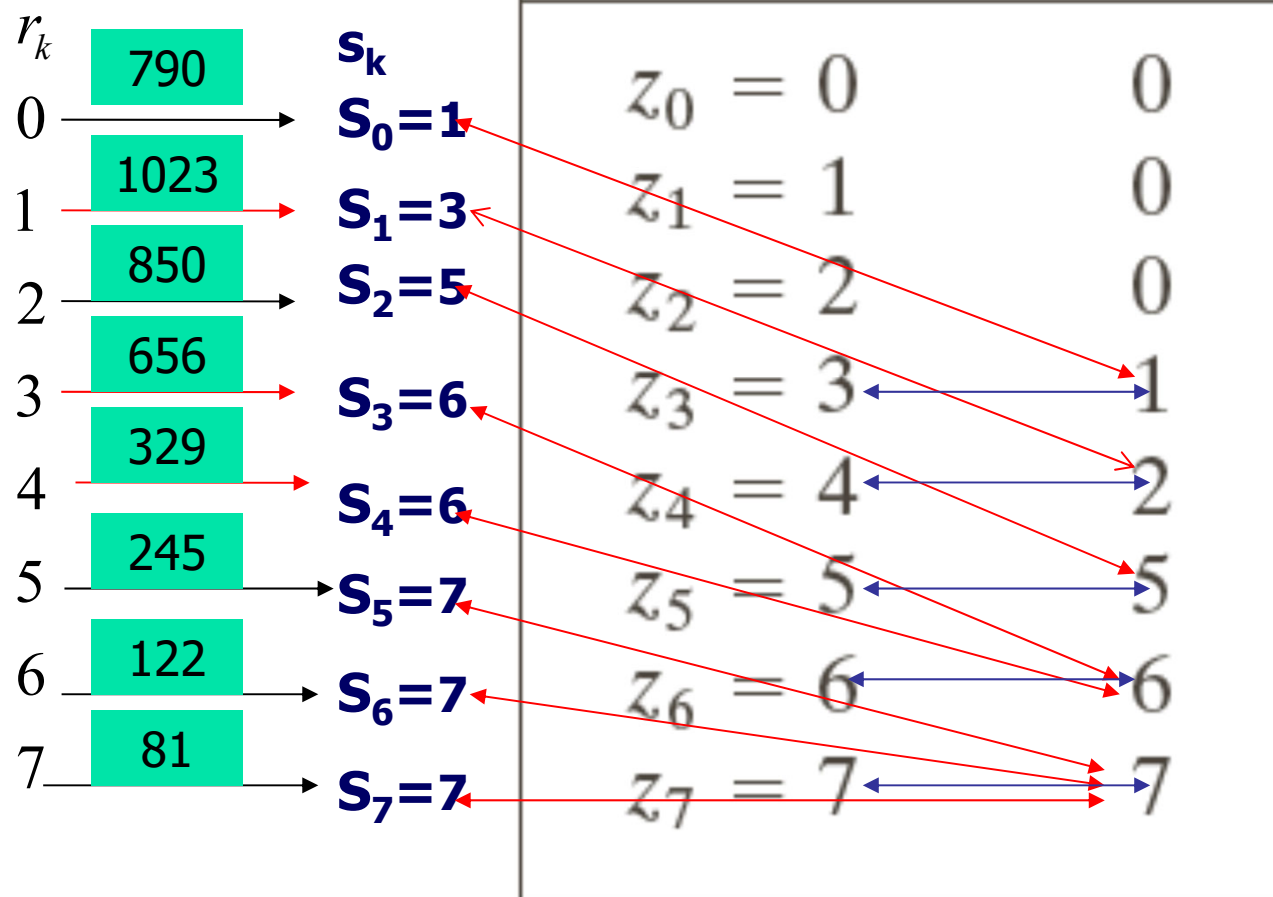


表3.3
变换函数G的所有可能值都是按z缩放，四舍五入，并有序的。

790
1023
850
656+329
245+122+81



示例：直方图匹配

$$r_k \rightarrow z_q$$

790	$0 \rightarrow 3$	790
1023	$1 \rightarrow 4$	1023
850	$2 \rightarrow 5$	850
656	$3 \rightarrow 6$	985
329	$4 \rightarrow 6$	
245	$5 \rightarrow 7$	
122	$6 \rightarrow 7$	448
81	$7 \rightarrow 7$	
4096		

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

z_q	Specified $p_z(z_q)$	Actual $p_z(z_k)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19
$z_4 = 4$	0.20	0.25
$z_5 = 5$	0.30	0.21
$z_6 = 6$	0.20	0.24
$z_7 = 7$	0.15	0.11

1023/4096



示例：直方图匹配

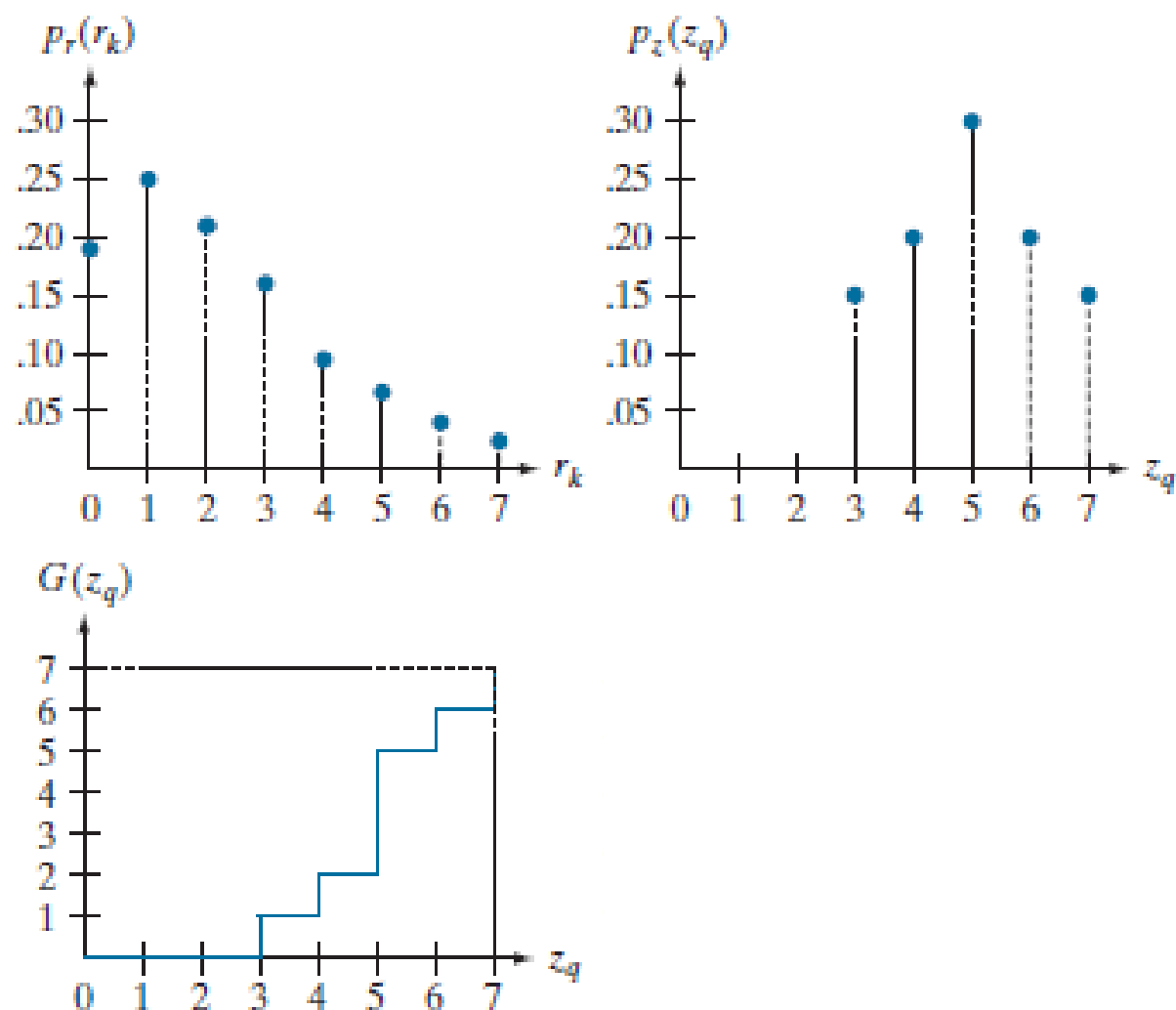


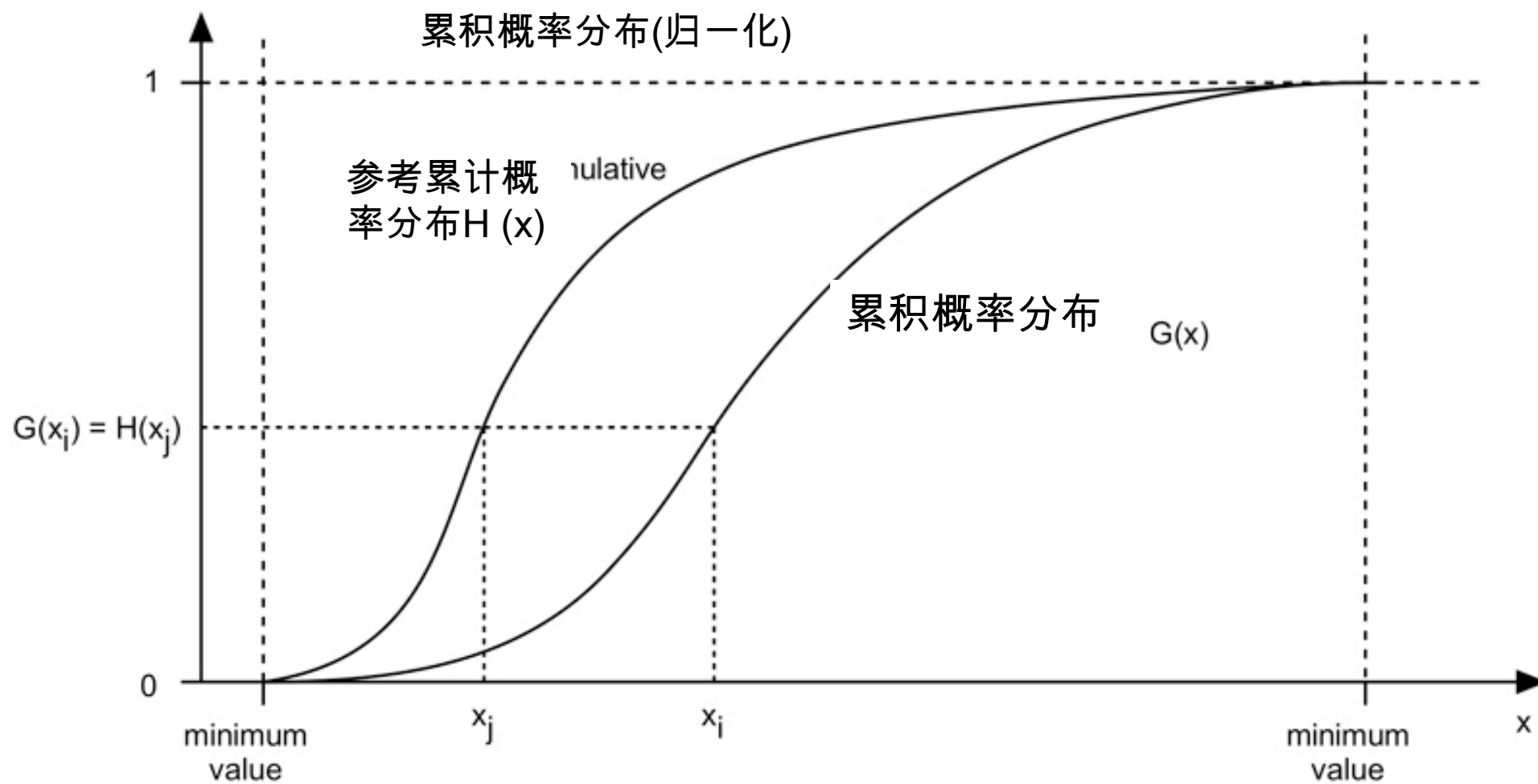
图3.22

(a) 3位bit图像的直方图。

(b) 指定直方图。

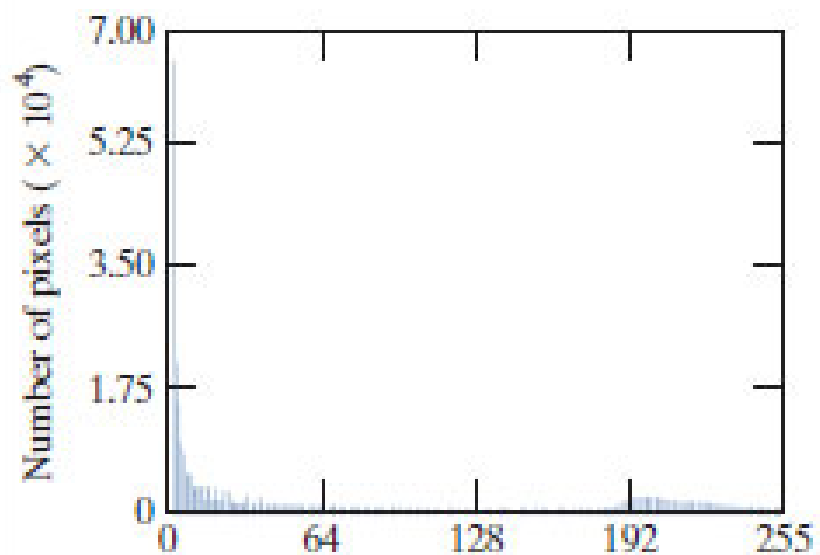
(c) 由指定直方图得到的变换函数。

(d) 直方图结果规范。比较b和d中的直方图。





示例：直方图匹配



a b

图3.23

(a)火星卫星火卫一的图像美国宇航局的火星全球的验船师。

(b)直方图。(原始图片由NASA提供。)



示例：直方图匹配

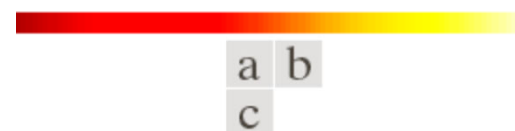
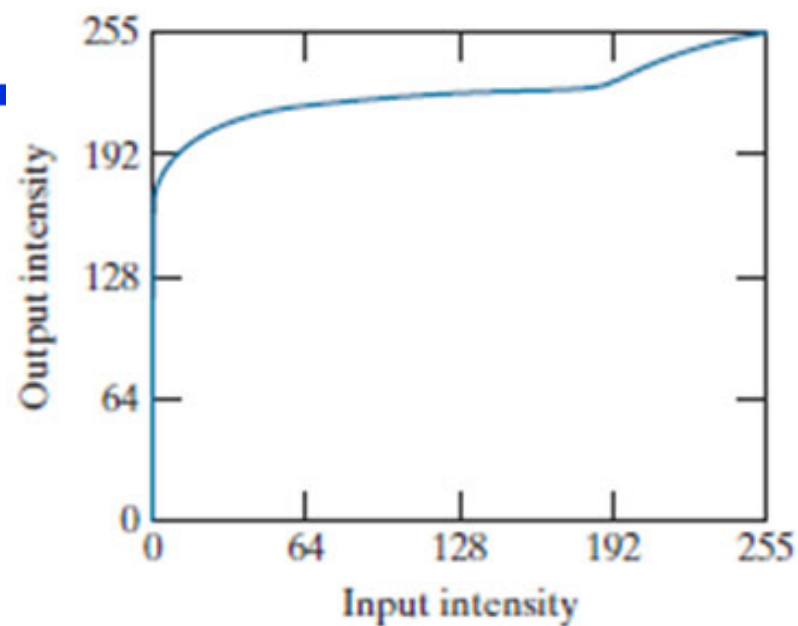
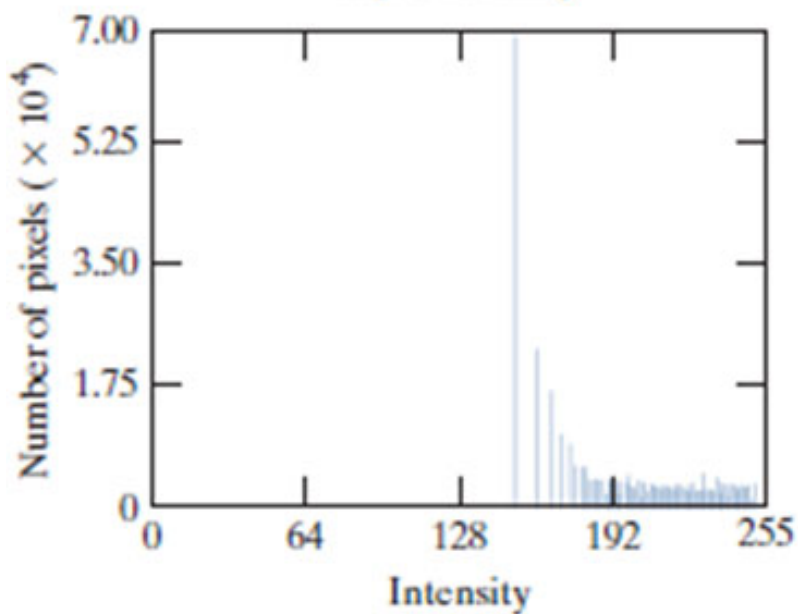


图3.24

(a)直方图变换函数
均衡。

(b)直方图均衡化图
像。

(c) b的直方图。



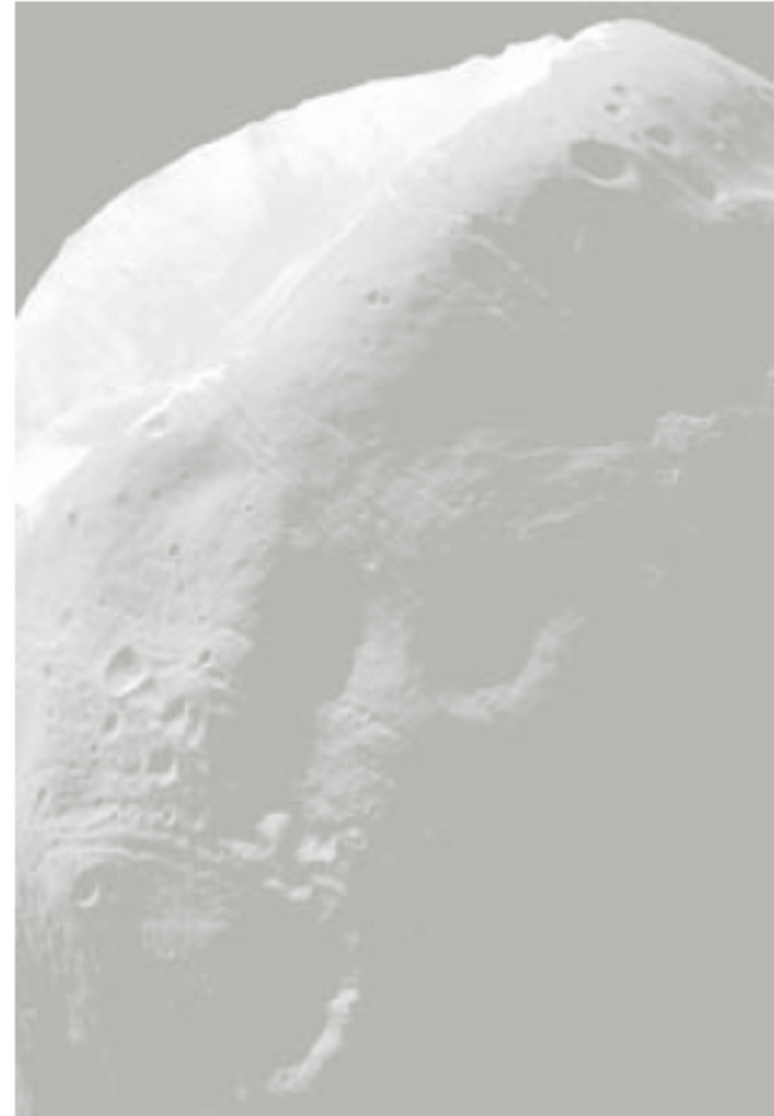


示例：直方图匹配

原始图像

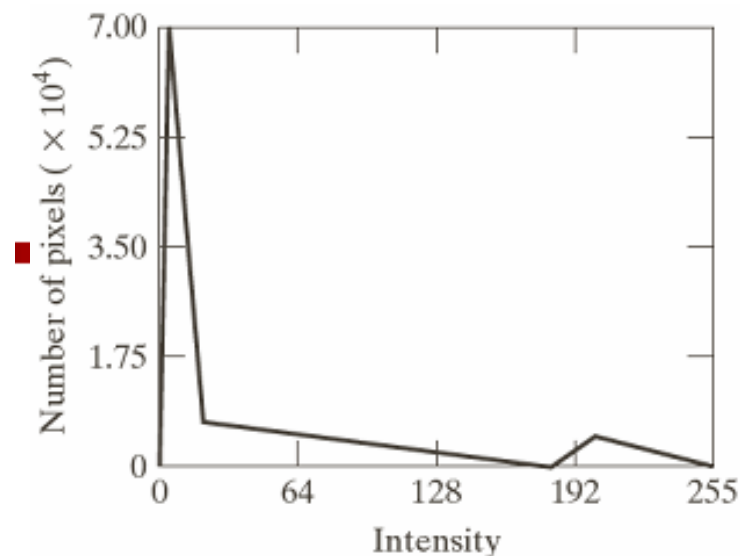
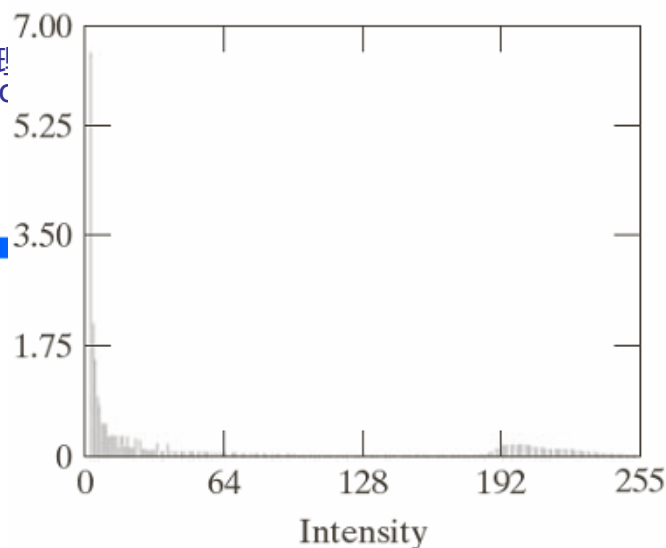


均衡图像

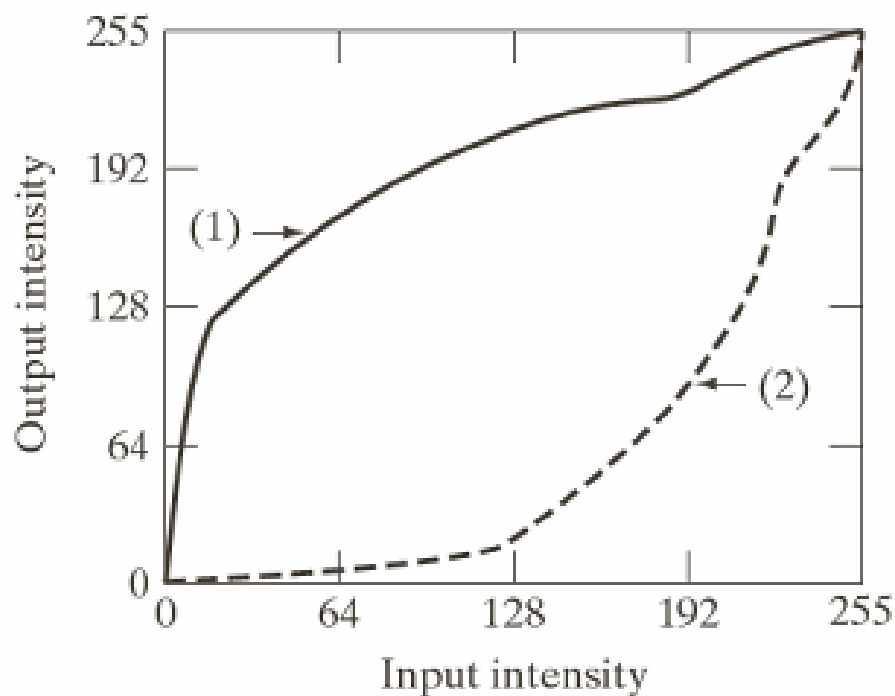




示例：直方图匹配



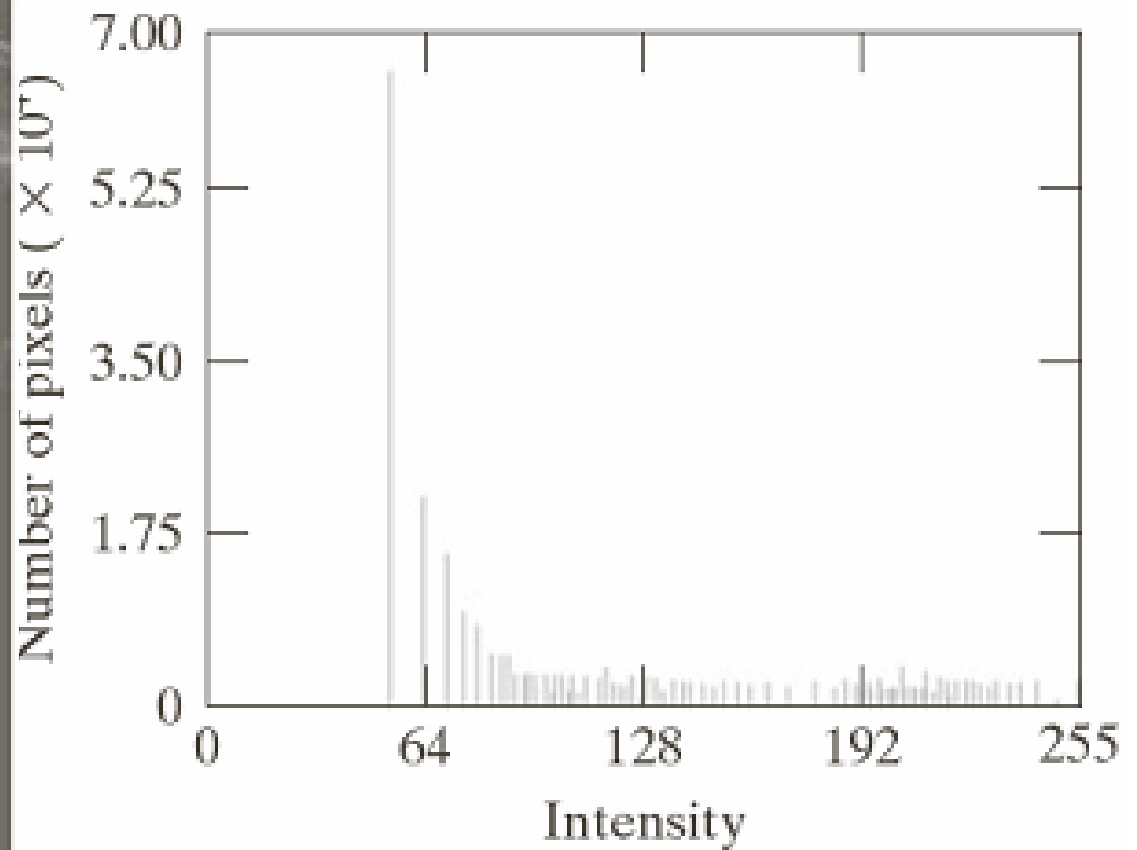
$$G(z_q) = (L-1) \sum_{i=0}^q p_z(z_i) = s_k$$



$$z_q = G^{-1}(s_k)$$

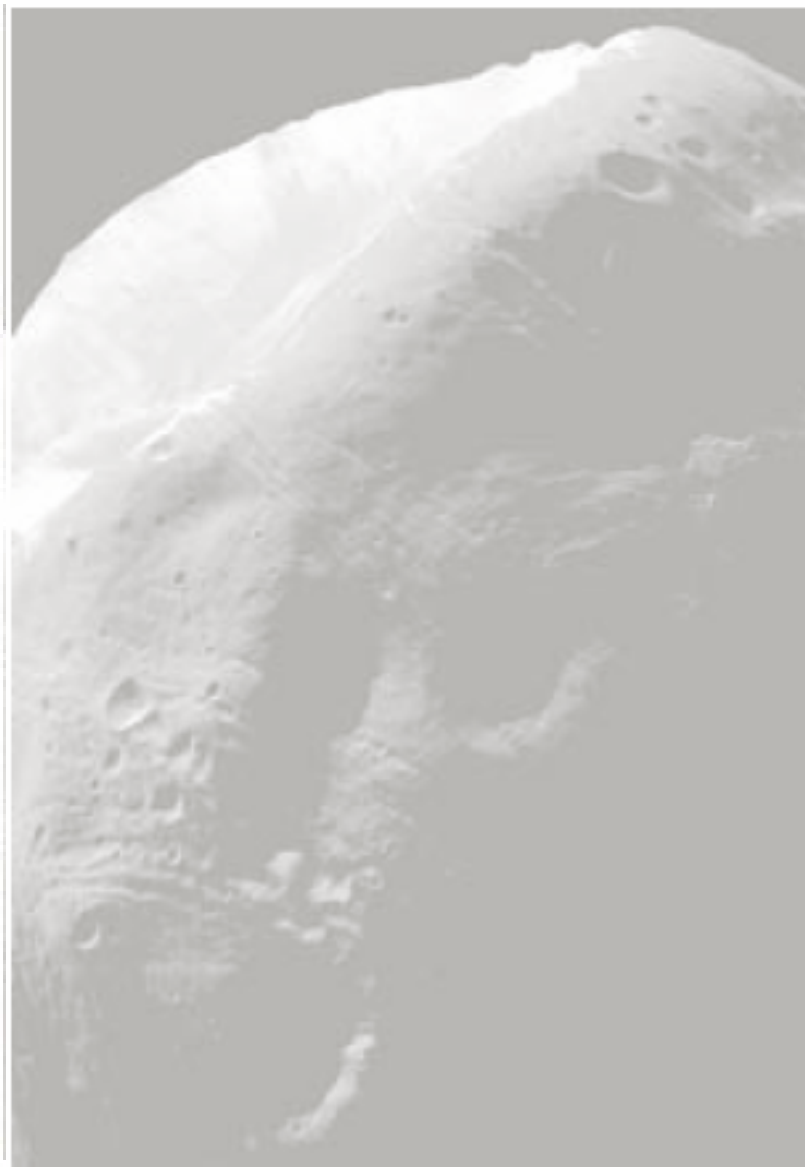


示例：直方图匹配





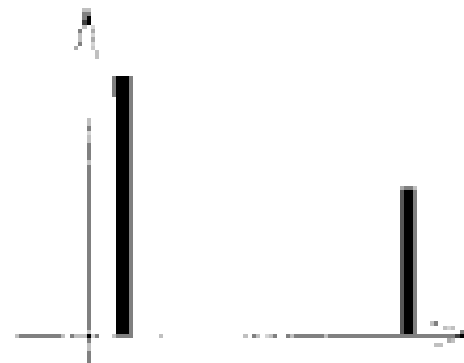
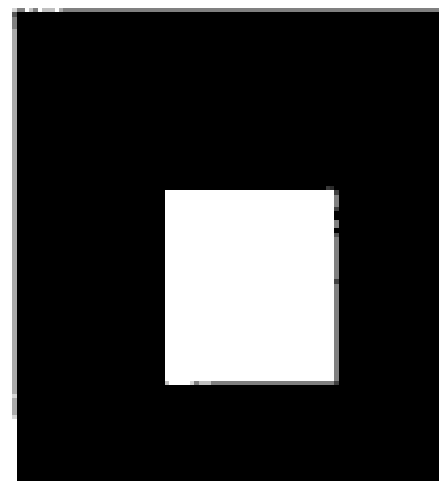
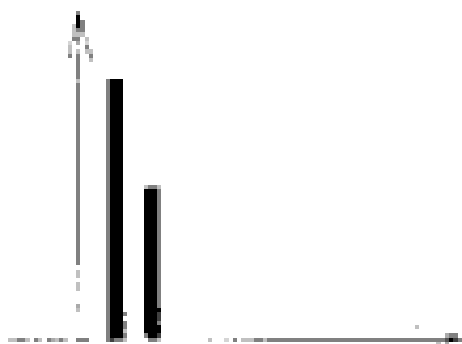
示例：直方图匹配





图像与视频处理
Image and Video Processing

直方图规格





图像与视频处理
Image and Video Processing

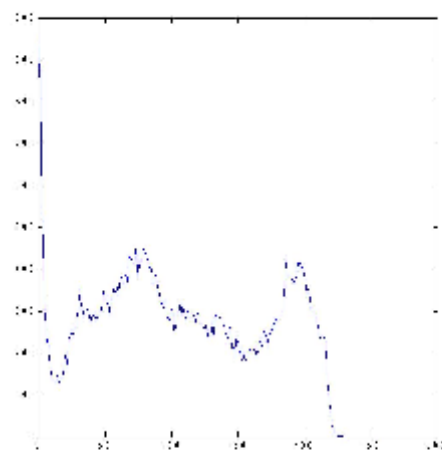
直方图规格



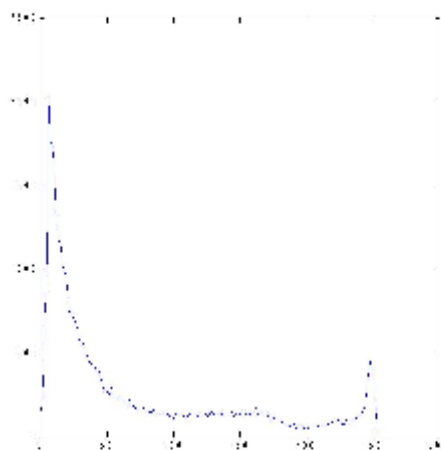
原始图像



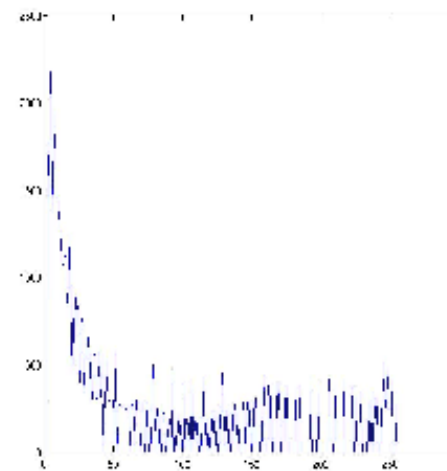
直方图修改后的图像



原始直方图



指定的直方图



修正后得到的直方图



目录

- 强度变换
 - 图像负片
 - 对数变换
 - 伽马变换
 - 分段线性变换
- 直方图处理
 - 直方图均衡
 - 直方图匹配
 - 局部直方图处理
- 空间过滤器
 - 空间相关性和卷积
 - 平滑线性过滤器和空间遮罩
 - 非线性滤波器
 - 锐化过滤器



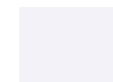
局部直方图处理

定义邻域并将其中心从一个像素移动到另一个像素

在每个位置，计算邻域中点的直方图。获得直方图均衡或直方图规范转换函数

映射以邻域为中心的像素的强度

移动到下一个位置并重复该过程





使用直方图统计进行图像增强

平均强度

$$m = \sum_{i=0}^{L-1} r_i p(r_i) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)$$

$$u_n(r) = \sum_{i=0}^{L-1} (r_i - m)^n p(r_i)$$

方差

$$\sigma^2 = u_2(r) = \sum_{i=0}^{L-1} (r_i - m)^2 p(r_i)$$

明显的差异

$$= \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x, y) - m]^2$$



使用直方图统计进行图像增强

区域平均强度

$$m_{s_{xy}} = \sum_{i=0}^{L-1} r_i p_{s_{xy}}(r_i)$$

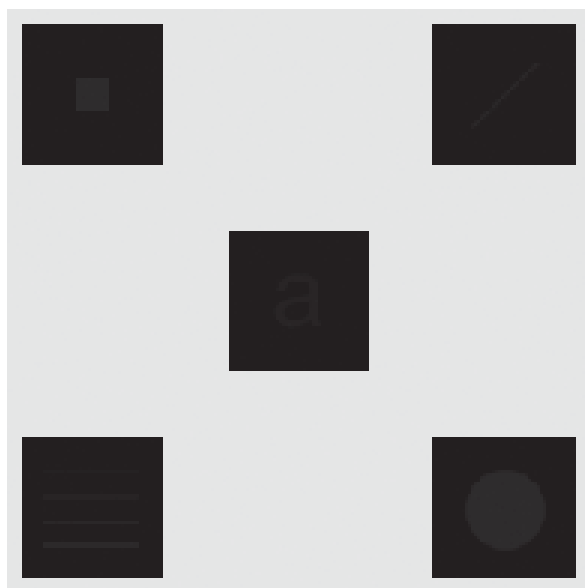
s_{xy} 表示邻域

局部方差

$$\sigma_{s_{xy}}^2 = \sum_{i=0}^{L-1} (r_i - m_{s_{xy}})^2 p_{s_{xy}}(r_i)$$



局部直方图处理：示例



a b c

FIGURE 3.26

(a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization.

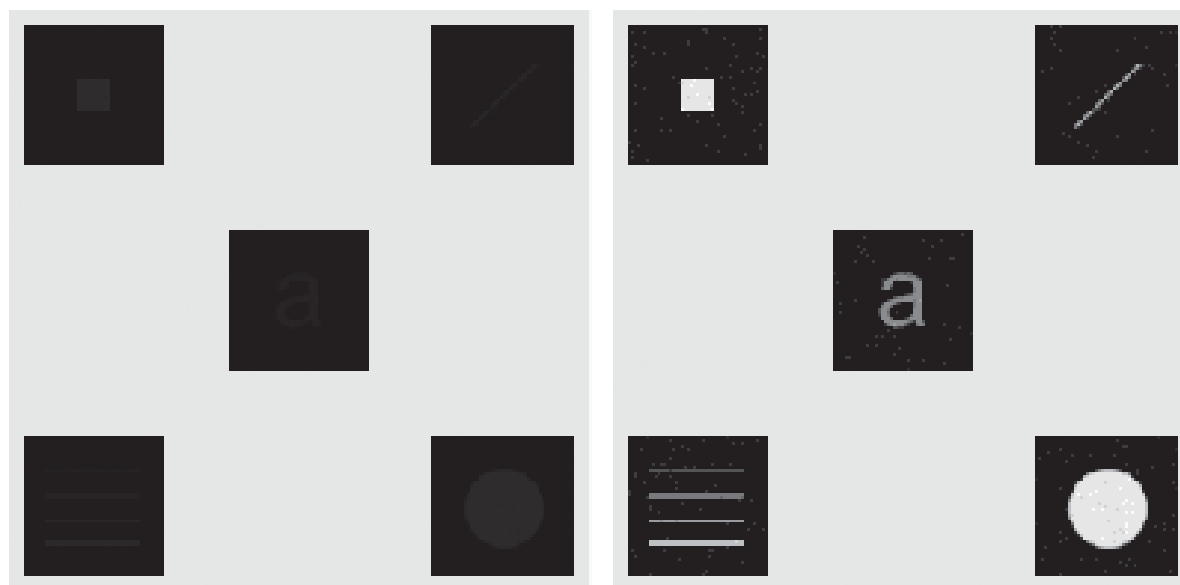
噪音，无更多细节

3*3 local

强度小，体积小
不影响全局HE



局部直方图处理：示例



a b c

a b

FIGURE 3.27
(a) Original image. (b) Result of local enhancement based on local histogram statistics. Compare (b) with Fig. 3.26(c).

局部增强

3*3 local

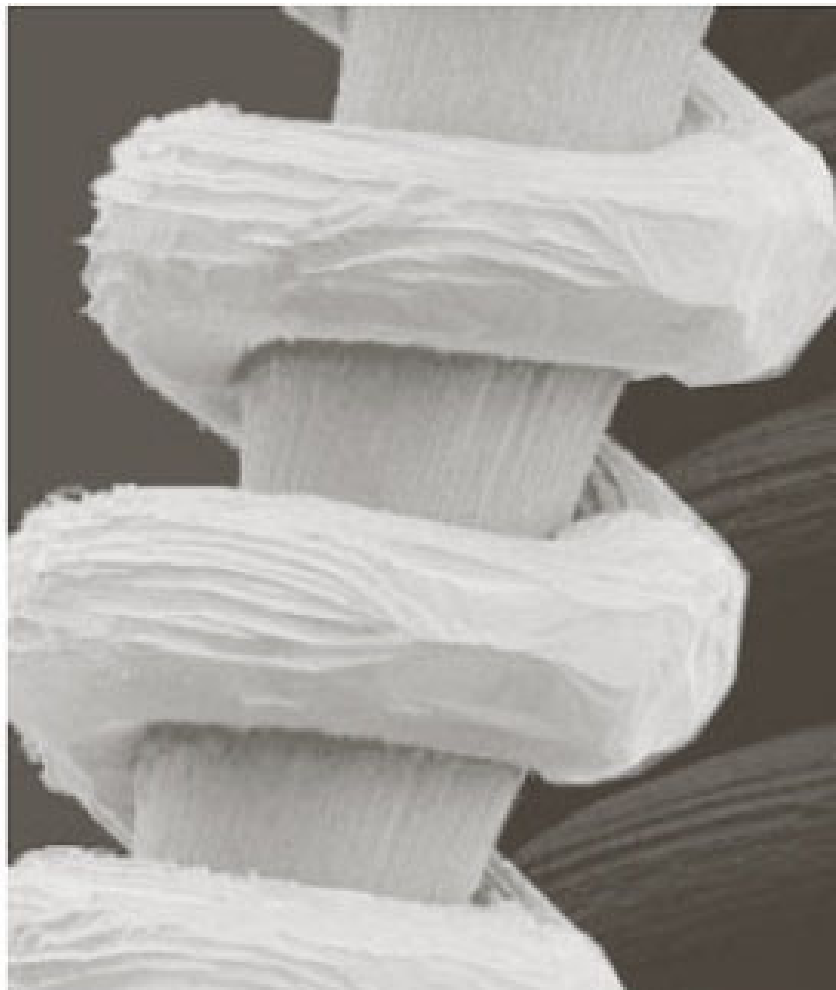
$$g(x, y) = \begin{cases} E \cdot f(x, y), & \text{if } k_0 m_G \leq m_{s_{xy}} \leq k_1 m_G \text{ and } k_2 \sigma_G \leq \sigma_{s_{xy}} \leq k_3 \sigma_G \\ f(x, y), & \text{otherwise} \end{cases}$$

m_G : 全局均值 σ_G : 全局标准差

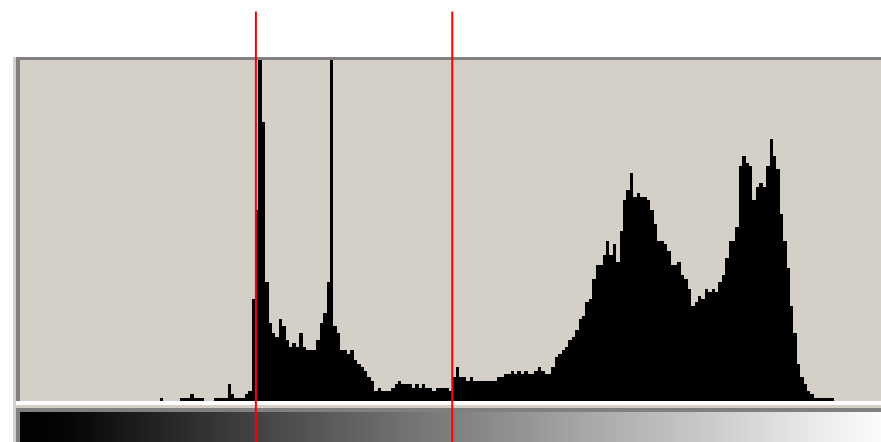
$k_0 = 0$; $k_1 = 0.1$; $k_2 = 0$; $k_3 = 0.1$; $E = 22.8$



使用直方图统计进行图像增强：示例



原始



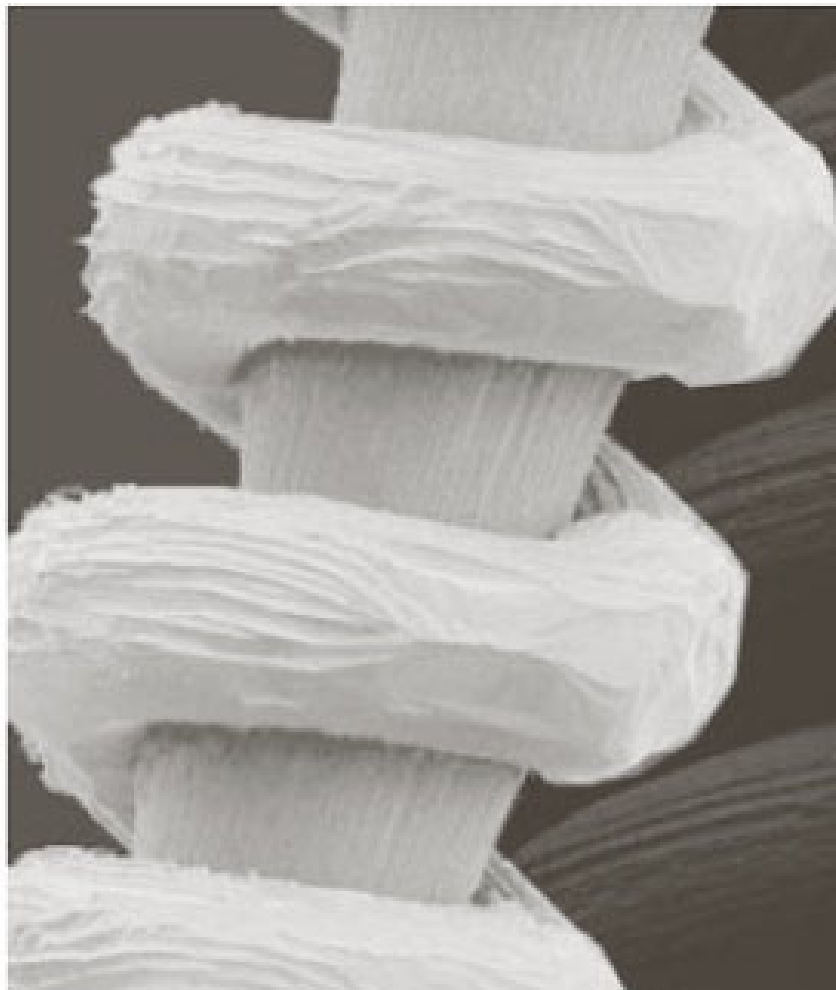
71

128

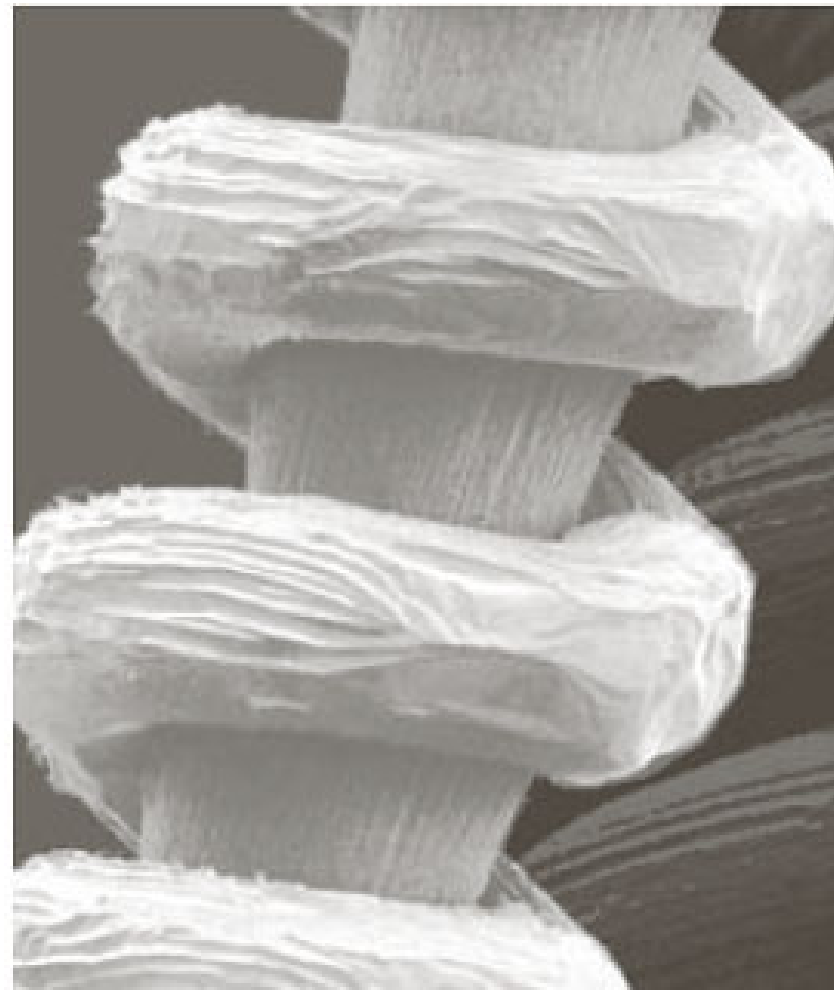
亮:不变
暗:变



使用直方图统计进行图像增强：示例



原始



全局



使用直方图统计进行图像增强：示例

$$g(x, y) = \begin{cases} E \cdot f(x, y), & \text{if } m_{s_{xy}} \leq k_0 m_G \text{ and } k_1 \sigma_G \leq \sigma_{s_{xy}} \leq k_2 \sigma_G \\ f(x, y), & \text{otherwise} \end{cases}$$

m_G : global mean; σ_G : global standard deviation

$k_0 = 0.4$; $k_1 = 0.02$; $k_2 = 0.4$; $E = 4$

$k_0 < 1, k_2 < 1$: 暗
 $k_0 > 1, k_2 > 1$: 亮
 $k_1 < k_2$

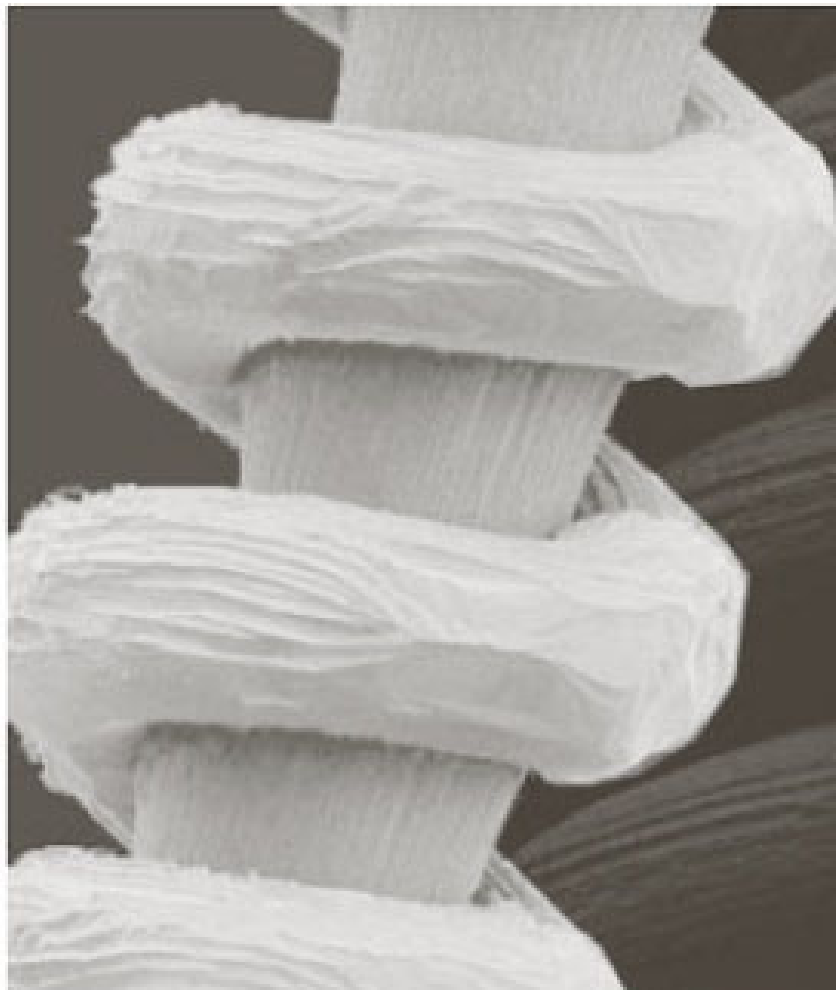
暗或亮:将局部平均强度
与图像平均强度进行比较



局部



使用直方图统计进行图像增强：示例



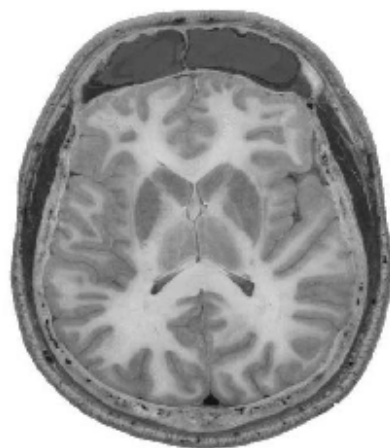
原始



全局-局部



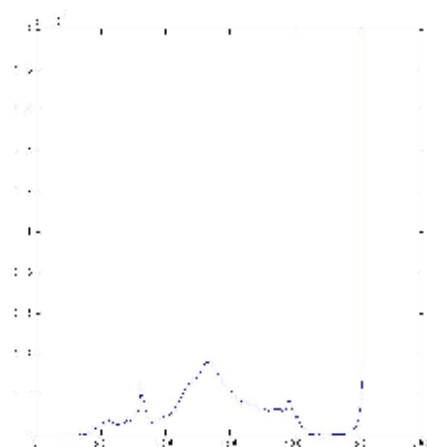
局部(自适应)直方图均衡化



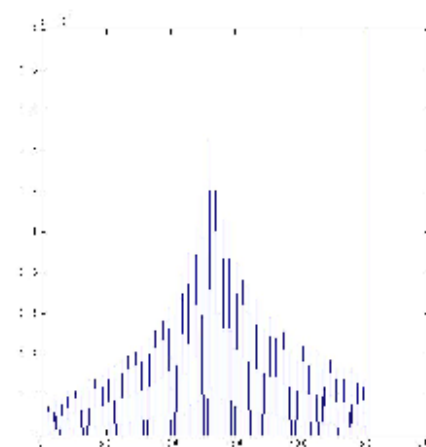
原始图像



7*7mask局部增强后的图像



原始直方图



局部增强后的直方图



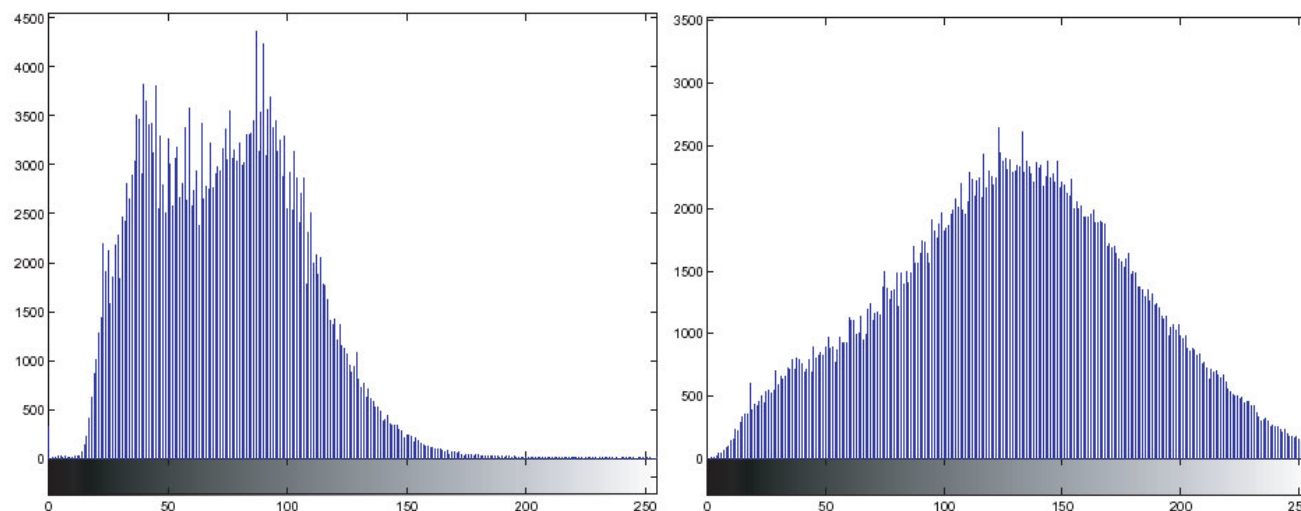
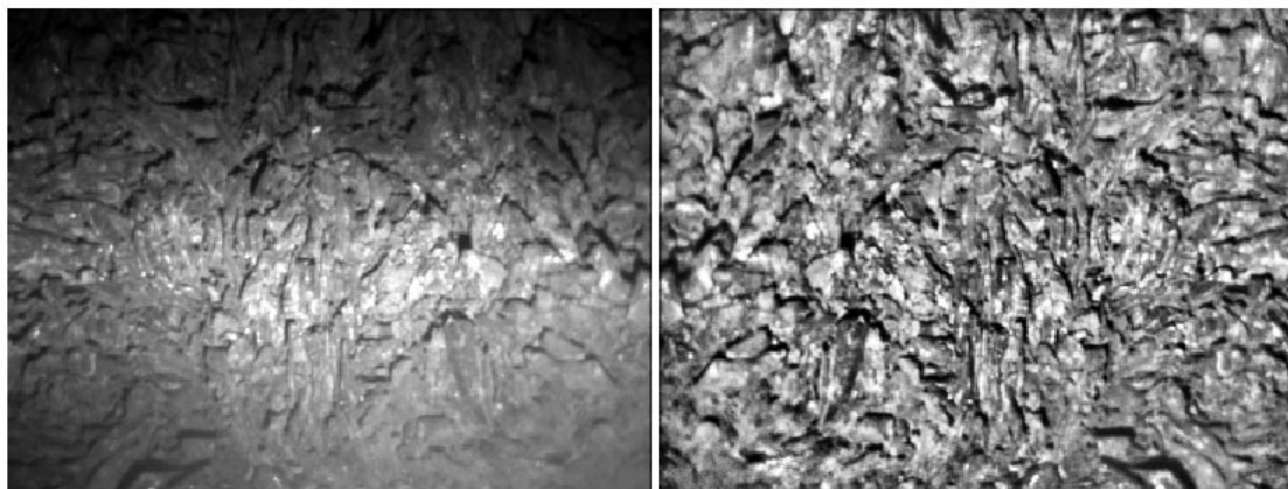
局部直方图均衡化





图像与视频处理
Image and Video Processing

局部直方图均衡化



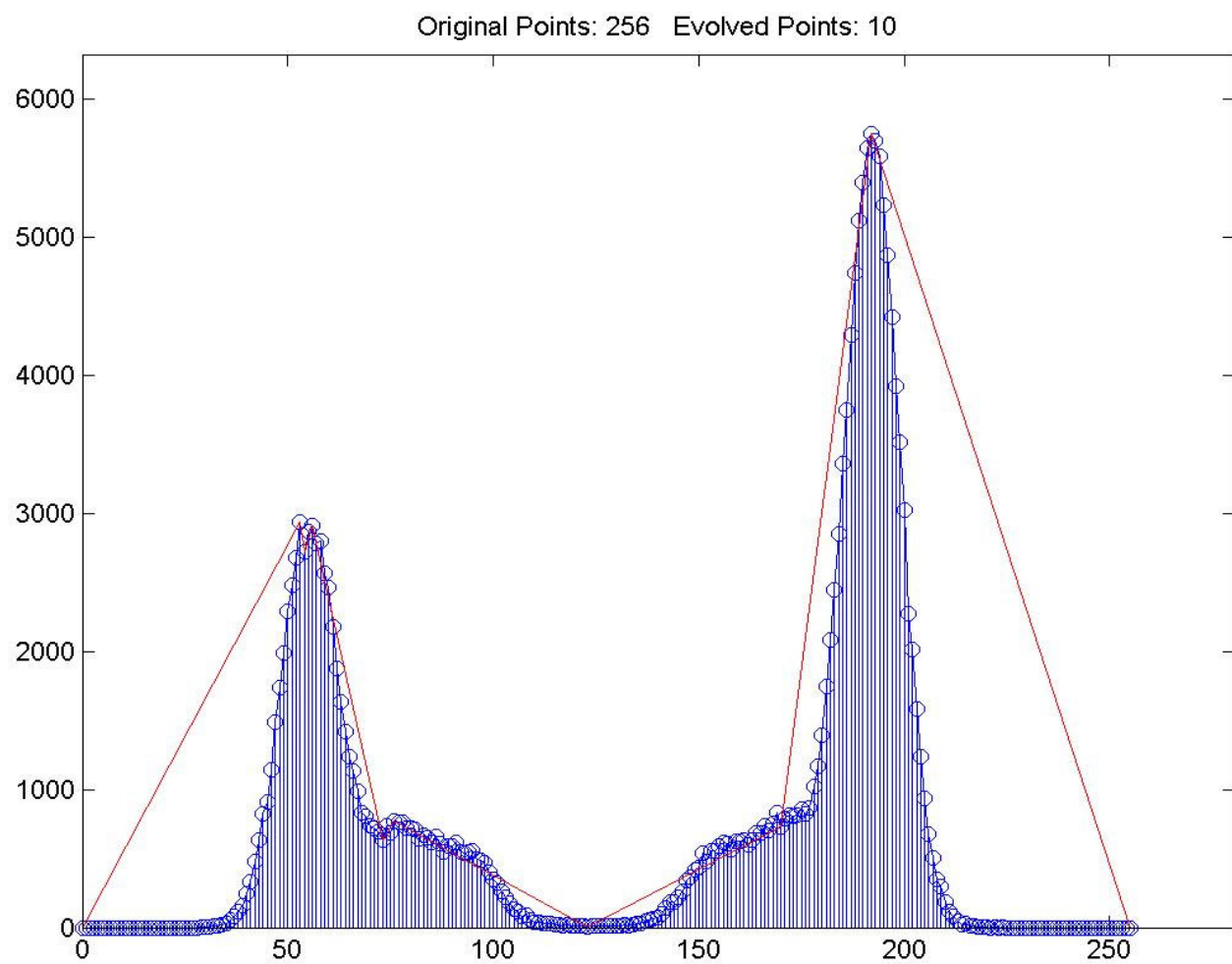


直方图:阈值分割-基础

- 假设灰度直方图对应于图像, $f(x,y)$,
- 由浅色背景中的深色物体组成
- 因此, 对象和背景像素的灰度级别分为两种主要模式
- 从背景中提取对象的一种明显方法是选择一个阈值 'T' 来分离这些模式
- 那么任意点 (x,y) 其中 $f(x,y) > T$ 为目标点, 否则, 此点称为背景点



例子





基础 (contd.)

- 如果图像直方图有两种主要模式，则称为双峰直方图。仅一个阈值就足以对图像进行分区
- 例如，如果一幅图像是由两种类型的浅色物体在黑暗背景上组成的，那么三种或三种以上的主导模式表征了图像直方图



基础 (contd.)

- 在这种情况下，直方图必须通过多个阈值进行划分.
- 多级阈值将一个点(x,y) 划分为一个对象类
如果 $T1 < f(x,y) \leq T2$,
到另一个对象类
如果 $f(x,y) > T2$
然后是背景
如果 $f(x,y) \leq T1$.



阈值处理

- 基本全局阈值法:

- 1) 选择一个初始估计的T
- 2) 使用T对图像进行分割。这将产生两组像素。G1由灰度值大于T的所有像素组成，G2由数值小于T的像素组成。
- 3) 计算区域G1和G2的像素的平均灰度值mean1和mean2。
- 4) 计算一个新的阈值
$$T = (1/2)(\text{mean1} + \text{mean2})$$
- 5) 重复步骤2到4，直到连续迭代中T的差异小于预定的参数T0。

- 基本自适应阈值法。具有不均匀光照的图像难以用直方图进行分割，这种方法是将原始图像分成子图像，并对每个子图像使用上述的阈值处理。



显示分割后的图像中的物体

- 通过给被阈值分隔的区域分配一个任意的像素值或平均像素值，可以区分出这些物体。



实验

- 使用的图像类型：
- 1) 具有双模态直方图结构的两幅灰度图像。
- 2) 具有多模态直方图结构的灰度图像。



图像与视频处理
Image and Video Processing

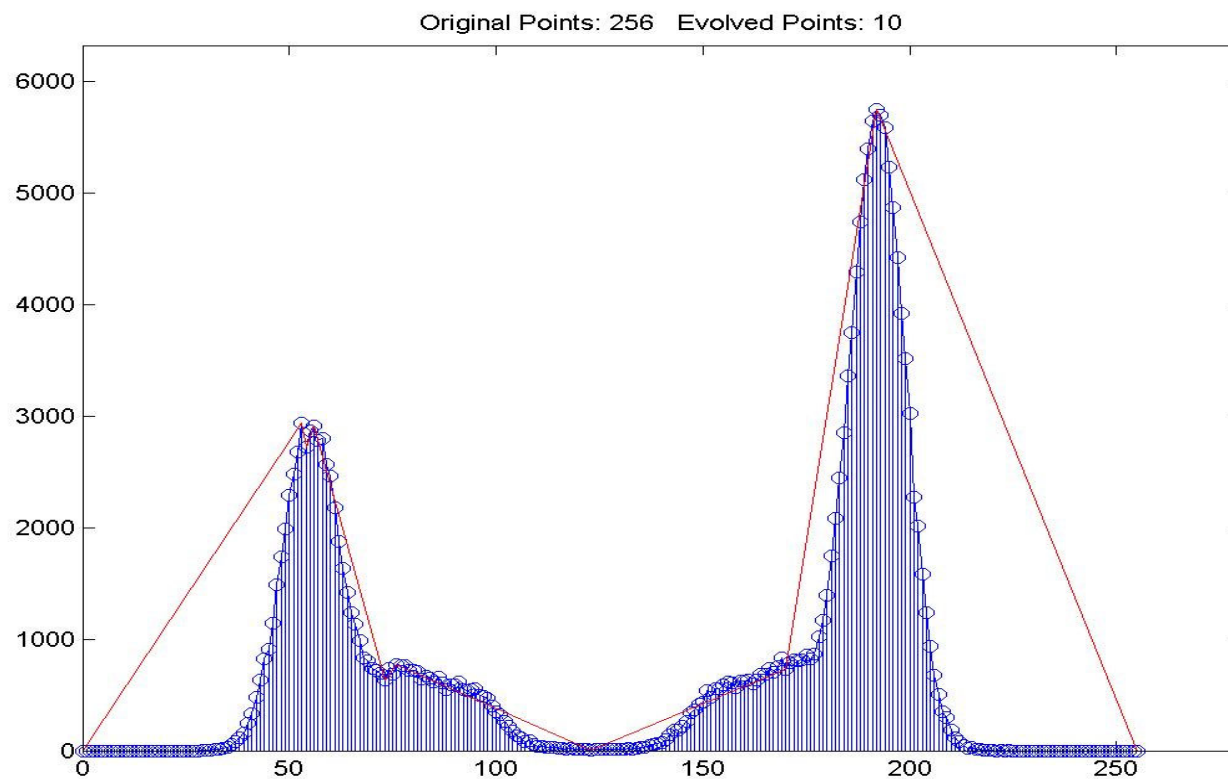
灰度图像--双峰型



浅色背景的指印图像



双峰 - 直方图



指纹的图像直方图



图像与视频处理
Image and Video Processing

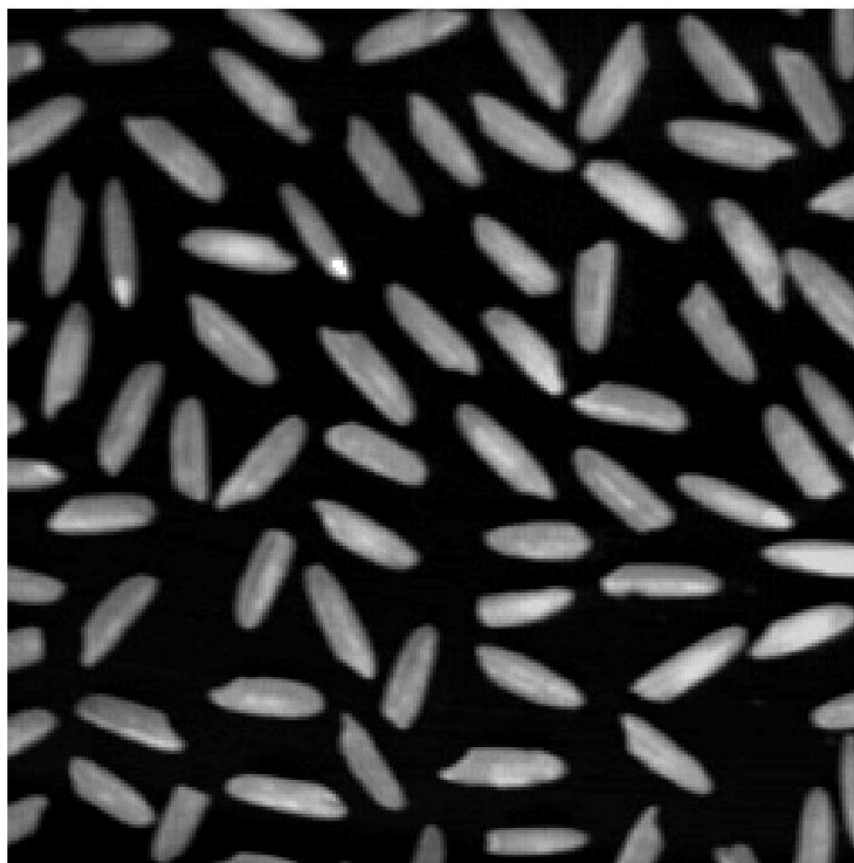
分割后的图像



分割后的图像



灰度图像 (2) -- 双模态

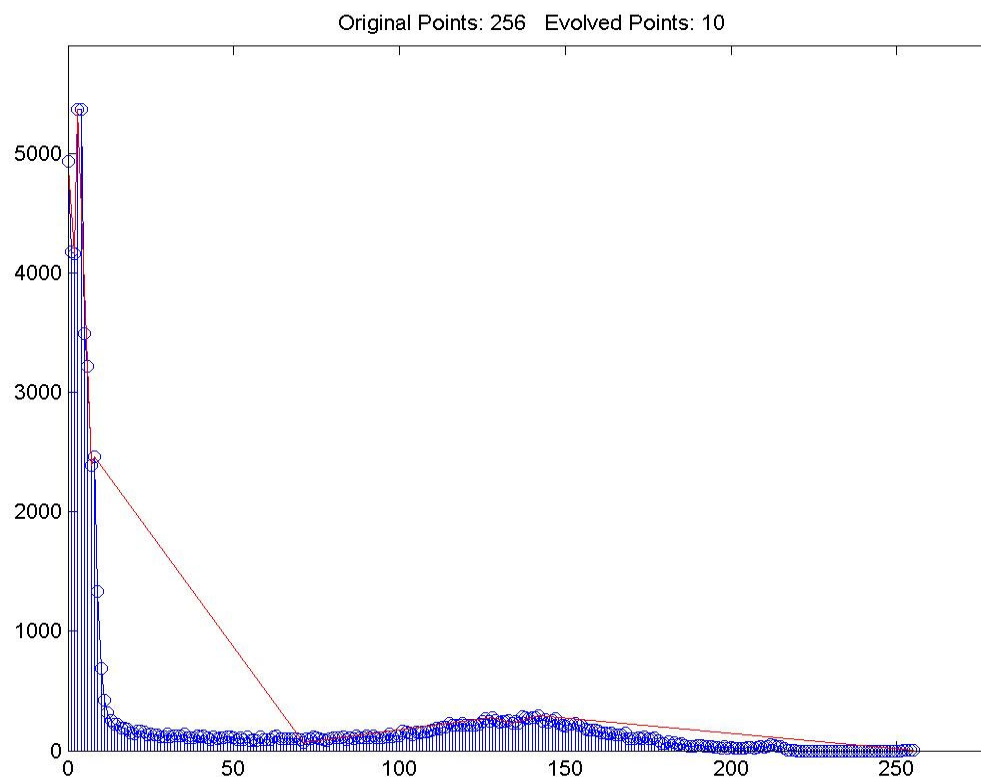


黑色背景的大米图像



图像与视频处理
Image and Video Processing

直方图

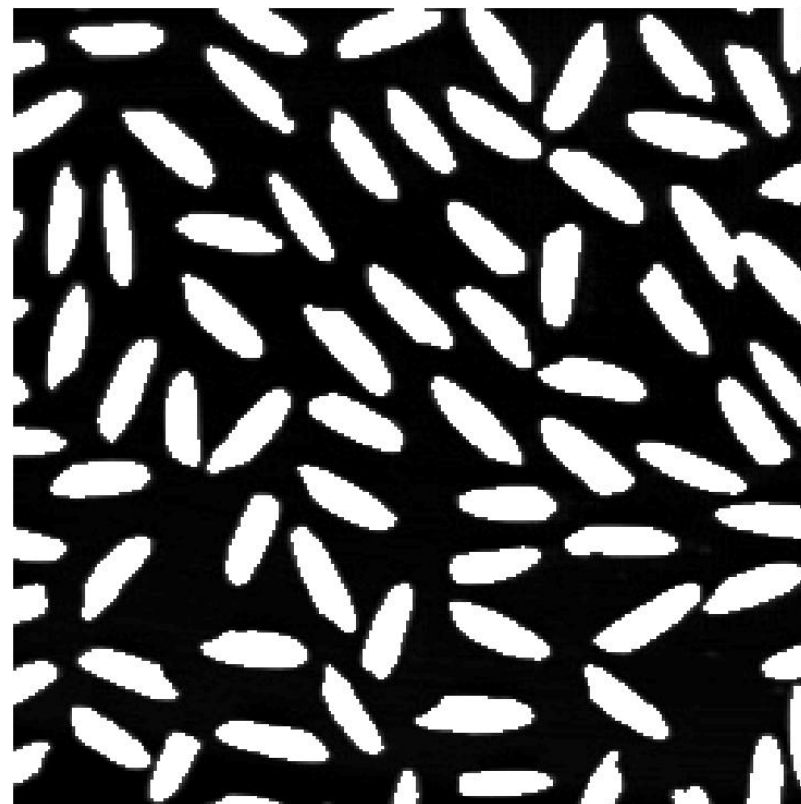
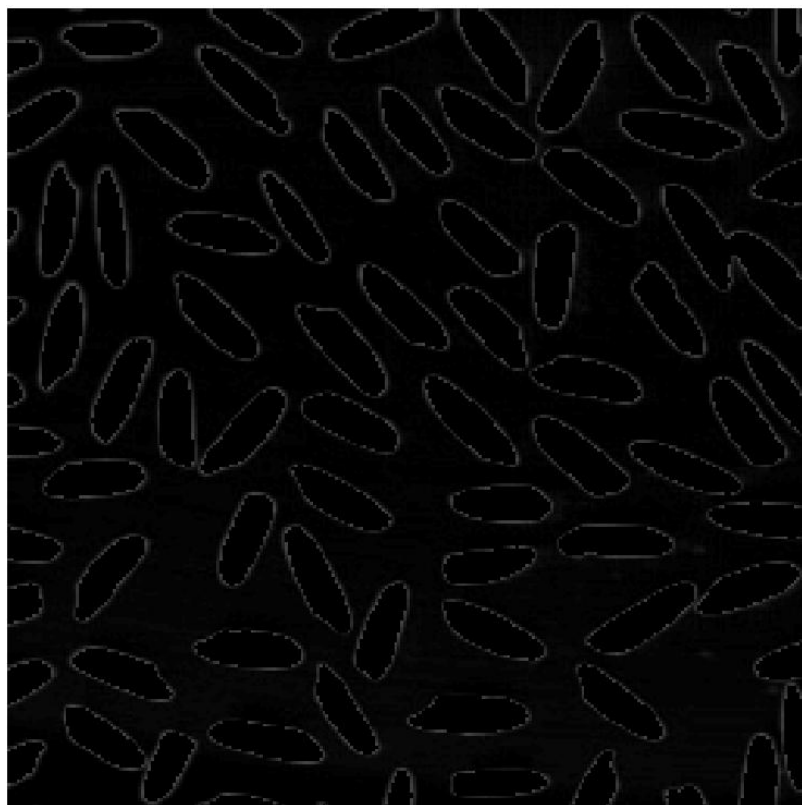


大米图像直方图



图像与视频处理
Image and Video Processing

分割后的图像



分割后的图像



图像与视频处理
Image and Video Processing

实验

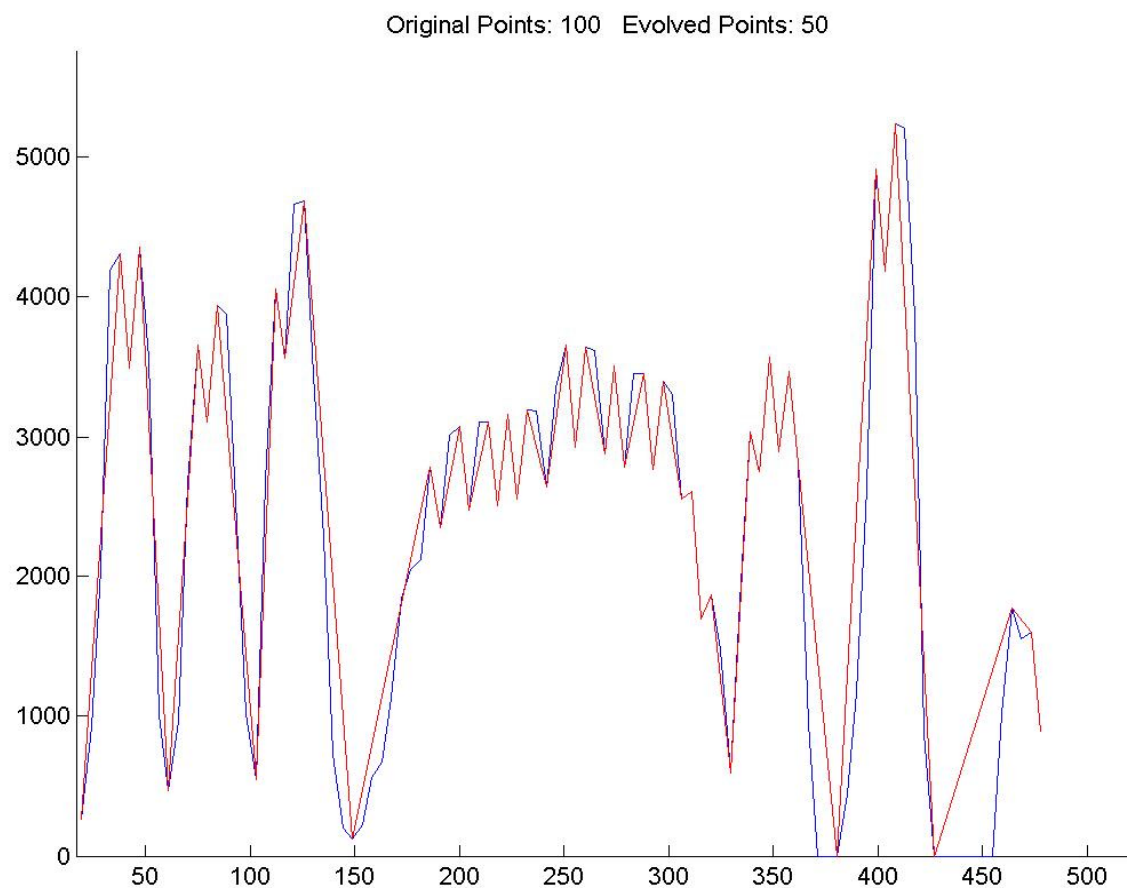


原始图像，目的是提取一行字



图像与视频处理
Image and Video Processing

直方图



通过绘制行数，直方图为我们提供了六个阈值点



图像与视频处理
Image and Video Processing

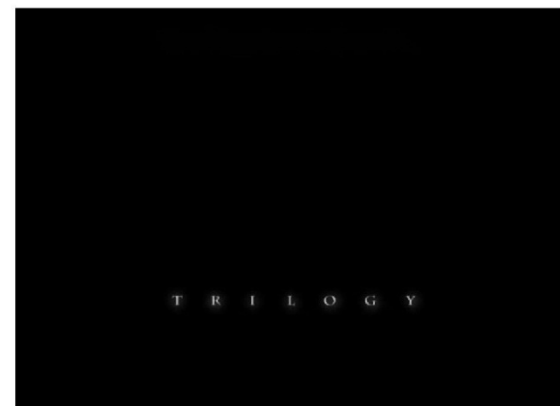
分割后的图像





图像与视频处理
Image and Video Processing

分割后的图像





总结

- 在对图像进行分割后，可以使用边缘检测技术来提取物体。
- 图像分割技术被广泛用于相似度搜索。



Content

- Intensity transformation
 - Image negative
 - Log transformation
 - Gamma transformation
 - Piecewise-linear transformation
- Histogram processing
 - Histogram equalization
 - Histogram matching
 - Local histogram processing
- Spatial filter
 - Spatial correlation and convolution
 - Smoothing linear filters&Spatial masks
 - Nonlinear filters
 - Sharpening filters



空间滤波

一个空间滤波器包括 (a) 一个邻域, 和 (b) 一个预定的操作

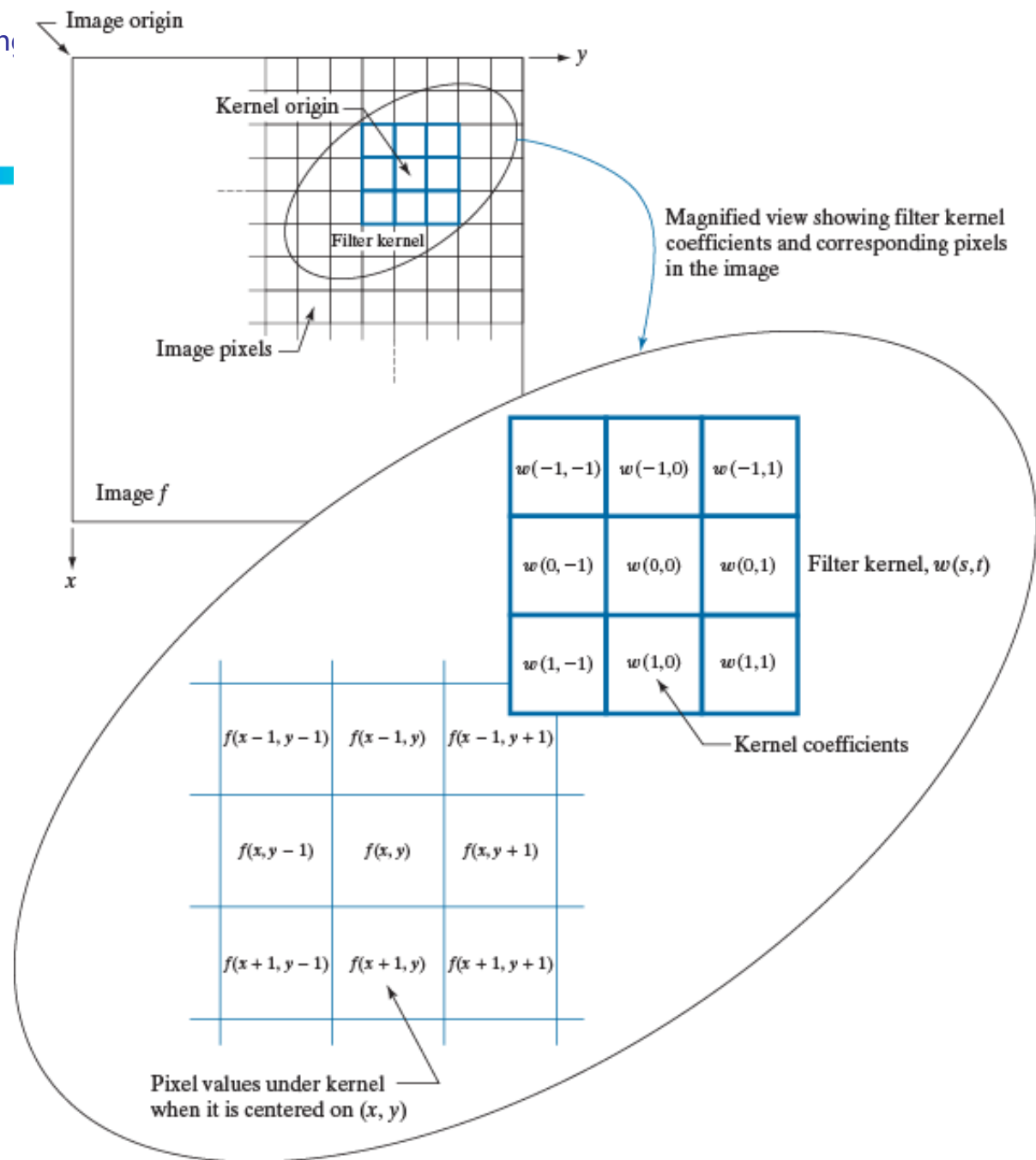
用大小为 $m * n$ 的滤波器对大小为 $M * N$ 的图像进行线性空间滤波, 其表达式为

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

一个新创建的像素, 其坐标等于邻域中心的坐标



空间滤波





空间关联

大小为 $m * n$ 的滤波器 $w(x,y)$ 与图像 $f(x,y)$ 的相关性，表示为

$$w(x, y) \star f(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

1. 相关性是过滤器位移的一个函数。
2. 与离散单位脉冲相关的函数的结果与函数相反；
3. 相同函数的相关性，在中心处产生最大值，用来寻找图像之间的匹配对



空间卷积

大小为 $m * n$ 的滤波器 $w(x,y)$ 与图像 $f(x,y)$ 的卷积，表示为

$$w(x, y) \star f(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x-s, y-t)$$

1. 与离散单位脉冲的函数卷积的结果产生一个函数的副本;
2. 将过滤器旋转180，使用相关性进行计算

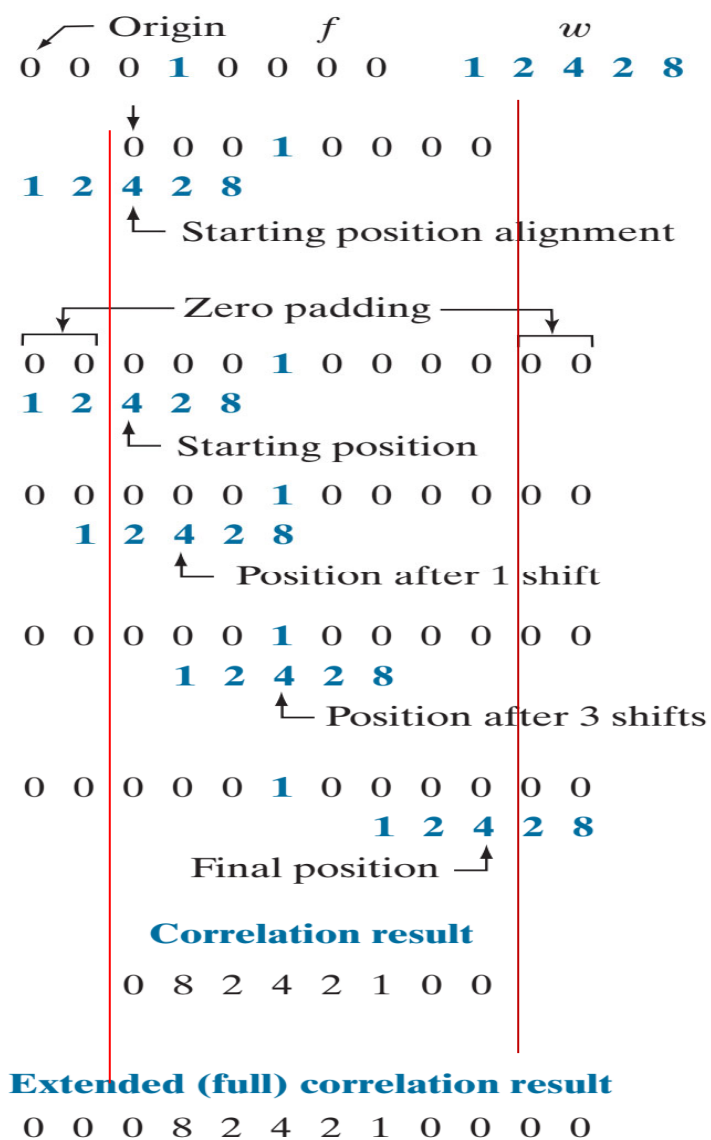


相关性

滤波器：1xm
(m-1)/2的零阵列

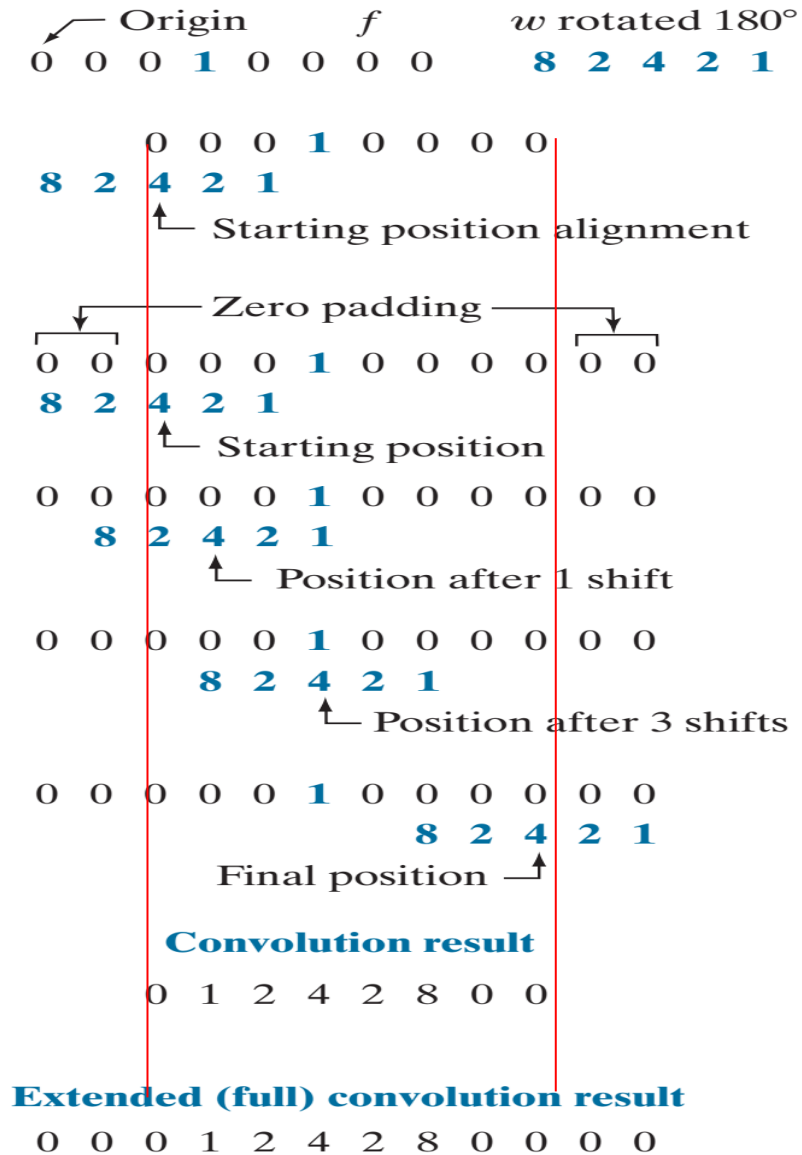
$$g(x) = \sum_{s=-a}^a w(s)f(x+s)$$

$$g(0) = \sum_{s=-2}^2 w(s)f(s+0) = 0$$





卷积





	Origin					f		
$\downarrow$	0	0	0	0	0			
	0	0	0	0	0			
	0	0	1	0	0	1	2	3
	0	0	0	0	0	4	5	6
	0	0	0	0	0	7	8	9
						w		

(a)

FIGURE 3.30

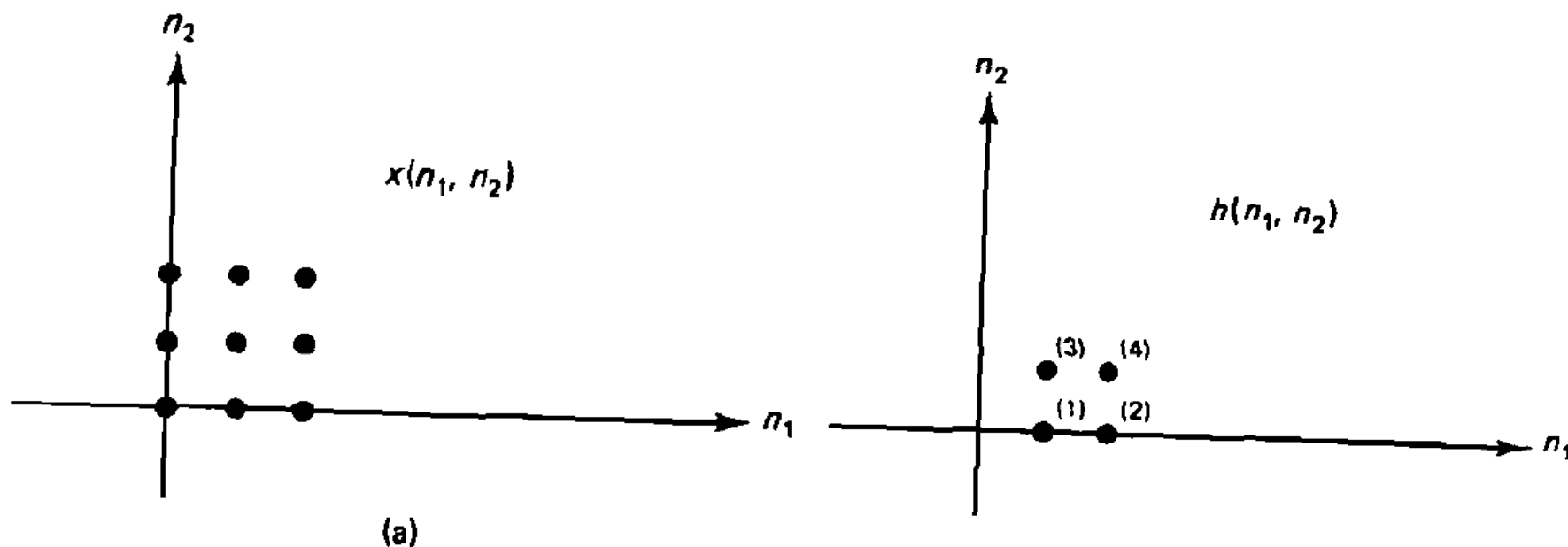
Correlation (middle row) and convolution (last row) of a 2-D filter with a 2-D discrete, unit impulse. The 0s are shown in gray to simplify visual analysis.



2维卷积

$$y[n_1, n_2] = x[n_1, n_2] * h[n_1, n_2]$$

$$= \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} x[k_1, k_2] \times h[n_1 - k_1, n_2 - k_2]$$

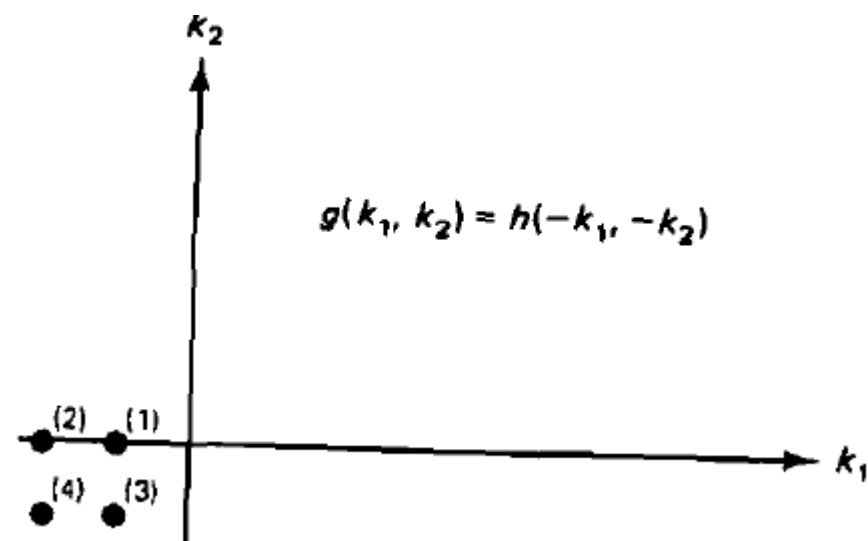
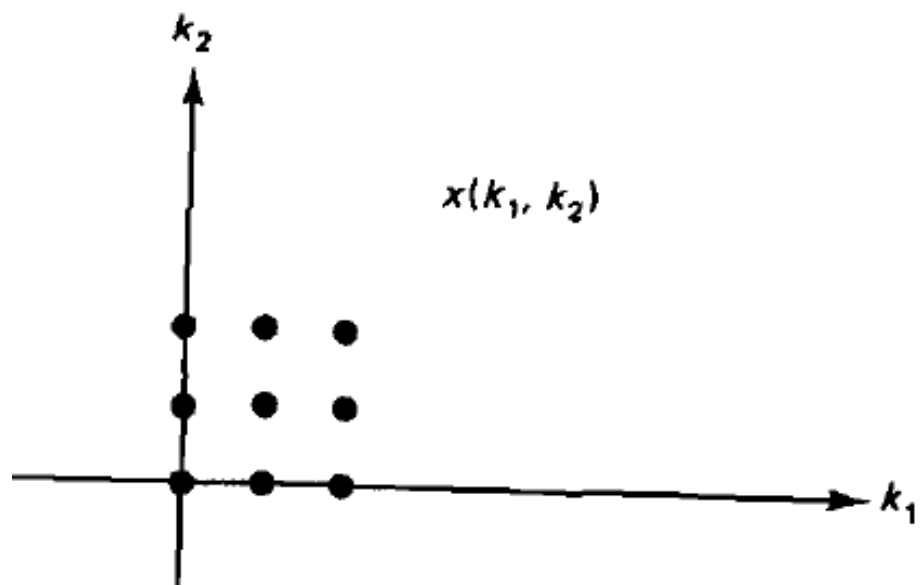




2维卷积

$$y[n_1, n_2] = x[n_1, n_2] * h[n_1, n_2]$$

$$= \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} x[k_1, k_2] \times h[n_1 - k_1, n_2 - k_2]$$

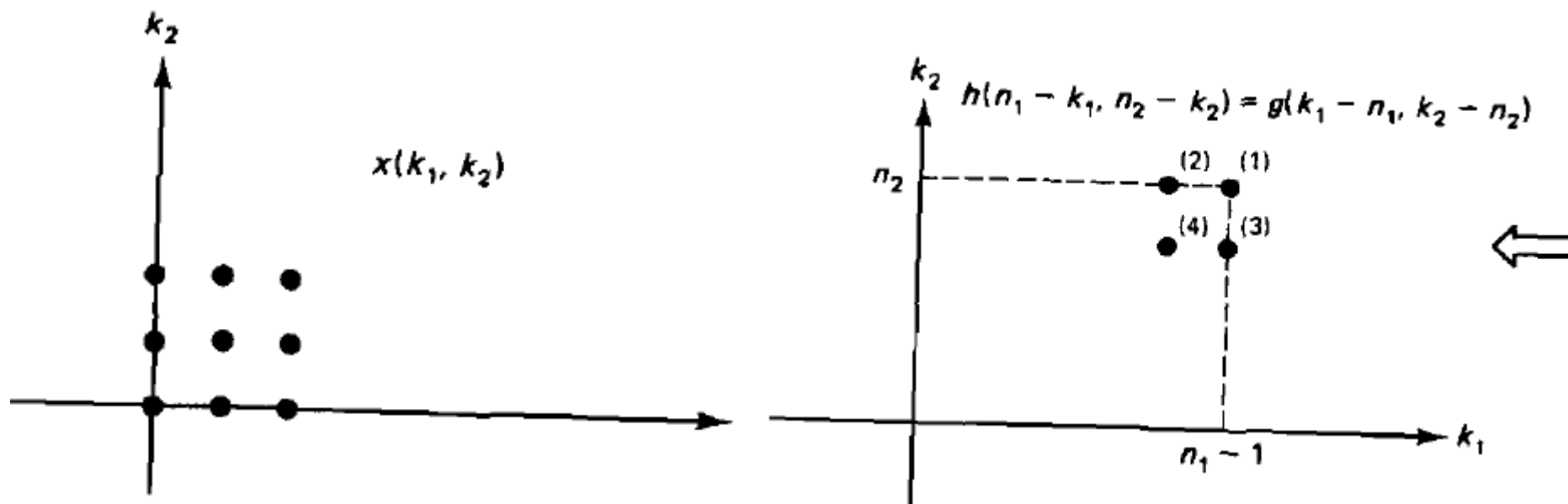


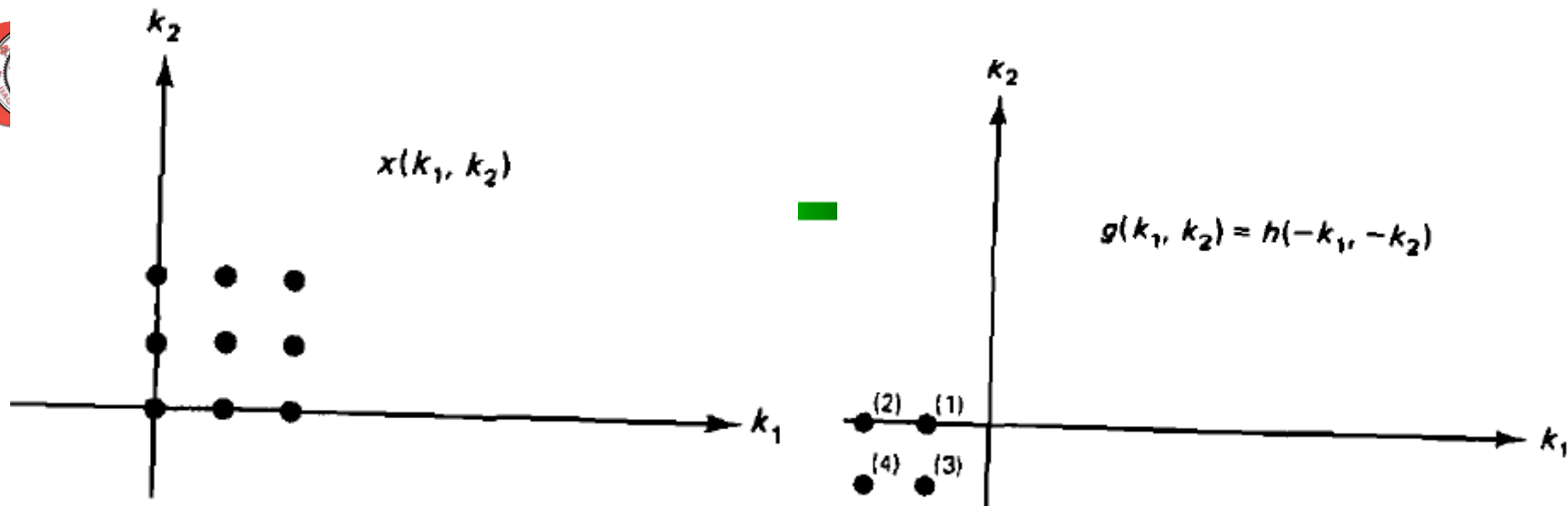


2维卷积

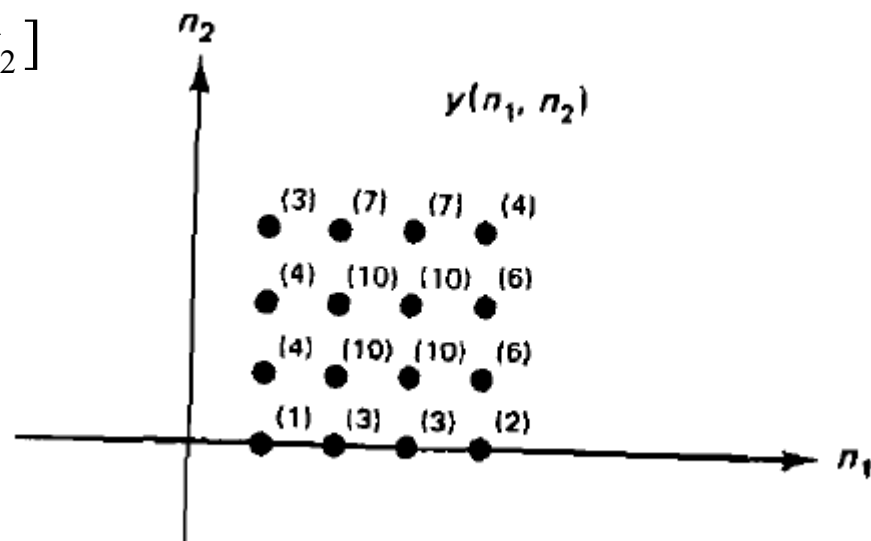
$$y[n_1, n_2] = x[n_1, n_2] * h[n_1, n_2]$$

$$= \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} x[k_1, k_2] \times h[n_1 - k_1, n_2 - k_2]$$





$$y[n_1, n_2] = x[n_1, n_2] * h[n_1, n_2]$$
$$= \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} x[k_1, k_2] \times h[n_1 - k_1, n_2 - k_2]$$





Content

- Intensity transformation
 - Image negative
 - Log transformation
 - Gamma transformation
 - Piecewise-linear transformation
- Histogram processing
 - Histogram equalization
 - Histogram matching
 - Local histogram processing
- Spatial filter
 - Spatial correlation and convolution
 - **Smoothing linear filters&Spatial masks**
 - Nonlinear filters
 - Sharpening filters



平滑空间滤波器

平滑过滤器用于模糊和降噪

模糊用于在提取物体之前去除小的细节，以及弥补线条或曲线中的小缝隙。

平滑的空间滤波器包括线性滤波器和非线性滤波器。



平滑空间滤波器

平均滤波器：低通滤波器

减少强度水平的尖锐过渡：降噪

平滑空间滤波器：模糊边缘（尖锐的强度过渡），平滑虚假轮廓。

减少不相关的细节，这意味着像素区域相对于滤波器掩码的大小来说是很小的。



空间平滑线性滤波器

用大小为 $m \times n$ 的加权平均滤波器过滤 $M \times N$ 的图像的一般实现方法为：

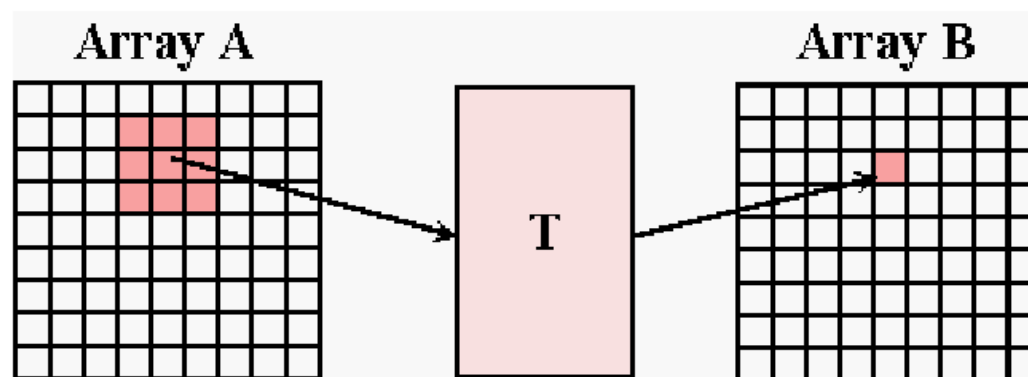
$$g(x, y) = \frac{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)}{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t)}$$

where $m = 2a + 1$, $n = 2b + 1$.



图像与视频处理
Image and Video Processing

空间掩码



		w_1	w_2	w_3
		w_4	w_5	w_6
		w_7	w_8	w_9



w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

FIGURE 3.31
Another
representation of
a general 3×3
filter mask.



两个平滑平均滤波器掩码

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$
$$\frac{1}{4.8976} \times \begin{bmatrix} 0.3679 & 0.6065 & 0.3679 \\ 0.6065 & 1.0000 & 0.6065 \\ 0.3679 & 0.6065 & 0.3679 \end{bmatrix}$$

$$\frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

加权平均

a b

FIGURE 3.32 Two 3×3 smoothing (averaging) filter masks. The constant multiplier in front of each mask is equal to 1 divided by the sum of the values of its coefficients, as is required to compute an average.

将系数值作为与原点距离增加的函数来减小
这只是减少模糊的一种尝试

主观题 10分

⚙️ 设置



为什么低通滤波器得系数平均为1

正常使用主观题需2.0以上版本雨课堂

作答



例子：空间低通掩码（局部平均）

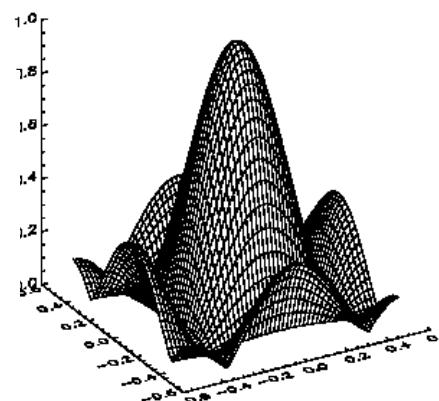
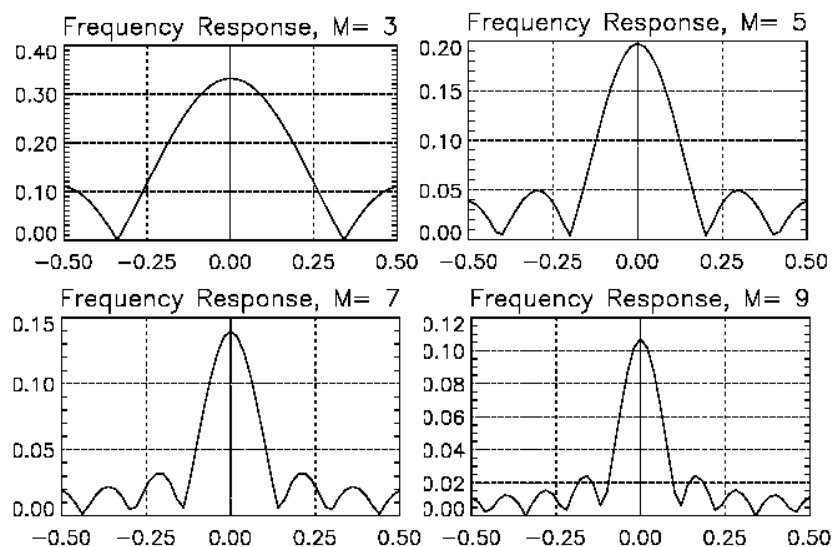
$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

$$\frac{1}{25} \times \begin{array}{|c|c|c|c|c|} \hline 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 \\ \hline \end{array}$$

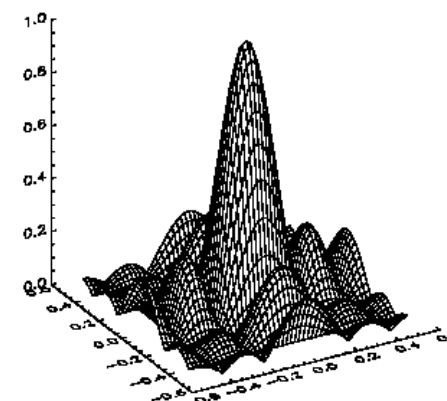
[illegible]



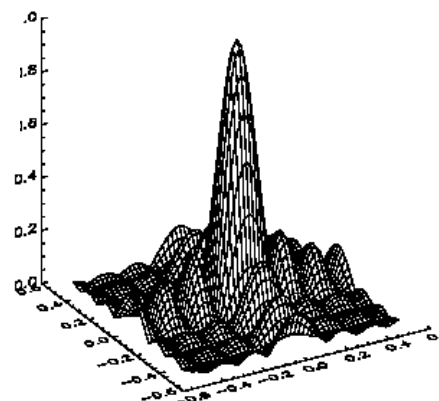
一维和二维局部平均掩码的反应



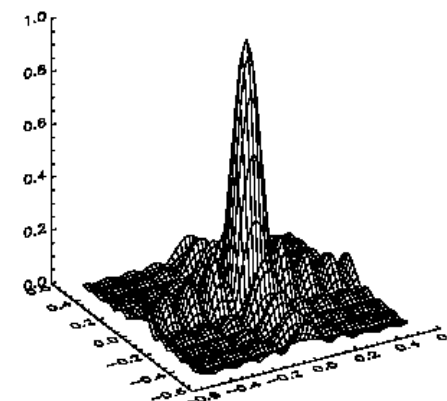
Frequency Response for $M=3$



Frequency Response for $M=5$



Frequency Response for $M=7$



Frequency Response for $M=9$



平滑平均



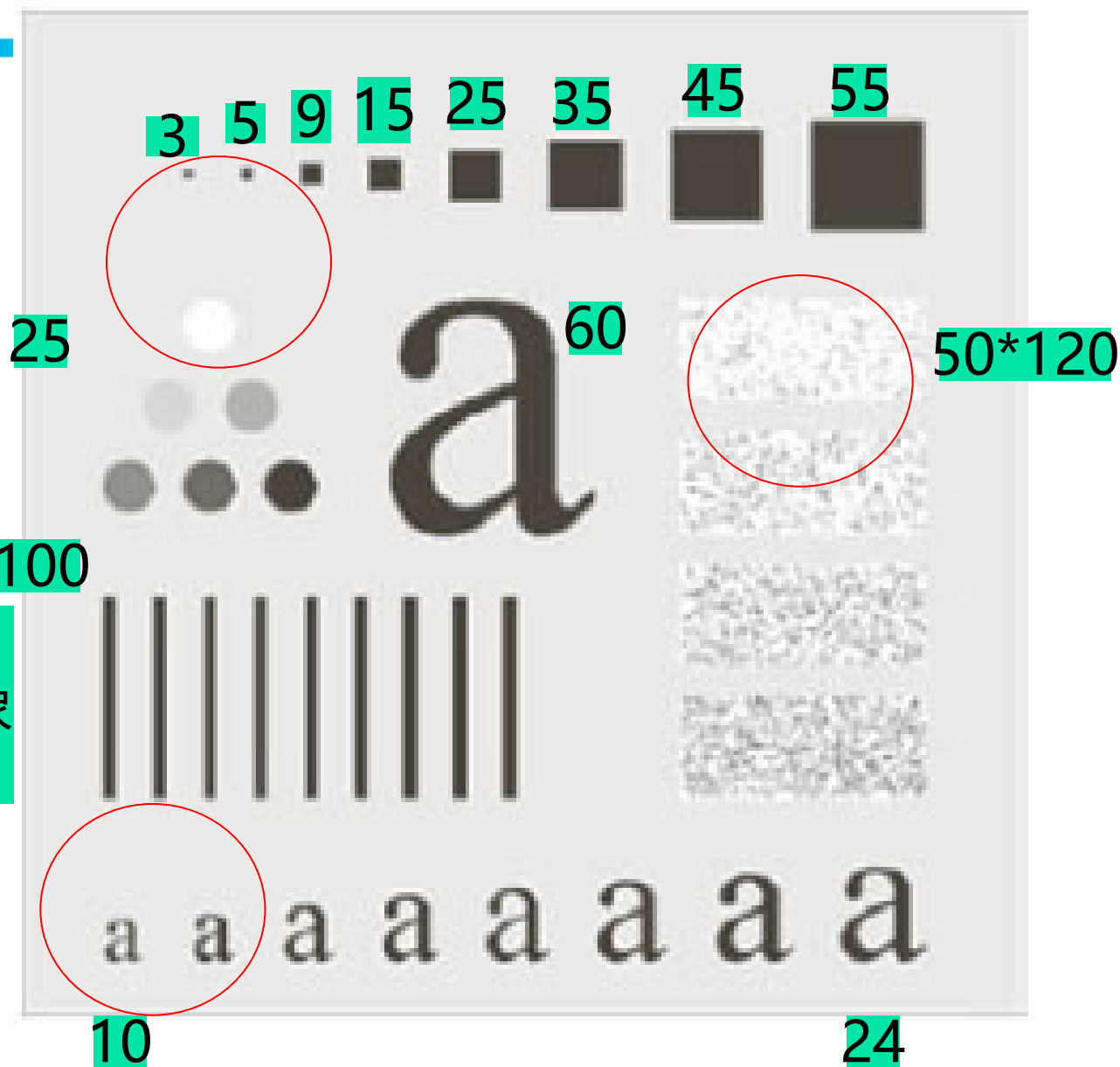


平滑平均

3*3

255-
204-153-
102-51-0

模糊化消除了小对象



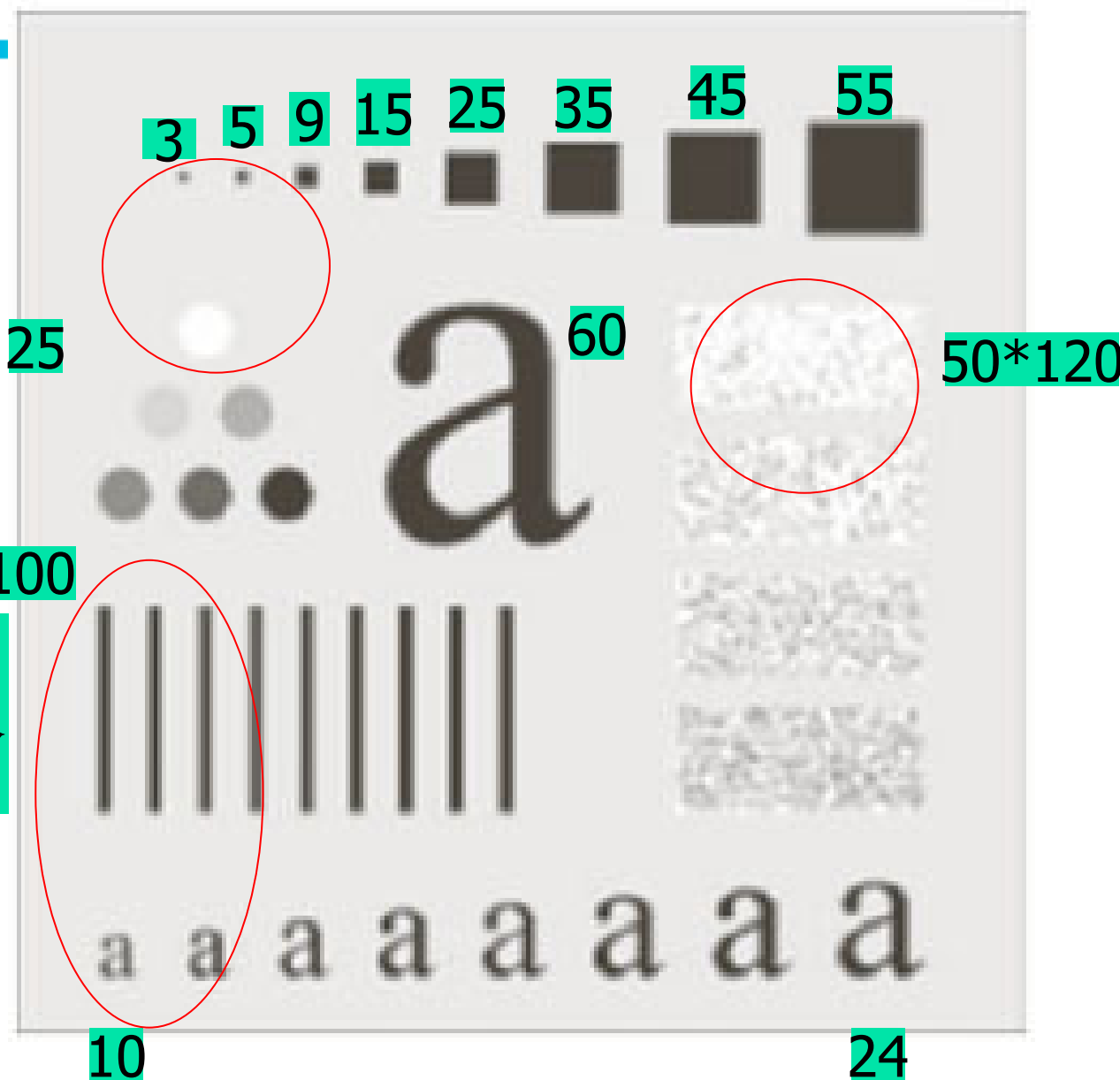


平滑平均

5*5

255-
204-153-
102-51-0

模糊化消除了小对象



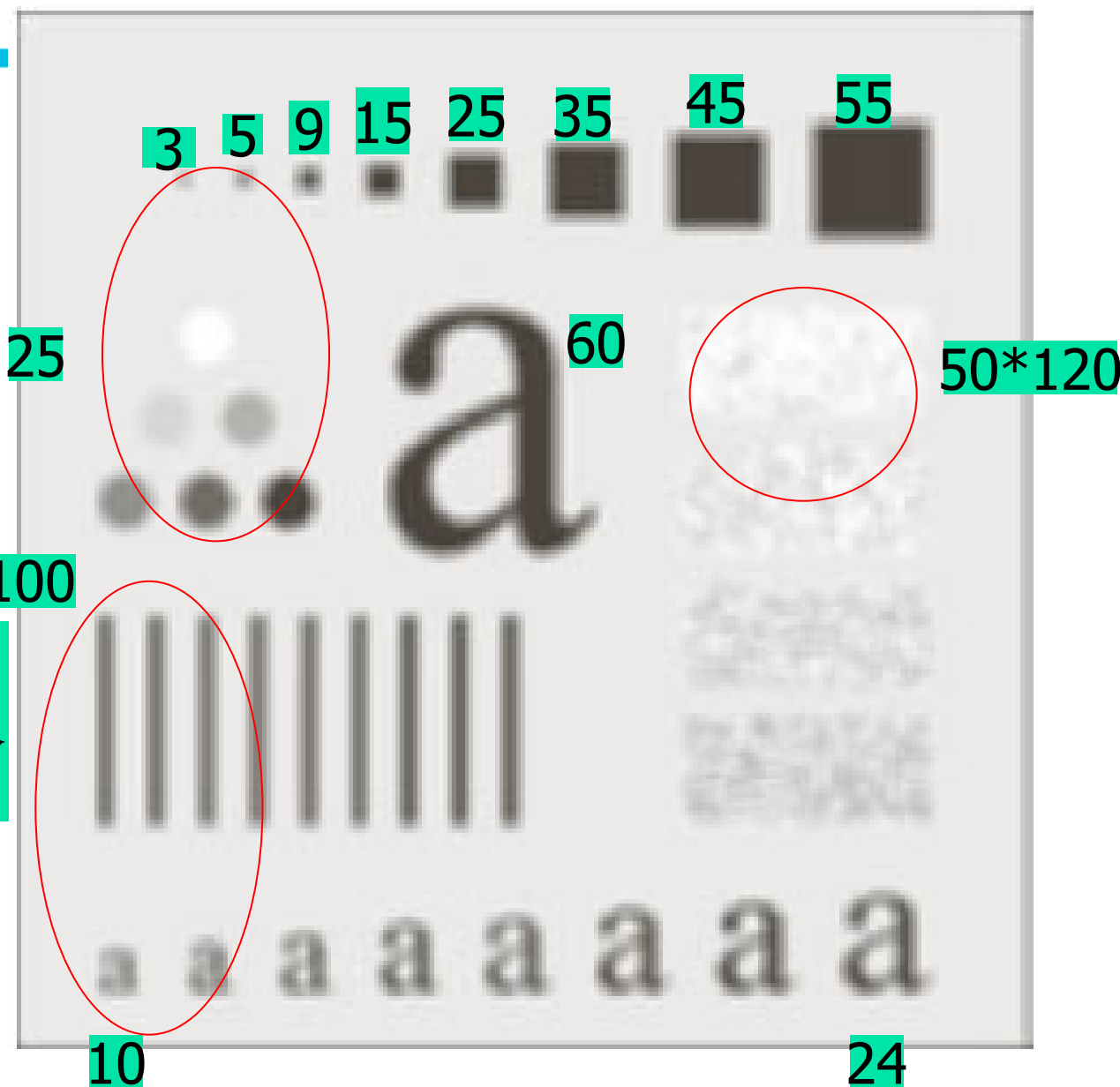


平滑平均

9*9

255-
204-153-
102-51-0

模糊化消除了小对象





平滑平均 15*15

255-
204-153-
102-51-0



模糊化消除了小对象

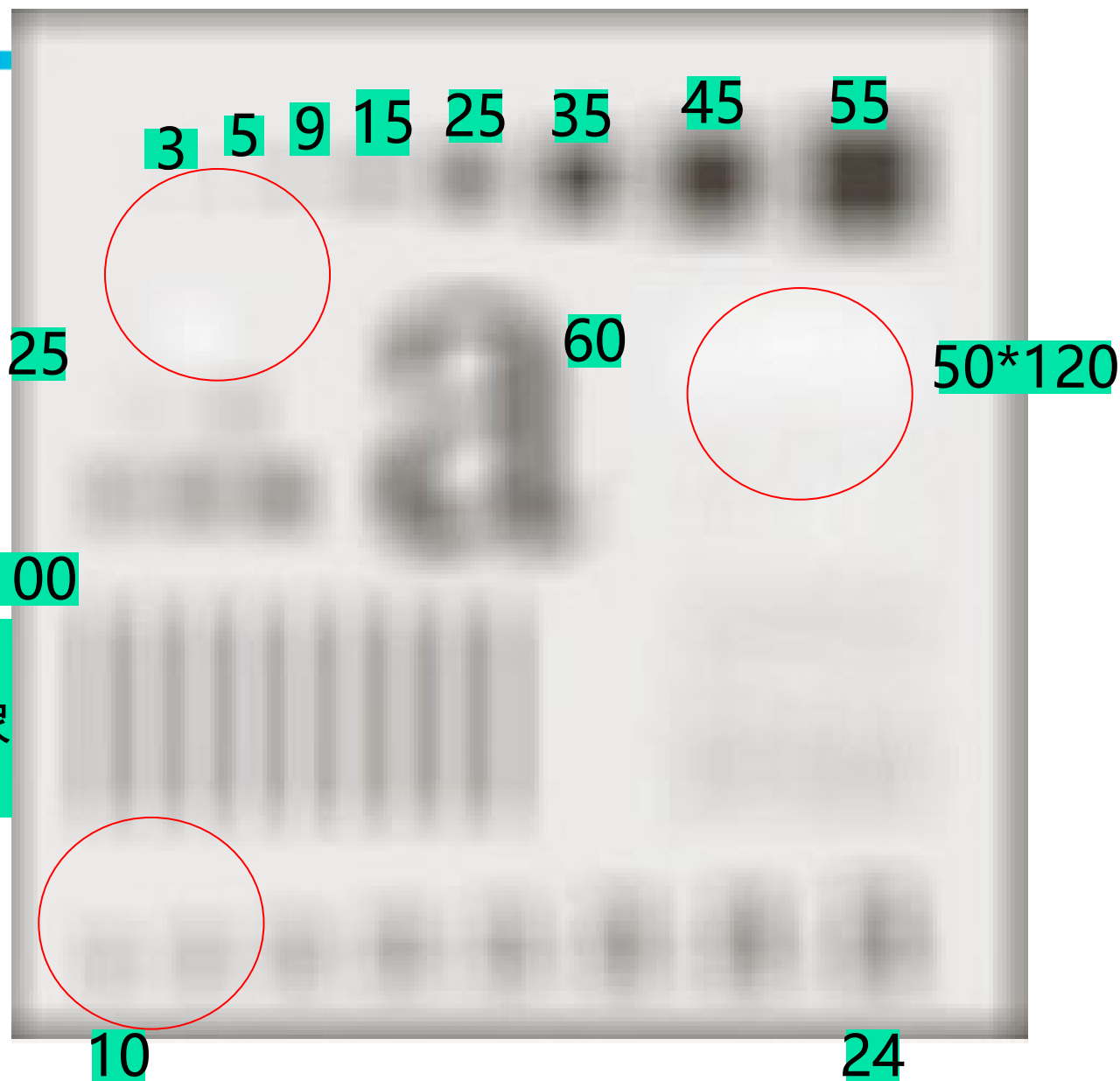


平滑平均 35×35

255-
204-153-
102-51-0

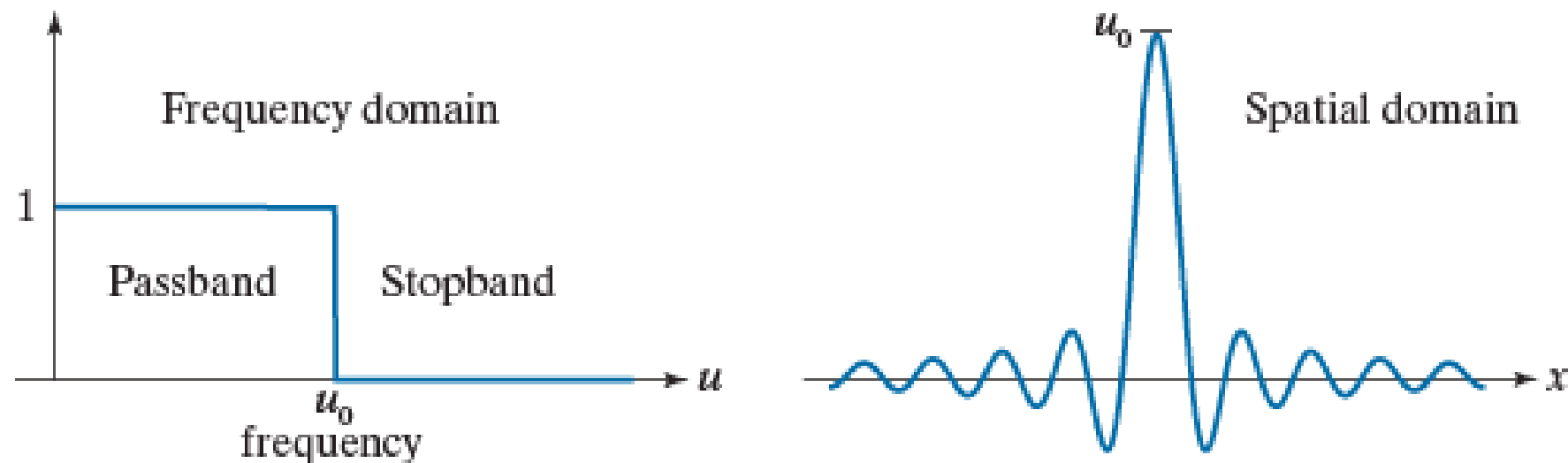
5×100

模糊化消除了小对象





模糊效应



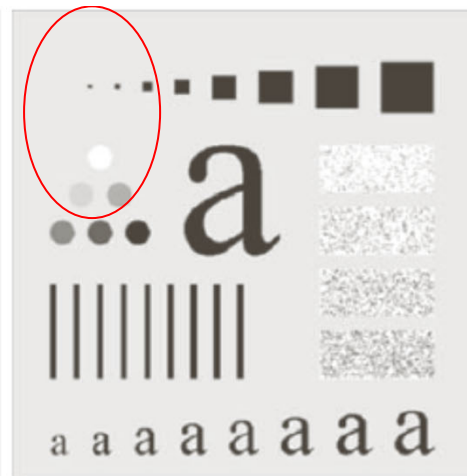
a b

FIGURE 3.32

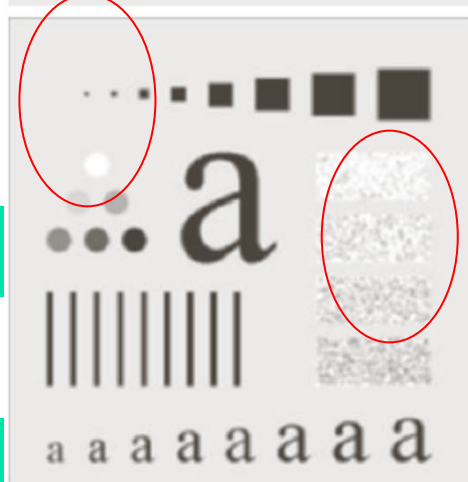
(a) Ideal 1-D low-pass filter transfer function in the frequency domain.
(b) Corresponding filter kernel in the spatial domain.



平滑平均



3



5

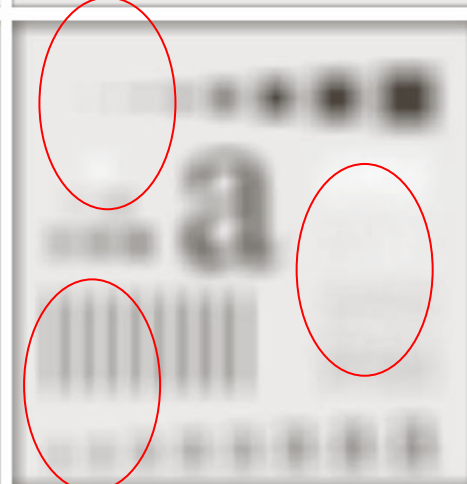


9

模糊化消除了小对象



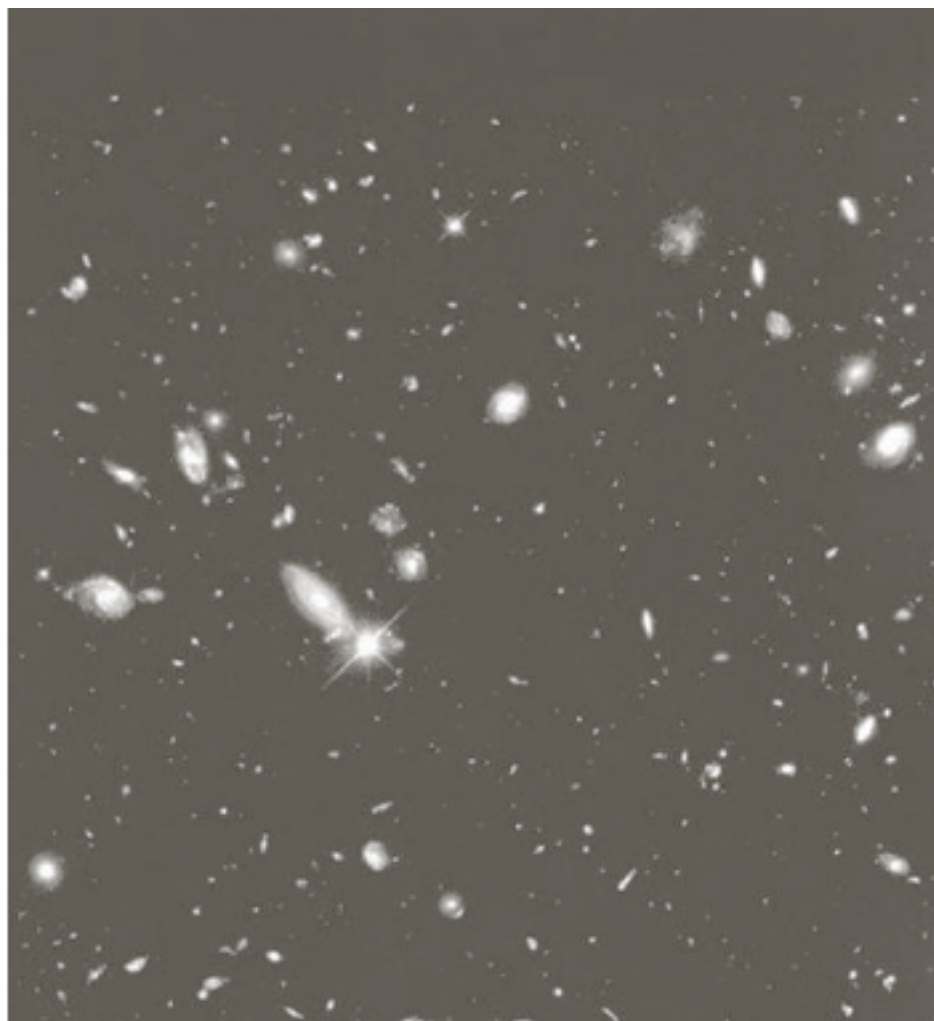
15



35



例子：物体的粗略表示法



a b c

FIGURE 3.34 (a) Image of size 528×485 pixels from the Hubble Space Telescope. (b) Image filtered with a 15×15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)



a b
c d

1024 × 1024

FIGURE 3.33

(a) Test pattern of size 1024 × 1024 pixels.

(b)-(d) Results of

lowpass filtering with box kernels of sizes 3 × 3,

11 × 11,

and 21 × 21, respectively.

When zero (black) padding is used, the net result of smoothing at or near the border is a **dark gray border** that arises from including black pixels in the averaging process.





a b
c d

1024 × 1024

FIGURE 3.33

(a) Test pattern of size 1024 × 1024 pixels.

(b)-(d) Results of lowpass filtering with box kernels of sizes 3 × 3, 11 × 11, and 21 × 21, respectively.

When zero (black) padding is used, the net result of smoothing at or near the border is a **dark gray border** that arises from including black pixels in the averaging process.





a b
c d

1024 × 1024

FIGURE 3.33

(a) Test pattern of size 1024 × 1024 pixels.

(b)-(d) Results of lowpass filtering with box kernels of sizes 3 × 3, 11 × 11, and 21 × 21, respectively.

When zero (black) padding is used, the net result of smoothing at or near the border is a **dark gray border** that arises from including black pixels in the averaging process.

11*11





a b
c d

1024 × 1024

FIGURE 3.33

(a) Test pattern of size 1024 × 1024 pixels.

(b)-(d) Results of lowpass filtering with box kernels of sizes 3 × 3, 11 × 11, and 21 × 21, respectively.

When zero (black) padding is used, the net result of smoothing at or near the border is a **dark gray border** that arises from including black pixels in the averaging process.



21*21



3*3

a b
c d

1024 × 1024

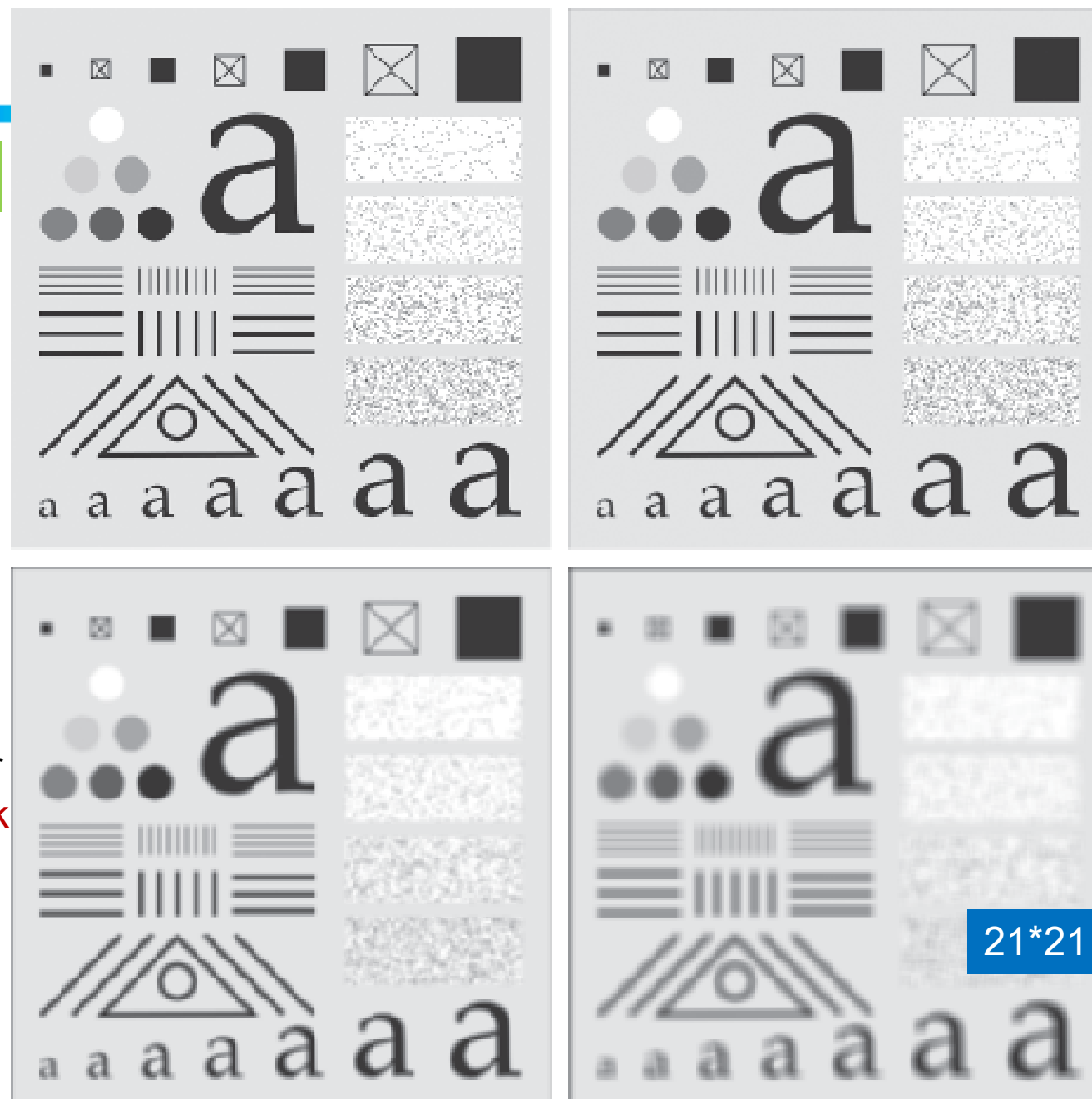
FIGURE 3.33

(a) Test pattern of size 1024 × 1024 pixels.

(b)-(d) Results of lowpass filtering with box kernels of sizes 3 × 3, 11 × 11, and 21 × 21, respectively.

When zero (black) padding is used, the net result of smoothing at or near the border is a **dark gray border** that arises from including black pixels in the averaging process.

11*11



21*21



低通高斯滤波器核

FIGURE 3.34

Distances from the center for various sizes of square kernels.

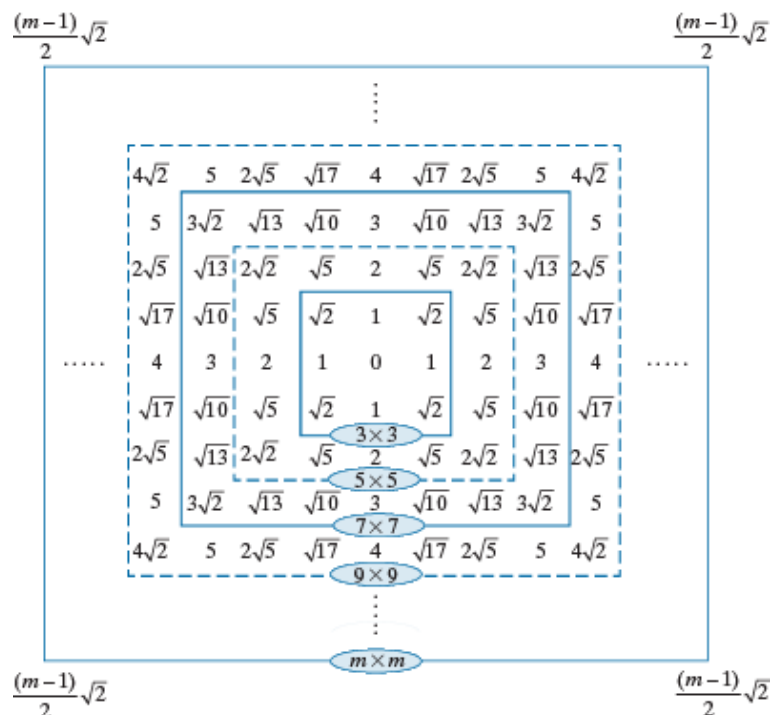
$$w(s, t) = G(s, t) = Ke^{-\frac{s^2 + t^2}{2\sigma^2}}$$

$$r = [s^2 + t^2]^{1/2}$$

$$r_{\max}^2 = \left[\frac{(m-1)}{2} \sqrt{2} \right]^2 = \frac{(m-1)^2}{2}$$

m 表示卷积核的宽度

$$\lceil 6\sigma \rceil \times \lceil 6\sigma \rceil \quad K=1 \text{ and } \sigma=1$$



$$\frac{1}{4.8976} \times$$

0.3679	0.6065	0.3679
0.6065	1.0000	0.6065
0.3679	0.6065	0.3679



低通高斯滤波器核

a b

FIGURE 3.35

(a) Sampling a Gaussian function to obtain a discrete Gaussian kernel. The values shown are for $K = 1$ and $\sigma = 1$. (b) Resulting 3×3 kernel [this is the same as Fig. 3.31(b)].

$$\begin{bmatrix} (-1, -1) & (-1, 0) & (-1, 1) \\ (0, -1) & (0, 0) & (0, 1) \\ (1, -1) & (1, 0) & (1, 1) \end{bmatrix}$$

$$\frac{1}{2\pi\sigma^2} \begin{bmatrix} \exp(-\frac{2}{2\sigma^2}) & \exp(-\frac{1}{2\sigma^2}) & \exp(-\frac{2}{2\sigma^2}) \\ \exp(-\frac{2}{2\sigma^2}) & 1 & \exp(-\frac{2}{2\sigma^2}) \\ \exp(-\frac{2}{2\sigma^2}) & \exp(-\frac{1}{2\sigma^2}) & \exp(-\frac{2}{2\sigma^2}) \end{bmatrix}$$

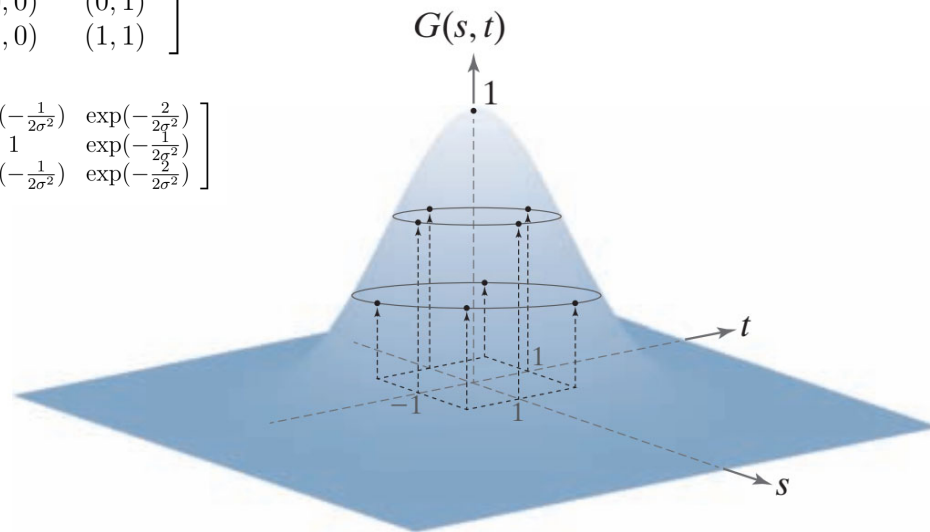
$$w(s, t) = G(s, t) = Ke^{-\frac{s^2 + t^2}{2\sigma^2}}$$

$$r = [s^2 + t^2]^{1/2}$$

$$r_{\max}^2 = \left[\frac{(m-1)}{2} \sqrt{2} \right]^2 = \frac{(m-1)^2}{2}$$

m 表示卷积核的宽度

$$\lceil 6\sigma \rceil \times \lceil 6\sigma \rceil \quad K = 1 \text{ and } \sigma = 1$$



$$\frac{1}{4.8976} \times$$

0.3679	0.6065	0.3679
0.6065	1.0000	0.6065
0.3679	0.6065	0.3679

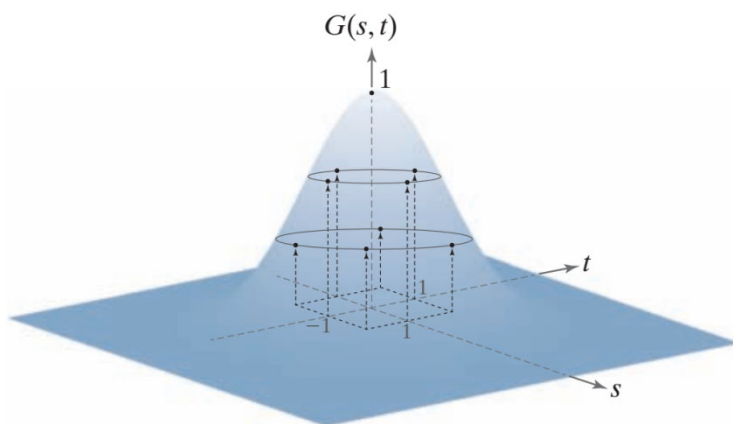


低通高斯滤波器核



a b c

FIGURE 3.36 (a) A test pattern of size 1024×1024 . (b) Result of lowpass filtering the pattern with a Gaussian kernel of size 21×21 , with standard deviations $\sigma = 3.5$. (c) Result of using a kernel of size 43×43 , with $\sigma = 7$. This result is comparable to Fig. 3.33(d). We used $K = 1$ in all cases.



$$\lceil 6\sigma \rceil \times \lceil 6\sigma \rceil$$

$$6 * 3.5 = 21$$

$$6 * 7 = 42$$



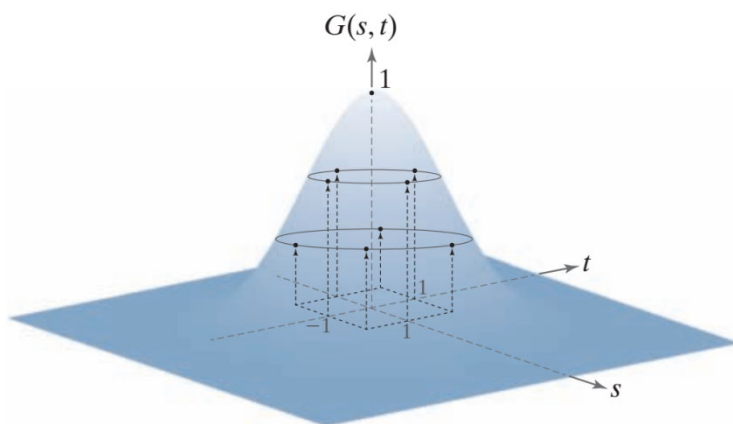
低通高斯滤波器核

21*21 delta=3.5

a b c



FIGURE 3.36 (a) A test pattern of size 1024×1024 . (b) Result of lowpass filtering the pattern with a Gaussian kernel of size 21×21 , with standard deviations $\sigma = 3.5$. (c) Result of using a kernel of size 43×43 , with $\sigma = 7$. This result is comparable to Fig. 3.33(d). We used $K = 1$ in all cases.



$$\lceil 6\sigma \rceil \times \lceil 6\sigma \rceil$$

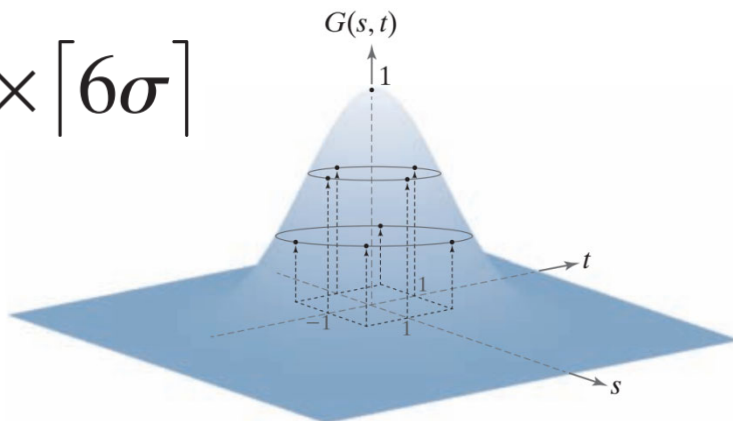
$$6 * 3.5 = 21$$

$$6 * 7 = 42$$



FIGURE 3.36 (a) A test pattern of size 1024×1024 . (b) Result of lowpass filtering the pattern with a Gaussian kernel of size 21×21 , with standard deviations $\sigma = 3.5$. (c) Result of using a kernel of size 43×43 , with $\sigma = 7$. This result is comparable to Fig. 3.33(d). We used $K = 1$ in all cases.

$$\lceil 6\sigma \rceil \times \lceil 6\sigma \rceil$$



$$\lceil 6\sigma \rceil \times \lceil 6\sigma \rceil$$

$$6 * 3.5 = 21$$

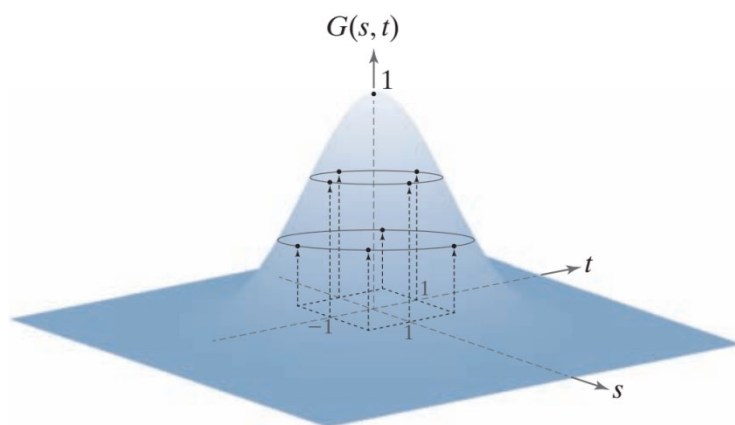
$$6 * 7 = 42$$



低通高斯滤波器核



FIGURE 3.36 (a) A test pattern of size 1024×1024 . (b) Result of lowpass filtering the pattern with a Gaussian kernel of size 21×21 , with standard deviations $\sigma = 3.5$. (c) Result of using a kernel of size 43×43 , with $\sigma = 7$. This result is comparable to Fig. 3.33(d). We used $K = 1$ in all cases.



$$\lceil 6\sigma \rceil \times \lceil 6\sigma \rceil$$

$$6 * 3.5 = 21$$

$$6 * 7 = 42$$



低通高斯滤波器核

43*43 delta=7

1024 × 1024

a b c



FIGURE 3.37 (a) Result of filtering Fig. 3.36(a) using a Gaussian kernels of size 43×43 , with $\sigma = 7$. (b) Result of using a kernel of 85×85 , with the same value of σ . (c) Difference image.



85*85 delta=7

1024 × 1024

a b c



FIGURE 3.37 (a) Result of filtering Fig. 3.36(a) using a Gaussian kernels of size 43×43 , with $\sigma = 7$. (b) Result of using a kernel of 85×85 , with the same value of σ . (c) Difference image.



低通高斯滤波器核

1024 × 1024

a b c

FIGURE 3.37 (a) Result of filtering Fig. 3.36(a) using a Gaussian kernels of size 43×43 , with $\sigma = 7$. (b) Result of using a kernel of 85×85 , with the same value of σ . (c) Difference image.



低通高斯滤波器核



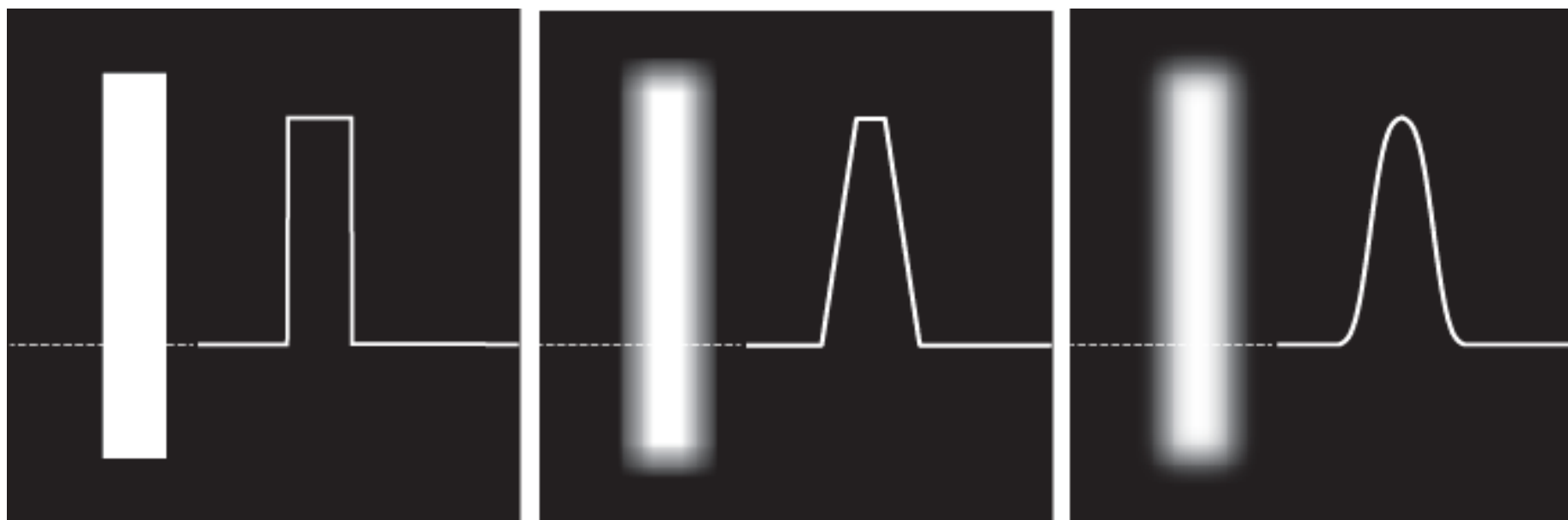
1024 × 1024

a b c

FIGURE 3.37 (a) Result of filtering Fig. 3.36(a) using a Gaussian kernels of size 43×43 , with $\sigma = 7$. (b) Result of using a kernel of 85×85 , with the same value of σ . (c) Difference image.



低通高斯滤波器核



Box 71×71

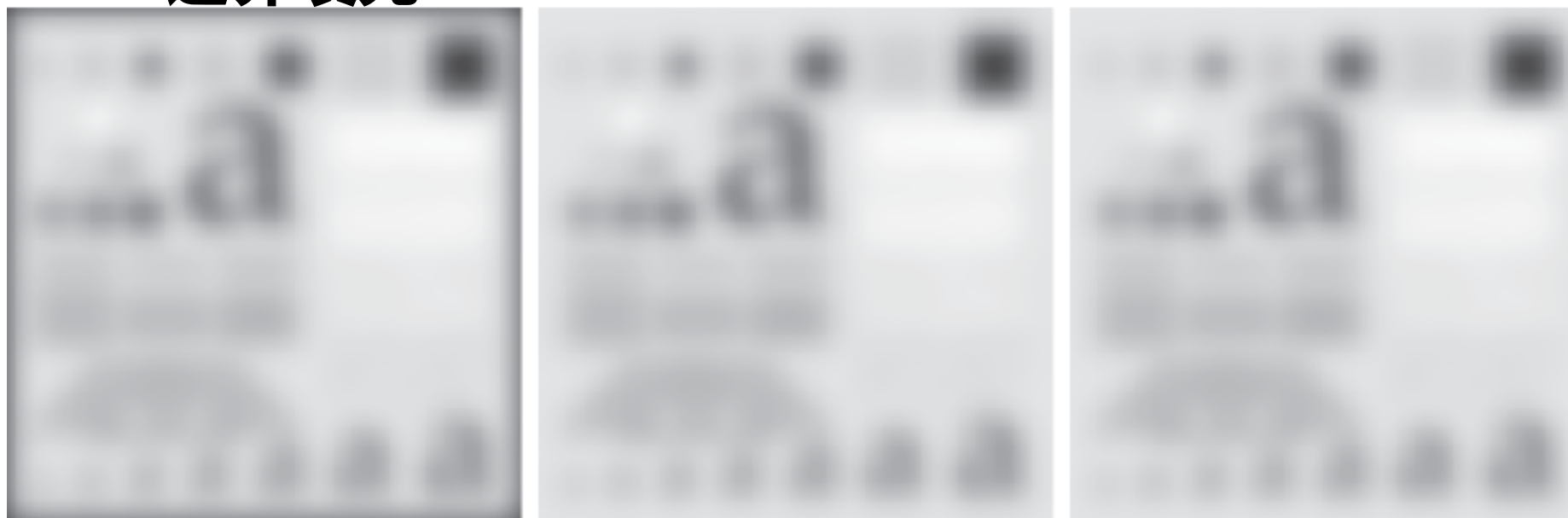
Gaussian: 151×151
 $K = 1$ and $\sigma = 25$,

a b c

FIGURE 3.38 (a) Image of a white rectangle on a black background, and a horizontal intensity profile along the scan line shown dotted. (b) Result of smoothing this image with a box kernel of size 71×71 , and corresponding intensity profile. (c) Result of smoothing the image using a Gaussian kernel of size 151×151 , with $K = 1$ and $\sigma = 25$. Note the smoothness of the profile in (c) compared to (b). The image and rectangle are of sizes 1024×1024 and 768×128 pixels, respectively.



边界填充



a b c

1024×1024

FIGURE 3.39 Result of filtering the test pattern in Fig. 3.36(a) using (a) zero padding, (b) mirror padding, and (c) replicate padding. A Gaussian kernel of size 187×187 , with $K = 1$ and $\sigma = 31$ was used in all three cases.

1. **填充0**
2. **镜像 (也称对称) 填充** : 图像边界外的数值是通过镜像反射图像的边界而获得的。
3. **复制填充** : 边界外的值被设置为等于最近的图像边界值。



边界填充

由一个给定大小的平滑核产生的相对模糊量直接取决于图像大小

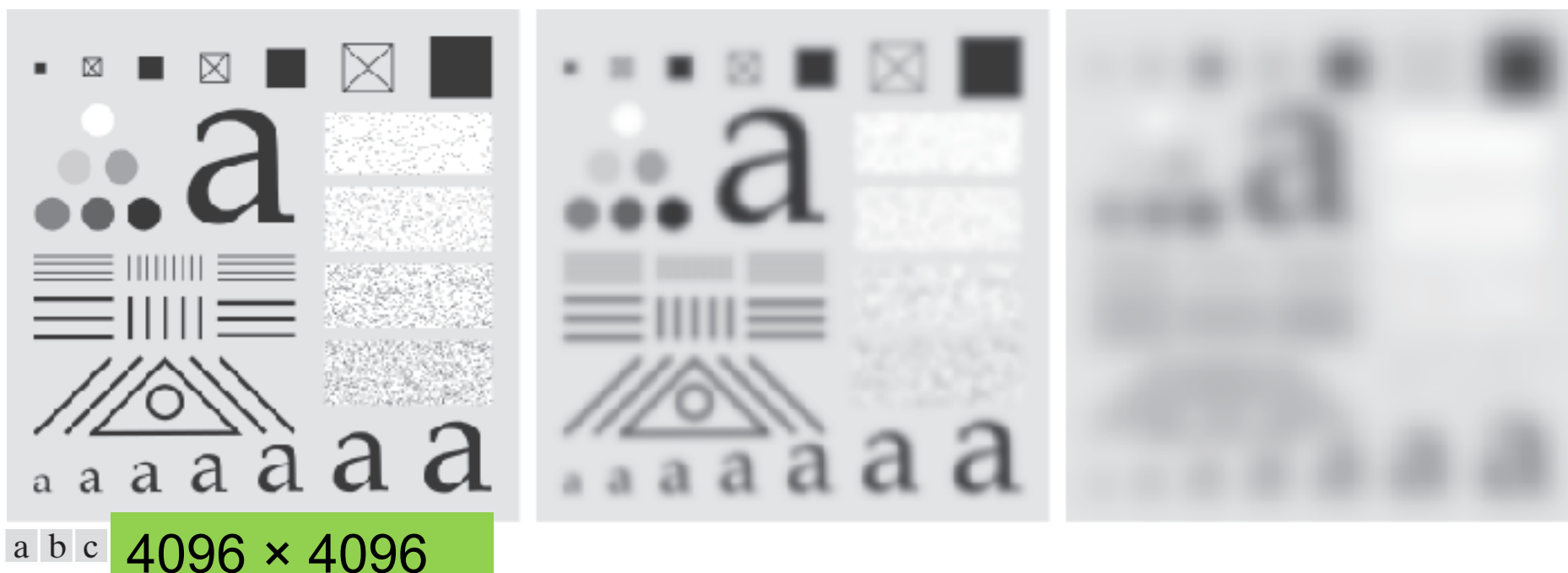
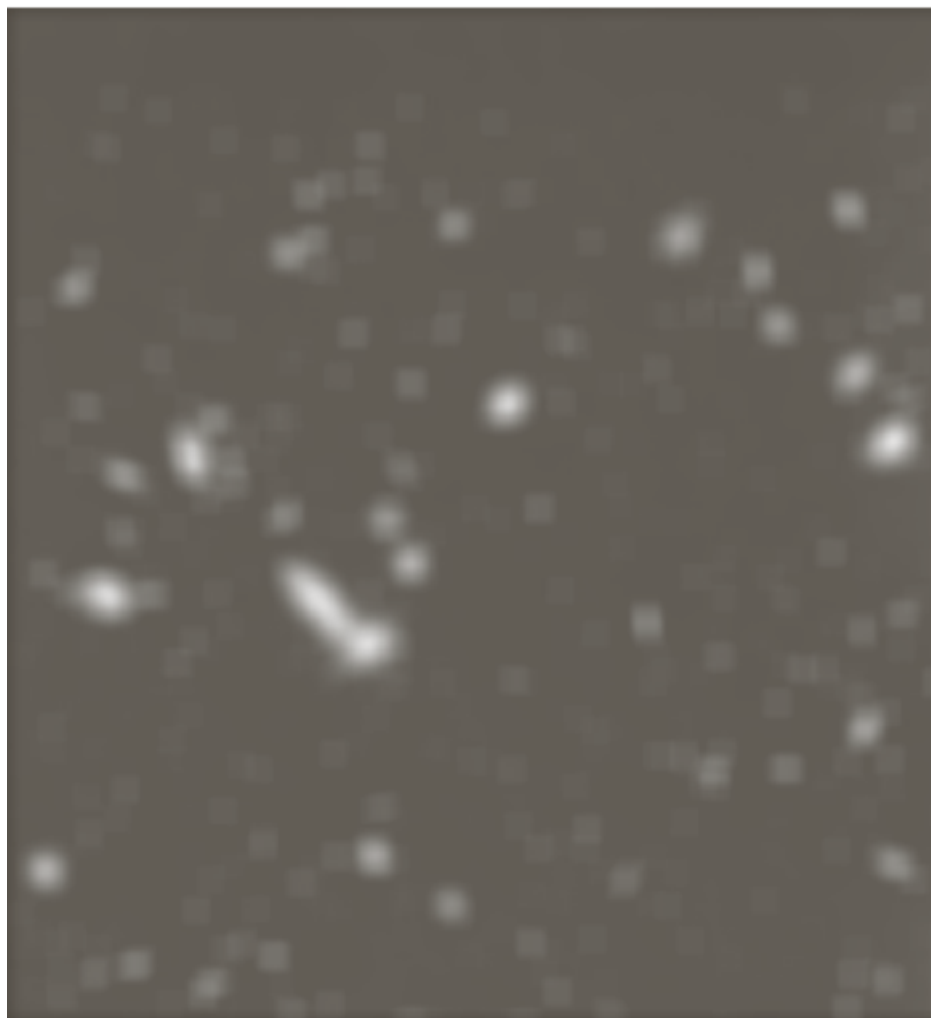


FIGURE 3.40 (a) test pattern of size 4096×4096 pixels. (b) Result of filtering the test pattern with the same Gaussian kernel used in Fig. 3.39. (c) Result of filtering the pattern using a Gaussian kernel of size 745×745 elements, with $K = 1$ and $\sigma = 124$. Mirror padding was used throughout.



例子：物体的粗略表示法

用 15×15
模糊化以
消除
小物体



a b c

图3.34 (a) 哈勃太空望远镜拍摄的528X485像素的图像。 (b) 用 15×15 平均掩模滤波的图像。 (c) 阈值化的结果 (b)。(原始图片由美国宇航局提供)



例子：物体的粗略表示法

用 15×15
模糊化以
消除
小物体

最高强度
阈值=25%

对象利益

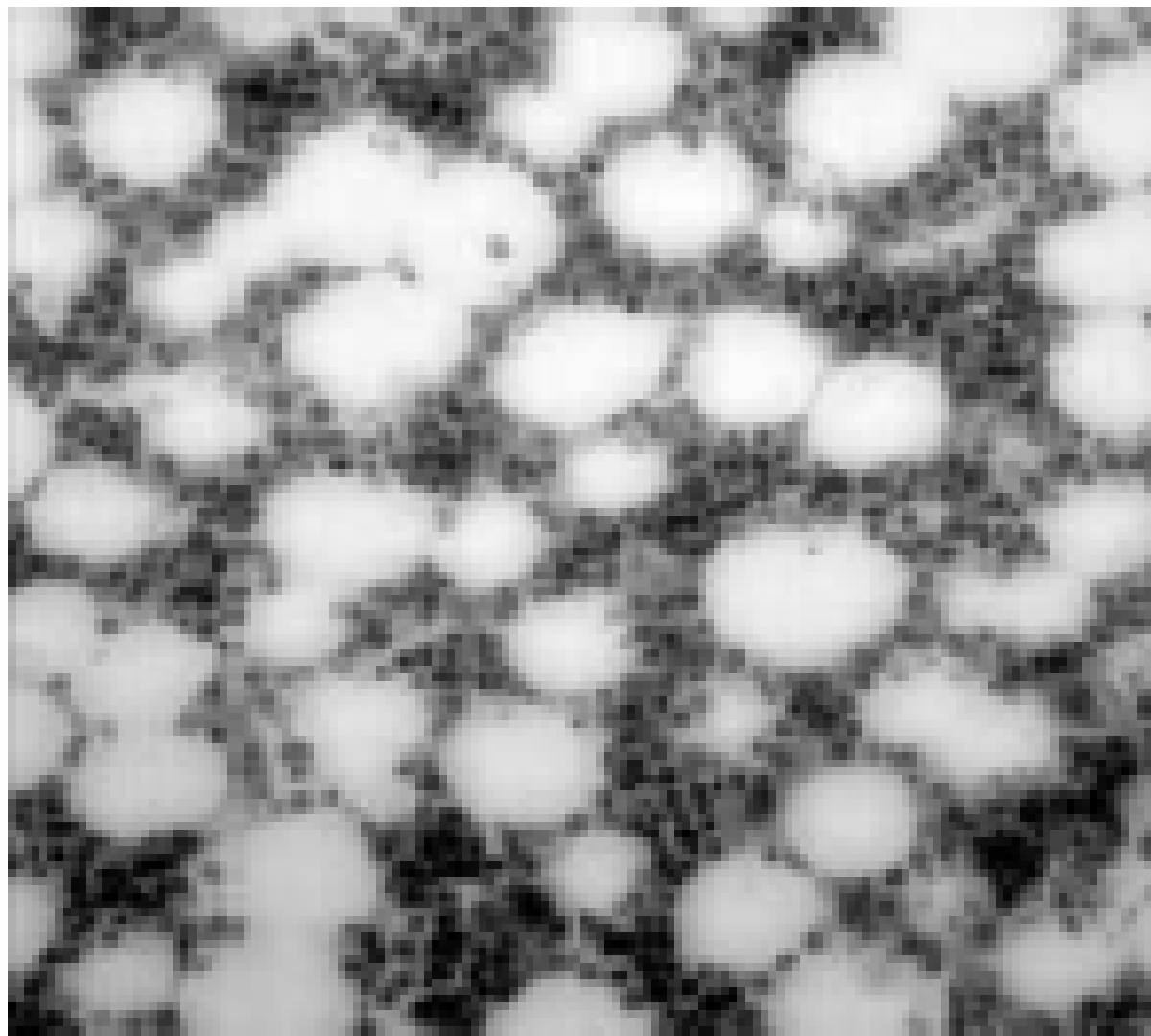


图3.34 (a) 哈勃太空望远镜拍摄的528X485像素的图像。(b) 用 15×15 平均掩模滤波的图像。(c) 阈值化的结果 (b)。(原始图片由美国宇航局提供)



图像平均化实例

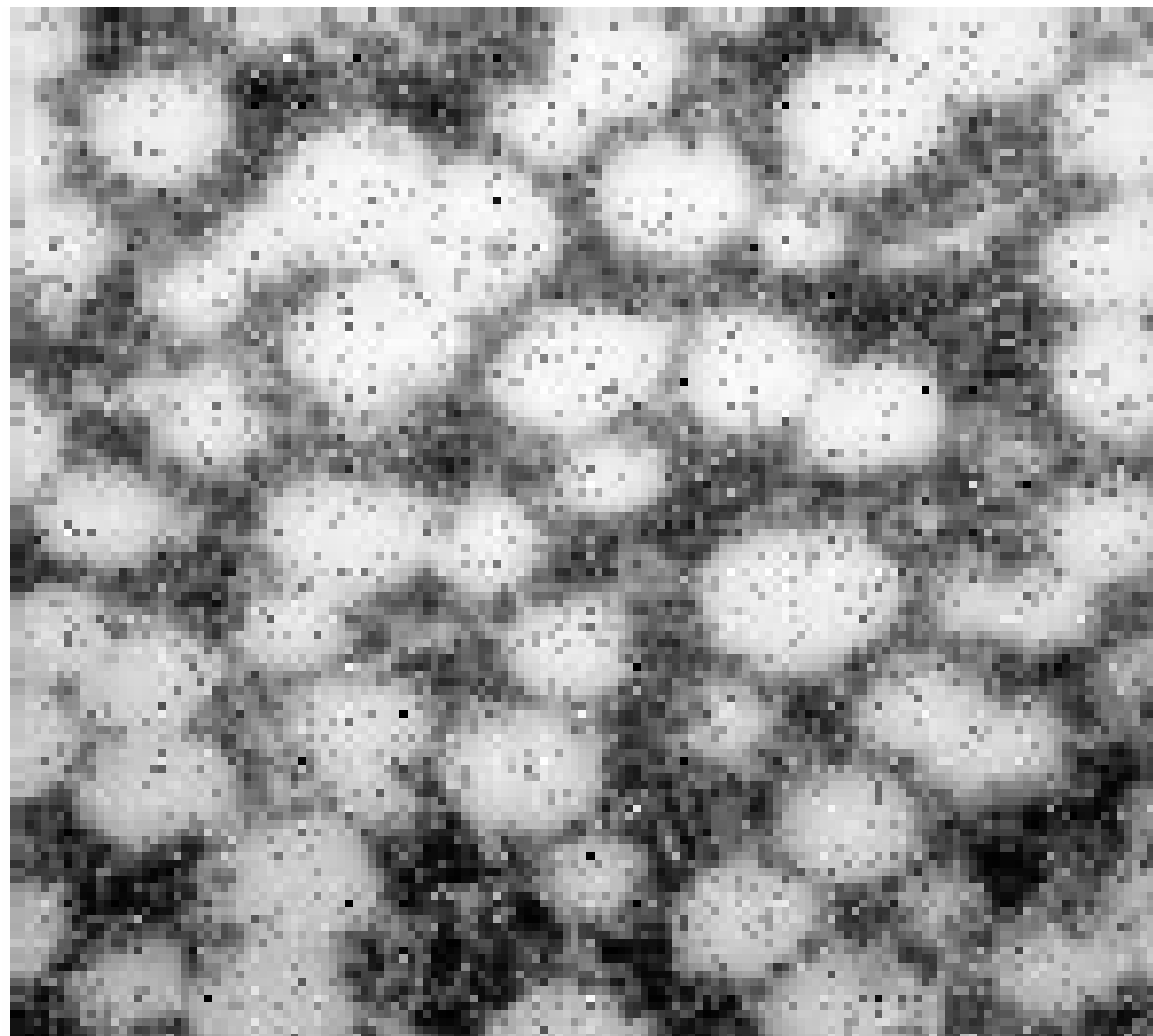
无噪音





图像平均化实例

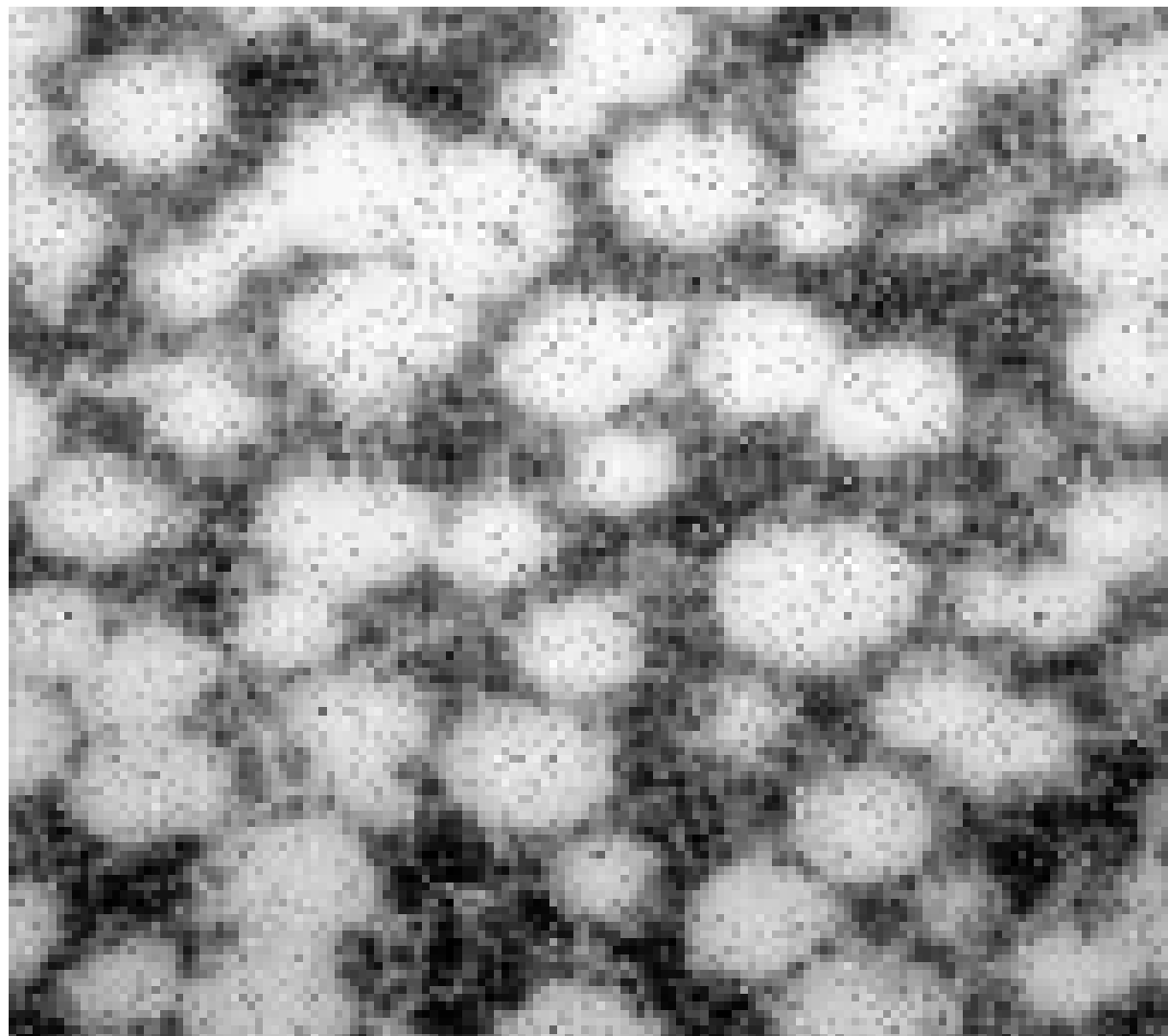
噪声方差0.05





图像平均化实例

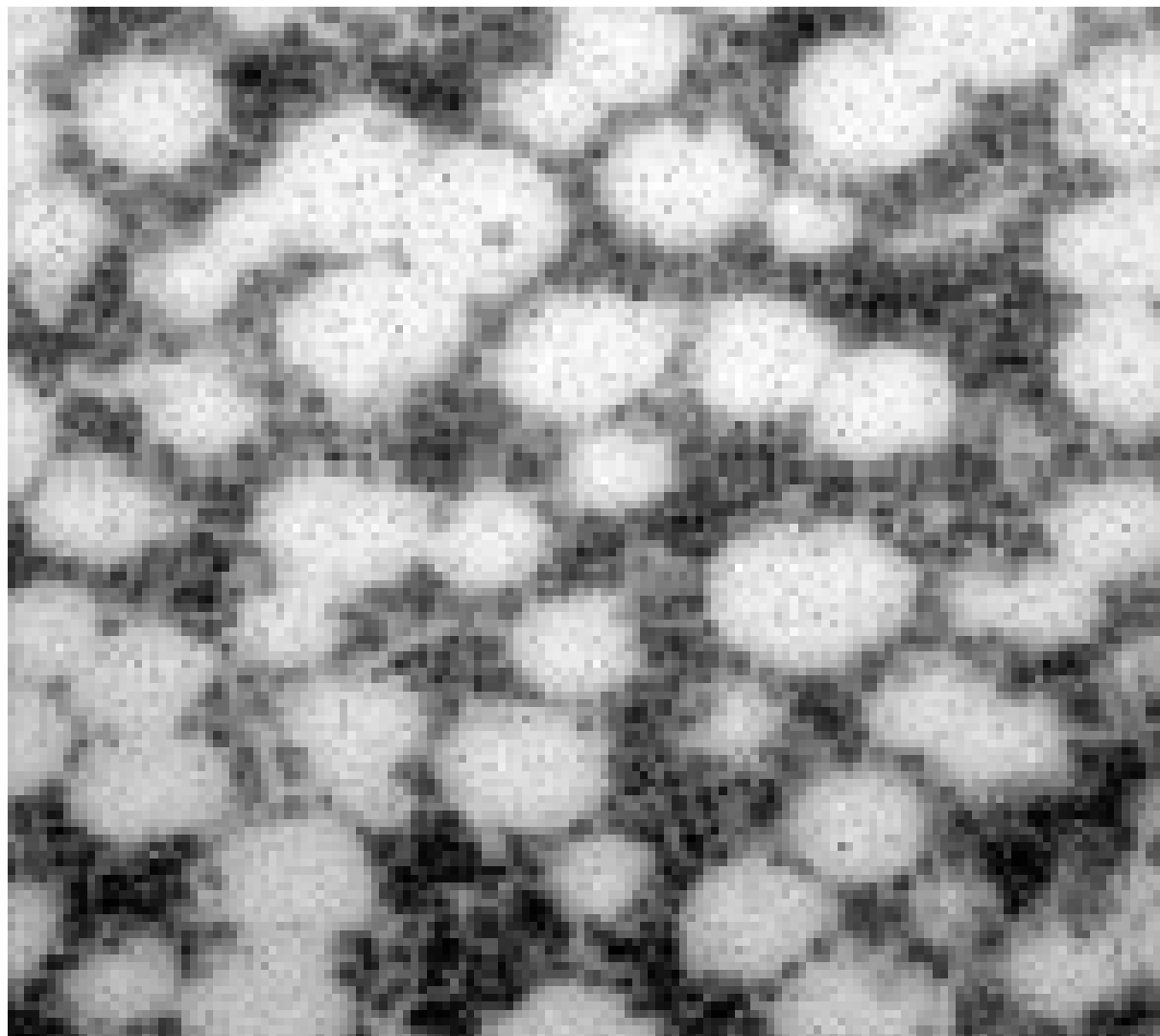
3×3





图像平均化实例

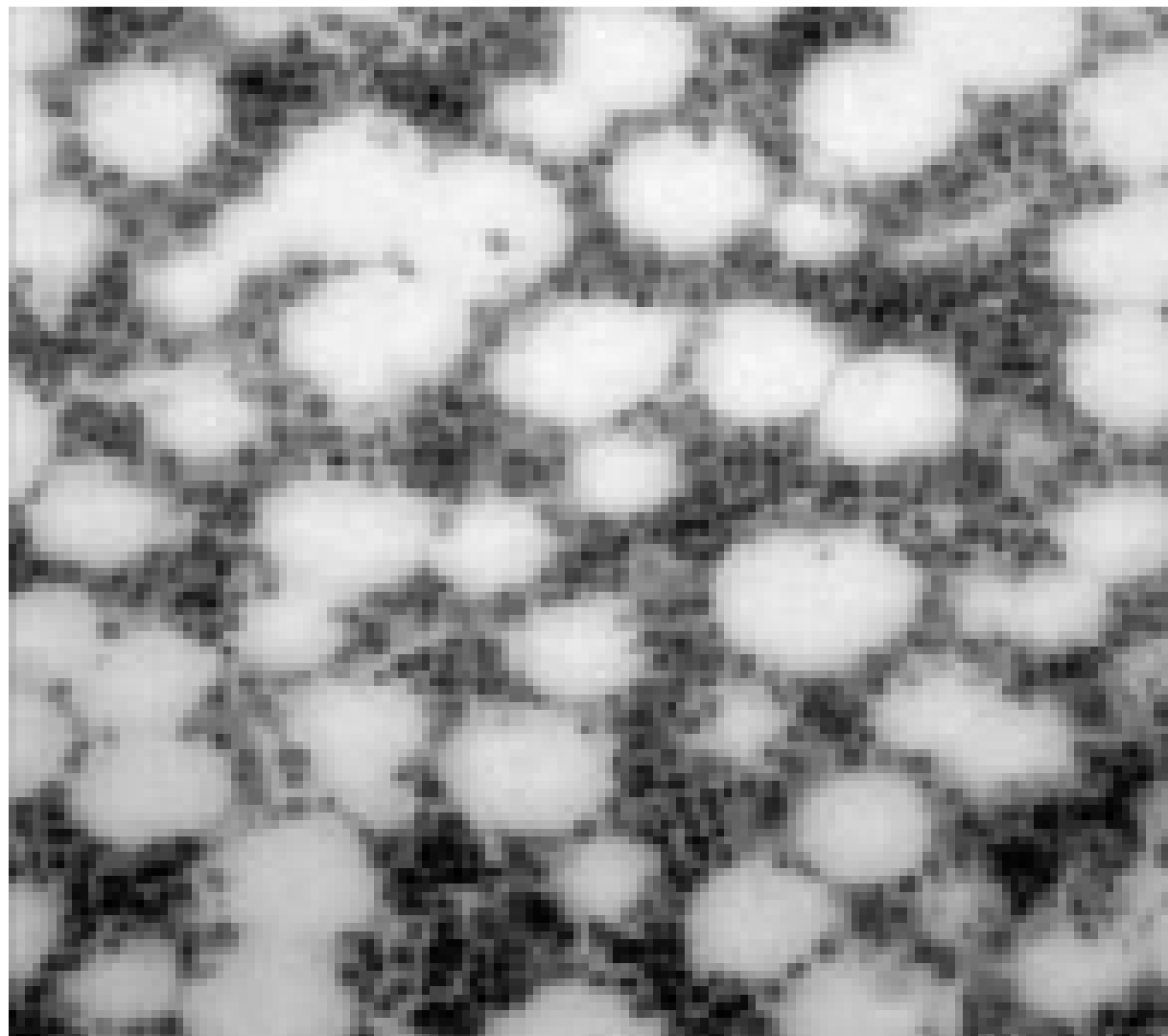
5×5





图像平均化实例

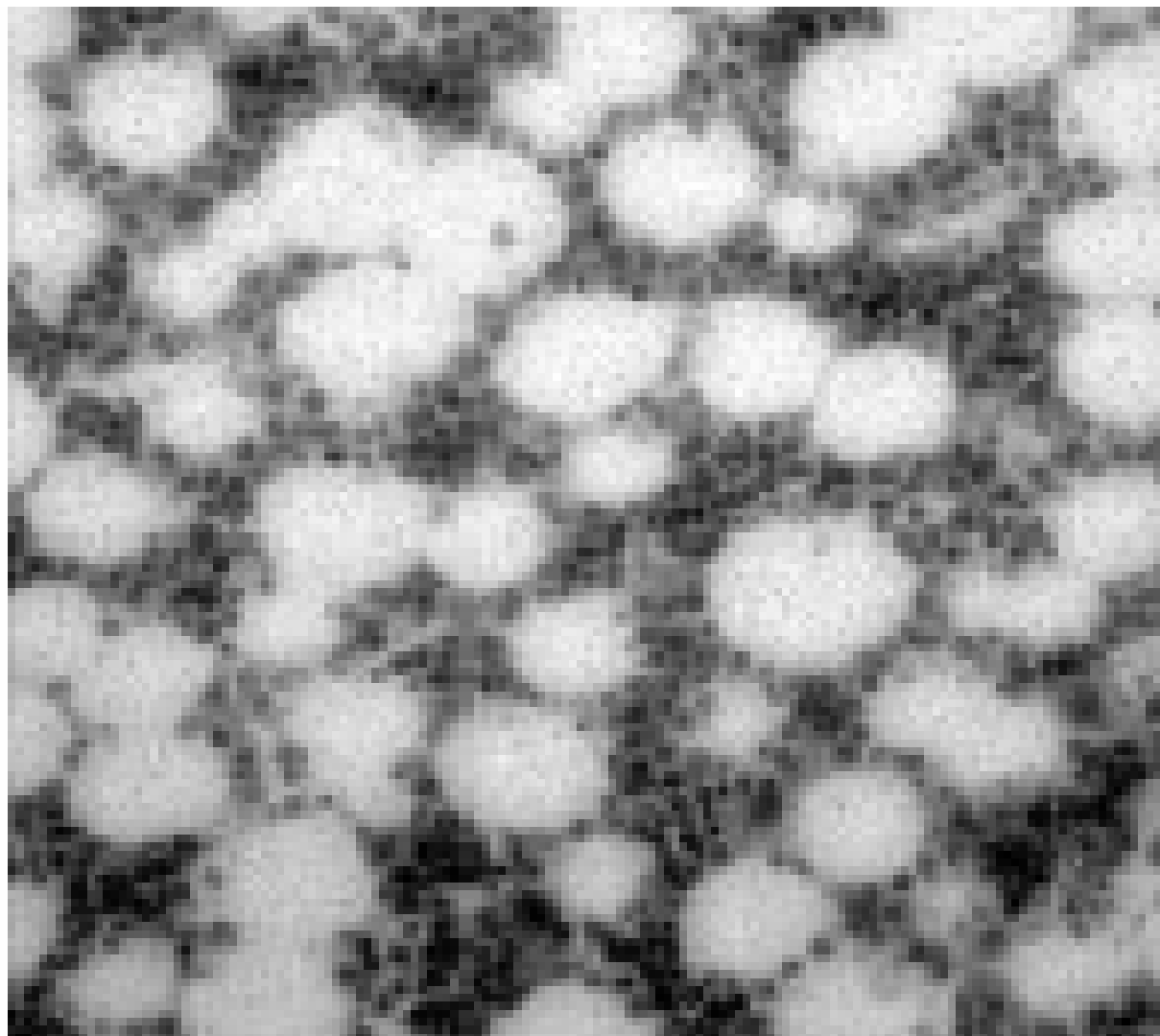
11×11





图像平均化实例

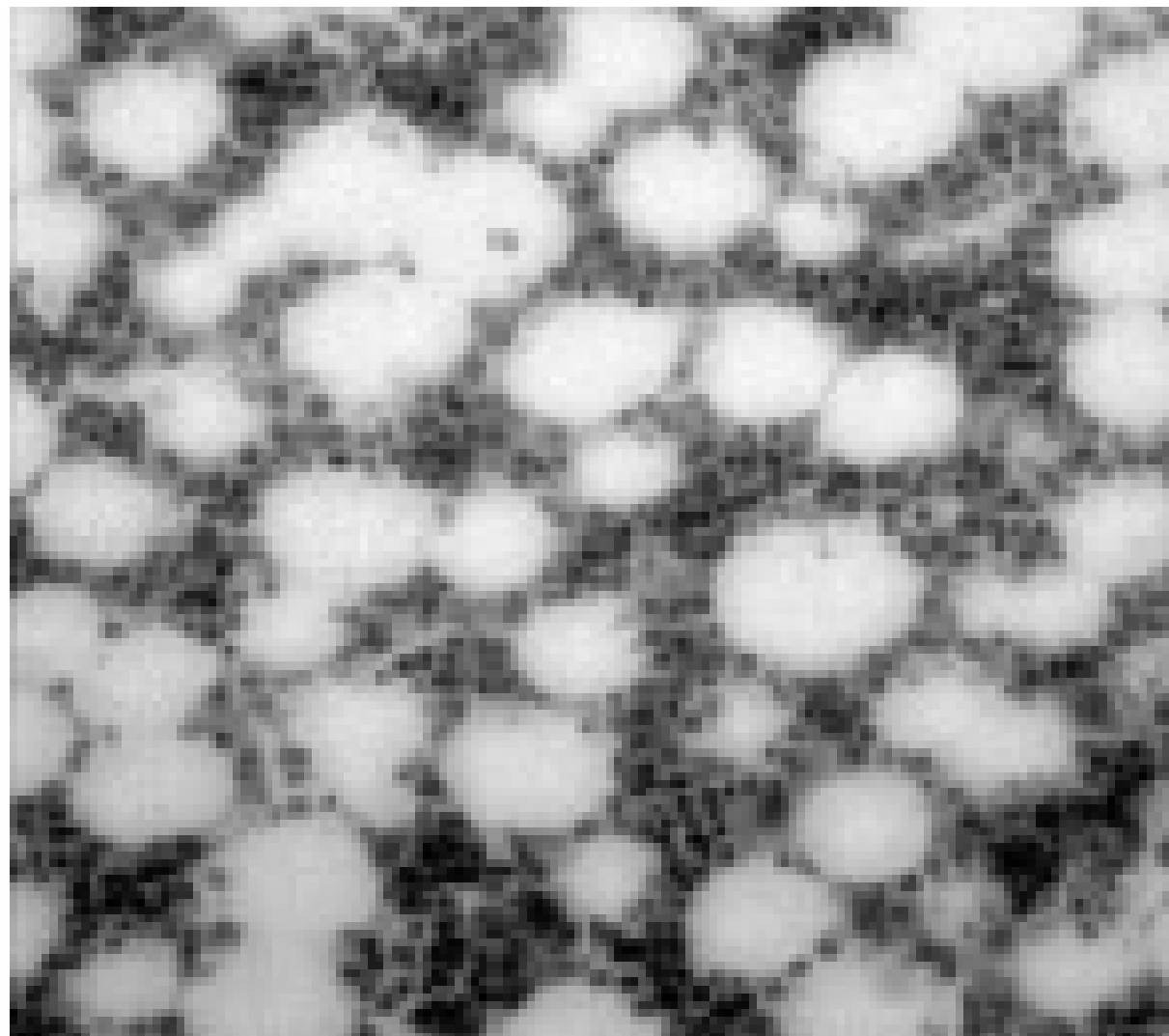
25×25





图像平均化实例

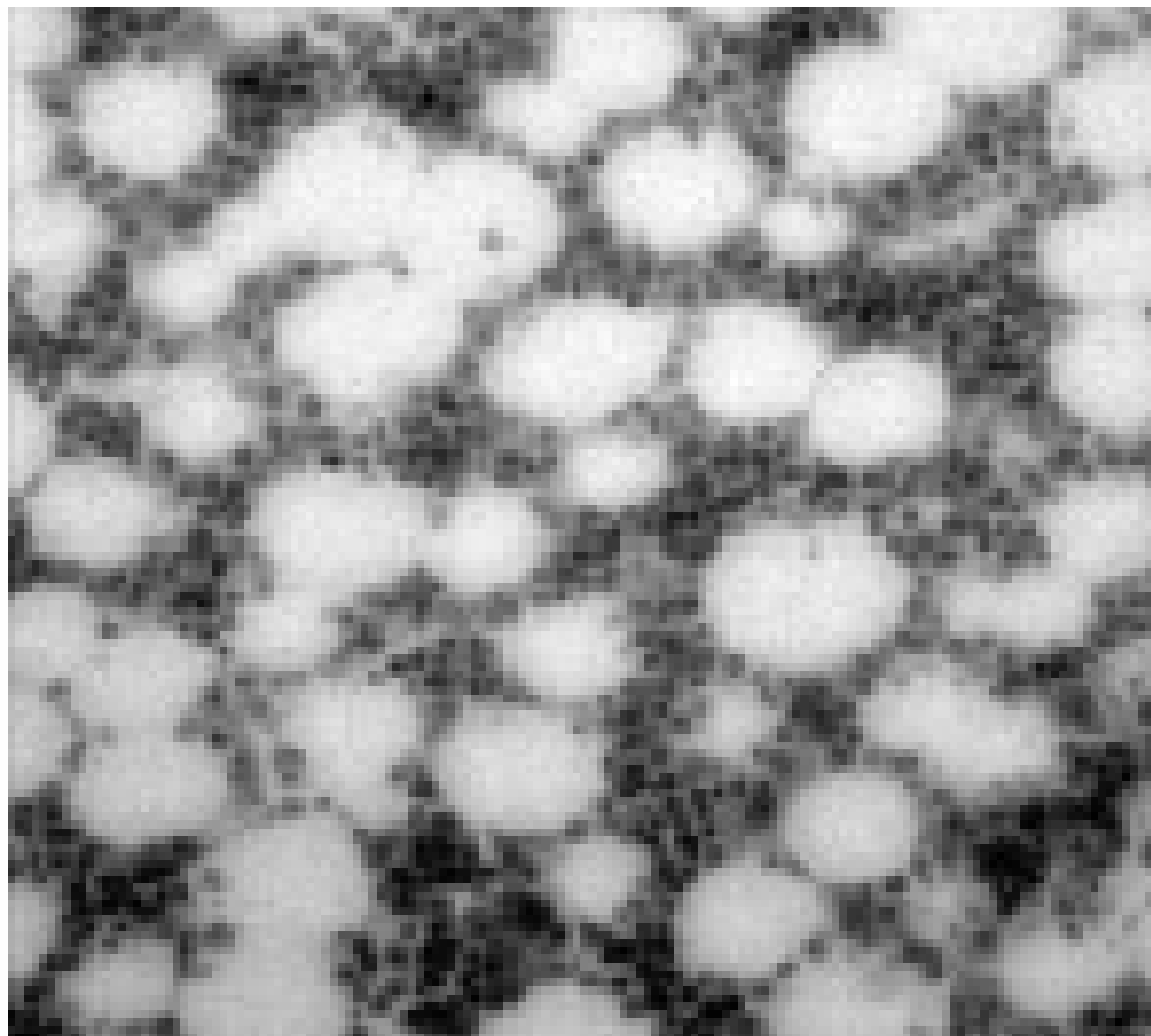
51×51





图像平均化实例

101×101

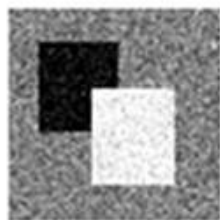




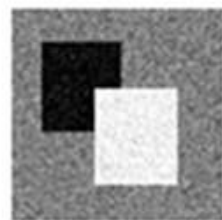
图像平均化实例



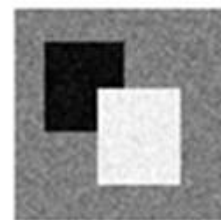
1张图片



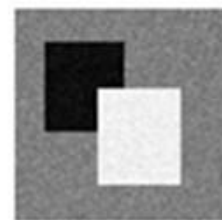
2



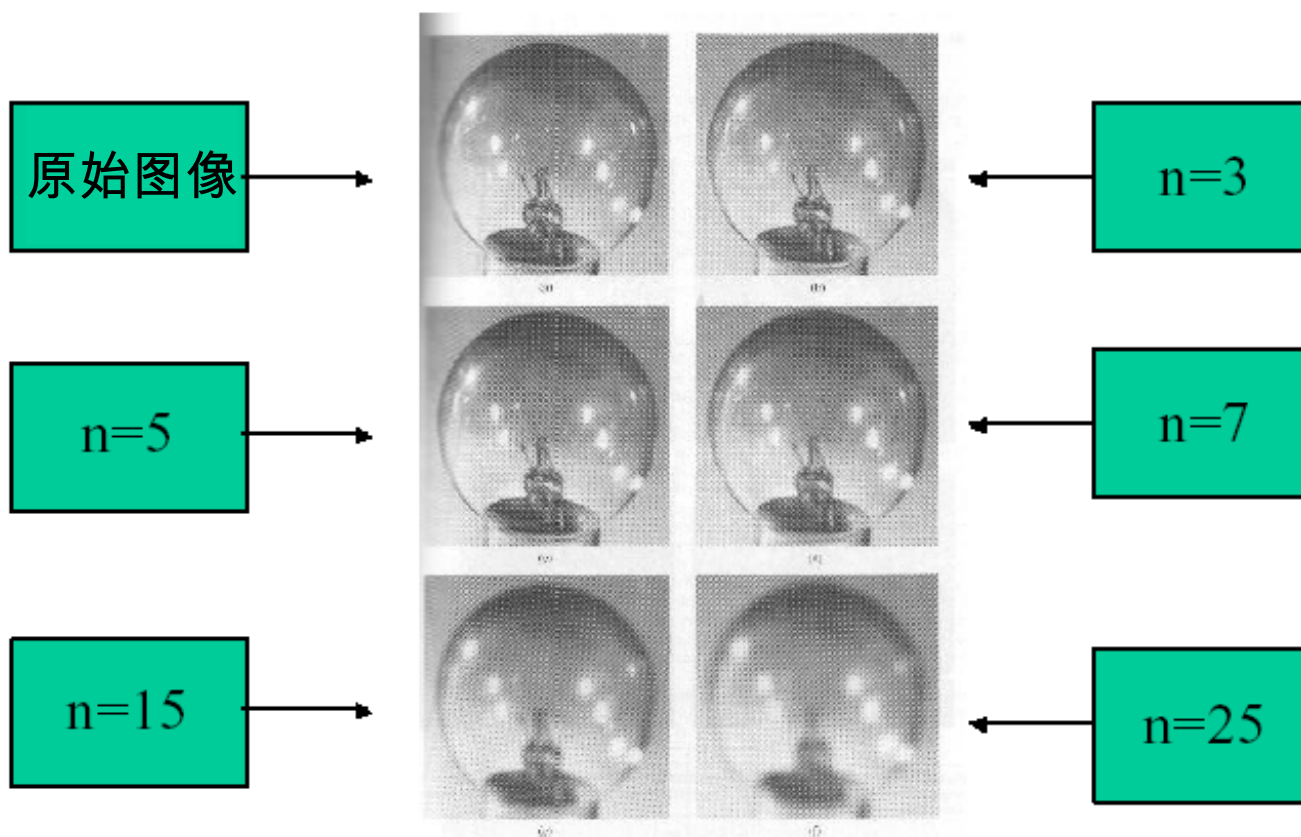
5



10



20

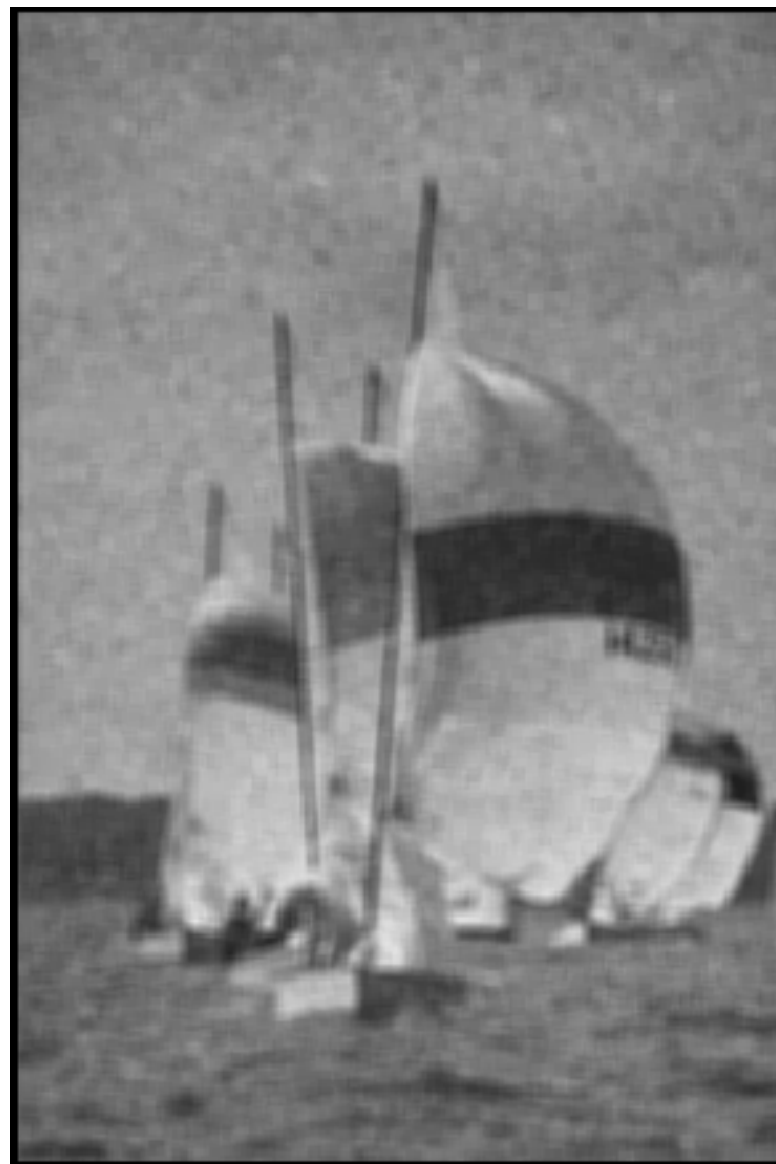
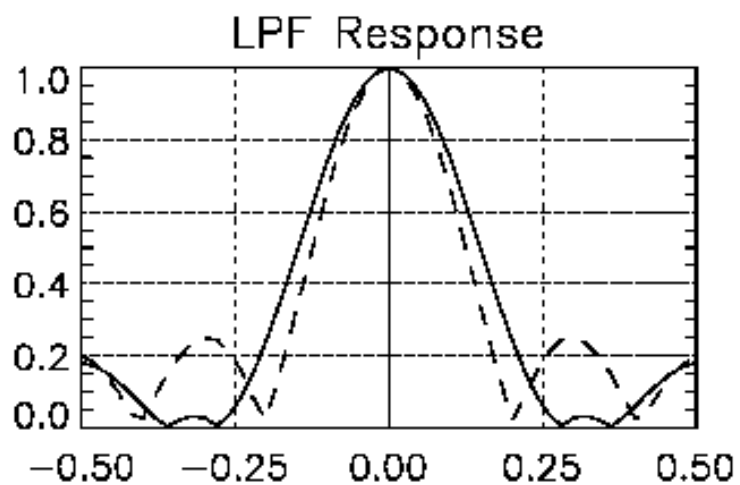




其他低通掩码 (加权滤波器)

$$\frac{1}{25} \times \begin{bmatrix} 1 & 1 & 2 & 1 & 1 \\ 1 & 2 & 3 & 2 & 1 \\ 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 3 & 2 & 1 \\ 1 & 1 & 2 & 1 & 1 \end{bmatrix}$$

平均

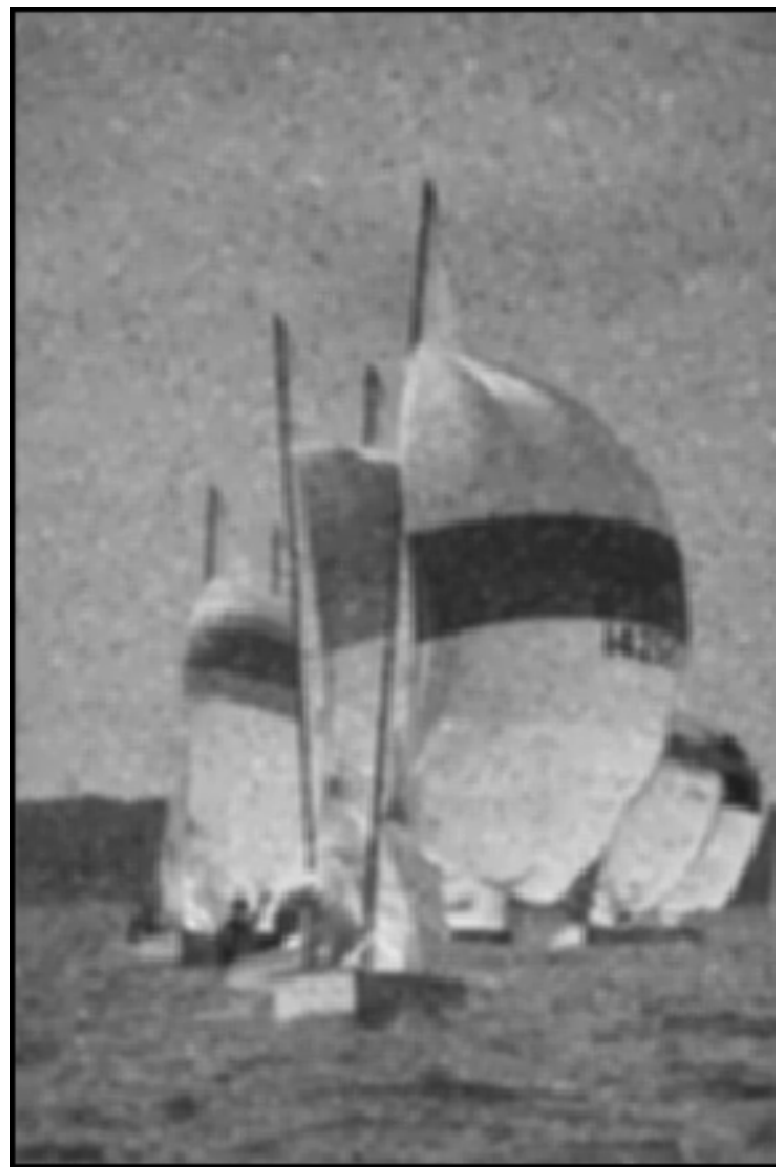
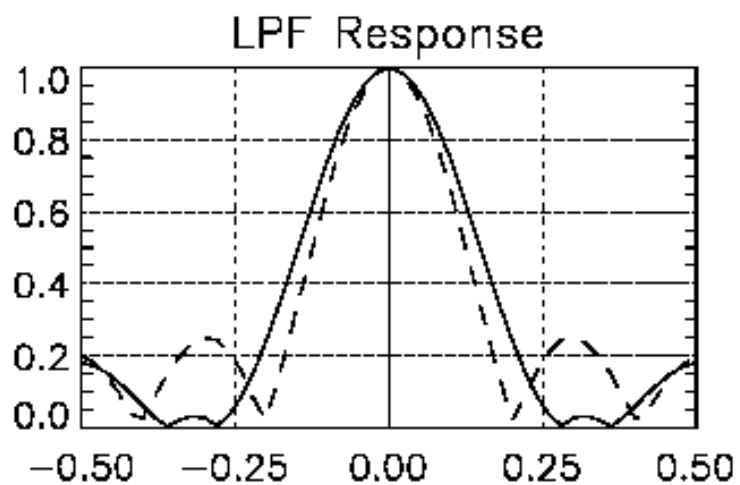




其他低通掩码（加权滤波器）

$$\frac{1}{25} \times \begin{bmatrix} 1 & 1 & 2 & 1 & 1 \\ 1 & 2 & 3 & 2 & 1 \\ 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 3 & 2 & 1 \\ 1 & 1 & 2 & 1 & 1 \end{bmatrix}$$

加权





例子：局部平均法

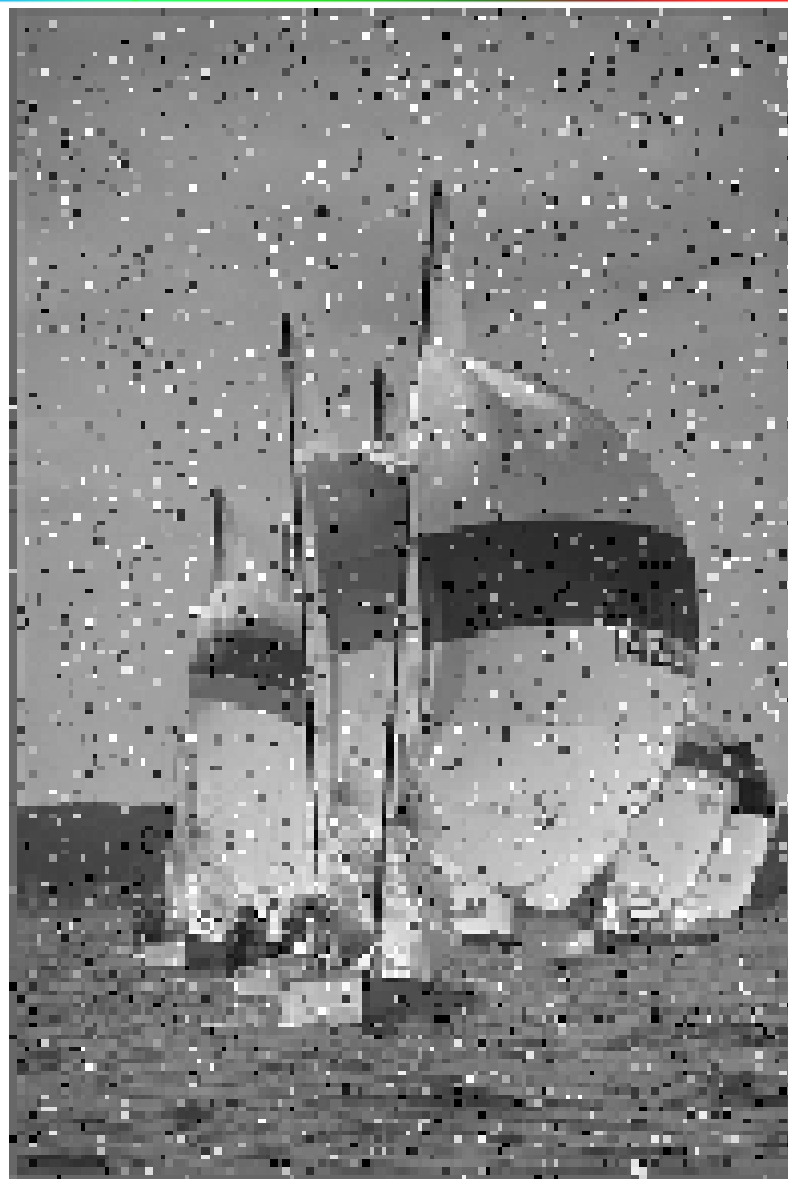
原始图像





例子：局部平均法

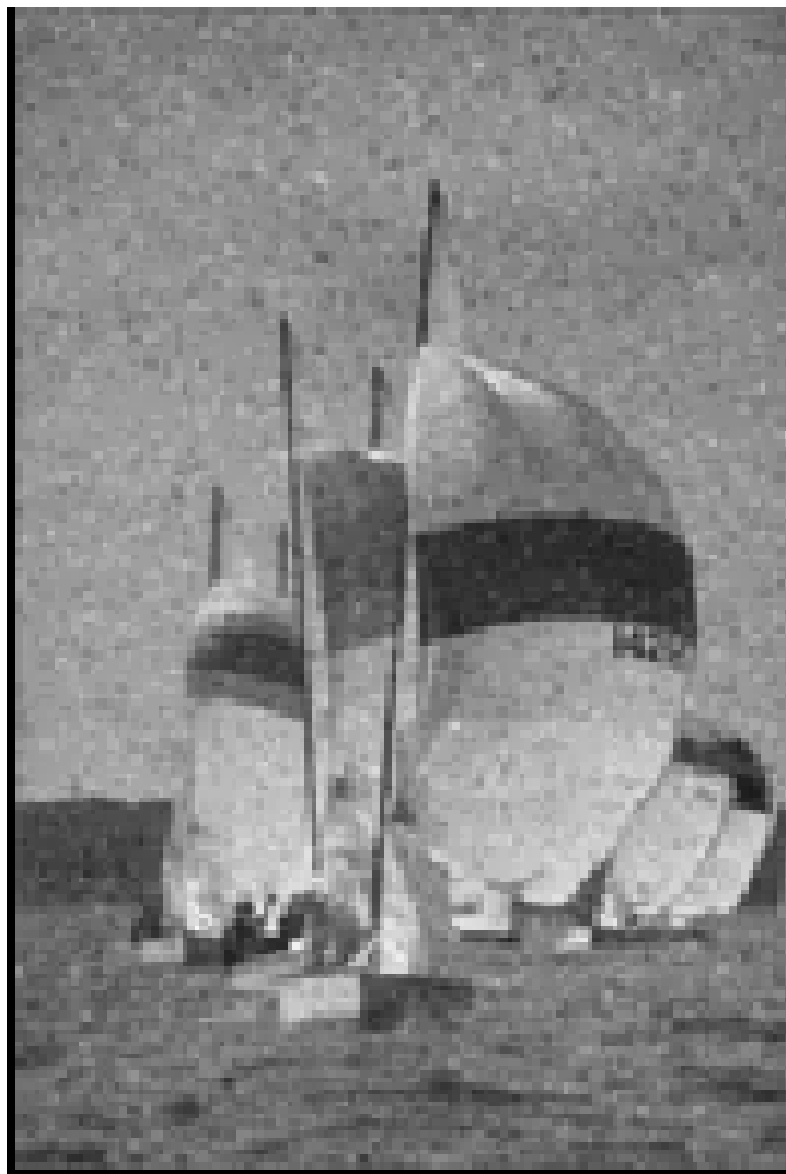
加噪声图像





例子：局部平均法

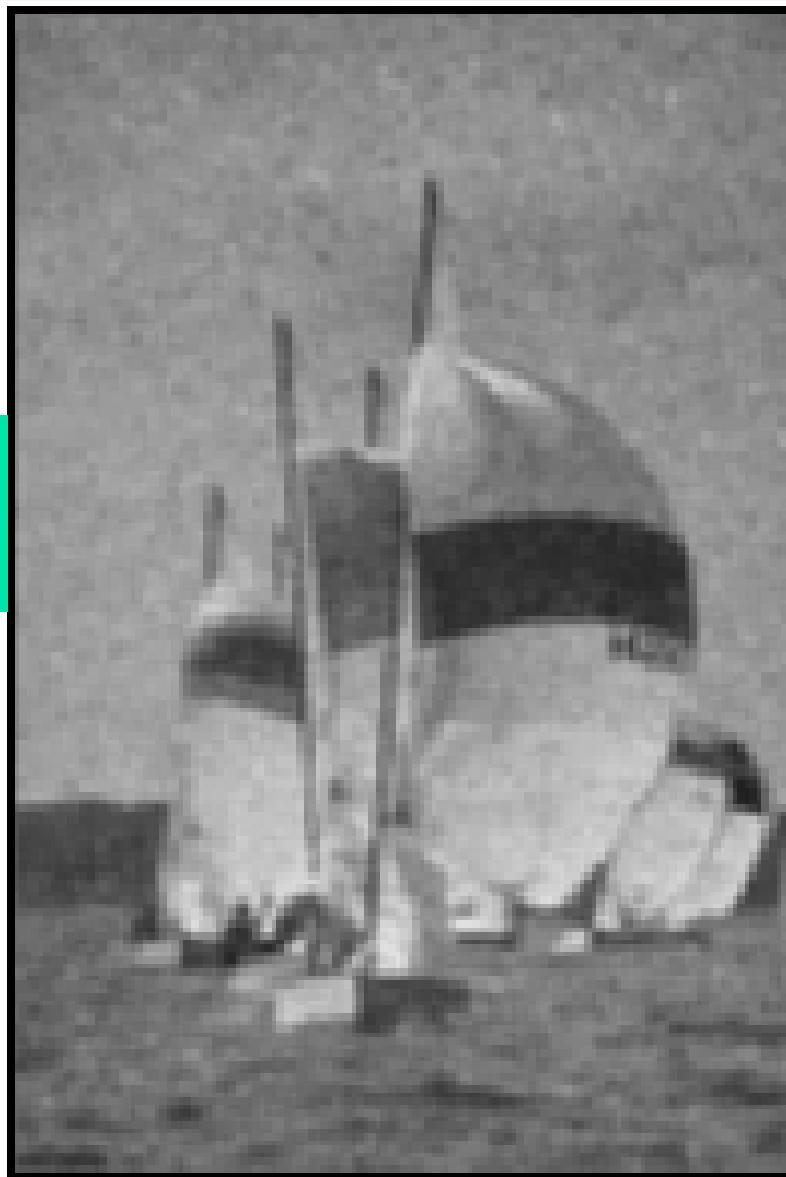
3×3平滑的图像





例子：局部平均法

5×5 平滑的图像





使用低通滤波的阴影校正

- ✓ 2048 2048 × 棋盘式图像
- ✓ 内部方块的大小为128×128像素
- ✓ 低通滤波器512×512高斯核（四倍于方块的大小）， $K=1$ ， $s=128$ （等于方块的大小）。



a b c

图3.42 (a) 由-45°方向的阴影图案阴影的图像。(b) 估计使用低通滤波获得的阴影图案。(c) (a) 除以 (b) 的结果。（见第9.8节阴影校正的形态学方法）。



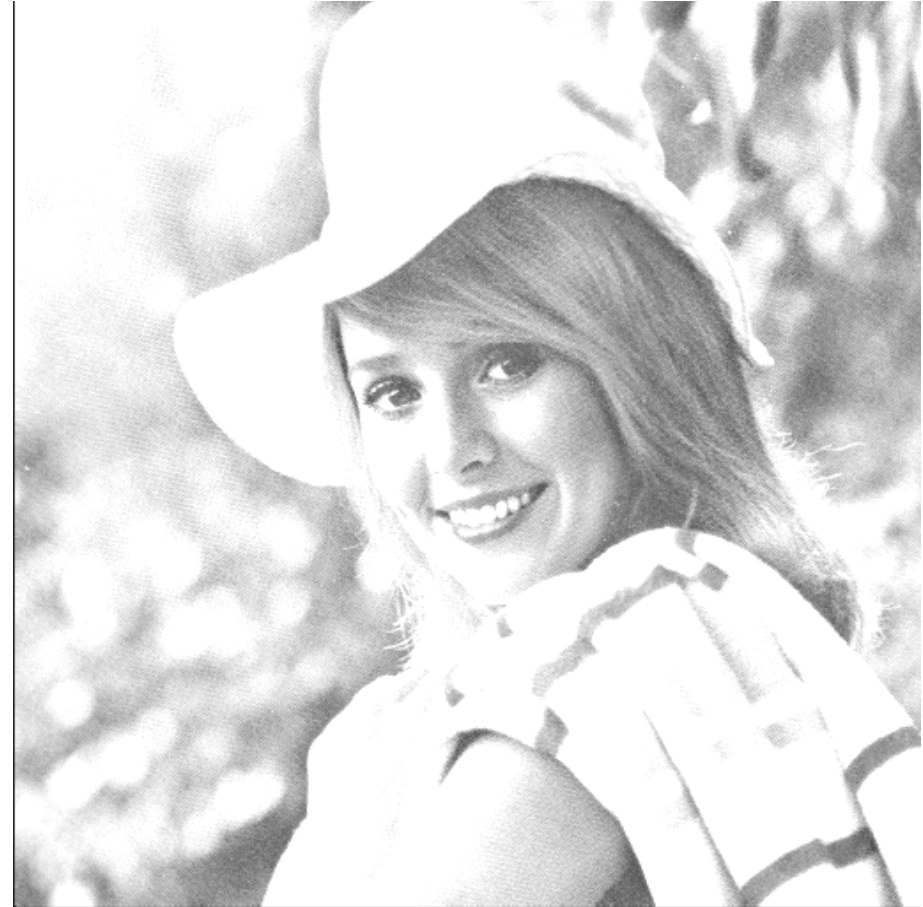
目录

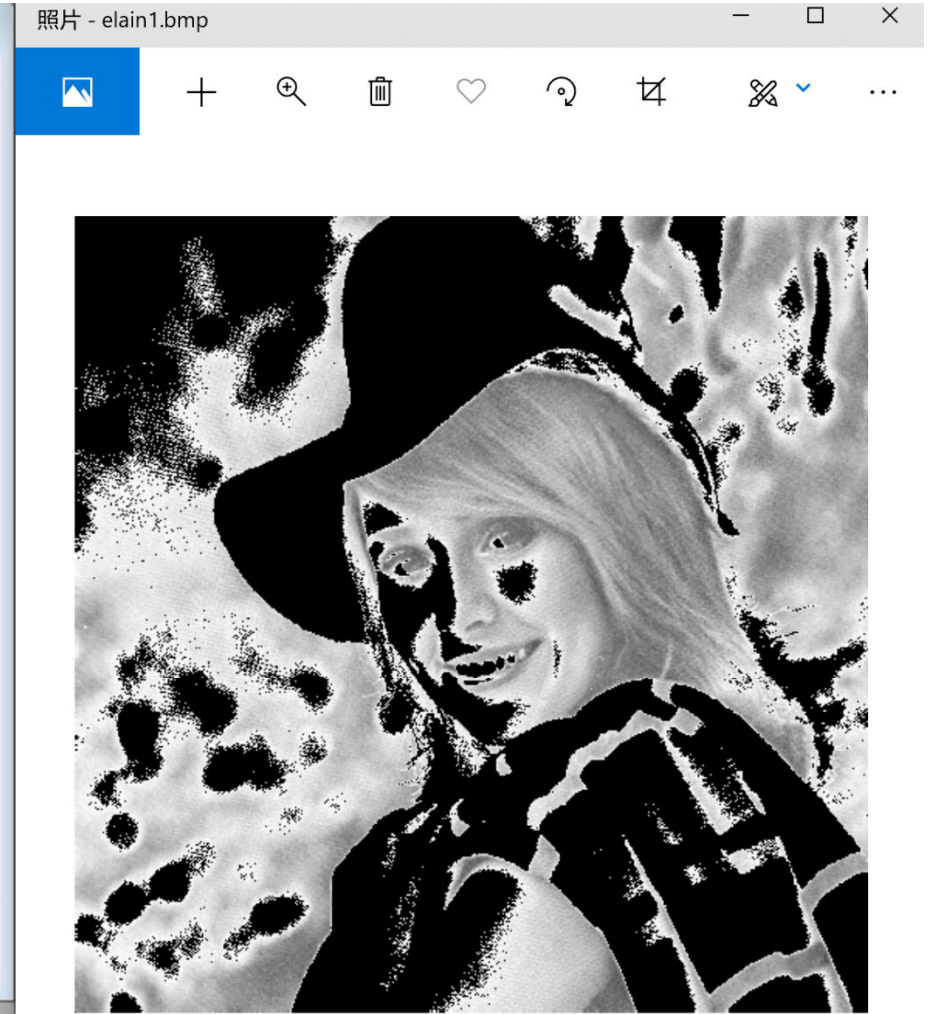
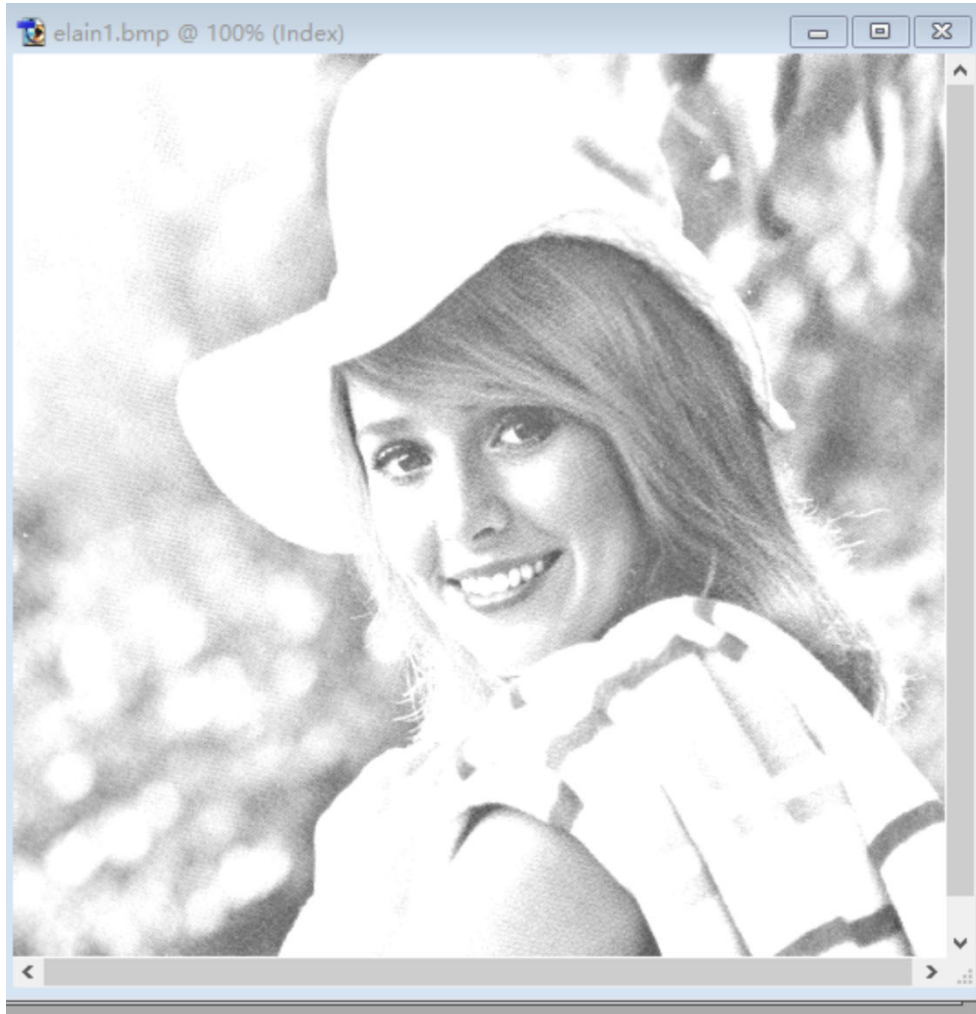
- 强度转换
 - 图像负片
 - 对数转换
 - 伽马变换
 - 碎片化线性变换
- 直方图处理
 - 直方图均衡化
 - 直方图匹配
 - 局部直方图处理
- 空间滤波器
 - 空间相关性和卷积
 - 平滑线性滤波器和空间掩码
 - 非线性滤波器
 - 锐化滤波器





一副图像, bmp和photoshop







位图文件的组成	结构名称	符号
位图文件头(bitmap-file header)	BITMAPFILEHEADER	bmfh
位图信息头(bitmap-information header)	BITMAPINFOHEADER	bmih
彩色表(color table)	RGBQUAD	aColors[]
图象数据阵列字节	BYTE	aBitmapBits[]



	0032h	Important Colors	1 dword	指定重要的颜色数。当该域的值等于颜色数时（或者等于0时），表示所有颜色都一样重要
--	-------	------------------	---------	--

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f	
00000000h:	42	4D	38	04	04	00	00	00	00	00	36	04	00	00	28	00	; BM8.....6...(.
00000010h:	00	00	00	02	00	00	00	02	00	00	01	00	08	00	00	00	;
00000020h:	00	00	00	00	00	00	12	0B	00	00	12	0B	00	00	00	00	;
00000030h:	00	00	00	00	00	00	00	00	00	01	01	01	00	02	02		;
00000040h:	02	00	03	03	03	00	04	04	04	00	05	05	05	00	06	06	;
00000050h:	06	00	07	07	07	00	08	08	08	00	09	09	09	00	0A	0A	;
00000060h:	0A	00	0B	0B	0B	00	0C	0C	0C	00	0D	0D	0D	00	0E	0E	;
00000070h:	0E	00	0F	0F	0F	00	10	10	10	00	11	11	11	00	12	12	;
00000080h:	12	00	13	13	13	00	14	14	14	00	15	15	15	00	16	16	;
00000090h:	16	00	17	17	17	00	18	18	18	00	19	19	19	00	1A	1A	;
000000a0h:	1A	00	1B	1B	1B	00	1C	1C	1C	00	1D	1D	1D	00	1E	1E	;
000000b0h:	1E	00	1F	1F	1F	00	20	20	20	00	21	21	21	00	22	22	; "!!!."
000000c0h:	22	00	23	23	23	00	24	24	24	00	25	25	25	00	26	26	; ".###.\$\$\$.%%%.&&
000000d0h:	26	00	27	27	27	00	28	28	28	00	29	29	29	00	2A	2A	; &.''.(((.))).**
000000e0h:	2A	00	2B	2B	2B	00	2C	2C	2C	00	2D	2D	2D	00	2E	2E	; *.+++.,,,.---...
000000f0h:	2E	00	2F	2F	2F	00	30	30	30	00	31	31	31	00	32	32	; ..///.000.111.22
00000100h:	32	00	33	33	33	00	34	34	34	00	35	35	35	00	36	36	; 2.333.444.555.66
00000110h:	36	00	37	37	37	00	38	38	38	00	39	39	39	00	3A	3A	; 6.777.888.999.::
00000120h:	3A	00	3B	3B	3B	00	3C	3C	3C	00	3D	3D	3D	00	3E	3E	; :.;;;.<<<===.>>
00000130h:	3E	00	3F	3F	3F	00	40	40	40	00	41	41	41	00	42	42	; >.???.@@@.AAA.BB
00000140h:	42	00	43	43	43	00	44	44	44	00	45	45	45	00	46	46	; B.CCC.DDD.EEE.FF
00000150h:	46	00	47	47	47	00	48	48	48	00	49	49	49	00	4A	4A	; F.GGG.HHH.III.JJ
00000160h:	4A	00	4B	4B	4B	00	4C	4C	4C	00	4D	4D	4D	00	4E	4E	; J.KKK.LLL.MMM.NN
00000170h:	4E	00	4F	4F	4F	00	50	50	50	00	51	51	51	00	52	52	; N.OOO.PPP.QQQ.RR
00000180h:	52	00	53	53	53	00	54	54	54	00	55	55	55	00	56	56	; R.SSS.TTT.UUU.VV
00000190h:	56	00	57	57	57	00	58	58	58	00	59	59	59	00	5A	5A	; V.WWW.XXX.YYY.ZZ
000001a0h:	5A	00	5B	5B	5B	00	5C	5C	5C	00	5D	5D	5D	00	5E	5E	; Z.[[.\\ .]]).^
000001b0h:	5E	00	5F	5F	5F	00	60	60	60	00	61	61	61	00	62	62	; ^.___.'``.aaa.bb
000001c0h:	62	00	63	63	63	00	64	64	64	00	65	65	65	00	66	66	; b.ccc.ddd.eee.ff
000001d0h:	66	00	67	67	67	00	68	68	68	00	69	69	69	00	6A	6A	; f.ggg.hhh.iii.jj

54bytes



	0032h	Important Colors	1 dword	指定重要的颜色数。当该域的值等于颜色数时（或者等于0时），表示所有颜色都一样重要
--	-------	------------------	---------	--

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f	
00000000h:	42	4D	6C	02	04	00	00	00	00	00	6A	02	00	00	28	00	; Bm1.....j...(. .
00000010h:	00	00	00	02	00	00	00	02	00	00	01	00	08	00	00	00	;
00000020h:	00	00	00	00	00	00	12	0B	00	00	12	0B	00	00	8D	00	;?
00000030h:	00	00	8D	00	00	00	34	34	34	00	36	36	36	00	37	37	; ..?...444.666.77
00000040h:	37	00	39	39	39	00	3A	3A	3A	00	3C	3C	3C	00	3D	3D	; 7.999.:::.<<<.=
00000050h:	3D	00	3F	3F	3F	00	40	40	40	00	41	41	41	00	43	43	; =.???.@@@.AAA.CC
00000060h:	43	00	44	44	44	00	46	46	46	00	47	47	47	00	49	49	; C.DDD.FFF.GGG.II
00000070h:	49	00	4A	4A	4A	00	4C	4C	4C	00	4D	4D	4D	00	4E	4E	; I.JJJ.LLL.MMM.NN
00000080h:	4E	00	50	50	50	00	51	51	51	00	53	53	53	00	54	54	; N.PPP.QQQ.SSS.TT
00000090h:	54	00	56	56	56	00	57	57	57	00	59	59	59	00	5A	5A	; T.VVV.WWW.YYY.ZZ
000000a0h:	5A	00	5C	5C	5C	00	5D	5D	5D	00	5E	5E	5E	00	60	60	; Z.\\\.]^..`
000000b0h:	60	00	61	61	61	00	63	63	63	00	64	64	64	00	66	66	; `.aaa.ccc.ddd.ff
000000c0h:	66	00	67	67	67	00	69	69	69	00	6A	6A	6A	00	6B	6B	; f.ggg.iii.jjj.kk
000000d0h:	6B	00	6D	6D	6D	00	6E	6E	6E	00	70	70	70	00	71	71	; k.mmm.nnn.ppp.qq
000000e0h:	71	00	73	73	73	00	74	74	74	00	76	76	76	00	77	77	; q.sss.ttt.vvv.ww
000000f0h:	77	00	79	79	79	00	7A	7A	7A	00	7B	7B	7B	00	7D	7D	; w.yyy.zzz.{{{.}}
00000100h:	7D	00	7E	7E	7E	00	80	80	80	00	81	81	81	00	83	83	; }.~vvv.€€€.三?傻
00000110h:	83	00	84	84	84	00	86	86	86	00	87	87	87	00	88	88	; ?剌?俺?噍?盗
00000120h:	88	00	8A	8A	8A	00	8B	8B	8B	00	8D	8D	8D	00	8E	8E	; ?婀?娼?咽?帛
00000130h:	8E	00	90	90	90	00	91	91	91	00	93	93	93	00	94	94	; ?悖?憫?搗?敏
00000140h:	94	00	95	95	95	00	97	97	97	00	98	98	98	00	9A	9A	; ?俺?棗?槁?殮
00000150h:	9A	00	9B	9B	9B	00	9D	9D	9D	00	9E	9E	9E	00	A0	A0	; ?涑?溟?濂?臄
00000160h:	A0	00	A1	A1	A1	00	A3	A3	A3	00	A4	A4	A4	00	A5	A5	; ? ?#?い?ウ
00000170h:	A5	00	A7	A7	A7	00	A8	A8	A8	00	AA	AA	AA	00	AB	AB	; ?è?è? ?驱
00000180h:	AB	00	AD	AD	AD	00	AE	AE	AE	00	B0	B0	B0	00	B1	B1	; ?綜?簪?鞍?北
00000190h:	B1	00	B2	B2	B2	00	B4	B4	B4	00	B5	B5	B5	00	B7	B7	; ?膊?创?档?贩
000001a0h:	B7	00	B8	B8	B8	00	BA	BA	BA	00	BB	BB	BB	00	BD	BD	; ?父?汉?换?浇
000001b0h:	BD	00	BE	BE	BE	00	C0	C0	C0	00	C1	C1	C1	00	C2	C2	; ?揪?览?亮?侣
000001c0h:	C2	00	C4	C4	C4	00	C5	C5	C5	00	C7	C7	C7	00	C8	C8	; ?哪?排?乔?热
000001d0h:	C8	00	CA	CA	CA	00	CB	CB	CB	00	CD	CD	CD	00	CE	CE	; ?适?怂?屯?挝

8D=141



图像与视频处理 Image and Video Processing

```
0 1 2 3 4 5 6 7 8 9 a b c d e f
00000000h: 42 4D 6C 02 04 00 00 00 00 00 6A 02 00 00 28 00 ; BM1.....j...(.
00000010h: 00 00 00 02 00 00 00 02 00 00 01 00 08 00 00 00 ; .....
00000020h: 00 00 00 00 00 00 12 0B 00 00 12 0B 00 00 8D 00 ; .....?
00000030h: 00 00 8D 00 00 00 34 34 34 00 36 36 36 00 37 37 ; ..?...444.666.77
00000040h: 37 00 39 39 39 00 3A 3A 3A 00 3C 3C 3C 00 3D 3D ; 7.999.:::.<<<==
00000050h: 3D 00 3F 3F 3F 00 40 40 40 00 41 41 41 00 43 43 ; =.???.@@@.AAA.CC
00000060h: 43 00 44 44 44 00 46 46 46 00 47 47 47 00 49 49 ; C.DDD.FFF.GGG.II
00000070h: 49 00 4A 4A 4A 00 4C 4C 4C 00 4D 4D 4D 00 4E 4E ; I.JJJ.LLL.MMM.NN
00000080h: 4E 00 50 50 50 00 51 51 51 00 53 53 53 00 54 54 ; N.PPP.QQQ.SSS.TT
00000090h: 54 00 56 56 56 00 57 57 57 00 59 59 59 00 5A 5A ; T.VVV.WWW.YYY.ZZ
000000a0h: 5A 00 5C 5C 5C 00 5D 5D 5D 00 5E 5E 5E 00 60 60 ; Z.\\\.]]].^^.``
000000b0h: 60 00 61 61 61 00 63 63 63 00 64 64 64 00 66 66 ; `.aaa.ccc.ddd.ff
000000c0h: 66 00 67 67 67 00 69 69 69 00 6A 6A 6A 00 6B 6B ; f.ggg.iii.jjj.kk
000000d0h: 6B 00 6D 6D 6D 00 6E 6E 6E 00 70 70 70 00 71 71 ; k.mmm.nnn.ppp.qq
000000e0h: 71 00 73 73 73 00 74 74 74 00 76 76 76 00 77 77 ; q.sss.ttt.vvv.ww
000000f0h: 77 00 79 79 79 00 7A 7A 7A 00 7B 7B 7B 00 7D 7D ; w.yyy.zzz.{{.}}
00000100h: 7D 00 7E 7E 7E 00 80 80 80 00 81 81 81 00 83 83 ; }.nnn.€€€.三?傻
00000110h: 83 00 84 84 84 00 86 86 86 00 87 87 87 00 88 88 ; ?劉?噯?噯?盜
00000120h: 88 00 8A 8A 8A 00 8B 8B 8B 00 8D 8D 8D 00 8E 8E ; ?婀?嫗?嶗?帟
00000130h: 8E 00 90 90 90 00 91 91 91 00 93 93 93 00 94 94 ; ?佇?憫?搗?敏
00000140h: 94 00 95 95 95 00 97 97 97 00 98 98 98 00 9A 9A ; ?噯?策?檣?殮
00000150h: 9A 00 9B 9B 9B 00 9D 9D 9D 00 9E 9E 9E 00 A0 A0 ; ?漑?溟?濤?濂
00000160h: A0 00 A1 A1 A1 00 A3 A3 A3 00 A4 A4 A4 00 A5 A5 ; ? ?#?い?ヶ
00000170h: A5 00 A7 A7 A7 00 A8 A8 A8 00 AA AA AA 00 AB AB ; ?E?è? ?驱
00000180h: AB 00 AD AD AD 00 AE AE AE 00 B0 B0 B0 00 B1 B1 ; ?綜?谿?鞍?北
00000190h: B1 00 B2 B2 B2 00 B4 B4 B4 00 B5 B5 B5 00 B7 B7 ; ?膊?创?档?販
000001a0h: B7 00 B8 B8 B8 00 BA BA BA 00 BB BB BB 00 BD BD ; ?父?汉?換?澆
000001b0h: BD 00 BE BE BE 00 C0 C0 C0 00 C1 C1 C1 00 C2 C2 ; ?揪?览?亮?侶
000001c0h: C2 00 C4 C4 C4 00 C5 C5 C5 00 C7 C7 C7 00 C8 C8 ; ?哪?排?乔?热
000001d0h: C8 00 CA CA CA 00 CB CB CB 00 CD CD CD 00 CE CE ; ?适?怱?屯?挝
000001e0h: CE 00 CF CF CF 00 D1 D1 D1 00 D2 D2 D2 00 D4 D4 ; ?舷?艇?乙?裁
000001f0h: D4 00 D5 D5 D5 00 D7 D7 D7 00 D8 D8 D8 00 DA DA ; ?照?鬃?劓?潛
00000200h: DA 00 DB DB DB 00 DC DC DC 00 DE DE DE 00 DF DF ; ?圪?尙?擲?哌
00000210h: DF 00 E1 E1 E1 00 E2 E2 E2 00 E4 E4 E4 00 E5 E5 ; ?後?付?滢?邁
00000220h: E5 00 E7 E7 E7 00 E8 E8 E8 00 EA EA EA 00 EB EB ; ?珑?挺?覲?脰
00000230h: EB 00 EC EC EC 00 EE EE EE 00 EF EF EF 00 F1 F1 ; ?祆?铤?鑣?耨
00000240h: F1 00 F2 F2 F2 00 F4 F4 F4 00 F5 F5 F5 00 F7 F7 ; ?蚪?趙?貂?鼯
00000250h: F7 00 F8 F8 F8 00 F9 F9 F9 00 FB FB FB 00 FC FC ; ? ? ? ?
00000260h: FC 00 FE FE FE 00 FF FF FF 00 01 49 46 3E 37 2C ; ? ? ..IF>7,
```



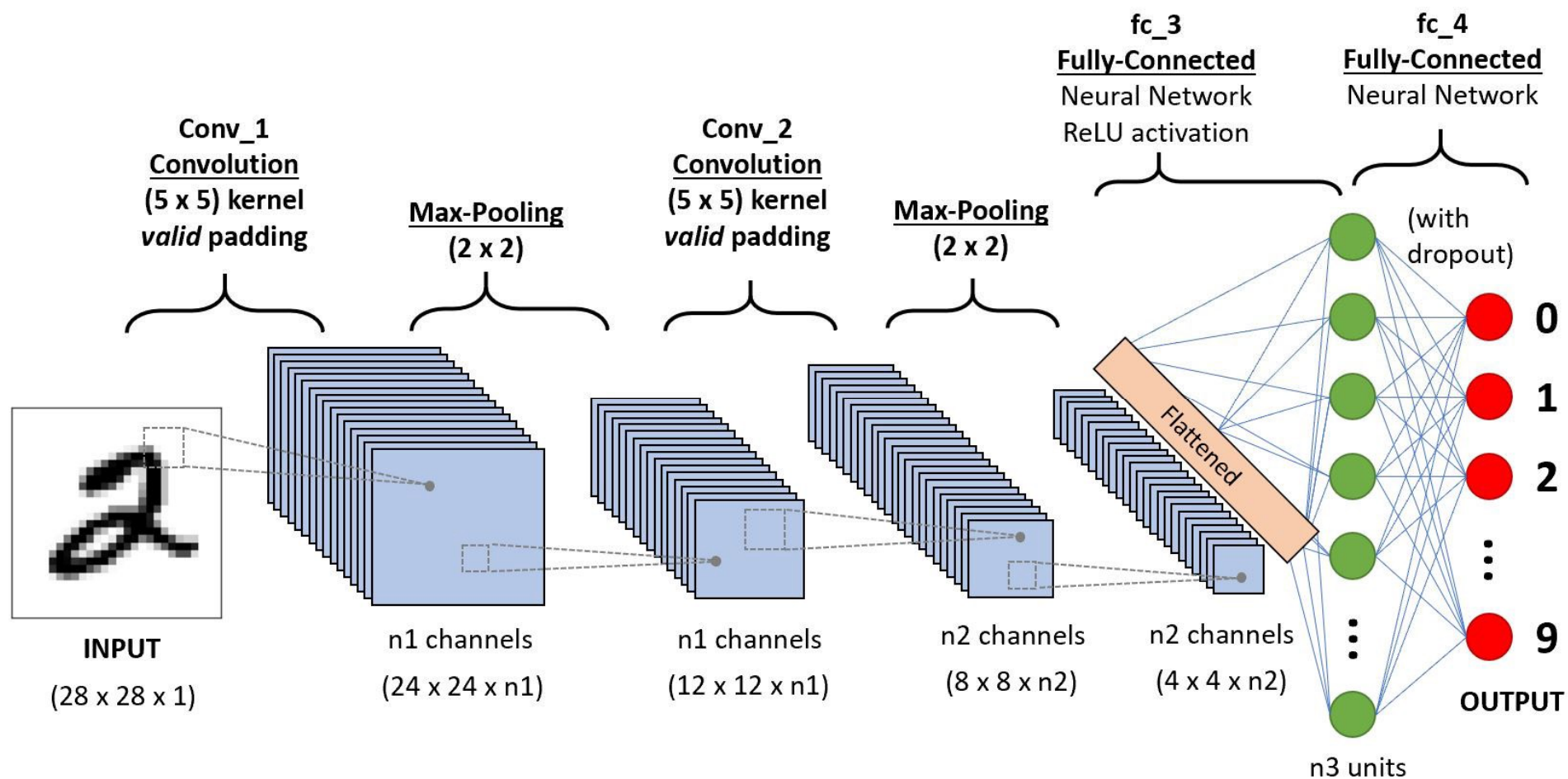

图像与视频处理 Image and Video Processing



	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
00000000h:	42	4D	38	04	04	00	00	00	00	00	36	04	00	00	28	00	; BM8.....6...(. 00000010h:	00	00	00	02	00	00	00	02	00	01	00	08	00	00	00	; 00000020h:	00	00	00	00	00	00	00	12	08	00	00	12	08	00	00	00	; 00000030h:	00	00	00	00	00	00	00	00	00	01	01	01	01	01	02	02	; 00000040h:	02	00	03	03	03	00	04	04	04	00	05	05	05	00	06	06	; 00000050h:	06	00	07	07	07	00	08	08	08	00	09	09	09	00	0A	0A	; 00000060h:	0A	00	0B	0B	0B	00	0C	0C	0C	00	0D	0D	0D	00	0E	0E	; 00000070h:	0E	00	0F	0F	0F	00	10	10	10	00	11	11	11	00	12	12	; 00000080h:	12	00	13	13	13	00	14	14	14	00	15	15	15	00	16	16	; 00000090h:	16	00	17	17	17	00	18	18	18	00	19	19	19	00	1A	1A	; 000000a0h:	1A	00	1B	1B	1B	00	1C	1C	1C	00	1D	1D	1D	00	1E	1E	; 000000b0h:	1E	00	1F	1F	1F	00	20	20	20	00	21	21	21	00	22	22	; !!!."" 000000c0h:	22	00	23	23	23	00	24	24	24	00	25	25	25	00	26	26	; ".###.\$\$\$.%%%.&& 000000d0h:	26	00	27	27	27	00	28	28	28	00	29	29	29	00	2A	2A	; &.''.(((.))).** 000000e0h:	2A	00	2B	2B	2B	00	2C	2C	2C	00	2D	2D	2D	00	2E	2E	; *.+++,---... 000000f0h:	2E	00	2F	2F	2F	00	30	30	30	00	31	31	31	00	32	32	; ..///.000.111.22 00000100h:	32	00	33	33	33	00	34	34	34	00	35	35	35	00	36	36	; 2.333.444.555.66 00000110h:	36	00	37	37	37	00	38	38	38	00	39	39	39	00	3A	3A	; 6.777.888.999.:: 00000120h:	3A	00	3B	3B	3B	00	3C	3C	3C	00	3D	3D	3D	00	3E	3E	; :.;;.<<<==>> 00000130h:	3E	00	3F	3F	3F	00	40	40	40	00	41	41	41	00	42	42	; >.>??.@@@.AAA.BB 00000140h:	42	00	43	43	43	00	44	44	44	00	45	45	45	00	46	46	; B.CCC.DDD.EEE.FF 00000150h:	46	00	47	47	47	00	48	48	48	00	49	49	49	00	4A	4A	; F.GGG.HHH.III.JJ 00000160h:	4A	00	4B	4B	4B	00	4C	4C	4C	00	4D	4D	4D	00	4E	4E	; J.KKK.LLL.MMM.NN 00000170h:	4E	00	4F	4F	4F	00	50	50	50	00	51	51	51	00	52	52	; N.OOO.PPP.QQQ.RR 00000180h:	52	00	53	53	53	00	54	54	54	00	55	55	55	00	56	56	; R.SSS.TTT.UUU.VV 00000190h:	56	00	57	57	57	00	58	58	58	00	59	59	59	00	5A	5A	; V.WWW.XXX.YYY.ZZ 000001a0h:	5A	00	5B	5B	5B	00	5C	5C	5C	00	5D	5D	5D	00	5E	5E	; Z.[[.\\\\.]]).^ 000001b0h:	5E	00	5F	5F	5F	00	60	60	60	00	61	61	61	00	62	62	; ^._.''.aaa.bb 000001c0h:	62	00	63	63	63	00	64	64	64	00	65	65	65	00	66	66	; b.ccc.ddd.eee.ff 000001d0h:	66	00	67	67	67	00	68	68	68	00	69	69	69	00	6A	6A	; f.ggg.hhh.iii.jj 000001e0h:	6A	00	6B	6B	6B	00	6C	6C	6C	00	6D	6D	6D	00	6E	6E	; j.kkk.lll.mmm.nn 000001f0h:	6E	00	6F	6F	6F	00	70	70	70	00	71	71	71	00	72	72	; n.ooo.ppp.qqq.rr 00000200h:	72	00	73	73	73	00	74	74	74	00	75	75	75	00	76	76	; r.sss.ttt.uuu.vv 00000210h:	76	00	77	77	77	00	78	78	78	00	79	79	79	00	7A	7A	; v.www.xxx.yyy.zz 00000220h:	7A	00	7B	7B	7B	00	7C	7C	7C	00	7D	7D	7D	00	7E	7E	; z.{{.} .}}}.~w 00000230h:	7E	00	7F	7F	7F	00	80	80	80	00	81	81	81	00	82	82	; ~.w.€€€.三?倭 00000240h:	82	00	83	83	83	00	84	84	84	00	85	85	85	00	86	86	; ?倭?剌?厖?俺 00000250h:	86	00	87	87	87	00	88	88	88	00	89	89	89	00	8A	8A	; ?俺?盗?嗷?婀 00000260h:	8A	00	8B	8B	8B	00	8C	8C	8C	00	8D	8D	8D	00	8E	8E	; ?娼?窟?咽?唇 00000270h:	8E	00	8F	8F	8F	00	90	90	90	00	91	91	91	00	92	92	; ?零?停?憫?掀 00000280h:	92	00	93	93	93	00	94	94	94	00	95	95	95	00	96	96	; ?塿?敬?醺?枫 00000290h:	96	00	97	97	97	00	98	98	98	00	99	99	99	00	9A	9A	; ?棗?梅?樞?殮 000002a0h:	9A	00	9B	9B	9B	00	9C	9C	9C	00	9D	9D	9D	00	9E	9E	; ?瀌?潜?溟?滢 000002b0h:	9E	00	9F	9F	9F	00	A0	A0	A0	00	A1	A1	A1	00	A2	A2	; ?煙?醴? ?ii 000002c0h:	A2	00	A3	A3	A3	00	A4	A4	A4	00	A5	A5	A5	00	A6	A6	; ?#?{? ?Z 000002d0h:	A6	00	A7	A7	A7	00	A8	A8	A8	00	A9	A9	A9	00	AA	AA	; ?È?è?---? 000002e0h:	AA	00	AB	AB	AB	00	AC	AC	AC	00	AD	AD	AD	00	AE	AE	; ?驱?郅?縗?簪 000002f0h:	AE	00	AF	AF	AF	00	B0	B0	B0	00	B1	B1	B1	00	B2	B2	; ?绝?鞍?北?脾 00000300h:	B2	00	B3	B3	B3	00	B4	B4	B4	00	B5	B5	B5	00	B6	B6	; ?吵?创?档?抖 00000310h:	B6	00	B7	B7	B7	00	B8	B8	B8	00	B9	B9	B9	00	BA	BA	; ?贩?父?构?汉 00000320h:	BA	00	BB	BB	BB	00	BC	BC	BC	00	BD	BD	BD	00	BE	BE	; ?换?技?浇?揪 00000330h:	BE	00	BF	BF	BF	00	C0	C0	C0	00	C1	C1	C1	00	C2	C2	; ?靠?览?亮?侶 00000340h:	C2	00	C3	C3	C3	00	C4	C4	C4	00	C5	C5	C5	00	C6	C6	; ?妹?哪?排?破 00000350h:	C6	00	C7	C7	C7	00	C8	C8	C8	00	C9	C9	C9	00	CA	CA	; ?乔?热?缮?适 00000360h:	CA	00	CB	CB	CB	00	CC	CC	CC	00	CD	CD	CD	00	CE	CE	; ?忒?烫?屯?掇 00000370h:	CE	00	CF	CF	CF	00	D0	D0	D0	00	D1	D1	D1	00	D2	D2	; ?舷?行?艇?乙 00000380h:	D2	00	D3	D3	D3	00	D4	D4	D4	00	D5	D5	D5	00	D6	D6	; ?佑?裁?照?种 00000390h:	D6	00	D7	D7	D7	00	D8	D8	D8	00	D9	D9	D9	00	DA	DA	; ?髻?刳?儗?潜 000003a0h:	DA	00	DB	DB	DB	00	DC	DC	DC	00	DD	DD	DD	00	DE	DE	; ?圻?苜?萁?鄒 000003b0h:	DE	00	DF	DF	DF	00	E0	E0	E0	00	E1	E1	E1	00	E2	E2	; ?哌?嘈?後?付 000003c0h:	E2	00	E3	E3	E3	00	E4	E4	E4	00	E5	E5	E5	00	E6	E6	; ?泮?滢?邁?骀 000003d0h:	E6	00	E7	E7	E7	00	E8	E8	E8	00	E9	E9	E9	00	EA	EA	; ?珑?挺?輝?觀 000003e0h:	EA	00	EB	EB	EB	00	EC	EC	EC	00	ED	ED	ED	00	EE	EE	; ?脰?沃?吗?铽 000003f0h:	EE	00	EF	EF	EF	00	F0	F0	F0	00	F1	F1	F1	00	F2	F2	; ?鑊?痧?穉?蜩 00000400h:	F2	00	F3	F3	F3	00	F4	F4	F4	00	F5	F5	F5	00	F6	F6	; ?篴?越?貂?蠅 00000410h:	F6	00	F7	F7	F7	00	F8	F8	F8	00	F9	F9	F9	00	FA	FA	; ?駟? ? ? 00000420h:	FA	00	FB	FB	FB	00	FC	FC	FC	00	FD	FD	FD	00	FE	FE	; ? ? ? ? 00000430h:	FE	00	FF	FF	FF	00	01	49	46	3E	37	2C	38	37	46	44	; ? ..IF>7,87FD

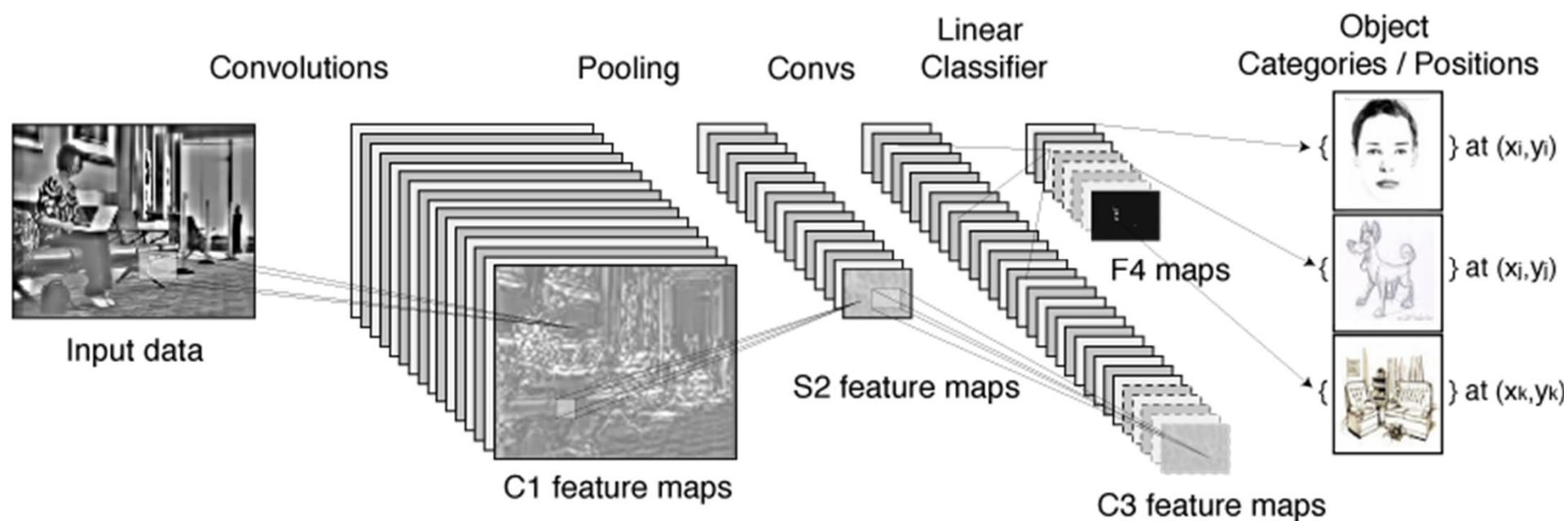


卷积神经网络





特征

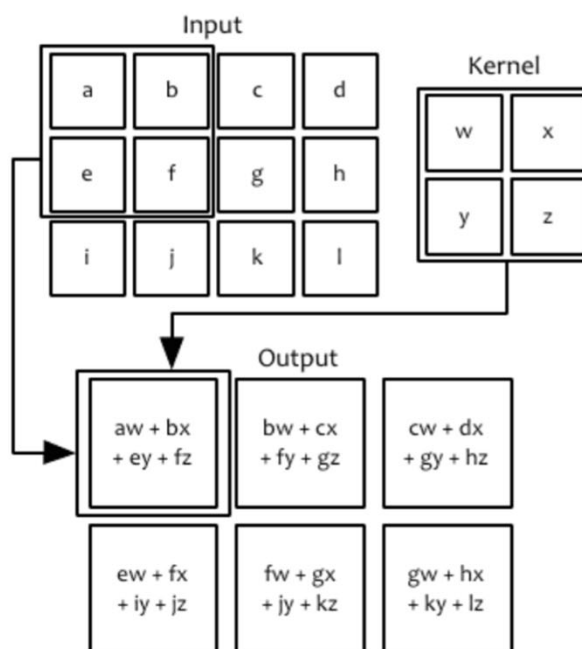




CNN中的卷积操作

- 输入：一个图像（二维数组） x
- 卷积核/运算器（可学习参数的二维阵列）： w
- 特征图（处理数据的二维数组）： S
- 在二维领域的卷积操作。

$$s[i, j] = (x * w)[i, j] = \sum_{m=-M}^M \sum_{n=-N}^N x[i + m, j + n] w[m, n]$$





空间卷积

0	0	0	0	0	0	0
0	1	0	0	0	1	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

输入图像



0	0	1
1	0	0
0	1	1

特征检测器

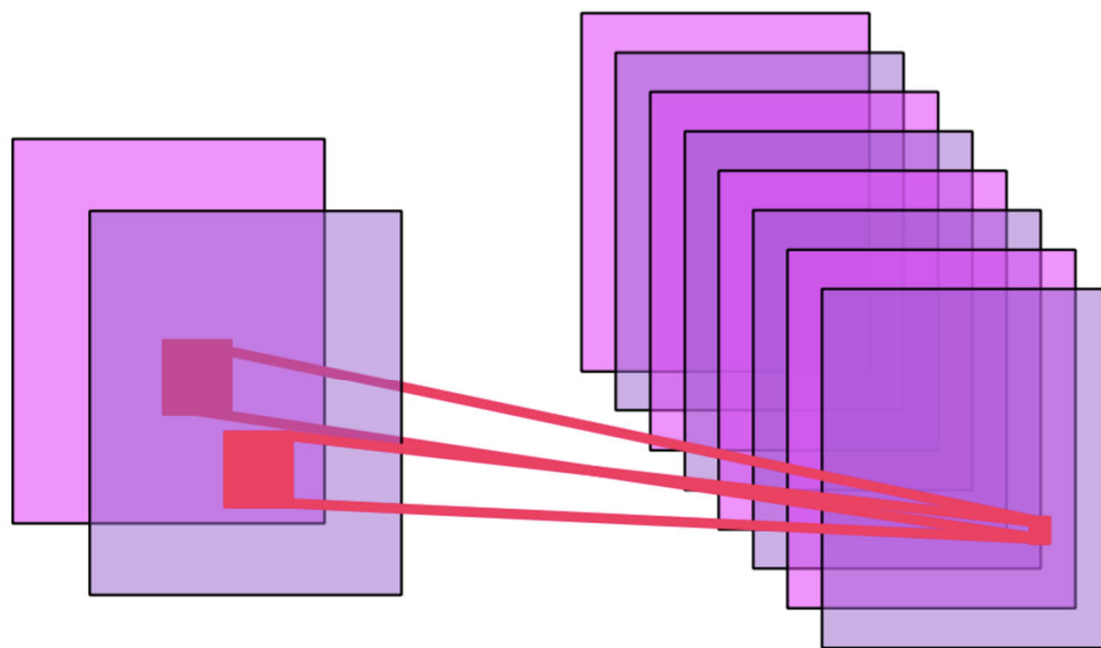


0	1	0	0	0
0	1	1	1	0
1	0	1	2	1
1	4			

特征图



多卷积



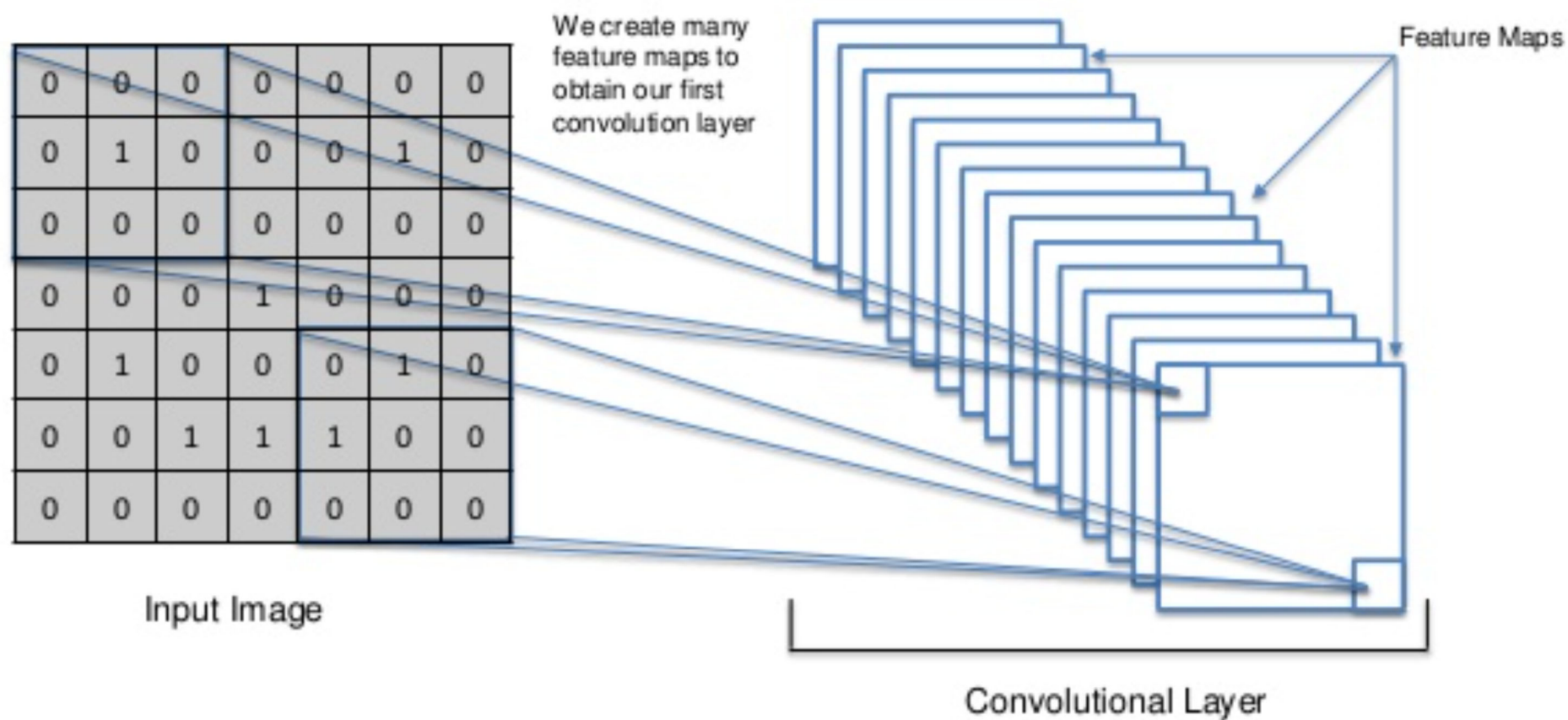
多卷积

通常有多个特征图，每个卷积运算符都有一个。



图像与视频处理
Image and Video Processing

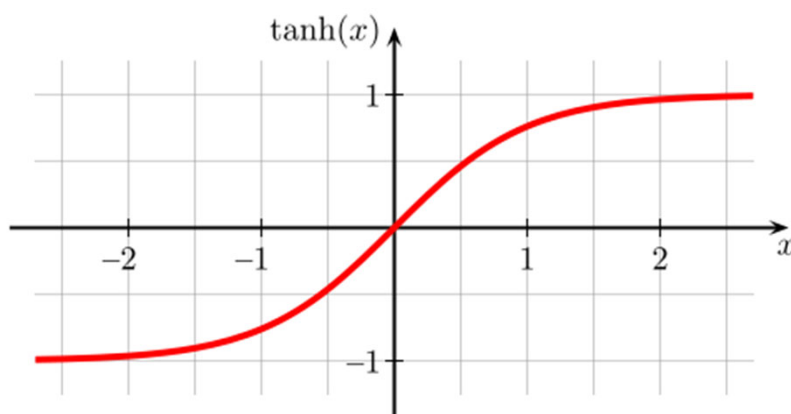
多特征图





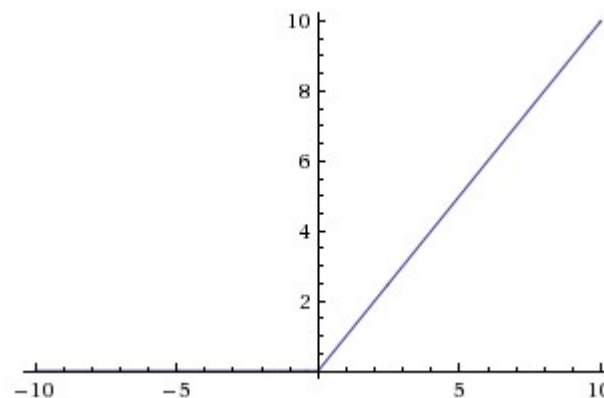
非线性

■ Tanh(x)



$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

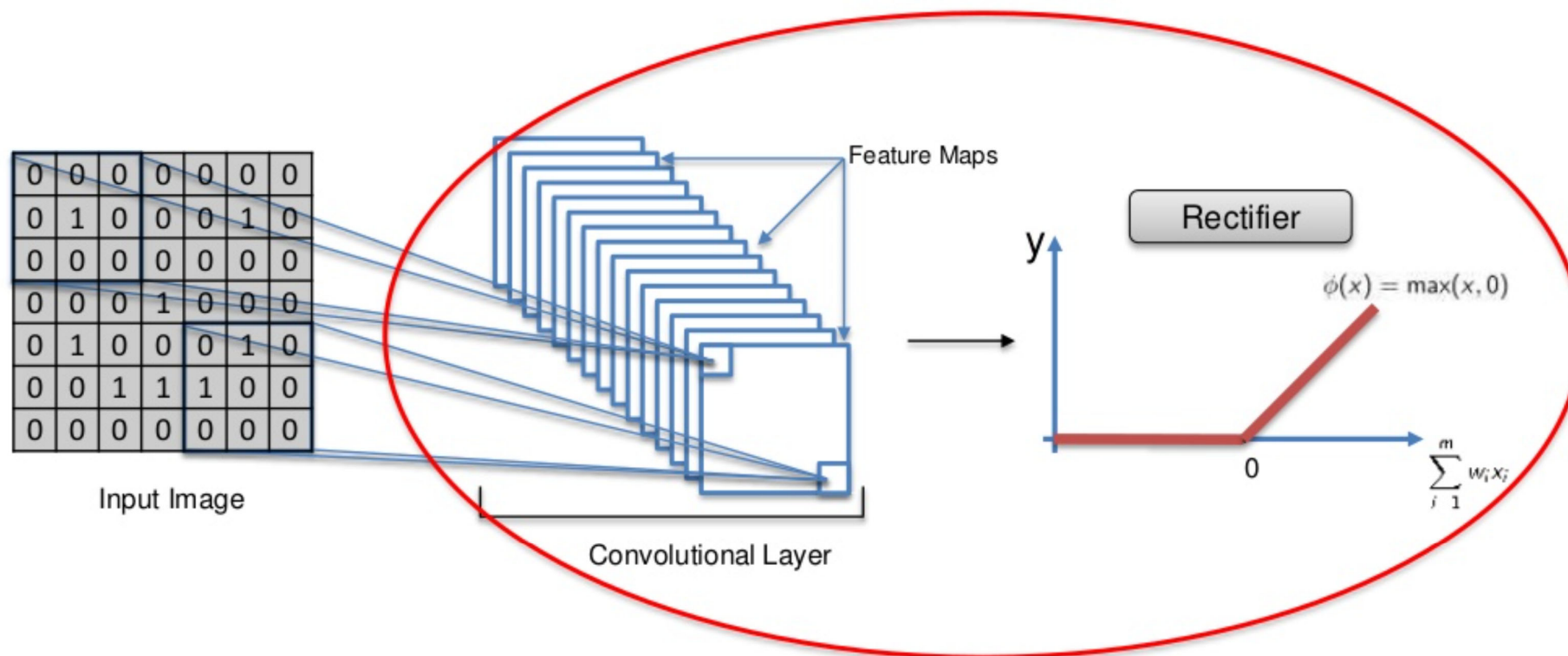
■ ReLU



$$f(x) = \max(0, x)$$



ReLU



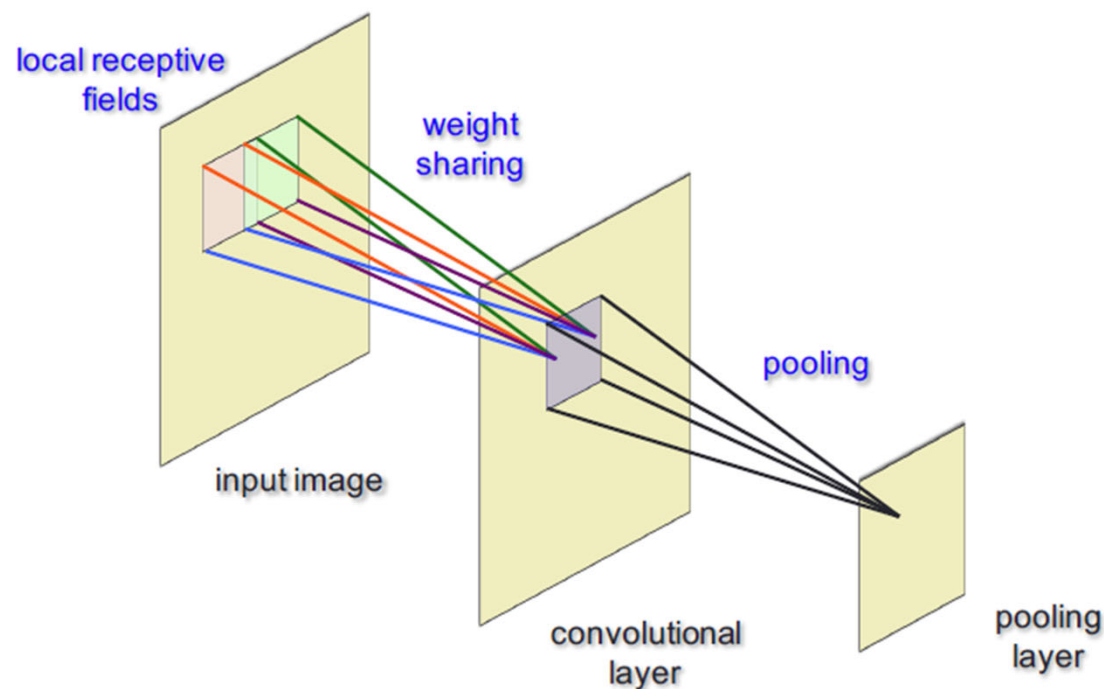


<https://www.slideshare.net/KirillEremenko/de>



池化

- 常见的集合操作。
 - **Max pooling (最大池化)** :报告一个矩形邻域内的最大输出。
 - **Average pooling (平均池化)** :报告一个矩形邻域的平均输出 (可能按与中心像素的距离加权) 。





Max pooling (最大池化)

0	1	0	0	0
0	1	1	1	0
1	0	1	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

Max Pooling



1	1	0
4	2	

Pooled Feature Map

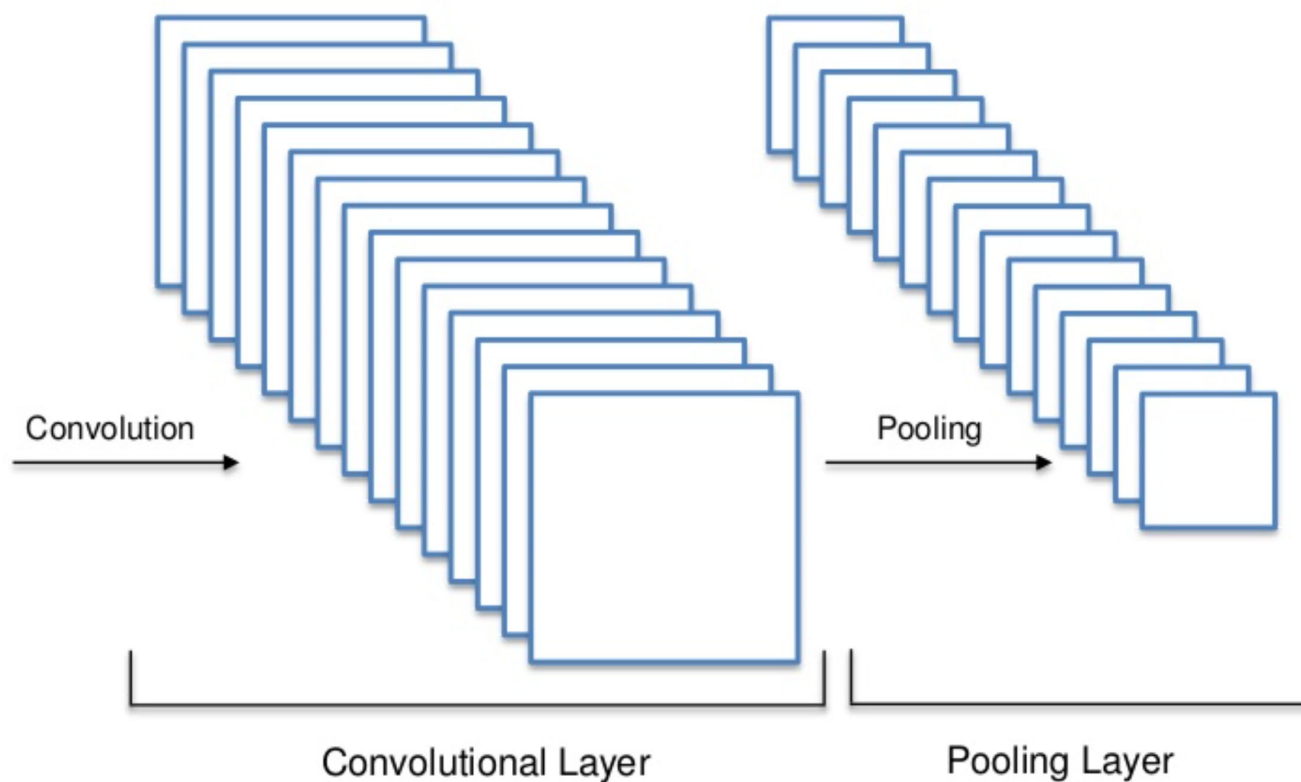


图像与视频处理
Image and Video Processing

池化层

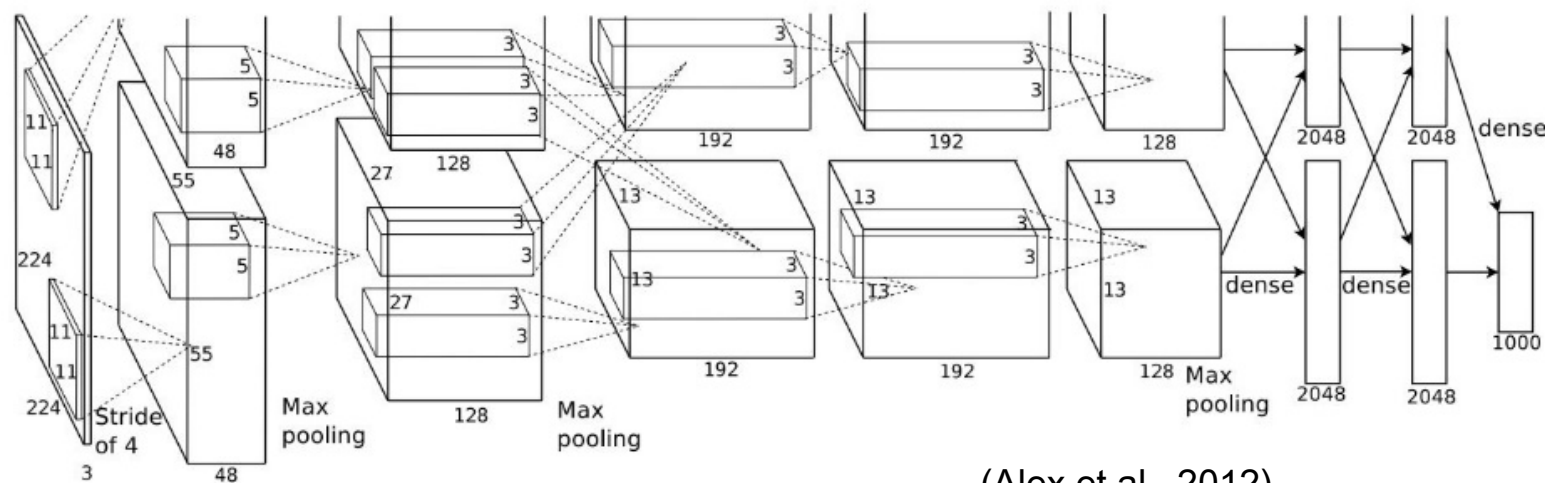
0	0	0	0	0	0	0
0	1	0	0	0	1	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image





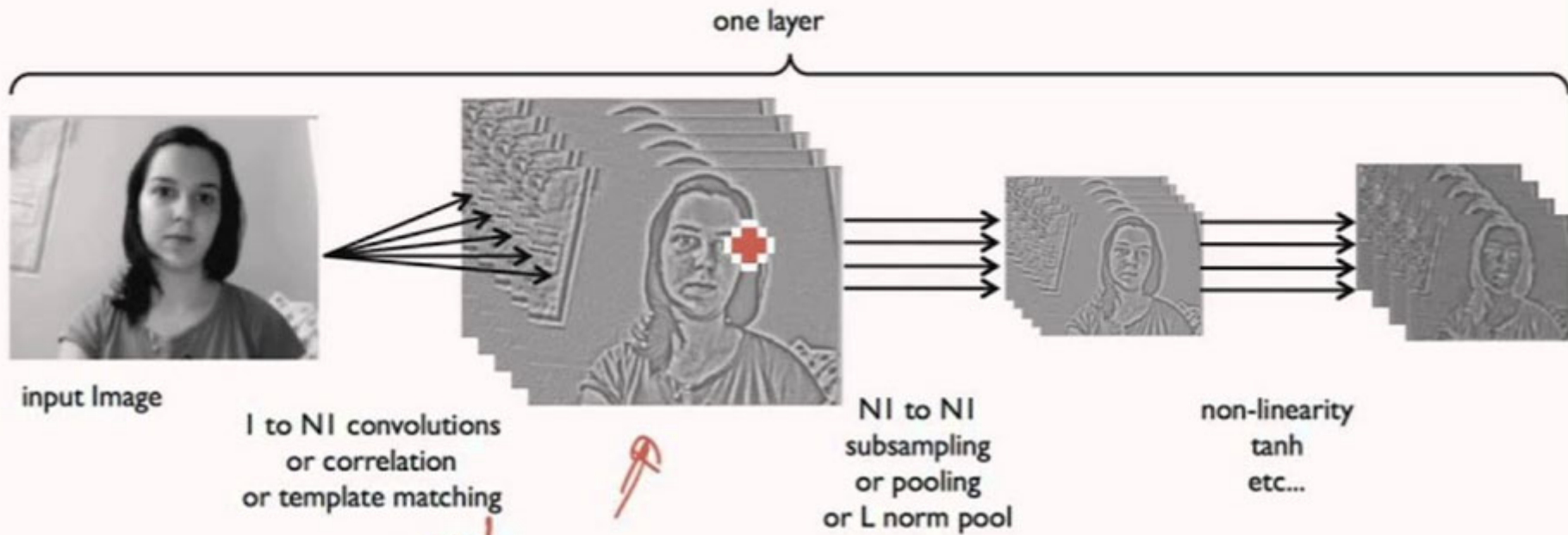
Deep CNN: ImageNet 2012 赢家



(Alex et al., 2012)

- 每个卷积层有多个特征图。
- 多个卷积层用于提取不同层次的特征。
- 高层将低层的特征图作为输入。

Convnets (Fukushima, LeCun, Hinton)





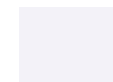
阶次统计学（非线性）滤波器

- 非线性

- 基于过滤掩码中包含的像素的排序（排名）。

- 用排序结果决定的值替换中心像素的值

例如，中值过滤器（脉冲噪声-椒盐噪声），最大过滤器，最小过滤器





中位数过滤

123	125	126	130	140
122	124	126	127	135
118	120	150	125	134
119	115	119	123	133
111	116	110	120	130

Neighbourhood values:

115, 119, 120, 123, 124,
125, 126, 127, 150

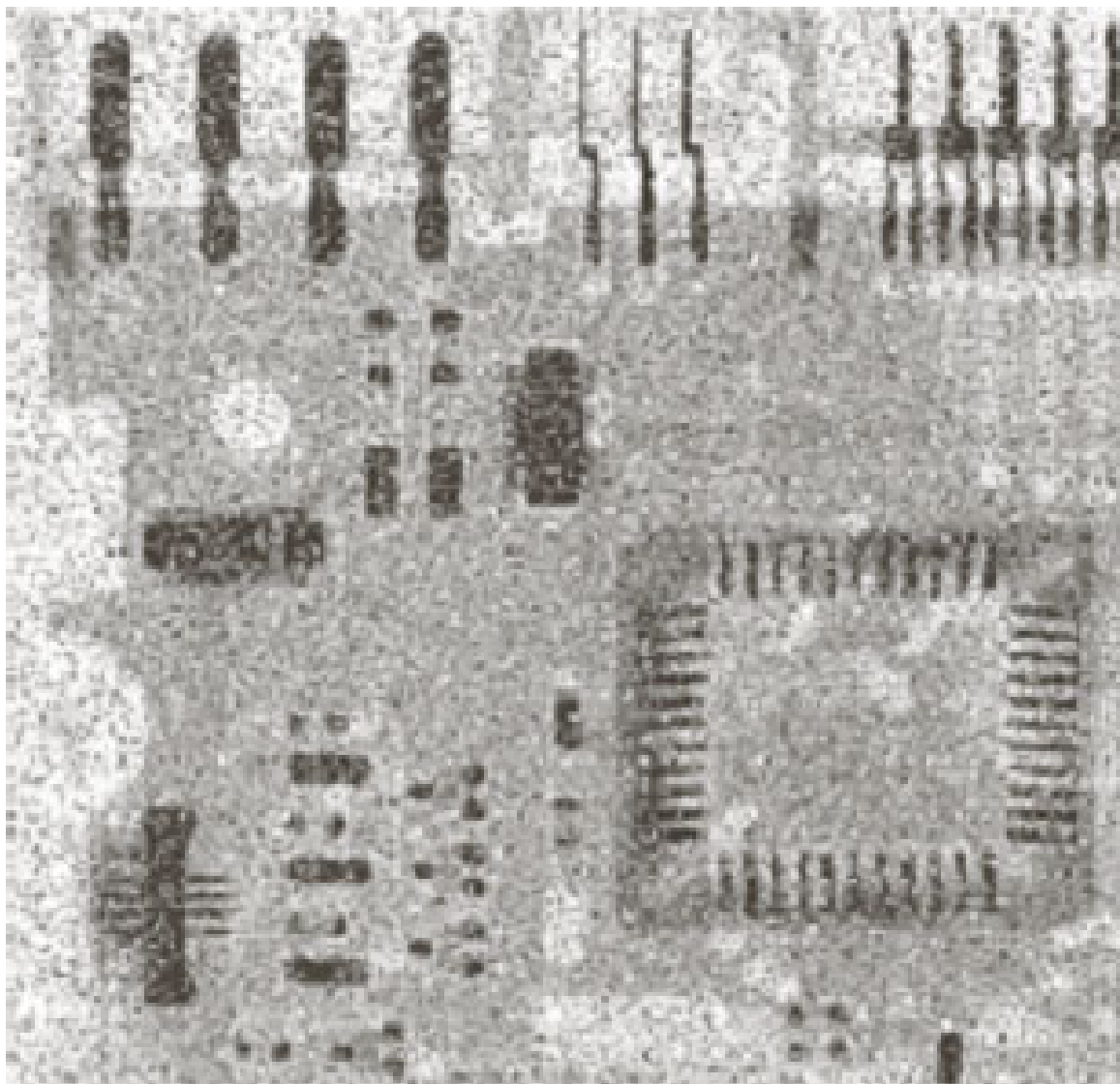
Median value: 124

强制具有明显强度的点更像它们的邻居
孤立的像素群相对于它们的邻居来说是亮的或暗的。



例子：使用中值滤波进行降噪

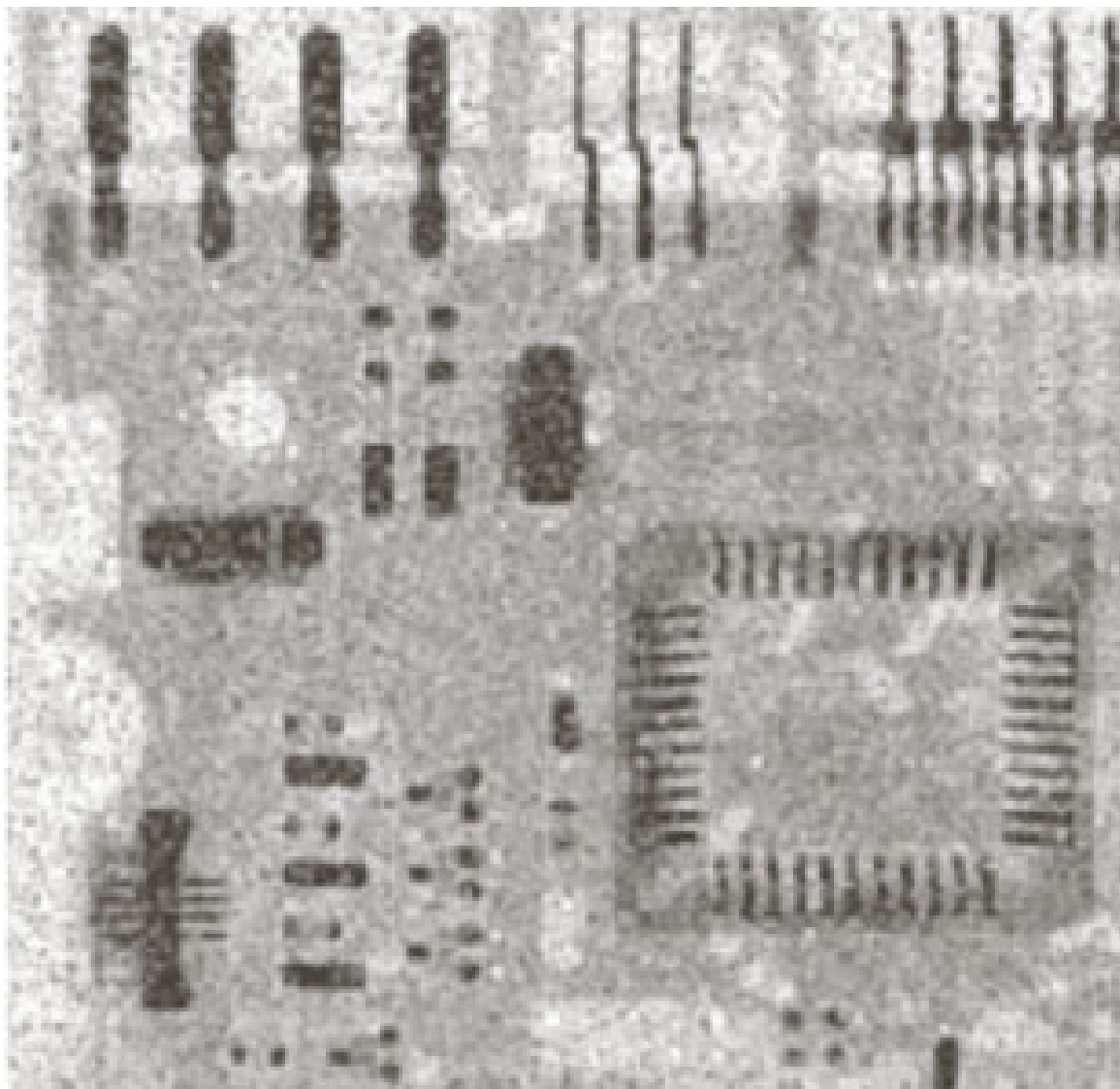
椒盐噪声





例子：使用中值滤波进行降噪

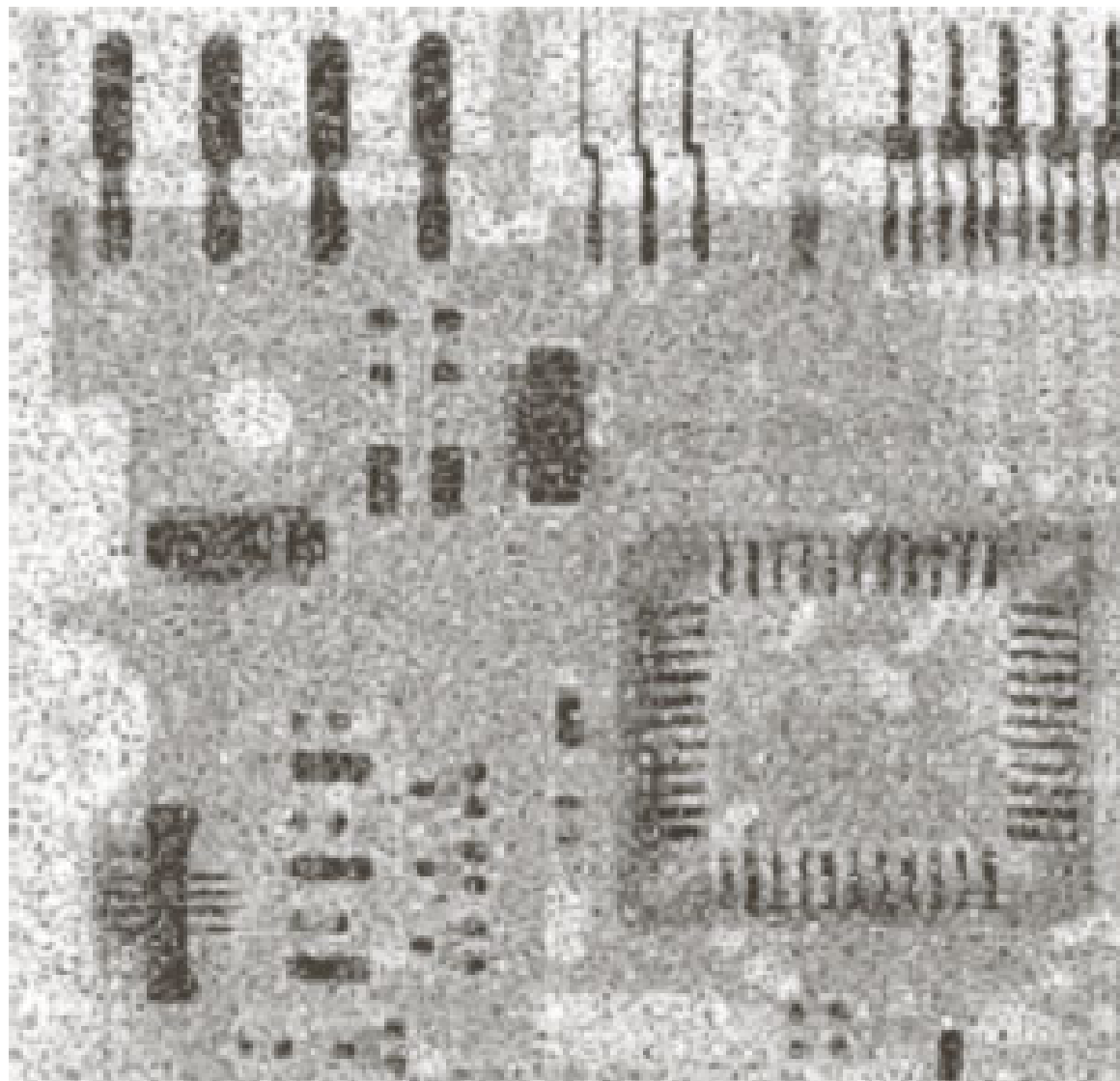
3×3平均掩膜





例子：使用中值滤波进行降噪

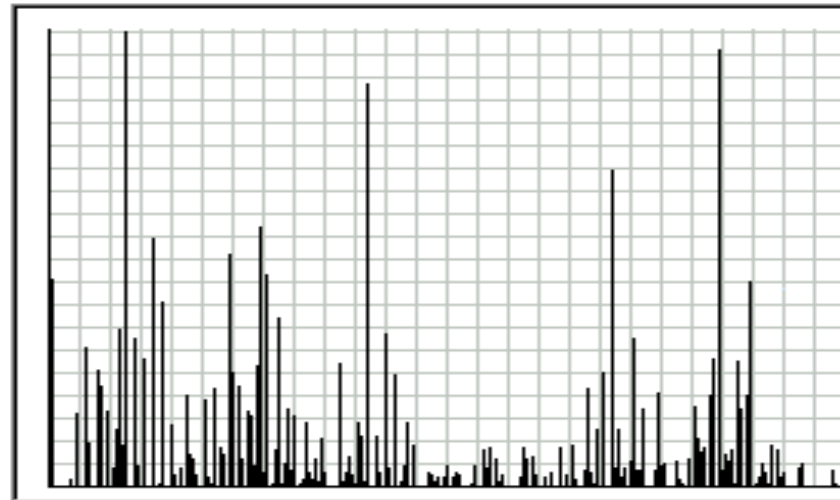
3×3 中值





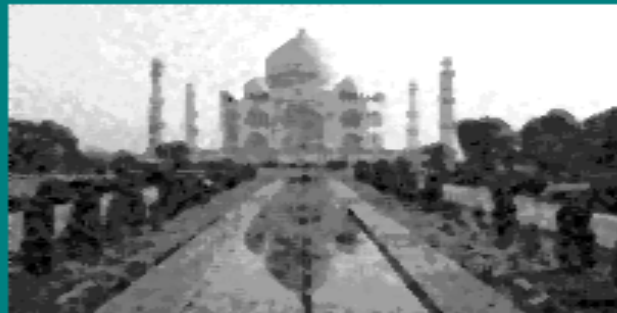
1366

0



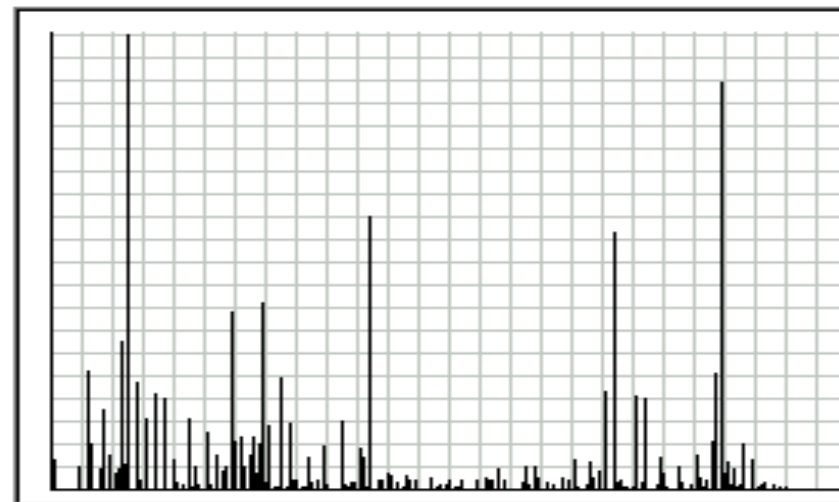
255

Fig. 4.12(a) Original Image



2509

0



255

Fig 4.12(b) Median Filtered Image



例子：使用中值滤波进行降噪

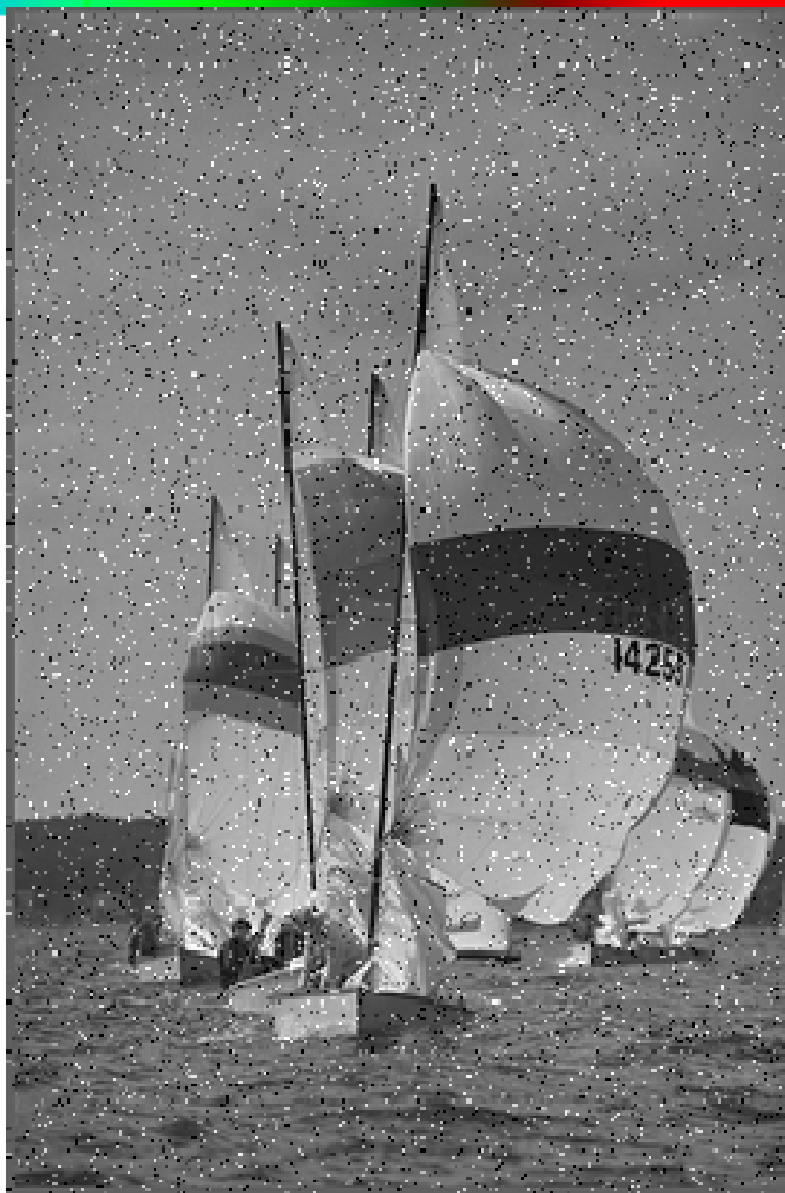
原始图像





例子：使用中值滤波进行降噪

椒盐噪声图像





例子：使用中值滤波进行降噪

中值 3×3
平滑化图像

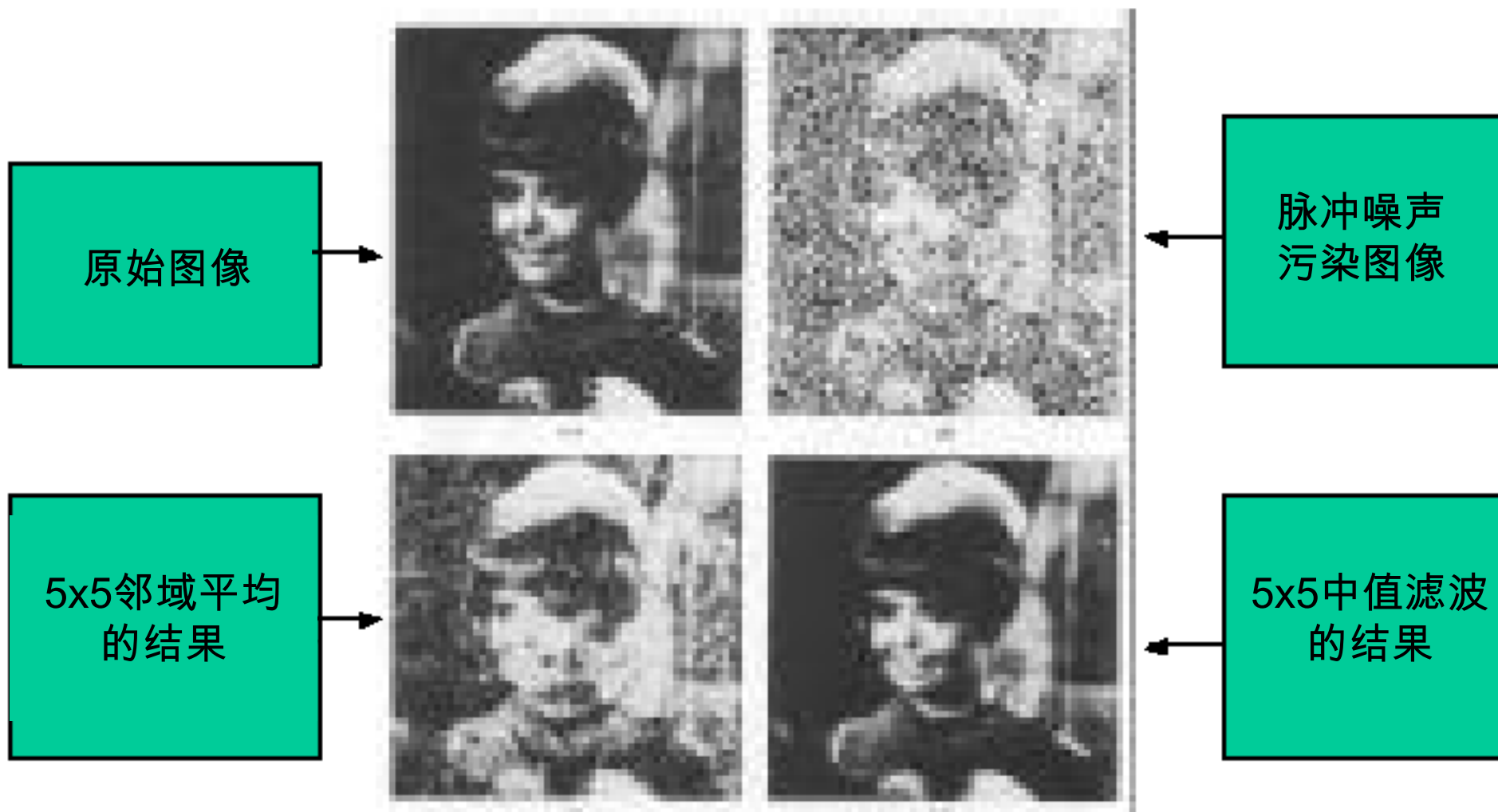




例子：使用中值滤波进行降噪

中值 5×5
平滑化图像







目录

- 强度转换
 - 图像负片
 - 对数转换
 - 伽马变换
 - 碎片化线性变换
- 直方图处理
 - 直方图均衡化
 - 直方图匹配
 - 局部直方图处理
- 空间滤波器
 - 空间相关性和卷积
 - 平滑线性滤波器和空间掩码
 - 非线性滤波器
 - 锐化滤波器



作业

Project3:直方图图像增强;

共10幅经变亮或者变暗处理的源图像;

要求:

- 1.把附件图像的直方图画出;
- 2.把所有图像进行直方图均衡; 输出均衡后的图像和源图像进行比对; 分析改善内容;
- 3.进一步把图像按照对源图像直方图的观察, 各自指定不同源图像的直方图, 进行直方图匹配, 进行图像增强;
- 4.利用直方图对图像elain和woman进行分割;

要求: 按照作业报告格式提交报告, 下一周三提交;

另附原始图像4幅:



Project 4

- 空域低通滤波器：分别用高斯滤波器和中值滤波器去平滑测试图像test1和2，模板大小分别是 3×3 ， 5×5 ， 7×7 ；
 - 利用固定方差 $\sigma=1.5$ 产生高斯滤波器. 附件有产生高斯滤波器的方法；
- 利用高通滤波器滤波测试图像test3,4：包括 unsharp masking, Sobel edge detector, and Laplace edge detection;
- 按标准格式提交报告；